Physics in Topological Insulator Film

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Abstract

Strong topological insulator (STI) is a unique topological phase in three spatial dimensions, which hosts single Dirac cone on each surface. Exploiting film geometry would greatly influence the behaviour of both surface and bulk states. Quantum spin Hall effect (QSHE) would emerge with thin-film limit due to finite-size gap, while a novel half quantum mirror Hall effect (HQMHE) as an analogy to 2-D QSHE but with half-quantized nature emerges when the gap is suppressed by making film thick. Combining the film with other elements like magnetic doping and heterostructure with other platforms would lead to numerous phenomena, like quantum anomalous Hall effect (QAHE) aka Chern insulator (CI), together with half-quantized version, so-called axion insulator (AI) and topological superconductor (TSC). When magnetic field directly steps in, quantum Hall effect (QHE) by Landau levels (LL) would reveal. What is more, if we adopt the synergistic effect generated by two approaches above, we may have a chance to realize high Chern number in real materials.

Besides the novel phenomena mentioned, STI film is a natural platform to realize stable 'quantum anomaly' in condensed matter, as it offers naturally a $2+\varepsilon$ geometry. And the anomaly contribution reveals itself mainly in HQMHE $(1/2-1/2)$, half-quantized QAHE $(1/2+0)$ aka axion metal (AM), and a novel QAHE but without chiral edge state (CES) $(1/2+1/2)$. Investigating into those novel phenomena would greatly enhance our understanding of how an ultraviolet (UV) complete theory would exist and impact the physical world non-trivially.

In this review we report on these phenomena within a unified frame by exact solution of STI film.

Chapter 0

Three Theorems

0.1 Fermion Doubling Theorem

Fermion Doubling theorem is a generalization of Nielsen-Ninomiya theorem[39], which states that for a quadratic Hamiltonian on lattice

$$\mathscr{H} = \sum_{m,n} \Psi_m^\dagger H(\boldsymbol{r}_m, \boldsymbol{r}_n) \Psi_n, \tag{0.1}$$

there is an equal number of chiral/parity symmetric fermions with opposite handedness in odd/even spatial dimensions if we require simultaneous standing of general statements below:

• Translational invariance: $H(\boldsymbol{r}_m, \boldsymbol{r}_n) = H(\boldsymbol{r}_m - \boldsymbol{r}_n)$;
• Locality: $H(\boldsymbol{r}_m, \boldsymbol{r}_n)$ must vanish fast enough to have a Fourier transform with continuous derivatives;
• Hermiticity: $H(\boldsymbol{r}_m, \boldsymbol{r}_n)$ is Hermitian;
• Local charge quantization and conservation (gauge invariance).

Due to this no-go theorem, we generally meet with either paired massless fermions like in graphene and Weyl semimetal, or we admit completeness of lattice system (especially condensed matter system) and introduce high-energy regulator that breaks chiral/parity symmetry explicitly like in boundary states of robust topological phases.

0.2 Quantization Theorem

Quantization theorem is a generalized version of Thouless-Kohmoto-Nightingale-den Nijs theorem[55] (see C.2). And it states that nth winding/Chern number in $(2n-1)/(2n)$ spatial dimensions is quantized into integers for a band insulator, where sublattice symmetry is required for odd case.

0.3 Half-Quantization Theorem

Half-quantization theorem is a stabilized version of strict-dimension quantum anomalous semimetals[15], and is a generalized version of parity invariant case[69]. This theorem states that partial integral with the same form as nth winding/Chern number

in $(2n-1+\epsilon)/(2n+\epsilon)$ spatial dimensions is quantized into half-integers for a metal, where sublattice symmetry is required for odd case. Partial here means integrated charts bounded by chemical potential

• stay away from low energy/Dirac point; • stop inside chiral/parity invariant regime for odd/even spatial dimensions.

And the introduction of ϵ naturally leads to film geometry as a stabilizer of chiral/parity invariant regime, inside which corresponding symmetry is preserved as a local one in parameter-space.

Chapter 1

Exact Solution

In this chapter we present an exact solution of 3-D topological insulator (TI) film. Whose Hamiltonian with nearest-neighbour hopping on cubic lattice is[19, 60, 24]

$$\mathscr{H}_{TI} = \sum_i \Psi_i^\dagger \mathscr{M}_0 \Psi_i + \sum_{i,\mu} \left(\Psi_i^\dagger \mathscr{T}_\mu \Psi_{i+\mu} + \text{h.c.} \right), \tag{1.1}$$

where energy and hopping matrices are $\mathscr{M}_0 = (m_0 - 2\sum_\mu t_\mu)\beta$, $\mathscr{T}_\mu = t_\mu \beta - i\frac{\lambda_\mu}{2}\alpha_\mu$, with i and μ denoting site locations and three spatial directions while $\{\beta, \alpha_\mu\}$ denoting Dirac matrices under standard Dirac representation

$$\begin{cases} \beta = \sigma_0 \otimes \tau_z \\ \alpha_\mu = \sigma_\mu \otimes \tau_x \end{cases},$$

where Pauli matrices σ_μ and τ_μ represent different degrees of freedom (DoF), respectively. For instance, one could choose they to represent spin and pseudo-spin (orbital, valley, $\cdots$) DoFs. Notice that when adopting a full Fourier transformation upon all three spatial dimensions, i.e., an infinite bulk system, the Hamiltonian is transformed into the standard modified Dirac's equation[52] on lattice $\mathscr{H}_{TI} = \sum_{\boldsymbol{k}} \Psi_{\boldsymbol{k}}^\dagger H(\boldsymbol{k}) \Psi_{\boldsymbol{k}}$ where

$$H(\boldsymbol{k}) = \sum_\mu \lambda_\mu \sin(k_\mu a_\mu)\alpha_\mu + \left[m_0 - 4t_\mu \sin^2\left(\frac{k_\mu a_\mu}{2}\right) \right] \beta, \tag{1.2}$$

whose continuum model is written into an elegant way as $H \sim \lambda k \cdot \alpha + (m_0 - tk^2)\beta$.

Consider such a thin film with limited number of sites along z direction, by Fourier transformation in $x - y$ plane, one has

$$\begin{aligned} \mathscr{H}_{\text{Film}} = &\sum_{i_z,\boldsymbol{k}} \left(\Psi_{i_z,\boldsymbol{k}}^\dagger \mathscr{M}_0(\boldsymbol{k}) \Psi_{i_z,\boldsymbol{k}} + \Psi_{i_z,\boldsymbol{k}}^\dagger \mathscr{T}_z \Psi_{i_z+1,\boldsymbol{k}} + \Psi_{i_z+1,\boldsymbol{k}}^\dagger \mathscr{T}_z^\dagger \Psi_{i_z,\boldsymbol{k}} \right) \\ &+ \sum_{i_z,\boldsymbol{k}} \Psi_{i_z,\boldsymbol{k}}^\dagger H_\parallel \Psi_{i_z,\boldsymbol{k}} \end{aligned} \tag{1.3}$$

with

$$H_\parallel = \lambda_\parallel [\sin(k_x a)\sigma_x \otimes \tau_x + \sin(k_y b)\sigma_y \otimes \tau_x],$$

and $\mathscr{M}_0(\boldsymbol{k}) = M_0(\boldsymbol{k})\sigma_0 \otimes \tau_z = [m_0(\boldsymbol{k}) - 2t_\perp]\sigma_0 \otimes \tau_z$, where

$$m_0(\boldsymbol{k}) = m_0 - 4t_\parallel \left(\sin^2 \frac{k_x a}{2} + \sin^2 \frac{k_y b}{2} \right).$$

Note that we have set $t_x = t_y = t_\parallel$, $t_z = t_\perp$, $\lambda_x = \lambda_y = \lambda_\parallel$, $\lambda_z = \lambda_\perp, a = b$.

Now, we would like to find an effective model grasping the physical nature of up and down surface states. Notice that we not only want to show the surface Dirac cone which is usually regraded as a signature of 3-D TI[19, 24, 60, 62, 23], but explore more extensive area in plane k space where significant high-energy physics emerges, and to this concern we may separate Hamiltonian at $\boldsymbol{k}$ as

$$\mathscr{H}_{\text{Film}}(\boldsymbol{k}) = \mathscr{H}_{1d}(\boldsymbol{k}) + \mathscr{H}_S(\boldsymbol{k})$$

with

$$\begin{aligned} \mathscr{H}_{1d}(\boldsymbol{k}) &= \sum_{i_z} \left(\Psi^\dagger_{i_z,\boldsymbol{k}} \mathscr{M}_0(\boldsymbol{k}) \Psi_{i_z,\boldsymbol{k}} + \Psi^\dagger_{i_z,\boldsymbol{k}} \mathscr{T}_z \Psi_{i_z+1,\boldsymbol{k}} + \text{h.c.} \right), \\ \mathscr{H}_S(\boldsymbol{k}) &= \sum_{i_z} \Psi^\dagger_{i_z,\boldsymbol{k}} H_\parallel \Psi_{i_z,\boldsymbol{k}}. \end{aligned} \tag{1.4}$$

Clearly, $\mathscr{M}_0(\boldsymbol{k})$ in $\mathscr{H}_{1d}(\boldsymbol{k})$ would be the main element dominating whether it has zero modes solution, under suitable boundary condition. In the numerical simulation, we set the model parameters as $\lambda_\parallel = 0.41$ eV, $\lambda_\perp = 0.44$ eV, $t_\parallel = 0.566$ eV, $t_\perp = 0.4$ eV, $m_0 = 0.28$ eV, $a = b = 1$ nm, $c = 0.5$ nm and $L_z = 19$ if with no specific indication[62].

1.1 Continuum Model

Firstly we deal with the problem in a simpler way under continuous limit. The procedure here mainly follows [68, 32, 51]. In such limit the Hamiltonian for one 3-D TI could be written as

$$\begin{aligned} H_{TI}(\boldsymbol{k}, k_z) &= \lambda_\parallel (\boldsymbol{k} \cdot \boldsymbol{\sigma}) \otimes \tau_x + \lambda_\perp k_z \sigma_z \otimes \tau_x + (m_0(\boldsymbol{k}) - t_\perp k_z^2) \sigma_0 \otimes \tau_z \\ &= H_{1d}(\boldsymbol{k}, k_z) + H_\parallel(\boldsymbol{k}), \end{aligned} \tag{1.5}$$

where $H_\parallel(\boldsymbol{k}) = \lambda_\parallel (\boldsymbol{k} \cdot \boldsymbol{\sigma}) \otimes \tau_x$ and by substituting $k_z \longmapsto -i\partial_z$,

$$\begin{aligned} H_{1d}(\boldsymbol{k}, z) &= -i\lambda_\perp \partial_z \sigma_z \otimes \tau_x + (m_0(\boldsymbol{k}) + t_\perp \partial_z^2) \sigma_0 \otimes \tau_z \\ &= \begin{pmatrix} h(s) & \\ & h(-s) \end{pmatrix}, \end{aligned} \tag{1.6}$$

inside which two sub-blocks

$$h(s) = -is\lambda_\perp \partial_z \tau_x + (m_0(\boldsymbol{k}) + t_\perp \partial_z^2)\tau_z = \begin{pmatrix} m_0(\boldsymbol{k}) + t_\perp \partial_z^2 & -is\lambda_\perp \partial_z \\ -is\lambda_\perp \partial_z & -m_0(\boldsymbol{k}) - t_\perp \partial_z^2 \end{pmatrix}, \tag{1.7}$$

with s defined by eigenvalue of σ_z. All parameters are real with $m_0(\boldsymbol{k}) = m_0 - t_\parallel k^2 > 0$ as the criterion for the region where surface states emerge. For the purpose of keeping consistence with lattice model above, one in fact needs to substitute parameters as

$$\lambda_\perp \to c\lambda_\perp, \ \lambda_\parallel \to a\lambda_\parallel, \ t_\perp \to c^2 t_\perp, t_\parallel \to a^2 t_\parallel.$$

However, we would not write in that way explicitly for simplicity. Also, to make discussion pithy, we shall omit s in wavefunction below.

Now we solve the eigen-problem of $h(s)$, which turns out to be a second-order differential equation and allows us to set solutions with trial function $\phi = \phi_\xi e^{i\xi z}$. Using $\partial_z \phi = i\xi\phi$, $\partial_z^2 \phi = -\xi^2 \phi$, one has equation below:

$$\begin{pmatrix} m_0(\boldsymbol{k}) - t_\perp \xi^2 & s\lambda_\perp \xi \\ s\lambda_\perp \xi & -m_0(\boldsymbol{k}) + t_\perp \xi^2 \end{pmatrix} \phi = E\phi, \tag{1.8}$$

which readily leads to

$$E^2 - (m_0(\boldsymbol{k}) - t_\perp \xi^2)^2 - \lambda_\perp^2 \xi^2 = 0, \tag{1.9}$$

and gives

$$\xi_\alpha^p = p\xi_\alpha = p\sqrt{-\frac{F}{D} + (-1)^{\alpha-1}\frac{\sqrt{R}}{D}},\ p = \pm,\ \alpha = 1, 2, \tag{1.10}$$

where

$$D = 2t_\perp^2,\ F = -2m_0(\boldsymbol{k})t_\perp + s\lambda_\perp^2,\ R = F^2 - 2D(m_0^2(\boldsymbol{k}) - E^2).$$

For each ξ_α^s, one has

$$\phi_{\alpha p} = \begin{pmatrix} s\lambda_\perp p\xi_\alpha \\ E - m_0(\boldsymbol{k}) + t_\perp \xi_\alpha^2 \end{pmatrix}, \tag{1.11}$$

and the general solution would be

$$\Phi = \sum_{\alpha p} C_{\alpha p}\phi_{\alpha p} e^{ip\xi_\alpha z}. \tag{1.12}$$

Now considering finite size along z direction with top and bottom surfaces located at $\pm\frac{L}{2}$, respectively, one would have boundary condition

$$\Phi(\pm\frac{L}{2}) = 0, \tag{1.13}$$

applying which one would get four linear equations for coefficients

$$P(C_{1+}, C_{1-}, C_{2+}, C_{2-})^T = 0, \tag{1.14}$$

which leads to equation below for non-vanishing Cs by defining $m_\alpha = E - m_0(\boldsymbol{k}) + t_\perp \xi_\alpha^2$:

$$\begin{aligned}
0 &= \det(P) \\
&= \det \begin{pmatrix} s\lambda_\perp \xi_1 e^{i\xi_1 L/2} & -s\lambda_\perp \xi_1 e^{-i\xi_1 L/2} & s\lambda_\perp \xi_2 e^{i\xi_2 L/2} & -s\lambda_\perp \xi_2 e^{-i\xi_2 L/2} \\ s\lambda_\perp \xi_1 e^{-i\xi_1 L/2} & -s\lambda_\perp \xi_1 e^{i\xi_1 L/2} & s\lambda_\perp \xi_2 e^{-i\xi_2 L/2} & -s\lambda_\perp \xi_2 e^{i\xi_2 L/2} \\ m_1 e^{i\xi_1 L/2} & m_1 e^{-i\xi_1 L/2} & m_2 e^{i\xi_2 L/2} & m_2 e^{-i\xi_2 L/2} \\ m_1 e^{-i\xi_1 L/2} & m_1 e^{i\xi_1 L/2} & m_2 e^{-i\xi_2 L/2} & m_2 e^{i\xi_2 L/2} \end{pmatrix} \\
&= -s\lambda_\perp^2 \xi_1^2 e^{i\xi_1 L} m_2^2 (e^{i\xi_2 L} - e^{-i\xi_2 L}) - s\lambda_\perp^2 \xi_1 \xi_2 e^{i(\xi_1 - \xi_2)L/2} m_1 m_2 [e^{-i(\xi_1 - \xi_2)L/2} - e^{i(\xi_1 - \xi_2)L/2}] \\
&\quad + [-\lambda_\perp^2 \xi_1 \xi_2 e^{i(\xi_1 + \xi_2)L/2}] m_1 m_2 [e^{-i(\xi_1 + \xi_2)L/2} - e^{i(\xi_1 + \xi_2)L/2}] \\
&\quad - \big\{ -s\lambda_\perp^2 \xi_1^2 e^{-i\xi_1 L} m_2^2 (e^{i\xi_2 L} - e^{-i\xi_2 L}) - s\lambda_\perp^2 \xi_1 \xi_2 e^{i(\xi_1 - \xi_2)L/2} m_1 m_2 [e^{-i(\xi_1 - \xi_2)L/2} - e^{i(\xi_1 - \xi_2)L/2}] \\
&\quad - [-\lambda_\perp^2 \xi_1 \xi_2 e^{-i(\xi_1 + \xi_2)L/2}] m_1 m_2 [e^{-i(\xi_1 + \xi_2)L/2} - e^{i(\xi_1 + \xi_2)L/2}] \big\} + (1 \leftrightarrow 2) \\
&= 4\lambda_\perp^2 \xi_1^2 m_2^2 \sin(\xi_1 L) \sin(\xi_2 L) - 4\lambda_\perp^2 \xi_1 \xi_2 m_1 m_2 \sin^2((\xi_1 - \xi_2)L/2) \\
&\quad - 4\lambda_\perp^2 \xi_1 \xi_2 m_1 m_2 \sin^2((\xi_1 + \xi_2)L/2) + (1 \leftrightarrow 2) \\
&= 4\lambda_\perp^2 \xi_1^2 m_2^2 4 \sin(\xi_1 L/2) \cos(\xi_1 L/2) \sin(\xi_2 L/2) \cos(\xi_2 L/2) \\
&\quad - 4\lambda_\perp^2 \xi_1 \xi_2 m_1 m_2 4 \sin^2(\xi_1 L/2) \cos^2(\xi_2 L/2) + (1 \leftrightarrow 2) \\
&= 4\lambda_\perp^2 \xi_1 \xi_2 m_1 m_2 4 \sin(\xi_1 L/2) \cos(\xi_1 L/2) \sin(\xi_2 L/2) \cos(\xi_2 L/2) \cdot \\
&\quad \left\{ \frac{\xi_1 m_2}{\xi_2 m_1} + \frac{\xi_2 m_1}{\xi_1 m_2} - \frac{\tan(\xi_1 L/2)}{\tan(\xi_2 L/2)} - \frac{\tan(\xi_2 L/2)}{\tan(\xi_1 L/2)} \right\},
\end{aligned}$$

which equals solving

$$a + \frac{1}{a} = b + \frac{1}{b},\ \text{i.e.,}\ a^2 + (b + \frac{1}{b})a + 1 = 0,$$

from which one gets two transcendental equations

$$\frac{m_1\xi_2}{m_2\xi_1} = \frac{\tan\xi_2 L/2}{\tan\xi_1 L/2} \tag{1.15a}$$

$$\frac{m_1\xi_2}{m_2\xi_1} = \frac{\tan\xi_1 L/2}{\tan\xi_2 L/2} \tag{1.15b}$$

which gives two energies varying with $\boldsymbol{k}$, designated as E_+ and E_-, respectively. To be clearer,

$$E_+ = m_0(\boldsymbol{k}) - t_\perp \frac{\xi_1^2 g^+(\xi_1) - \xi_2^2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)},\ g^+(\xi) = \frac{\tan(\xi L/2)}{\xi}, \tag{1.16a}$$

$$E_- = m_0(\boldsymbol{k}) - t_\perp \frac{\xi_1^2 g^-(\xi_1) - \xi_2^2 g^-(\xi_2)}{g^-(\xi_1) - g^-(\xi_2)},\ g^-(\xi) = \frac{1}{\tan(\xi L/2)\xi}. \tag{1.16b}$$

In common sense, it is time taking $E_\pm$ into expressions of ξs, together with the coefficients equations again and solve them. However, that not only is tricky but lacks of physical insight, and we shall change our perspective.

Notice that under parity operation $z \leftrightarrow -z$, $\tau_x \leftrightarrow -\tau_x$ and $h(s) \leftrightarrow h(s)$, then both $h(s)$ and H_{1d} has parity symmetry and the general solution should contain two factors below considering the boundary condition:

$$\begin{cases} f_+(z) = \dfrac{\cos(\xi_1 z)}{\cos(\xi_1 L/2)} - \dfrac{\cos(\xi_2 z)}{\cos(\xi_2 L/2)} \\ f_-(z) = \dfrac{\sin(\xi_1 z)}{\sin(\xi_1 L/2)} - \dfrac{\sin(\xi_2 z)}{\sin(\xi_2 L/2)} \end{cases}, \tag{1.17}$$

where the subscripts refer to even or odd parity. Now we could assume that for energy E, $h(s)$ has solution

$$\phi = \tilde{c} f_+ + \tilde{d} f_- = \begin{pmatrix} \tilde{c}_1 f_+ + \tilde{d}_1 f_- \\ \tilde{c}_2 f_+ + \tilde{d}_2 f_- \end{pmatrix}. \tag{1.18}$$

By simply writing $c_{1/2} = \dfrac{\cos\big(\xi_{1/2} z\big)}{\cos\big(\xi_{1/2} L/2\big)}$, $s_{1/2} = \dfrac{\sin\big(\xi_{1/2} z\big)}{\sin\big(\xi_{1/2} L/2\big)}$ and using

$$\begin{cases} \partial_z f_+ = -\xi_1 \dfrac{\sin(\xi_1 z)}{\cos(\xi_1 L/2)} + \xi_2 \dfrac{\sin(\xi_2 z)}{\cos(\xi_2 L/2)} = -\xi_1 s_1 \tan(\xi_1 L/2) + \xi_2 s_2 \tan(\xi_2 L/2) \\ \partial_z f_- = \xi_1 \dfrac{\cos(\xi_1 z)}{\sin(\xi_1 L/2)} - \xi_2 \dfrac{\cos(\xi_2 z)}{\sin(\xi_2 L/2)} = \xi_1 c_1 \cot(\xi_1 L/2) - \xi_2 c_2 \cot(\xi_2 L/2) \\ \partial_z^2 f_+ = -\xi_1^2 c_1 + \xi_2^2 c_2,\ \partial_z^2 f_- = -\xi_1^2 s_1 + \xi_2^2 s_2 \end{cases},$$

the stationary Schrödinger equation $h(s)\phi = E\phi$ would give, for the first line:

$$\begin{aligned} 0 &= \tilde{c}_1[(m_0(\boldsymbol{k}) - E) f_+ + t_\perp(-\xi_1^2 c_1 + \xi_2^2 c_2)] + \tilde{d}_1[(m_0(\boldsymbol{k}) - E) f_- + t_\perp(-\xi_1^2 s_1 + \xi_2^2 s_2)] \\ &\quad - i\tilde{c}_2 \lambda_\perp(-\xi_1 s_1 \tan(\xi_1 L/2) + \xi_2 s_2 \tan(\xi_2 L/2)) \\ &\quad - i\tilde{d}_2 \lambda_\perp(\xi_1 c_1 \cot(\xi_1 L/2) - \xi_2 c_2 \cot(\xi_2 L/2)) \\ &= c_1[-(-m_0(\boldsymbol{k}) + E + t_\perp \xi_1^2)\tilde{c}_1 - i\tilde{d}_2 \lambda_\perp \xi_1 \cot(\xi_1 L/2)] \\ &\quad + c_2[(-m_0(\boldsymbol{k}) + E + t_\perp \xi_2^2)\tilde{c}_1 + i\tilde{d}_2 \lambda_\perp \xi_2 \cot(\xi_2 L/2)] \\ &\quad + s_1[-(m_0(\boldsymbol{k}) + E + t_\perp \xi_1^2)\tilde{d}_1 + i\tilde{c}_2 \lambda_\perp \xi_1 \tan(\xi_1 L/2)] \\ &\quad + s_2[(-m_0(\boldsymbol{k}) + E + t_\perp \xi_2^2)\tilde{d}_1 - i\tilde{c}_2 \lambda_\perp \xi_2 \tan(\xi_2 L/2)], \end{aligned}$$

which holds for any z and thus readily gives two pairs of equations for coefficients:

$$\begin{cases} -(-m_0(\boldsymbol{k}) + E + t_\perp \xi_1^2)\tilde{c}_1 - i\tilde{d}_2 \lambda_\perp \xi_1 \cot(\xi_1 L/2) = 0 \\ +(-m_0(\boldsymbol{k}) + E + t_\perp \xi_2^2)\tilde{c}_1 + i\tilde{d}_2 \lambda_\perp \xi_2 \cot(\xi_2 L/2) = 0 \end{cases}, \tag{1.19a}$$

$$\begin{cases} -(-m_0(\boldsymbol{k}) + E + t_\perp \xi_1^2)\tilde{d}_1 + i\tilde{c}_2 \lambda_\perp \xi_1 \tan(\xi_1 L/2) = 0 \\ +(-m_0(\boldsymbol{k}) + E + t_\perp \xi_2^2)\tilde{d}_1 - i\tilde{c}_2 \lambda_\perp \xi_2 \tan(\xi_2 L/2) = 0 \end{cases}. \tag{1.19b}$$

Similarly, the second line equation would yield four equations for vanishing coefficients of $c_{1/2}$ and $s_{1/2}$

$$\begin{cases} +(-m_0(\boldsymbol{k}) - E + t_\perp \xi_1^2)\tilde{c}_2 - i\tilde{d}_1 \lambda_\perp \xi_1 \cot(\xi_1 L/2) = 0 \\ -(-m_0(\boldsymbol{k}) - E - t_\perp \xi_2^2)\tilde{c}_2 + i\tilde{d}_1 \lambda_\perp \xi_2 \cot(\xi_2 L/2) = 0 \end{cases}, \tag{1.20a}$$

$$\begin{cases} +(-m_0(\boldsymbol{k}) - E + t_\perp \xi_1^2)\tilde{d}_2 + i\tilde{c}_1 \lambda_\perp \xi_1 \tan(\xi_1 L/2) = 0 \\ -(-m_0(\boldsymbol{k}) - E + t_\perp \xi_2^2)\tilde{d}_2 - i\tilde{c}_1 \lambda_\perp \xi_2 \tan(\xi_2 L/2) = 0 \end{cases}. \tag{1.20b}$$

Notice that Eq. (1.19)(a) and Eq. (1.20)(b) should stand simultaneously, since they describe the same set of coefficients within two lines of one equation. The same is also true for Eq. (1.19)(b) and Eq. (1.20)(a).

Now for Eq. (1.19), we add up two equations in each pair to eliminate E and get

$$-t_\perp(\xi_1^2 - \xi_2^2)\tilde{c}_1 - i\tilde{d}_2 \lambda_\perp [\xi_1 \cot(\xi_1 L/2) - \xi_2 \cot(\xi_2 L/2)] = 0, \tag{1.21a}$$

$$-t_\perp(\xi_1^2 - \xi_2^2)\tilde{d}_1 + i\tilde{c}_2 \lambda_\perp [\xi_1 \tan(\xi_1 L/2) - \xi_2 \tan(\xi_2 L/2)] = 0. \tag{1.21b}$$

Then by defining two coefficients

$$\eta_1 = \frac{\xi_1^2 - \xi_2^2}{\xi_1 \cot(\xi_1 L/2) - \xi_2 \cot(\xi_2 L/2)}, \tag{1.22a}$$

$$\eta_2 = \frac{\xi_1^2 - \xi_2^2}{\xi_1 \tan(\xi_1 L/2) - \xi_2 \tan(\xi_2 L/2)}, \tag{1.22b}$$

one acquires

$$\tilde{d}_2 = i t_\perp \eta_1 \tilde{c}_1 / \lambda_\perp, \tag{1.23a}$$

$$\tilde{c}_2 = -i t_\perp \eta_2 \tilde{d}_1 / \lambda_\perp. \tag{1.23b}$$

At this time, if one brings this result back into pair functions, one would get that for Eq. (1.19)(a)

$$\begin{aligned} E &= m_0(\boldsymbol{k}) - t_\perp \xi_1^2 + t_\perp \eta_1 \xi_1 \cot(\xi_1 L/2) = m_0(\boldsymbol{k}) - t_\perp \xi_1 \xi_2 \frac{\xi_2 \cot(\xi_1 L/2) - \xi_1 \cot(\xi_2 L/2)}{\xi_1 \cot(\xi_1 L/2) - \xi_2 \cot(\xi_2 L/2)} \\ &= m_0(\boldsymbol{k}) - t_\perp \frac{\xi_2^2 \tan(\xi_2 L/2)/\xi_2 - \xi_1^2 \tan(\xi_1 L/2)/\xi_1}{\tan(\xi_2 L/2)/\xi_2 - \tan(\xi_1 L/2)/\xi_1} \\ &= m_0(\boldsymbol{k}) - t_\perp \frac{\xi_1^2 g^+(\xi_1) - \xi_2^2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)} = E_+, \end{aligned}$$

and for Eq. (1.19)(b)

$$E = m_0(\boldsymbol{k}) - t_\perp \xi_1^2 + t_\perp \eta_2 \xi_1 \tan(\xi_1 L/2) = m_0(\boldsymbol{k}) - t_\perp \frac{\xi_1^2 g^-(\xi_1) - \xi_2^2 g^-(\xi_2)}{g^-(\xi_1) - g^-(\xi_2)} = E_-,$$

with corresponding eigenfunctions

$$\phi_1^+ = C_1^+ \begin{pmatrix} -i\lambda_\perp f_+ \\ t_\perp \eta_1 f_- \end{pmatrix}, \ E = E_+, \tag{1.24a}$$

$$\phi_1^- = C_1^- \begin{pmatrix} i\lambda_\perp f_- \\ t_\perp \eta_2 f_+ \end{pmatrix}, \ E = E_-, \tag{1.24b}$$

where C is the norm, and super and lower indices represent $E_\pm$ and line index, respectively.

Similarly, for Eq. (1.20), we add up two equations in each pair and get

$$\tilde{d}_1 = -it_\perp \eta_1 \tilde{c}_2/\lambda_\perp, \tag{1.25a}$$

$$\tilde{c}_1 = it_\perp \eta_2 \tilde{d}_2/\lambda_\perp, \tag{1.25b}$$

substituting which into pair equations would give

$$E = -m_0(\boldsymbol{k}) + t_\perp \frac{\xi_1^2 g^+(\xi_1) - \xi_2^2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)} = -E_+, \tag{1.26a}$$

$$E = -m_0(\boldsymbol{k}) + t_\perp \frac{\xi_1^2 g^-(\xi_1) - \xi_2^2 g^-(\xi_2)}{g^-(\xi_1) - g^-(\xi_2)} = -E_-, \tag{1.26b}$$

with corresponding eigenfunctions

$$\phi_2^+ = C_2^+ \begin{pmatrix} t_\perp \eta_1 f_- \\ i\lambda_\perp f_+ \end{pmatrix}, \ E = -E_+ \tag{1.27a}$$

$$\phi_2^- = C_2^- \begin{pmatrix} -t_\perp \eta_2 f_+ \\ i\lambda_\perp f_- \end{pmatrix}, \ E = -E_- \tag{1.27b}$$

and clearly, one has $C_1^\iota = C_2^\iota$ and $\phi_1^\iota = i\tau_y \phi_2^\iota$ as they are chiral partners ($\iota = \pm$).

From the above discussion we seemingly have four solutions, mathematical restriction, however, tells that equations from different lines for the same set of coefficients must stand simultaneously, which gives us two relations as

$$1 = \left| \frac{it_\perp \eta_1}{s\lambda_\perp} \cdot \frac{it_\perp \eta_2}{s\lambda_\perp} \right| \Longrightarrow |\eta_1 \eta_2| = \frac{\lambda_\perp^2}{t_\perp^2}, \tag{1.28a}$$

$$E_+ = -E_-, \tag{1.28b}$$

and the latter is also a physical result from Dirac equation. Then, we only have two independent solutions for one $h(s)$ sub-block, say Eq. (1.24)(a) and Eq. (1.27)(a). As stated above, Eq. (1.19)(a) and Eq. (1.20)(b) should be recognized to give the same solution, then combining their two first equations gives

$$\begin{cases} -(-m_0(\boldsymbol{k}) + E + t_\perp \xi_1^2)\tilde{c}_1 - i\tilde{d}_2 \lambda_\perp \xi_1 \cot(\xi_1 L/2) = 0 \\ +(-m_0(\boldsymbol{k}) - E + t_\perp \xi_1^2)\tilde{d}_2 + i\tilde{c}_1 \lambda_\perp \xi_1 \tan(\xi_1 L/2) = 0 \end{cases}, \tag{1.29}$$

which again leads to

$$E^2 - (m_0(\boldsymbol{k}) - t_\perp \xi_1^2)^2 - s\lambda_\perp^2 \xi_1^2 = 0, \tag{1.30}$$

and the same could be done for ξ_2. Then we see that the guessing solution not only satisfies the boundary condition, but also satisfies all $E - \xi$ equations, thus it is indeed our solution.

Furthermore, we may argue for some property of factors defined above. Firstly we notice that by Eq (1.10), ξ_α are both complex or not complex for a given energy, where complex means both real and imaginary parts of ξ are non-vanishing, determined by the sign of R. When it is not the case, we first claim that the combination $\xi\tan\xi$ and $\xi\cot\xi$ are always real, and the only staff needed proof here is when ξ is imaginary, $\tan\xi$ is imaginary, which is true since

$$\begin{cases} \sin\xi = \sin(i\,\mathrm{Im}\,\xi) = \dfrac{e^{-\mathrm{Im}\,\xi} - e^{\mathrm{Im}\,\xi}}{2i} \text{ is imaginary} \\ \cos\xi = \cos(i\,\mathrm{Im}\,\xi) = \dfrac{e^{-\mathrm{Im}\,\xi} + e^{\mathrm{Im}\,\xi}}{2} \text{ is real} \end{cases},$$

and $\tan\xi = \sin\xi/\cos\xi$ would be imaginary. Under such consideration, f and η above are real without doubt. On the other hand, when it is the case that ξ_α are complex, one could see $\xi_2 = \xi_1^*$, and we may let $\xi_1 = x + iy$. Then one would have

$$\begin{aligned} \cos(\xi_1 t) &= \frac{e^{i\xi_1 t} + e^{-i\xi_1 t}}{2} = \frac{e^{ixt}e^{-yt} + e^{-ixt}e^{yt}}{2} = \cos(xt)\cosh(yt) - i\sin(xt)\sinh(yt) \\ \cos(\xi_2 t) &= \frac{e^{i\xi_2 t} + e^{-i\xi_2 t}}{2} = \frac{e^{ixt}e^{yt} + e^{-ixt}e^{-yt}}{2} = \cos(xt)\cosh(yt) + i\sin(xt)\sinh(yt), \end{aligned}$$

which leads to the conclusion that $f_+(z) = \dfrac{\cos(\xi_1 z)}{\cos(\xi_1 L/2)} - \dfrac{\cos(\xi_2 z)}{\cos(\xi_2 L/2)}$ would have its real part cancelled and become purely imaginary. The same argument is true for $f_-(z)$. Then any quadratic form of f would give a real factor. As for η, one notices that both numerator and denominator would only have imaginary part left, and their division would give a real coefficient. The discussion here is general and stand for the lattice model, though more cases might need to be specified.

Now, we would like to extract

$$\varphi(s) = \phi_1^{s,+},\ \chi(s) = \phi_2^{s,+}$$

as two solutions for $h(s)$ as basis construction. Then by defining

$$\begin{cases} m = E_+ = m_0(\boldsymbol{k}) - t_\perp \dfrac{\xi_1^2 g^+(\xi_1) - \xi_2^2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)}, \\ g^+(\xi) = \dfrac{\tan(\xi L/2)}{\xi}, \\ C = C_1^+ = C_2^+ \end{cases} \tag{1.31}$$

one obtains four projecting basis in certain sequence as

$$\begin{aligned} \Phi_1 &= \begin{pmatrix} \varphi(+) \\ 0 \end{pmatrix} = C \begin{pmatrix} -i\lambda_\perp f_+ \\ t_\perp \eta_1 f_- \\ 0 \\ 0 \end{pmatrix}, \Phi_2 = \begin{pmatrix} 0 \\ \chi(-) \end{pmatrix} = C \begin{pmatrix} 0 \\ 0 \\ t_\perp \eta_1 f_- \\ -i\lambda_\perp f_+ \end{pmatrix}, \\ \Phi_3 &= \begin{pmatrix} \chi(+) \\ 0 \end{pmatrix} = C \begin{pmatrix} t_\perp \eta_1 f_- \\ i\lambda_\perp f_+ \\ 0 \\ 0 \end{pmatrix}, \Phi_4 = \begin{pmatrix} 0 \\ \varphi(-) \end{pmatrix} = C \begin{pmatrix} 0 \\ 0 \\ i\lambda_\perp f_+ \\ t_\perp \eta_1 f_- \end{pmatrix}, \end{aligned} \tag{1.32}$$

with energy $(m,-m,-m,m)$, respectively. Notice that $\Phi_{3,4}$ are chiral partners of $\Phi_{1,2}$ by $i\tau_y$, respectively. To obtain m, a set of closed equations need to be solved

$$m = m_0(\boldsymbol{k}) - t_\perp \frac{\xi_1^2 g^+(\xi_1) - \xi_2^2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)}, \tag{1.33a}$$

$$\xi_\alpha = \sqrt{-\frac{F}{D} + (-1)^{\alpha-1}\frac{\sqrt{R}}{D}},\ \alpha = 1,2, \tag{1.33b}$$

where

$$\begin{cases} g^+(\xi) = \tan(\xi L/2)/\xi \\ D = 2t_\perp^2 \\ F = -2m_0(\boldsymbol{k})t_\perp + s\lambda_\perp^2 \\ R = F^2 - 2D(m_0^2(\boldsymbol{k}) - m^2) \end{cases}. \tag{1.34}$$

Basically, there are three variables (m,ξ_1,ξ_2) with three equations, then they could be determined exactly.

The projection would give effective Hamiltonian as $H_{\text{EFF}} = \langle\Phi|\, H_{TI}(\boldsymbol{k})\,|\Phi\rangle$, with

$$(H_{\text{EFF}})_{ij} = \langle\Phi_i|\, H_{TI}(\boldsymbol{k})\,|\Phi_j\rangle = |C|^2 \int \mathrm{d}z\ \Phi_i^\dagger(z) H_{TI}(\boldsymbol{k},z)\Phi_j(z), \tag{1.35}$$

where the integral is done from $-L/2$ to $L/2$. Clearly, projection on H_{1d} would give $\mathrm{diag}(m,-m,-m,m)$, then we only need to deal with $H_\parallel(\boldsymbol{k}) = \lambda_\parallel(\boldsymbol{k}\cdot\boldsymbol{\sigma})\otimes\tau_x = \lambda_\parallel(k_x\sigma_x + k_y\sigma_y)\otimes\tau_x$ term. Since $H_\parallel(\boldsymbol{k})$ is purely off-diagonal, it is easy to conclude that

$$\begin{aligned} \langle\Phi_i|H_\parallel|\Phi_i\rangle &= 0,\ i = 1,2,3,4 \\ \langle\Phi_1|H_\parallel|\Phi_3\rangle &= 0 = \langle\Phi_2|H_\parallel|\Phi_4\rangle\,. \end{aligned}$$

Then only four terms need consideration by hermicity, among which

$$\begin{aligned} \langle\Phi_1|H_\parallel|\Phi_4\rangle &= \int_{-L/2}^{L/2} \mathrm{d}z\ \Phi_1^\dagger H_\parallel \Phi_4 = \int_{-L/2}^{L/2} \mathrm{d}z\ \lambda_\parallel(k_x - ik_y)\varphi^\dagger(\lambda_\perp)\tau_x\varphi(-\lambda_\perp) \\ &= \lambda_\parallel(k_x - ik_y)\int_{-L/2}^{L/2} \mathrm{d}z\ |C|^2 i\lambda_\perp t_\perp \eta_1 (f_+^* f_- + f_-^* f_+) = 0, \\ \langle\Phi_2|H_\parallel|\Phi_3\rangle &= \int_{-L/2}^{L/2} \mathrm{d}z\ \Phi_2^\dagger H_\parallel \Phi_3 = \int_{-L/2}^{L/2} \mathrm{d}z\ \lambda_\parallel(k_x - ik_y)\chi^\dagger(\lambda_\perp)\tau_x\chi(-\lambda_\perp) \\ &= \lambda_\parallel(k_x - ik_y)\int_{-L/2}^{L/2} \mathrm{d}z\ |C|^2 i\lambda_\perp t_\perp \eta_1 (f_-^* f_+ + f_+^* f_-) = 0, \end{aligned}$$

as $f_- f_+$ is odd to z. Then, the only remaining terms are

$$\begin{aligned} \langle\Phi_1|H_\parallel|\Phi_2\rangle &= \int_{-L/2}^{L/2} \mathrm{d}z\ \lambda_\parallel(k_x - ik_y)\varphi^\dagger(\lambda_\perp)\tau_x\chi(-\lambda_\perp) \\ &= \lambda_\parallel(k_x - ik_y)|C|^2 \int_{-L/2}^{L/2} \mathrm{d}z\ (\lambda_\perp^2|f_+|^2 + t_\perp^2\eta_1^2|f_-|^2) = \lambda_\parallel(k_x - ik_y), \\ \langle\Phi_3|H_\parallel|\Phi_4\rangle &= \int_{-L/2}^{L/2} \mathrm{d}z\ \lambda_\parallel(k_x - ik_y)\varphi^\dagger(\lambda_\perp)\tau_x\chi(-\lambda_\perp) = \lambda_\parallel(k_x - ik_y), \end{aligned}$$

where the normalization condition is used. And finally we arrive at the effective four-band Hamiltonian

$$H_{\text{EFF}} = \begin{pmatrix} m & \lambda_\parallel(k_x - ik_y) & & \\ \lambda_\parallel(k_x + ik_y) & -m & & \\ & & -m & \lambda_\parallel(k_x - ik_y) \\ & & \lambda_\parallel(k_x + ik_y) & m \end{pmatrix} \tag{1.36}$$
$$= \lambda_\parallel \tau_0(\boldsymbol{k}\cdot\boldsymbol{\sigma}) + m(\boldsymbol{k})\tau_z\sigma_z.$$

1.1.1 Two Limits in Continuum Model

It is predicted that when our system is thick enough so that finite size effect could be neglected, $E(k) = m(k)$ for surface state could be written as[69]

$$\lim_{L\to+\infty} E(k) = \Theta(-m_0(k))m_0(k). \tag{1.37}$$

As shown in Fig. 1.1(a), correspondence among mass m obtained from our closed equa-

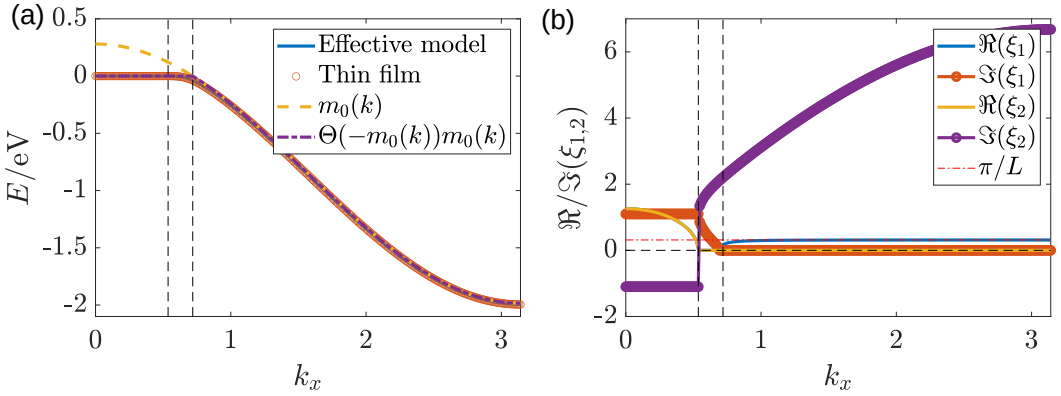


Figure 1.1: (a) m as mass term from closed equations of effective model, diagonalization of H_{1d}, and compare with guessed analytic form. (b) $\text{Re}(\xi_{1,2})$ and $\text{Im}(\xi_{1,2})$.

tions for effective model, mass from direct diagonalization of H_{1d} and the guessed analytic form reveals unambiguously.

Assuming that $m_0 > 0,\ \lambda > 0,\ t > 0$, we then define two k_cs by

$$\begin{cases} 4m_0(k_{c1})t_\perp - \lambda_\perp^2 = 0 \\ m_0(k_{c2}) = 0 \end{cases}. \tag{1.38}$$

Inside k_{c1}, $4m_0(k)t_\perp - \lambda_\perp^2 > 0$ while inside k_{c2}, $m_0(k) > 0$.

As shown in Fig. 1.1(b), when $k > k_{c2}$, $\xi_1 \approx \pi/L$, which would lead to

$$\tan\left(\xi\frac{L}{2}\right) \approx \tan\left(\frac{\pi}{2}\right) \to +\infty.$$

Then we see

$$E = m_0(k) - t_\perp \frac{\xi_1^2 g^+(\xi_1) - \xi_2^2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)}$$
$$\to m_0(k) - t_\perp\xi_1^2 \approx m_0(k) - t_\perp\frac{\pi^2}{L^2},$$

and one sees that

$$\lim_{L\to+\infty} E(k > k_{c2}) = m_0(k),$$

together with the lowest-order correction

$$E(k > k_{c2}) \approx m_0(k) - t_\perp \frac{\pi^2}{L^2}. \tag{1.39}$$

To obtain correct formula above needs to show

$$\lim_{L\to+\infty} \xi_1^2 g^+(\xi_1) = \lim_{L\to+\infty} \xi_1 \tan(\xi_1 L/2) = +\infty, \tag{1.40}$$

which is down by introducing

$$\begin{aligned}
\lim_{L\to+\infty,\ \Delta\to0^+} \frac{\pi}{L+\Delta} \tan\left(\frac{\pi}{L+\Delta}\frac{L}{2}\right) &= \lim_{L\to+\infty,\ \Delta\to0} \pi \frac{1}{\cos^2(\frac{\pi}{2}\frac{L}{L+\Delta})} \frac{\Delta}{(L+\Delta)^2} \\
&= \lim_{L\to+\infty,\ \Delta\to0^+} \pi \frac{1}{\sin^2(\frac{\pi}{2}\frac{\Delta}{L+\Delta})} \frac{\Delta}{(L+\Delta)^2} \\
&= \lim_{L\to+\infty,\ \Delta\to0^+} \frac{4}{\pi} \frac{(L+\Delta)^2}{\Delta^2} \frac{\Delta}{(L+\Delta)^2} \\
&= \lim_{L\to+\infty,\ \Delta\to0^+} \frac{4}{\pi\Delta} = +\infty.
\end{aligned}$$

On the other hand, when $k < k_{c2}$, we could firstly see from Fig. 1.1(b) that ξ maintain both real and imaginary parts, and discussion involving complex field is necessary. To be clear, we shall define the main branch as $(-\pi, \pi]$, compatible with numerical calculation. However, note that the need of fixed main branch is due to our explicit choice of form of ξ_α by the complex $\sqrt{A+\sqrt{B}}$ function. What is more, we figure out relations below from the picture that

$$\begin{cases} \xi_1 = \xi_2^*, \ \ \mathrm{Im}(\xi_1) > 0, & k < k_{c1} \\ \mathrm{Im}(\xi_1) > 0, \ \ \mathrm{Im}(\xi_2) > 0, & k_{c1} < k < k_{c2} \end{cases}, \tag{1.41}$$

then both two cases need study.

By $\beta = e^{i\xi}$, from

$$\tan(\xi L/2) = \frac{1}{i}\frac{\beta^{L/2} - \beta^{-L/2}}{\beta^{L/2} + \beta^{-L/2}},$$

while

$$\beta^t = e^{i\xi t} = e^{i\,\mathrm{Re}\,\xi t} e^{-\,\mathrm{Im}\,\xi t},$$

we say that

$$\lim_{L\to+\infty} \beta^{L/2} = \begin{cases} 0, & \mathrm{Im}(\xi) > 0 \\ +\infty, & \mathrm{Im}(\xi) < 0 \end{cases},$$

and the inverse is also true for $\beta^{-L/2}$. Then one concludes that

$$\lim_{L\to+\infty} \tan(\xi L/2) = \begin{cases} i, & \mathrm{Im}(\xi) > 0 \\ -i, & \mathrm{Im}(\xi) < 0 \end{cases}.$$

Now for the key discussion, we come back to Eq. (1.33)(a) and observe that when $k < k_{c1}$

$$\begin{aligned}\lim_{L\to+\infty} E(k<k_{c1}) &= \lim_{L\to+\infty}\left[m_0(k) - t_\perp \frac{\xi_1^2 g^+(\xi_1) - \xi_2^2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)}\right] \\ &= m_0(k) - t_\perp \frac{i\xi_1 - (-i)\xi_2}{i/\xi_1 - (-i)/\xi_2} = m_0(k) - t_\perp \xi_1\xi_2,\end{aligned} \tag{1.42}$$

where we have used observed result as $\mathrm{Im}(\xi_1) > 0 > \mathrm{Im}(\xi_2)$, which, however, does not actually depend on 1 or 2, as long as we have the conjugate relation between ξs. Meanwhile, when $k_{c1} < k < k_{c2}$, and since both $\mathrm{Im}(\xi_{1,2}) > 0$, we have

$$\begin{aligned}\lim_{L\to+\infty} E(k_{c1}<k<k_{c2}) &= \lim_{L\to+\infty}\left[m_0(k) - t_\perp \frac{\xi_1^2 g^+(\xi_1) - \xi_2^2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)}\right] \\ &= m_0(k) - t_\perp \frac{i\xi_1 - i\xi_2}{i/\xi_1 - i/\xi_2} = m_0(k) + t_\perp \xi_1\xi_2.\end{aligned} \tag{1.43}$$

Then, we need to exploit Eq. (1.33)(b) which reads

$$\begin{aligned}\xi_\alpha &= \sqrt{-\frac{F}{D} + (-1)^{\alpha-1}\frac{\sqrt{R}}{D}} = \frac{1}{\sqrt{D}}\sqrt{-F + (-1)^{\alpha-1}\sqrt{F^2 - 2D(m_0^2(k) - E^2)}} \\ &= \frac{1}{\sqrt{2}t_\perp}\sqrt{(2m_0(k)t_\perp - \lambda_\perp^2) + (-1)^{\alpha-1}\lambda_\perp\sqrt{\lambda_\perp^2 - 4m_0(k)t_\perp + \frac{t_\perp^2 E^2}{\lambda_\perp^2}}} \\ &= \frac{\lambda_\perp}{\sqrt{2}t_\perp}\sqrt{(2\gamma-1) + (-1)^{\alpha-1}\sqrt{1-4\gamma+4\delta^2}},\end{aligned} \tag{1.44}$$

where two dimensionless values

$$\begin{cases}\gamma = m_0(k)t_\perp/\lambda_\perp^2 \\ \delta = Et_\perp/\lambda_\perp^2\end{cases} \tag{1.45}$$

are defined. Notice the definition

$$0 < \gamma(k_{c1} < k < k_{c2}) < \frac{1}{4} < \gamma(k < k_{c1}). \tag{1.46}$$

One should also be aware that thanks to the double-sqrt form, ξ_α defined here are properly fixed into the main branch unambiguously.

Now let us come back to $\xi_{1,2}$. In order to prove that $E(k < k_{c2}) = 0$, i.e., $\delta(k < k_{c2}) = 0$, we may firstly take a step back by assuming its correctness, then we need to verify that this is consistent with equations and situations about $\mathrm{Im}(\xi_{1,2})$ we have; and if this works, the existence of such a solution would survive at least. Now let us assume that

$$E(k < k_{c2}) = 0 = \delta(k < k_{c2}),$$

which leads to equations

$$\begin{cases}0 = m_0(k) - t_\perp\xi_1\xi_2 \\ \xi_\alpha = \dfrac{\lambda_\perp}{\sqrt{2}t_\perp}\sqrt{(2\gamma-1) + (-1)^{\alpha-1}\sqrt{1-4\gamma}}\end{cases}, \quad k < k_{c1}, \tag{1.47}$$

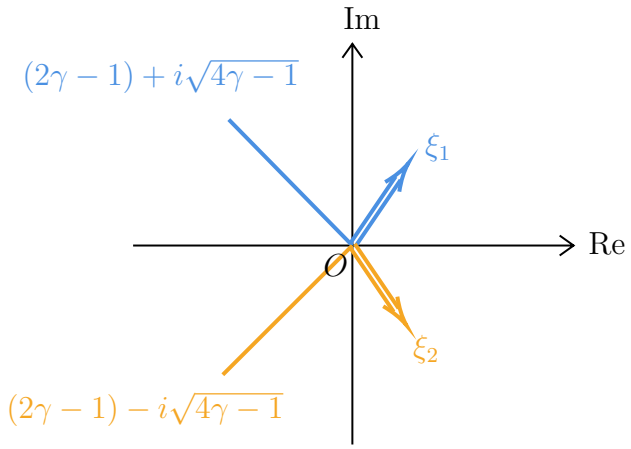
and

$$\begin{cases} 0 = m_0(k) + t_\perp \xi_1 \xi_2 \\ \xi_\alpha = \dfrac{\lambda_\perp}{\sqrt{2}t_\perp}\sqrt{(2\gamma-1) + (-1)^{\alpha-1}\sqrt{1-4\gamma}} \end{cases}, \quad k_{c1} < k < k_{c2}. \tag{1.48}$$

Now γ determines everything, and what we need to verify is consistence between choice of γ value and standing of equations above.

($k < k_{c1}$) By definition $\gamma > 1/4$ now, then we have

$$\xi_\alpha = \frac{\lambda_\perp}{\sqrt{2}t_\perp}\sqrt{(2\gamma-1) + (-1)^{\alpha-1} i\sqrt{4\gamma-1}}, \tag{1.49}$$



and according to the main branch $(-\pi, \pi]$ we have chosen above, we shall obtain $\xi_{1,2}$ similar to picture above (here we choose $(2\gamma-1) < 0$, which does not matter), and we indeed see $\mathrm{Im}(\xi_1) > 0 > \mathrm{Im}(\xi_2)$ as assumed; meanwhile, now

$$\begin{aligned} m_0(k) - t_\perp \xi_1 \xi_2 &= m_0(k) - t_\perp \frac{\lambda_\perp^2}{2t_\perp^2}\sqrt{(2\gamma-1)^2 + (4\gamma-1)} \\ &= m_0(k) - t_\perp \frac{\lambda_\perp^2}{2t_\perp^2}\sqrt{4\gamma^2} = m_0(k) - \frac{\lambda_\perp^2}{t_\perp}\gamma \\ &= m_0(k) - \frac{\lambda_\perp^2}{t_\perp}\frac{m_0(k)t_\perp}{\lambda_\perp^2} = 0, \end{aligned}$$

which serves as self-consistent zero-energy condition.

($k_{c1} < k < k_{c2}$) By definition $0 < \gamma < 1/4$ now. Notice that

$$(2\gamma-1)^2 - (\sqrt{1-4\gamma})^2 = 4\gamma^2 > 0,$$

we then have

$$\xi_\alpha = i\frac{\lambda_\perp}{\sqrt{2}t_\perp}\sqrt{1-2\gamma + (-1)^\alpha\sqrt{1-4\gamma}}, \tag{1.50}$$

and both $\mathrm{Im}(\xi_{1,2}) > 0$ consistent with assumption before. Meanwhile, now

$$\begin{aligned} m_0(k) + t_\perp \xi_1 \xi_2 &= m_0(k) + i^2 t_\perp \frac{\lambda_\perp^2}{2t_\perp^2} \sqrt{(1-2\gamma)^2 - (1-4\gamma)} \\ &= m_0(k) - t_\perp \frac{\lambda_\perp^2}{2t_\perp^2} \sqrt{4\gamma^2} = 0, \end{aligned}$$

which serves as self-consistent zero-energy condition.

$k = k_{c1}$ case could be put into the second case. Our two cases study shows unambiguously that

$$\lim_{L \to +\infty} E(k < k_{c2}) = 0, \tag{1.51}$$

then together with $k > k_{c2}$ case, we say

$$E(k) = \Theta(-m_0(k)) m_0(k) = \begin{cases} 0, & m_0(\boldsymbol{k}) > 0 \\ m_0(\boldsymbol{k}), & m_0(\boldsymbol{k}) \leq 0 \end{cases}, \tag{1.52}$$

is the solution to set of closed equations Eq. (1.33) when the system has large L along z.

We could in fact conserve lowest order correction to see the finite size gap when L is not that large. For ξ_1 and ξ_2, one could approximately get lowest order correction for $\tan(\xi L/2)$ by treating $\beta^{\pm L/2}$ as small quantity (depend on sign of $\mathrm{Im}(\xi)$)

$$\tan(\xi L/2) \approx \begin{cases} i(1 - 2\beta^L), & \mathrm{Im}(\xi) > 0 \\ -i(1 - 2\beta^{-L}), & \mathrm{Im}(\xi) < 0 \end{cases}. \tag{1.53}$$

Also notice that from the original $E - \xi$ equation

$$E^2 - (m_0(\boldsymbol{k}) - t_\perp \xi^2)^2 - \lambda_\perp^2 \xi^2 = 0,$$

which could be further split into (when $m = 0$ as zeroth-order)

$$t_\perp \xi^2 \pm i\lambda_\perp \xi - m_0(k) = 0, \tag{1.54}$$

one solves

$$\xi = \frac{s_1 i\lambda_\perp + s_2 \sqrt{4m_0(k)t_\perp - \lambda_\perp^2}}{2t_\perp} = \frac{\lambda_\perp}{2t_\perp}(is_1 + s_2\sqrt{4\gamma - 1}), \tag{1.55}$$

where $s_1, s_2 = \pm$ without restriction. Notice that in real calculation, one needs to specify which branch $\xi_{1,2}$ lie in, but such choice will not affect the final result as long as chosen $\xi_{1,2}$ satisfy zeroth-order solution. Now again we have two cases below:

• When $k < k_{c1}$, i.e., $\gamma > 1/4$, we have

$$\begin{cases} \xi_1 = \xi_2^* \\ \mathrm{Im}(\xi_1) > 0 > \mathrm{Im}(\xi_2) \\ \mathrm{Re}(\xi_1) = \mathrm{Re}(\xi_2) > 0 \end{cases}, \ k < k_{c1}, \tag{1.56}$$

as main branch condition, then

$$\begin{cases} \tan(\xi_1 L/2) \approx i(1 - 2\beta_1^L) \\ \tan(\xi_2 L/2) \approx -i(1 - 2\beta_2^{-L}) \end{cases}, \tag{1.57}$$

and

$$\begin{aligned}
E(k<k_{c1}) &\approx m_0(k) - t_\perp \frac{\xi_1 i(1-2\beta_1^L) - \xi_2(-i)(1-2\beta_2^{-L})}{\frac{i(1-2\beta_1^L)}{\xi_1} - \frac{(-i)(1-2\beta_2^{-L})}{\xi_2}} \\
&= m_0(k) - t_\perp \frac{\xi_1+\xi_2-2(\xi_1\beta_1^L+\xi_2\beta_2^{-L})}{\frac{1}{\xi_1}+\frac{1}{\xi_2}-2(\frac{\beta_1^L}{\xi_1}+\frac{\beta_2^{-L}}{\xi_2})} \\
&\approx m_0(k) - t_\perp \frac{\xi_1+\xi_2}{\frac{1}{\xi_1}+\frac{1}{\xi_2}}\left[1-2\frac{\xi_1\beta_1^L+\xi_2\beta_2^{-L}}{\xi_1+\xi_2}\right]\left[1+\frac{2}{\frac{1}{\xi_1}+\frac{1}{\xi_2}}\left(\frac{\beta_1^L}{\xi_1}+\frac{\beta_2^{-L}}{\xi_2}\right)\right] \\
&\approx m_0(k) - t_\perp\xi_1\xi_2\left[1-2\frac{\xi_1\beta_1^L+\xi_2\beta_2^{-L}}{\xi_1+\xi_2}+\frac{2}{\frac{1}{\xi_1}+\frac{1}{\xi_2}}\left(\frac{\beta_1^L}{\xi_1}+\frac{\beta_2^{-L}}{\xi_2}\right)\right] \\
&= (m_0(k) - t_\perp\xi_1\xi_2) + 2t_\perp\xi_1\xi_2\frac{\xi_1-\xi_2}{\xi_1+\xi_2}(e^{i\xi_1 L}-e^{-i\xi_2 L}).
\end{aligned}$$

Notice that first term in bracket is zeroth order as $E \approx 0$. Now, it is time to utilize four solutions in Eq. (1.55). By main branch condition above, accordingly we choose

$$\begin{cases} \xi_1 = \dfrac{\lambda_\perp}{2t_\perp}(\sqrt{4\gamma-1}+i) \\ \xi_2 = \dfrac{\lambda_\perp}{2t_\perp}(\sqrt{4\gamma-1}-i) \end{cases}, \tag{1.58}$$

considering that $\gamma > 1/4$ in this zone. Afterwards, one obtains

$$\begin{cases} \xi_1\xi_2 = \left(\dfrac{\lambda_\perp}{2t_\perp}\right)^2 4\gamma \\ \xi_1+\xi_2 = \dfrac{\lambda_\perp}{t_\perp}\sqrt{4\gamma-1} \\ \xi_1-\xi_2 = \dfrac{\lambda_\perp}{t_\perp}i \\ e^{i\xi_1 L}-e^{-i\xi_2 L} = 2i\sin\big(\lambda_\perp\sqrt{4\gamma-1}L/2t_\perp\big)e^{-\lambda_\perp L/2t_\perp} \end{cases},$$

and

$$\begin{aligned}
E(k<k_{c1}) &\approx 2t_\perp\xi_1\xi_2\frac{\xi_1-\xi_2}{\xi_1+\xi_2}(e^{i\xi_1 L}-e^{-i\xi_2 L}) \\
&= 2t_\perp\frac{\lambda_\perp^2}{t_\perp^2}\gamma\frac{i}{\sqrt{4\gamma-1}}2i\sin\Big(\lambda_\perp\sqrt{4\gamma-1}L/2t_\perp\Big)e^{-\lambda_\perp L/2t_\perp} \\
&= -\frac{4m_0(k)}{\sqrt{4\gamma-1}}\sin\Big(u\sqrt{4\gamma-1}L\Big)e^{-uL},
\end{aligned} \tag{1.59}$$

where $u = \lambda_\perp/2t_\perp$ is defined. Low energy surface state mass shows both exponentially decay and oscillating behavior.

• When $k_{c1} < k < k_{c2}$, i.e., $0 < \gamma < 1/4$, we have

$$\begin{cases} \mathrm{Im}(\xi_1) > 0 \\ \mathrm{Im}(\xi_2) > 0 \end{cases}, \ k_{c1} < k < k_{c2}, \tag{1.60}$$

as main branch condition, then

$$\begin{cases} \tan(\xi_1 L/2) \approx i(1-2\beta_1^L) \\ \tan(\xi_2 L/2) \approx i(1-2\beta_2^L) \end{cases}, \tag{1.61}$$

$$
\begin{aligned}
E(k_{c1}<k<k_{c2}) &\approx m_0(k) - t_\perp \frac{\xi_1 i(1-2\beta_1^L) - \xi_2 i(1-2\beta_2^L)}{\frac{i(1-2\beta_1^L)}{\xi_1} - \frac{i(1-2\beta_2^L)}{\xi_2}} \\
&= m_0(k) - t_\perp \frac{\xi_1 - \xi_2 - 2(\xi_1\beta_1^L - \xi_2\beta_2^L)}{\frac{1}{\xi_1} - \frac{1}{\xi_2} - 2(\frac{\beta_1^L}{\xi_1} - \frac{\beta_2^L}{\xi_2})} \\
&\approx m_0(k) - t_\perp \frac{\xi_1 - \xi_2}{\frac{1}{\xi_1} - \frac{1}{\xi_2}} \left[1 - 2\frac{\xi_1\beta_1^L - \xi_2\beta_2^L}{\xi_1 - \xi_2}\right] \left[1 + \frac{2}{\frac{1}{\xi_1} - \frac{1}{\xi_2}} \left(\frac{\beta_1^L}{\xi_1} - \frac{\beta_2^L}{\xi_2}\right)\right] \\
&\approx m_0(k) + t_\perp \xi_1\xi_2 \left[1 - 2\frac{\xi_1\beta_1^L - \xi_2\beta_2^L}{\xi_1 - \xi_2} + \frac{2}{\xi_2 - \xi_1}\left(\beta_1^L\xi_2 - \beta_2^L\xi_1\right)\right] \\
&= (m_0(k) + t_\perp\xi_1\xi_2) - 2t_\perp\xi_1\xi_2\frac{\xi_1+\xi_2}{\xi_1-\xi_2}(e^{i\xi_1 L} - e^{i\xi_2 L}),
\end{aligned}
$$

where first term in bracket is again zeroth order energy approaching zero. Again, utilizing four solutions in Eq. (1.55) with main branch condition above, we choose

$$
\begin{cases}
\xi_1 = i\dfrac{\lambda_\perp}{2t_\perp}(1+\sqrt{1-4\gamma}) \\
\xi_2 = i\dfrac{\lambda_\perp}{2t_\perp}(1-\sqrt{1-4\gamma})
\end{cases}, \tag{1.62}
$$

considering that $0 < \gamma < 1/4$ in this zone. The reason why $\mathrm{Im}(\xi_1) > \mathrm{Im}(\xi_2)$ comes from observation from Fig. 1.1(b), which, however, does not matter due to the dual form. Again, one obtains

$$
\begin{cases}
\xi_1\xi_2 = -\left(\dfrac{\lambda_\perp}{2t_\perp}\right)^2 4\gamma \\
\xi_1 + \xi_2 = i\lambda_\perp/t_\perp \\
\xi_1 - \xi_2 = i\lambda_\perp\sqrt{1-4\gamma}/t_\perp \\
e^{i\xi_1 L} - e^{i\xi_2 L} = -2\sinh\big(u\sqrt{1-4\gamma}L\big)e^{-uL}
\end{cases},
$$

and

$$
\begin{aligned}
E(k_{c1}<k<k_{c2}) &\approx -2t_\perp\xi_1\xi_2\frac{\xi_1+\xi_2}{\xi_1-\xi_2}(e^{i\xi_1 L} - e^{i\xi_2 L}) \\
&= -2t_\perp\left(-\frac{\lambda_\perp^2}{t_\perp^2}\gamma\right)\frac{1}{\sqrt{1-4\gamma}}\left[-2\sinh\Big(u\sqrt{1-4\gamma}L\Big)e^{-uL}\right] \\
&= -\frac{4m_0(k)}{\sqrt{1-4\gamma}}\sinh\Big(u\sqrt{1-4\gamma}L\Big)e^{-uL},
\end{aligned} \tag{1.63}
$$

where $u = \lambda_\perp/2t_\perp$ is defined again. Notice that

$$
\sin(ix) = i\sinh(x),
$$

we may have a unified expression for lowest order mass correction

$$
E(k<k_{c2}) = -\frac{4m_0(k)}{\sqrt{4\gamma-1}}\sin\Big(u\sqrt{4\gamma-1}L\Big)e^{-uL}. \tag{1.64}
$$

However, as a comment, in numerical calculation, m in zone $k_{c1}\sim k_{c2}$ is suppressed into zero in a much slower manner, which is caused by exponential cancellation between sinh and exp. However, since $\sqrt{1-4\gamma} < 1$ in the region, we conclude that the exponential increasing is always slower than the decaying, which finally pushes the state to zero energy for $L\to+\infty$.

Direct Criterion I — Zero Mode Solution

Here we directly solve 2×2 eigenvalue problem by assuming existence of zero-mode, and then we figure out when the assumption stands. Now by

$$h\phi = E\phi, \tag{1.65}$$

where

$$h = h(+) = -i\lambda_\perp \partial_z \tau_x + (m_0(\boldsymbol{k}) + t_\perp \partial_z^2)\tau_z. \tag{1.66}$$

Then by assuming

$$\phi = \eta_s e^{\xi z}\rho, \tag{1.67}$$

where

$$\tau_y \eta_s = s\eta_s \tag{1.68}$$

one finds that for zero-mode solution

$$i\tau_x H\phi = [\lambda_\perp \xi + s(m_0(\boldsymbol{k}) + t_\perp \xi^2)]\rho = 0.$$

Then requiring non-trivial ρ leads to

$$t_\perp \xi^2 + s\lambda_\perp \xi + m_0(\boldsymbol{k}) = 0, \tag{1.69}$$

which is equivalent to

$$\begin{cases} \xi_1 + \xi_2 = -\dfrac{s\lambda_\perp}{t_\perp} \\ \xi_1 \xi_2 = \dfrac{m_0(\boldsymbol{k})}{t_\perp} \end{cases}, \tag{1.70}$$

and solves

$$\xi = \frac{-s\lambda_\perp \pm \sqrt{\lambda_\perp^2 - 4m_0(\boldsymbol{k})t_\perp}}{2t_\perp} = u(-s \pm \sqrt{1-4\gamma}), \tag{1.71}$$

where

$$\begin{cases} \gamma = \dfrac{m_0(\boldsymbol{k})t_\perp}{\lambda_\perp^2} \\ u = \dfrac{\lambda_\perp}{2t_\perp} \end{cases}. \tag{1.72}$$

Notice that we have assumed $\lambda_\perp > 0$ in the simplification, without loss of generality. What is more, we may also assume $t_\perp > 0$ and thus $u > 0$, then ξ is almost only relevant to γ depending on $m_0(\boldsymbol{k})$.

Now boundary condition enters, and we assume that the system is a semi-infinite one along $z \in (-\infty, 0]$. For the sake of vanishing wavefunction when $z \to -\infty$, $\mathrm{Re}(\xi) > 0$ is required and we have three cases:

Case 1. $\gamma < 0$, which leads to $1 - 4\gamma > 1$ and one chooses $\xi = u(-s + \sqrt{1-4\gamma})$;

Case 2. $0 < \gamma < \dfrac{1}{4}$, which leads to $0 < 1 - 4\gamma < 1$ and one chooses $\xi = u(1 \pm \sqrt{1-4\gamma})$;

Case 3. $\gamma > 1/4$, which leads to $1 - 4\gamma < 0$ and one chooses $\xi = u(1 \pm i\sqrt{4\gamma - 1})$.

One sees two ξs are preserved in each case. On the other hand, $\phi(z = 0) = 0$ tells that

$$\phi \propto \sum_s \rho_s \eta_s (e^{\xi_s^+ z} - e^{\xi_s^- z}), \tag{1.73}$$

then case 1 above is excluded, and we find zero-mode condition

$$\gamma > 0 \Longleftrightarrow m_0(\boldsymbol{k})t_\perp > 0 \tag{1.74}$$

generally; and under our assumption, it becomes $m_0(\boldsymbol{k}) > 0$ in case study.

Direct Criterion II — Winding Number

1-D winding number is defined through

$$\omega_{1d} = \frac{1}{2\pi}\int \mathrm{d}k(\hat{d}\times\partial_k\hat{d})_y, \tag{1.75}$$

where reduced Hamiltonian $\hat{h} = \hat{d}\cdot\boldsymbol{\sigma}$ is defined with requirement of chiral symmetry, and in this definition, it is τ_y as $\{h, \tau_y\} = 0$. On the other band, by defining

$$h = \hat{d}_z + i\hat{d}_x, \tag{1.76}$$

one finds

$$\begin{aligned}\partial_k \ln h &= \frac{1}{h}\partial_k h = \frac{1}{\hat{d}_z + i\hat{d}_x}(\partial_k\hat{d}_z + i\partial_k\hat{d}_x)\\ &= (\hat{d}_z - i\hat{d}_x)(\partial_k\hat{d}_z + i\partial_k\hat{d}_x)\\ &= (\hat{d}_z\partial_k\hat{d}_z + \hat{d}_x\partial_k\hat{d}_x) + i(\hat{d}_z\partial_k\hat{d}_x - \hat{d}_x\partial_k\hat{d}_z)\\ &= i(\hat{d}\times\partial_k\hat{d})_y,\end{aligned}$$

then accordingly,

$$\omega_{1d} = \frac{1}{2\pi i}\int \mathrm{d}k\partial_k \ln h. \tag{1.77}$$

Since

$$\ln h = \ln\left[|h|e^{i\arg(h)}\right] = i\arg(h),$$

we find

$$\omega_{1d} = \frac{1}{2\pi}\arg(\hat{d}_z + i\hat{d}_x)\big|_{\mathrm{Dom}(k)}. \tag{1.78}$$

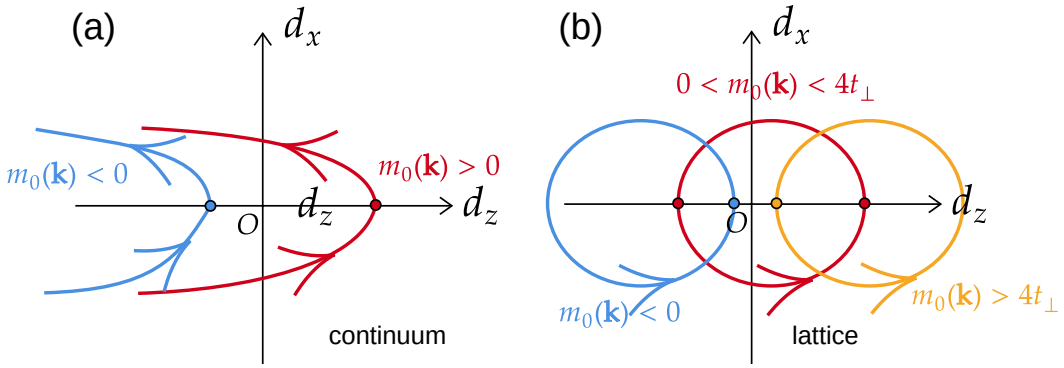


Figure 1.2: Winding for (a) continuum model, and (b) lattice model.

Again in the given case,

$$d(k_z) \equiv h(+, k_z) = h(+) = \lambda_\perp k_z\tau_x + (m_0(\boldsymbol{k}) - t_\perp k_z^2)\tau_z, \tag{1.79}$$

i.e., $\boldsymbol{d} = (\lambda_\perp k_z, 0, m_0(\boldsymbol{k}) - t_\perp k_z^2)$, and accordingly, as shown in Fig. 1.2(a), $m_0(\boldsymbol{k}) > 0$ gives one time of winding when k_z runs from $-\infty$ to $+\infty$ around original point where gap closes, then it offers a topologically distinct phase compared with $m_0(\boldsymbol{k}) < 0$ case. Meanwhile, when written as lattice model, we see

$$\boldsymbol{d}(k_z) = (\lambda_\perp \sin k_z, 0, m_0(\boldsymbol{k}) - 4\sin^2(\frac{k_z}{2})), \tag{1.80}$$

and non-trivial winding around original point yields condition

$$0 < m_0(\boldsymbol{k}) < 4t_\perp. \tag{1.81}$$

Complete Discussion for Thick Limit

Using original general $E-\xi$ relation, listing of equations are

$$\begin{cases} E^2-(m_0(\boldsymbol{k})-t_\perp\xi^2)^2-\lambda_\perp^2\xi^2=0 \\ E=m_0(\boldsymbol{k})-t_\perp\dfrac{\xi_1^2g^+(\xi_1)-\xi_2^2g^+(\xi_2)}{g^+(\xi_1)-g^+(\xi_2)} \end{cases},$$

or

$$\begin{cases} \xi_1^2+\xi_2^2=\dfrac{2m_0(\boldsymbol{k})t_\perp-\lambda_\perp^2}{t_\perp^2} \\ \xi_1^2\xi_2^2=\dfrac{m_0(\boldsymbol{k})^2-E^2}{t_\perp^2} \\ E=m_0(\boldsymbol{k})-t_\perp\dfrac{\xi_1^2g^+(\xi_1)-\xi_2^2g^+(\xi_2)}{g^+(\xi_1)-g^+(\xi_2)} \end{cases}, \tag{1.82}$$

where $g^+(\xi)=\tan(\xi L/2)/\xi$. Notice, however, that ξ_1 and ξ_2 here loses its main branch condition, and one has to fix that manually. Without lose of generality, we assume $\lambda_\perp>0,\ t_\perp>0$ in the following discussion, then the ultimate variable becomes $m_0(\boldsymbol{k})$.

By classification on $\tan(\xi L/2)$, one finds

$$\lim_{L\to+\infty}\tan(\xi L/2)=\begin{cases} i, & \mathrm{Im}(\xi)>0 \\ \text{N.A.}, & \mathrm{Im}(\xi)=0 \\ -i, & \mathrm{Im}(\xi)<0 \end{cases}. \tag{1.83}$$

Then accordingly, we separate three basic cases as

$$\begin{cases} \mathrm{Im}(\xi_1)>0>\mathrm{Im}(\xi_2) \\ \mathrm{Im}(\xi_{1,2})>0 \\ \mathrm{Im}(\xi_1)=0,\ \ \mathrm{Im}(\xi_2)>0 \end{cases}, \tag{1.84}$$

and due to the flexibility equations above serve, other cases could be obtained in a similar way.

Case I. ($\mathrm{Im}(\xi_1)>0>\mathrm{Im}(\xi_2)$)

Now $\tan(\xi_1L/2)=i=-\tan(\xi_2L/2)$ ($L\to+\infty$ ignored), and

$$\begin{cases} \xi_1^2+\xi_2^2=\dfrac{2m_0(\boldsymbol{k})t_\perp-\lambda_\perp^2}{t_\perp^2} \\ \xi_1^2\xi_2^2=\dfrac{m_0(\boldsymbol{k})^2-E^2}{t_\perp^2} \\ E=m_0(\boldsymbol{k})-t_\perp\xi_1\xi_2 \end{cases}, \tag{1.85}$$

then the second and third equations above leads to

$$m_0(\boldsymbol{k})^2-E^2=(m_0(\boldsymbol{k})-E)^2, \tag{1.86}$$

which gives us two possible solutions as $E=0$ or $E=m_0(\boldsymbol{k})$.

I. ($E = 0$) This condition leads to

$$\begin{cases} \xi_1\xi_2 = \dfrac{m_0(\boldsymbol{k})}{t_\perp} \\ \xi_1^2 + \xi_2^2 = \dfrac{2m_0(\boldsymbol{k})t_\perp - \lambda_\perp^2}{t_\perp^2} \end{cases}, \tag{1.87}$$

which leads to

$$\begin{cases} (\xi_1 + \xi_2)^2 = 4u^2(4\gamma - 1) \\ (\xi_1 - \xi_2)^2 = -4u^2 \end{cases}, \tag{1.88}$$

where

$$\begin{cases} \gamma = m_0(\boldsymbol{k})t_\perp/\lambda_\perp^2 \\ u = \lambda_\perp/2t_\perp \end{cases}. \tag{1.89}$$

Requirement $\mathrm{Im}(\xi_1) > 0 > \mathrm{Im}(\xi_2)$ then leads to

$$\begin{cases} \xi_1 + \xi_2 = \begin{cases} 2u\sqrt{4\gamma - 1}, & \gamma > 1/4 \\ 2ui\sqrt{1 - 4\gamma}, & \gamma < 1/4 \end{cases} \\ \xi_1 - \xi_2 = 2ui \end{cases}, \tag{1.90}$$

which offers:

- $\gamma > 1/4$:

$$\begin{cases} \xi_1 = u(\sqrt{4\gamma - 1} + i) \\ \xi_2 = u(\sqrt{4\gamma - 1} - i) \end{cases}; \tag{1.91}$$

- $\gamma < 1/4$:

$$\begin{cases} \xi_1 = iu(\sqrt{1 - 4\gamma} + 1) \\ \xi_2 = iu(\sqrt{1 - 4\gamma} - 1) \end{cases}. \tag{1.92}$$

Notice that latter condition stands only when $\gamma > 0$ so that $\mathrm{Im}(\xi_2) < 0$.

II. ($E = m_0(\boldsymbol{k})$) This condition leads to

$$\begin{cases} \xi_1\xi_2 = 0 \\ \xi_1^2 + \xi_2^2 = \dfrac{2m_0(\boldsymbol{k})t_\perp - \lambda_\perp^2}{t_\perp^2} \end{cases}, \tag{1.93}$$

and one finds one of $\xi_\alpha = 0$ is unavoidable, which fails the precondition and should be excluded, i.e., $E = m_0(\boldsymbol{k})$ is not a solution under the consideration.

Case II. ($\mathrm{Im}(\xi_{1,2}) > 0$)

Now $\tan(\xi_1 L/2) = i = \tan(\xi_2 L/2)$ ($L \to +\infty$ ignored), and

$$\begin{cases} \xi_1^2 + \xi_2^2 = \dfrac{2m_0(\boldsymbol{k})t_\perp - \lambda_\perp^2}{t_\perp^2} \\ \xi_1^2\xi_2^2 = \dfrac{m_0(\boldsymbol{k})^2 - E^2}{t_\perp^2} \\ E = m_0(\boldsymbol{k}) + t_\perp\xi_1\xi_2 \end{cases}, \tag{1.94}$$

then the second and third equations above leads to

$$m_0(\boldsymbol{k})^2 - E^2 = (m_0(\boldsymbol{k}) - E)^2, \tag{1.95}$$

which gives us two possible solutions as $E = 0$ or $E = m_0(\boldsymbol{k})$.

I. $(E = 0)$ This condition leads to

$$\begin{cases} \xi_1\xi_2 = -\dfrac{m_0(\boldsymbol{k})}{t_\perp} \\ \xi_1^2 + \xi_2^2 = \dfrac{2m_0(\boldsymbol{k})t_\perp - \lambda_\perp^2}{t_\perp^2} \end{cases}, \tag{1.96}$$

which leads to

$$\begin{cases} (\xi_1 + \xi_2)^2 = -4u^2 \\ (\xi_1 - \xi_2)^2 = 4u^2(4\gamma - 1) \end{cases}, \tag{1.97}$$

where

$$\begin{cases} \gamma = m_0(\boldsymbol{k})t_\perp/\lambda_\perp^2 \\ u = \lambda_\perp/2t_\perp \end{cases}. \tag{1.98}$$

Requirement $\mathrm{Im}(\xi_{1,2}) > 0$ then leads to

$$\begin{cases} \xi_1 + \xi_2 = 2ui \\ \xi_1 - \xi_2 = \begin{cases} 2u\sqrt{4\gamma - 1}, & \gamma > 1/4 \\ 2ui\sqrt{1 - 4\gamma}, & \gamma < 1/4 \end{cases} \end{cases}, \tag{1.99}$$

which offers:

- $\gamma > 1/4$:

$$\begin{cases} \xi_1 = u(\sqrt{4\gamma - 1} + i) \\ \xi_2 = u(-\sqrt{4\gamma - 1} + i) \end{cases}; \tag{1.100}$$

- $\gamma < 1/4$:

$$\begin{cases} \xi_1 = iu(\sqrt{1 - 4\gamma} + 1) \\ \xi_2 = iu(-\sqrt{1 - 4\gamma} + 1) \end{cases}. \tag{1.101}$$

Notice that latter condition stands only when $\gamma > 0$ so that $\mathrm{Im}(\xi_2) > 0$.

II. $(E = m_0(\boldsymbol{k}))$ This condition leads to

$$\begin{cases} \xi_1\xi_2 = 0 \\ \xi_1^2 + \xi_2^2 = \dfrac{2m_0(\boldsymbol{k})t_\perp - \lambda_\perp^2}{t_\perp^2} \end{cases}, \tag{1.102}$$

and one finds one of $\xi_\alpha = 0$ is unavoidable, which fails the precondition and should be excluded, i.e., $E = m_0(\boldsymbol{k})$ is not a solution under the consideration.

Case III. $(\mathrm{Im}(\xi_1) = 0,\ \mathrm{Im}(\xi_2) > 0)$

With guessed solution $E = m_0(\boldsymbol{k})$, we have

$$\begin{cases} \xi_1\xi_2 = 0 \\ \xi_1^2 + \xi_2^2 = \dfrac{2m_0(\boldsymbol{k})t_\perp - \lambda_\perp^2}{t_\perp^2} \end{cases}, \tag{1.103}$$

which leads to

$$\begin{cases} (\xi_1 + \xi_2)^2 = 4u^2(2\gamma - 1) \\ (\xi_1 - \xi_2)^2 = 4u^2(2\gamma - 1) \end{cases}, \tag{1.104}$$

and one is able to choose, for instance,

$$\begin{cases} \xi_1 = 0 \\ \xi_2 = 2ui\sqrt{1-2\gamma} \end{cases}, \tag{1.105}$$

which fulfills the requirement. Notice that we have assumed $\gamma < 1/2$, whose choice does not bother the self-consistent solution.

Notice that $\xi_1 = 0$ leads to degenerate eigenvalue $\pm\xi_1$, then generally one should assume another solution as

$$(A + Bz)e^{i\xi_1 z}\phi\big|_{\xi_1=0, E=m_0(\boldsymbol{k})},$$

which, however, only gives result that $B = 0$ while A is arbitrary.

Since $\gamma = m_0(\boldsymbol{k})t_\perp/\lambda_\perp^2$, we see that above discussion naturally leads to the conclusion that

$$E = \begin{cases} 0, & m_0(\boldsymbol{k}) > 0 \\ m_0(\boldsymbol{k}), & m_0(\boldsymbol{k}) < 0 \end{cases}. \tag{1.106}$$

1.2 Lattice Model

We could in fact make it more strict by solving the lattice Hamiltonian. Start from Eq (5.18) and notice that $[\mathscr{H}_{1d}(\boldsymbol{k}), \sigma_z] = 0$, from which we could let

$$\mathscr{H}_{1d}(\boldsymbol{k})\zeta_s \otimes |\phi^s(\boldsymbol{k})\rangle = \zeta_s \otimes \mathscr{H}_{1d}^s(\boldsymbol{k})\,|\phi^s(\boldsymbol{k})\rangle\,,\ \sigma_z\zeta_s = s\zeta_s,\ s = \pm, \tag{1.107}$$

where $\mathscr{H}_{1d}^s(\boldsymbol{k})$ is split Hamiltonian that only acts on one subspace. Under basis of $\{\Psi_{i_z,\boldsymbol{k}}\}_{i_z}$, $\mathscr{H}_{1d}^s(\boldsymbol{k})$ is in its matrix form denoted as $H_{1d}^s(\boldsymbol{k})$, which could be written explicitly as

$$H_{1d}^s(\boldsymbol{k}) = \begin{pmatrix} M_0(\boldsymbol{k})\tau_z & t_\perp\tau_z - i\frac{\lambda_\perp}{2}s\tau_x & & \\ t_\perp\tau_z + i\frac{\lambda_\perp}{2}s\tau_x & \ddots & \ddots & \\ & \ddots & \ddots & t_\perp\tau_z - i\frac{\lambda_\perp}{2}s\tau_x \\ & & t_\perp\tau_z + i\frac{\lambda_\perp}{2}s\tau_x & M_0(\boldsymbol{k}) \end{pmatrix}_{L_z\times L_z}, \tag{1.108}$$

with solution

$$H_{1d}^s(\boldsymbol{k})\phi^s(\boldsymbol{k}) = E^s(\boldsymbol{k})\phi^s(\boldsymbol{k}),\ \phi^s(\boldsymbol{k}) = \oplus_{i_z}\phi_{i_z}^s(\boldsymbol{k}). \tag{1.109}$$

To make discussion pithy, we shall omit $s, \boldsymbol{k}$ and let $M = M_0(\boldsymbol{k})$ below.

Eq (1.109) could be written in the recurrence form as

$$(t_\perp \tau_z + i\frac{\lambda_\perp}{2} s\tau_x)\phi_{i_z-1} + M\tau_z\phi_{i_z} + (t_\perp\tau_z - i\frac{\lambda_\perp}{2}s\tau_x)\phi_{i_z+1} = E\phi_{i_z}, \tag{1.110}$$

by observing which could we set trial function as $\phi_{i_z} = e^{i\xi i_z}\phi = \beta^{i_z}\phi$ where $\beta = e^{i\xi}$, and the equation would be reduced to

$$[(t_\perp\tau_z + i\frac{\lambda_\perp}{2}s\tau_x)\beta^{-1} + (M\tau_z - E) + (t_\perp\tau_z - i\frac{\lambda_\perp}{2}s\tau_x)\beta]\phi = 0, \tag{1.111}$$

i.e.,

$$\begin{pmatrix} t_\perp(\beta + \frac{1}{\beta}) + M - E & -i\frac{\lambda_\perp}{2}s(\beta - \frac{1}{\beta}) \\ -i\frac{\lambda_\perp}{2}s(\beta - \frac{1}{\beta}) & -t_\perp(\beta + \frac{1}{\beta}) - M - E \end{pmatrix}\phi = 0, \tag{1.112}$$

which firstly leads to

$$E^2 = [M + t_\perp(\beta + \frac{1}{\beta})]^2 - \frac{\lambda_\perp^2}{4}(\beta - \frac{1}{\beta})^2 = (M + 2t_\perp\cos\xi)^2 + \lambda_\perp^2\sin^2\xi, \tag{1.113}$$

requiring non-trivial ϕ. From (1.113) one could solve

$$\begin{aligned} \cos\xi_\alpha^p &= \frac{-Mt_\perp + (-1)^{\alpha-1}\sqrt{M^2t_\perp^2 - (t_\perp^2 - \lambda_\perp^2/4)(M^2 + \lambda_\perp^2 - E^2)}}{2(t_\perp^2 - \lambda_\perp^2/4)}, \\ \sin\xi_\alpha^p &= p\sqrt{1 - \cos^2\xi_\alpha},\ p = \pm, \alpha = 1, 2, \end{aligned} \tag{1.114}$$

which tells that

$$\beta_\alpha^p = e^{i\xi_\alpha^p} = \cos\xi_\alpha + ip\sqrt{1 - \cos^2\xi_\alpha}. \tag{1.115}$$

Here one thing to notice is that the sign change of $\sin\xi_\alpha^p$ could be caused by sign change of ξ, rather than shift a phase like $\xi \to \xi + \pi$, which would lead to the sign change of $\cos\xi$, too, and that is not our solution. This means that

$$\xi_\alpha^p = p\xi_\alpha. \tag{1.116}$$

For each ξ_α^p, one has

$$\phi_{\alpha p}^s = \begin{pmatrix} s\lambda_\perp p\sin\xi_\alpha \\ E - M - 2t_\perp\cos\xi_\alpha, \end{pmatrix} \tag{1.117}$$

and the general solution would be

$$\Phi^s = \sum_{\alpha p} C_{\alpha p}\phi_{\alpha p}^s e^{i\xi_\alpha^p i_z}. \tag{1.118}$$

Notice that we have adopted a slightly different form of general solution from continuum model, which has no impact on final result. Now consider boundary condition within the symmetric case in which the centre of 1-d chain sits at $z = 0$, then by denoting $l = L_z + 1$, which is just L/c in continuum model considering canonical boundary condition, we would have

$$\Phi^s(\pm\frac{l}{2}) = 0, \tag{1.119}$$

applying which one would get four linear equations for coefficients

$$P(C_{1+}, C_{1-}, C_{2+}, C_{2-})^T = 0.$$

This leads to equation below for non-vanishing Cs by defining $\lambda_\alpha = s\lambda_\perp \sin\xi_\alpha$ and $m_\alpha = E - M - 2t_\perp \cos\xi_\alpha$:

$$\begin{aligned}
0 &= \det(P) \\
&= \det \begin{pmatrix} \lambda_1 e^{i\xi_1 l/2} & -\lambda_1 e^{-i\xi_1 l/2} & \lambda_2 e^{i\xi_2 l/2} & -\lambda_2 e^{-i\xi_2 l/2} \\ \lambda_1 e^{-i\xi_1 l/2} & -\lambda_1 e^{i\xi_1 l/2} & \lambda_2 e^{-i\xi_2 l/2} & -\lambda_2 e^{i\xi_2 l/2} \\ m_1 e^{i\xi_1 l/2} & m_1 e^{-i\xi_1 l/2} & m_2 e^{i\xi_2 l/2} & m_2 e^{-i\xi_2 l/2} \\ m_1 e^{-i\xi_1 l/2} & m_1 e^{i\xi_1 l/2} & m_2 e^{-i\xi_2 l/2} & m_2 e^{i\xi_2 l/2} \end{pmatrix} \\
&= 4\lambda_\perp^2 m_1 m_2 \sin(\xi_1 l)\sin(\xi_2 l) \left\{ \frac{m_2 \sin\xi_1}{m_1 \sin\xi_2} + \frac{m_1 \sin\xi_2}{m_2 \sin\xi_1} - \frac{\tan(\xi_1 l/2)}{\tan(\xi_2 l/2)} - \frac{\tan(\xi_2 l/2)}{\tan(\xi_1 l/2)} \right\},
\end{aligned} \tag{1.120}$$

from which one gets two transcendental equations

$$\frac{m_1 \sin\xi_2}{m_2 \sin\xi_1} = \frac{\tan\xi_2 l/2}{\tan\xi_1 l/2} \tag{1.121a}$$

$$\frac{m_1 \sin\xi_2}{m_2 \sin\xi_1} = \frac{\tan\xi_1 l/2}{\tan\xi_2 l/2} \tag{1.121b}$$

which gives two energies varying with $\boldsymbol{k}$, designated as E_+ and E_-, respectively. Formally,

$$E_+ = M + 2t_\perp \frac{\cos\xi_1 g^+(\xi_1) - \cos\xi_2 g^+(\xi_2)}{g^+(\xi_1) - g^+(\xi_2)},\ g^+(\xi) = \frac{\tan(\xi(L_z+1)/2)}{\sin\xi}, \tag{1.122a}$$

$$E_- = M + 2t_\perp \frac{\cos\xi_1 g^-(\xi_1) - \cos\xi_2 g^-(\xi_2)}{g^-(\xi_1) - g^-(\xi_2)},\ g^-(\xi) = \frac{1}{\tan(\xi(L_z+1)/2)\sin\xi}. \tag{1.122b}$$

Now it is clear that by combination of Eq (1.121) with Eq (1.113) or Eq (1.114), both energy and characteristic penetration depth could be solved, at least numerically. One thing interesting to notice is that when ξ_α approach zero,

$$\begin{aligned}
E_\pm &\approx m_0(\boldsymbol{k}) - 2t_\perp + 2t_\perp \frac{(1-\xi_1^2/2)g^\pm(\xi_1) - (1-\xi_2^2/2)g^\pm(\xi_2)}{g^\pm(\xi_1) - g^\pm(\xi_2)} \\
&= m_0(\boldsymbol{k}) - t_\perp \frac{\xi_1^2 g^\pm(\xi_1) - \xi_2^2 g^\pm(\xi_2)}{g^\pm(\xi_1) - g^\pm(\xi_2)},
\end{aligned}$$

i.e., eigenenergy comes back to its form in continuum model. Meanwhile, it is essential that sites $i_z = \pm\dfrac{L_z+1}{2}$ are two fictitious points where boundary constraints take place, which fulfils the requirement of the lattice Hamiltonian, and the true lattice stops at $i_z = \pm\dfrac{L_z-1}{2}$ as we only have L_z sites. What is more, for compensation of unifying expression regardless of odevity of L_z, i_z would be forced to choose different ways to be taken out as follows

$$\begin{cases} i_z = 0, \pm1, \pm2, \cdots, \pm\dfrac{L_z+1}{2}, & \text{for } L_z \text{ odd}, \\ i_z = \pm\dfrac{1}{2}, \pm\dfrac{3}{2}, \cdots, \pm\dfrac{L_z+1}{2}, & \text{for } L_z \text{ even}, \end{cases} \tag{1.123}$$

which conforms mirror symmetry to $z = 0$.

To find higher energy bulk states from boundary condition, let $\lambda_\perp = 0$, so that the linear term, sometimes controlled by spin orbital coupling, which is viewed as a key factor in topological insulators[27, 4], is kicked out from the 1-D problem and all solved states here are viewed as trivial bulk states. Now the recurrence equation is reduced to

$$[t_\perp(\beta + \frac{1}{\beta}) + M]\tau_z\phi = E\phi,$$

with simple solutions as

$$E_\iota = \iota t_\perp(\beta + \frac{1}{\beta}) + M,\ \phi = \rho^\iota,\ \tau_z\rho^\iota = \iota\rho^\iota.$$

Now clearly ξ^p only have one superscript for its sign, and the general solution for some energy would become

$$\Psi = \sum_p C_p\rho e^{i\xi_p i_z},$$

then by similarly using boundary condition one gets $P'(C+, C_-)^T = 0$, which leads to

$$0 = \det(P') = \det\begin{pmatrix} e^{i\xi l/2} & e^{-i\xi l/2} \\ e^{-i\xi l/2} & e^{i\xi l/2} \end{pmatrix} = 2i\sin(\xi l).$$

And here we come to the situation as one simple quantum well, which is reflected in first part of Eq (1.120). This should be noted as it tells that our solution is unique, different and decoupled from purely normal quantum well states.

Let us come back to the solution. For the similar aim as to find eigenstates of H_{1d}, we could build several functions from β_α^p considering one-dimensional parity symmetry stated above. Denote

$$\begin{cases} E(\beta, i_z) = \dfrac{\beta^{i_z} + \beta^{-i_z}}{\beta^{(L_z+1)/2} + \beta^{-(L_z+1)/2}} = \dfrac{\cos(\xi i_z)}{\cos(\xi l/2)} \\ O(\beta, i_z) = \dfrac{\beta^{i_z} - \beta^{-i_z}}{\beta^{(L_z+1)/2} - \beta^{-(L_z+1)/2}} = \dfrac{\sin(\xi i_z)}{\sin(\xi l/2)} \end{cases}, \tag{1.124}$$

where 'E' and 'O', namely even and odd, represent the parity of two functions about z, from which we establish two sets of factors respecting boundary condition with even or odd parity

$$\begin{cases} f_+(i_z) = \sum_\alpha (-1)^{\alpha-1} E(\beta_\alpha^p, i_z) \\ f_-(i_z) = \sum_\alpha (-1)^{\alpha-1} O(\beta_\alpha^p, i_z) \end{cases}, \tag{1.125}$$

where the summation is over α without p since it only changes sign of ξ and thus does not influence the value of E or O. Before proceeding, let us find some special properties about those functions or factors. Let

$$\begin{cases} a = \beta + \dfrac{1}{\beta} = 2\cos\xi \\ b = \beta - \dfrac{1}{\beta} = 2i\sin\xi \end{cases}, \tag{1.126}$$

who weight as the *lattice differential operator*, which acts similarly to the partial operator in the continuum model, and notice that

$$\begin{cases} a_\alpha^p = 2\cos\xi_\alpha^p = 2\cos\xi_\alpha \equiv a_\alpha \\ b_\alpha^p = 2i\sin\xi_\alpha^p = 2ip\sin\xi_\alpha = pb_\alpha \end{cases}. \tag{1.127}$$

By left multiplying a or b on $E(\beta, i_z)$ or $O(\beta, i_z)$, one has

$$\begin{aligned}
aE(\beta, i_z) &= (\beta + \frac{1}{\beta})\frac{\beta^{i_z} + \beta^{-i_z}}{\beta^{(L_z+1)/2} + \beta^{-(L_z+1)/2}} = \frac{\beta^{i_z+1} + \beta^{-i_z-1} + \beta^{i_z-1} + \beta^{-i_z+1}}{\beta^{(L_z+1)/2} + \beta^{-(L_z+1)/2}} \\
&= E(\beta, i_z + 1) + E(\beta, i_z - 1), \\
bE(\beta, i_z) &= (\beta - \frac{1}{\beta})\frac{\beta^{i_z} + \beta^{-i_z}}{\beta^{(L_z+1)/2} + \beta^{-(L_z+1)/2}} = \frac{\beta^{i_z+1} - \beta^{-i_z-1} - \beta^{i_z-1} + \beta^{-i_z+1}}{\beta^{(L_z+1)/2} + \beta^{-(L_z+1)/2}} \\
&= [O(\beta, i_z + 1) - O(\beta, i_z - 1)]\frac{\beta^{l/2} - \beta^{-l/2}}{\beta^{l/2} + \beta^{-l/2}} \\
&= i[O(\beta, i_z + 1) - O(\beta, i_z - 1)] \tan(\xi l/2), \\
aO(\beta, i_z) &= O(\beta, i_z + 1) + O(\beta, i_z - 1), \\
bO(\beta, i_z) &= [E(\beta, i_z + 1) - E(\beta, i_z - 1)]\frac{\beta^{l/2} + \beta^{-l/2}}{\beta^{l/2} - \beta^{-l/2}} \\
&= -i[E(\beta, i_z + 1) - E(\beta, i_z - 1)] \cot(\xi l/2),
\end{aligned}$$

from which one has

$$\begin{cases} aE(\beta, i_z) \pm ib \tan(\xi l/2) O(\beta, i_z) = 2E(\beta, i_z \pm 1) \\ aO(\beta, i_z) \mp ib \cot(\xi l/2) E(\beta, i_z) = 2O(\beta, i_z \pm 1), \end{cases} \tag{1.128}$$

which leads to

$$\begin{aligned}
f_+(i_z \pm 1) &= \sum_\alpha (-1)^{\alpha-1} \frac{a_\alpha^p E(\beta_\alpha^p, i_z) \pm ib_\alpha^p \tan(\xi_\alpha^p l/2) O(\beta_\alpha^p, i_z)}{2} \\
&= \sum_\alpha (-1)^{\alpha-1} \frac{a_\alpha E(\beta_\alpha, i_z) \pm ib_\alpha \tan(\xi_\alpha l/2) O(\beta_\alpha, i_z)}{2} \equiv g_\pm, \qquad (1.129a) \\
f_-(i_z \pm 1) &= \sum_\alpha (-1)^{\alpha-1} \frac{a_\alpha^p O(\beta_\alpha^p, i_z) \mp ib_\alpha^p \cot(\xi_\alpha^p l/2) E(\beta_\alpha^p, i_z)}{2} \\
&= \sum_\alpha (-1)^{\alpha-1} \frac{a_\alpha O(\beta_\alpha, i_z) \mp ib_\alpha \cot(\xi_\alpha l/2) E(\beta_\alpha, i_z)}{2} \equiv h_\pm, \qquad (1.129b)
\end{aligned}$$

from which one could see that the iteration relation is also independent of p within our expectation.

Now we are able to come back and solve the chain problem. Let

$$\phi_{i_z} = cf_+(i_z) + df_-(i_z). \tag{1.130}$$

Bring this trial solution into Eq (1.110) and one has

$$\begin{aligned}
&\begin{pmatrix} t_\perp & i\frac{\lambda_\perp}{2}s \\ i\frac{\lambda_\perp}{2}s & -t_\perp \end{pmatrix} \begin{pmatrix} c_1 g_- + d_1 h_- \\ c_2 g_- + d_2 h_- \end{pmatrix} + \begin{pmatrix} M - E & \\ & -M - E \end{pmatrix} \begin{pmatrix} c_1 f_+ + d_1 f_- \\ c_2 f_+ + d_2 f_- \end{pmatrix} \\
&+ \begin{pmatrix} t_\perp & -i\frac{\lambda_\perp}{2}s \\ -i\frac{\lambda_\perp}{2}s & -t_\perp \end{pmatrix} \begin{pmatrix} c_1 g_+ + d_1 h_+ \\ c_2 g_+ + d_2 h_+ \end{pmatrix} = 0.
\end{aligned} \tag{1.131}$$

Where the first line would give

$$
\begin{aligned}
0 &= t_\perp[c_1(g_- + g_+) + d_1(h_- + h_+)] + i\frac{\lambda_\perp}{2}s[c_2(g_- - g_+) + d_2(h_- - h_+)] \\
&\quad + (M-E)(c_1 f_+ + d_1 f_-) \\
&= \sum_\alpha (-1)^{\alpha-1}\left\{ t_\perp\left[c_1\left(\frac{aE - ibtO}{2} + \frac{aE + ibtO}{2}\right) + d_1\left(\frac{aO + ibcE}{2} + \frac{aO - ibcE}{2}\right)\right]\right. \\
&\quad + (M-E)(c_1 E + d_1 O) \\
&\quad \left. + i\frac{\lambda_\perp}{2}s\left[c_2\left(\frac{aE - ibtO}{2} - \frac{aE + ibtO}{2}\right) + d_2\left(\frac{aO + ibcE}{2} - \frac{aO - ibcE}{2}\right)\right]\right\} \\
&= \sum_\alpha (-1)^{\alpha-1}\left\{ E(\beta_\alpha, i_z)\left[t_\perp c_1 a_\alpha + (M-E)c_1 - \frac{\lambda_\perp}{2}s d_2 b_\alpha \cot(\xi_\alpha l/2)\right]\right. \\
&\qquad \left. + O(\beta_\alpha, i_z)\left[t_\perp d_1 a_\alpha + (M-E)d_1 + \frac{\lambda_\perp}{2}s c_2 b_\alpha \tan(\xi_\alpha l/2)\right]\right\},
\end{aligned}
$$

and similarly, the second line would give

$$
\begin{aligned}
0 = \sum_\alpha (-1)^{\alpha-1}&\left\{ E(\beta_\alpha, i_z)\left[t_\perp c_2 a_\alpha + (M+E)c_2 + \frac{\lambda_\perp}{2}s d_1 b_\alpha \cot(\xi_\alpha l/2)\right]\right. \\
&\left. + O(\beta_\alpha, i_z)\left[t_\perp d_2 a_\alpha + (M+E)d_2 - \frac{\lambda_\perp}{2}s c_1 b_\alpha \tan(\xi_\alpha l/2)\right]\right\}.
\end{aligned}
$$

Thus, by requiring vanishing coefficients of $E(\beta_\alpha, i_z)$ and $O(\beta_\alpha, i_z)$, one obtains, after re-organization,

$$
\begin{cases}
(M - E + t_\perp a_\alpha)c_1 - \dfrac{\lambda_\perp}{2}s d_2 b_\alpha \cot(\xi_\alpha l/2) = 0 \\
-\dfrac{\lambda_\perp}{2}s c_1 b_\alpha \tan(\xi_\alpha l/2) + (M + E + t_\perp a_\alpha)d_2 = 0
\end{cases}, \tag{1.132a}
$$

$$
\begin{cases}
(M - E + t_\perp a_\alpha)d_1 + \dfrac{\lambda_\perp}{2}s c_2 b_\alpha \tan(\xi_\alpha l/2) = 0 \\
\dfrac{\lambda_\perp}{2}s d_1 b_\alpha \cot(\xi_\alpha l/2) + (M + E + t_\perp a_\alpha)c_2 = 0
\end{cases}, \tag{1.132b}
$$

for different α. Then considering different α one gains four solutions as pairs

$$
\begin{cases}
d_2 = \dfrac{i t_\perp \eta_1}{s\lambda_\perp} c_1, \ E = E_+ \\
c_1 = \dfrac{i t_\perp \eta_2}{s\lambda_\perp} d_2, \ E = -E_-
\end{cases}, \tag{1.133a}
$$

$$
\begin{cases}
c_2 = -\dfrac{i t_\perp \eta_2}{s\lambda_\perp} d_1, \ E = E_- \\
d_1 = -\dfrac{i t_\perp \eta_1}{s\lambda_\perp} c_2, \ E = -E_+
\end{cases}, \tag{1.133b}
$$

where two dimensionless factors

$$
\begin{aligned}
\eta_1 &= -2i\frac{a_1 - a_2}{b_1 \cot(\xi_1 l/2) - b_2 \cot(\xi_2 l/2)} = \frac{-2(\cos\xi_1 - \cos\xi_2)}{\sin\xi_1 \cot(\xi_1 l/2) - \sin\xi_2 \cot(\xi_2 l/2)}, \\
\eta_2 &= -2i\frac{a_1 - a_2}{b_1 \tan(\xi_1 l/2) - b_2 \tan(\xi_2 l/2)} = \frac{-2(\cos\xi_1 - \cos\xi_2)}{\sin\xi_1 \tan(\xi_1 l/2) - \sin\xi_2 \tan(\xi_2 l/2)}
\end{aligned} \tag{1.134}
$$

are introduced, and the additional sign could make sure that when ξ_α approach zero,

$$\eta_1 \to \frac{\xi_1^2 - \xi_2^2}{\xi_1 \cot(\xi_1 l/2) - \xi_2 \cot(\xi_2 l/2)},\ \eta_2 \to \frac{\xi_1^2 - \xi_2^2}{\xi_1 \tan(\xi_1 l/2) - \xi_2 \tan(\xi_2 l/2)},$$

again as the outcome in continuum model. Also, $E_\pm$ appeared here is just energy obtained in Eq (1.122). For the same reason as in the continuum model, one has

$$1 = \left|\frac{it_\perp \eta_1}{s\lambda_\perp} \cdot \frac{it_\perp \eta_2}{s\lambda_\perp}\right| \Longrightarrow |\eta_1 \eta_2| = \frac{\lambda_\perp^2}{t_\perp^2}, \tag{1.135a}$$

$$E_+ = -E_- \tag{1.135b}$$

With such belief, we could formally read from Eq. (1.132)

$$E^2 = (M + t_\perp a_\alpha)^2 - \lambda_\perp^2 b_\alpha^2/4 = (M + 2t_\perp \cos\xi_\alpha)^2 + \lambda_\perp^2 \sin^2 \xi_\alpha,$$

which is independent of α and matches the result of Eq (1.113). Similar property as whether the functions and coefficients defined here could be discussed in a similar way as in continuum model:

Case 1. When $\cos\xi$ is real, and

Subcase 1.1. $0 < |\cos\xi| < 1$, then $\sin\xi$ is also real and ξ is a true angle, then $\tan(\xi(L_z+1)/2)$ is real, so does $g(\xi)$, $f_\pm$ and η.

Subcase 1.2. $|\cos\xi| > 1$, now since $2\cos\xi = \beta + 1/\beta$, one has $\beta = \cos\xi \pm \sqrt{\cos^2\xi - 1}$ is real, and it is easy to see that $2\cos(\xi t) = \beta^t + \beta^{-t}$ is also real while $2\sin(\xi t) = -i(\beta^t - \beta^{-t})$ becomes imaginary; however, since $\sin(\xi t)$ come in pairs in $g(\xi)$, $f_\pm$ and η, they are still real.

Case 2. When $\cos\xi$ is complex with $\operatorname{Im}\cos\xi_1 = -\operatorname{Im}\cos\xi_2$, one is able to choose correspondingly $\xi_2 = \xi_1^*$, since when $\xi_1 = x + iy$, one has

$$\cos(\xi_1 t) = \frac{e^{i\xi_1 t} + e^{-i\xi_1 t}}{2} = \frac{e^{ixt}e^{-yt} + e^{-ixt}e^{yt}}{2} = \cos(xt)\cosh(yt) - i\sin(xt)\sinh(yt),$$
$$\cos(\xi_2 t) = \frac{e^{i\xi_2 t} + e^{-i\xi_2 t}}{2} = \frac{e^{ixt}e^{yt} + e^{-ixt}e^{-yt}}{2} = \cos(xt)\cosh(yt) + i\sin(xt)\sinh(yt),$$

and the relation for $\cos\xi$ is recovered when $t = 1$ (we ignore the case $\xi_2 = -x+iy = -(x-iy)$, changed by inverse choice of p). The equations above also lead to the conclusion that $f_+(l_z) = \dfrac{\cos(\xi_1 l_z)}{\cos(\xi_1 l/2)} - \dfrac{\cos(\xi_2 l_z)}{\cos(\xi_2 l/2)}$ would have its real part cancelled and become purely imaginary. The same argument is true for $f_-(l_z)$. Then any quadratic form of f would give a real factor. As for η, one notices that both numerator and denominator would only have imaginary part left, and their division would give a real coefficient.

In short, what we need solving to get all energy states m is the simultaneous equations below

$$m = M + 2t_\perp \frac{\cos\xi_1 g(\xi_1) - \cos\xi_2 g(\xi_2)}{g(\xi_1) - g(\xi_2)}, \tag{1.136a}$$

$$\cos\xi_\alpha = \frac{-Mt_\perp + (-1)^{\alpha-1}\sqrt{M^2t_\perp^2 - (t_\perp^2 - \lambda_\perp^2/4)(M^2 + \lambda_\perp^2 - m^2)}}{2(t_\perp^2 - \lambda_\perp^2/4)}, \tag{1.136b}$$

where

$$\begin{cases} M = M_0(\boldsymbol{k}) = m_0(\boldsymbol{k}) - 2t_\perp = m_0 - 4t_\parallel\left(\sin^2\frac{k_xa}{2} + \sin^2\frac{k_yb}{2}\right) - 2t_\perp, \\ g(\xi) = \dfrac{\tan(\xi(L_z+1))/2}{\sin\xi}, \end{cases} \tag{1.137}$$

and sign of ξ is fixed by $p = +$ so that

$$\sin\xi_\alpha = \sqrt{1 - \cos\xi_\alpha^2},\ \alpha = 1, 2. \tag{1.138}$$

Notice that we have re-defined m as eigenvalues of H_{1d} here, meaning mass term whose physical implication would be shown later. Basically, there are three variables ξ_1, ξ_2 and m, together with three equations above, then it is in a sense some *exact system of equations* but a non-linear transcendental version. From this set of equations, one may expect L_z solutions $m_n(\boldsymbol{k}), n = 1, 2, \cdots, L_z$ including one surface state and $L_z - 1$ purely trivial bulk states. And the other set of L_z solutions are just chiral partners with $-m_n(\boldsymbol{k})$. And $s = \pm$ indicates that there are in fact $4L_z$ solutions in total.

Here it comes to construct basis for projection, and we choose

$$\begin{cases} \varphi(s, i_z) = \begin{pmatrix} c_1 f_+ \\ d_2 f_- \end{pmatrix} = C\begin{pmatrix} -is\lambda_\perp f_+(i_z) \\ t_\perp\eta f_-(i_z) \end{pmatrix},\ E = m \\ \chi(s, i_z) = \begin{pmatrix} d_1 f_- \\ c_2 f_+ \end{pmatrix} = C\begin{pmatrix} t_\perp\eta f_-(i_z) \\ is\lambda_\perp f_+(i_z) \end{pmatrix},\ E = -m \end{cases},$$

where we have ignored lower index of η_1, and the norm is defined by

$$1 = \sum_{i_z}|\varphi(s, i_z)|^2 = |C|^2\sum_{i_z}(\lambda_\perp^2|f_+(i_z)|^2 + t_\perp^2|\eta|^2|f_-(i_z)|^2) = \sum_{i_z}|\chi(s, i_z)|^2.$$

Clearly, the two block basis functions share the same C. Then we have four basis in certain sequence as

$$\Phi_1 = \begin{pmatrix} \varphi(+) \\ 0 \end{pmatrix} = C\begin{pmatrix} -i\lambda_\perp f_+ \\ t_\perp\eta f_- \\ 0 \\ 0 \end{pmatrix},\ \Phi_2 = \begin{pmatrix} 0 \\ \chi(-) \end{pmatrix} = C\begin{pmatrix} 0 \\ 0 \\ t_\perp\eta f_- \\ -i\lambda_\perp f_+ \end{pmatrix},$$
$$\Phi_3 = \begin{pmatrix} \chi(+) \\ 0 \end{pmatrix} = C\begin{pmatrix} t_\perp\eta f_- \\ i\lambda_\perp f_+ \\ 0 \\ 0 \end{pmatrix},\ \Phi_4 = \begin{pmatrix} 0 \\ \varphi(-) \end{pmatrix} = C\begin{pmatrix} 0 \\ 0 \\ i\lambda_\perp f_+ \\ t_\perp\eta f_- \end{pmatrix}.$$

Now we turn to the projection, which is formally

$$\langle\Phi|H_{\text{Film}}|\Phi\rangle = \langle\Phi|H_{1d}|\Phi\rangle + \langle\Phi|H_\parallel|\Phi\rangle + \langle\Phi|V_{Zee}|\Phi\rangle,$$

where the first part, by eigenfunctions we got, is just $diag(m, -m, -m, m)$; while in the second part, since $H_\parallel = \lambda_\parallel(\sin(k_xa)\sigma_x \otimes \tau_x + \sin(k_yb)\sigma_y \otimes \tau_x)$ is purely off diagonal, it is easy to conclude that

$$\langle\Phi_i|H_\parallel|\Phi_i\rangle = 0,\ i = 1, 2, 3, 4$$
$$\langle\Phi_1|H_\parallel|\Phi_3\rangle = 0 = \langle\Phi_2|H_P|\Phi_4\rangle.$$

Then only four terms need consideration by hermicity, among which

$$\begin{aligned}
\langle\Phi_1|H_\parallel|\Phi_4\rangle &= \sum_{i_z}\Phi_1^\dagger H_\parallel\Phi_4 = \sum_{i_z}\lambda_\parallel(\sin(k_xa)-i\sin(k_yb))\varphi^\dagger(+)\tau_x\varphi(-)\\
&= \lambda_\parallel(\sin(k_xa)-i\sin(k_yb))\sum_{i_z}|C|^2i\lambda_\perp t_\perp\eta(f_+^*f_- + f_-^*f_+) = 0,\\
\langle\Phi_2|H_\parallel|\Phi_3\rangle &= \sum_{i_z}\Phi_2^\dagger H_\parallel\Phi_3 = \sum_{i_z}\lambda_\parallel(\sin(k_xa)-i\sin(k_yb))\chi^\dagger(+)\tau_x\chi(-)\\
&= \lambda_\parallel(\sin(k_xa)-i\sin(k_yb))\sum_{i_z}|C|^2i\lambda_\perp t_\perp\eta(f_-^*f_+ + f_+^*f_-) = 0,
\end{aligned}$$

as f_-f_+ is odd to z. Then, the only remaining terms are

$$\begin{aligned}
\langle\Phi_1|H_\parallel|\Phi_2\rangle &= \sum_{i_z}\Phi_1^\dagger H_\parallel\Phi_2 = \sum_{i_z}\lambda_\parallel(\sin(k_xa)-i\sin(k_yb))\varphi(+)^\dagger\tau_x\chi(-)\\
&= \lambda_\parallel(\sin(k_xa)-i\sin(k_yb))\sum_{i_z}|C|^2(t_\perp^2|\eta|^2|f_-|^2+\lambda_\perp^2|f_+|^2)\\
&= \lambda_\parallel(\sin(k_xa)-i\sin(k_yb)),\\
\langle\Phi_3|H_\parallel|\Phi_4\rangle &= \sum_{i_z}\Phi_3^\dagger H_\parallel\Phi_4 = \sum_{i_z}\lambda_\parallel(\sin(k_xa)-i\sin(k_yb))\chi(+)^\dagger\tau_x\varphi(-)\\
&= \lambda_\parallel(\sin(k_xa)-i\sin(k_yb))\sum_{i_z}|C|^2(\lambda_\perp^2|f_+|^2+t_\perp^2|\eta|^2|f_-|^2)\\
&= \lambda_\parallel(\sin(k_xa)-i\sin(k_yb)),
\end{aligned}$$

where normalization condition is used. And finally we arrive at the effective four-band Hamiltonian

$$\begin{aligned}
H_{\text{EFF}} &= \begin{pmatrix}
m & \lambda_\parallel(\sin(k_xa) & & \\
 & -i\sin(k_yb)) & & \\
\lambda_\parallel(\sin(k_xa) & -m & & \\
-i\sin(k_yb)) & & & \\
 & & -m & \lambda_\parallel(\sin(k_xa) \\
 & & & -i\sin(k_yb)) \\
 & & \lambda_\parallel(\sin(k_xa) & m \\
 & & -i\sin(k_yb)) &
\end{pmatrix}\\
&= \lambda_\parallel\tau_0(\sin(k_xa)\sigma_x+\sin(k_yb)\sigma_y)+m\tau_z\sigma_z.
\end{aligned} \tag{1.139}$$

Now by counting all m_n, we obtain the effective Hamiltonian

$$H_{\text{EFF}}(\boldsymbol{k}) = \bigoplus_{n=1}^{L_z}\left[\lambda_\parallel(\sin(k_xa)\sigma_x+\sin(k_yb)\sigma_y)+m_n\tau_z\sigma_z\right] = \bigoplus_{n,\chi}h_{n,\chi}(\boldsymbol{k}), \tag{1.140}$$

where every

$$h_{n,\chi} = \lambda_\parallel(\sin(k_xa)\sigma_x+\sin(k_yb)\sigma_y)+\chi m_n\sigma_z, \tag{1.141}$$

is a Dirac's fermion with χ the eigenvalue of τ_z, which would turn out to be the projected mirror operator as we will see later. Notice that our effective model contains n blocks and $2n$ sub-blocks, where each sub-block is a Dirac Fermion with k-dependent mass. Also

notice here that our effective model is equivalent to the original one, since by counting all $4L_z$ states, what we have done in the projection is just a unitary transformation that switches basis. Our effective model is exact in the sense that it is solved based on lattice model, containing all bands branches and suitable for all k values, at either low or high energy zones (by the latter I mean the zone away from Dirac point/mass gap). And if one has to say some difference, the only loss in the matrix form of effective Hamiltonian is the z-information, which, however, has been encoded in the new basis. As a general comment, the effective model is powerful in discovering and recognizing physics related to Dirac's mass term, while when it comes to physical quantity combined more closely to layer dependent property, it might be better to directly exploit on the original lattice model (but it would still be insightful to borrow Dirac's physics within each branch).

A rough group theoretical view may explain why we obtain those sub-blocks. Viewing $SO(1,3) \simeq SU(2) \otimes SU(2)$ with choosing representation dimension $2s+1 = 2$ ($s = 1/2$) for each $SU(2)$ group, together with irreducible decomposition $\mathbf{2} \otimes \mathbf{2} = \mathbf{2} \oplus \mathbf{2}$, the L_z layers of topological insulator film leaves us with tensor product space $SO(1,3) \otimes \mathbb{I}_{L_z} \to [\mathbf{2} \oplus \mathbf{2}] \otimes \mathbb{I}_{L_z}$, which gives us the sub-block diagonalized form.

To show the validity of the effective model, we calculated m_n to compare with eigenvalues of H_{1d} as shown in Fig. 1.3(a) (b), and they show perfect correspondence, which is expected since our solution is exact (notice however that eigenvalues of H^s_{1d} contain both $\pm m_n$). What is more, m_n shows a jumping behavior between positive and negative. The unusual behavior of m_2 near $k = 0$ reflect its close relation with zero-energy mode m_1 within our parameter choice.

As one more comment, our derivation could be generalized to arbitrary d-dimensional bulk with $(d-1)$-dimensional surface, and is not confined by topological triviality of the system. Say we have

$$H(\boldsymbol{k}) = \sum_{\mu} \lambda_\mu \sin(k_\mu a_\mu) \alpha_\mu + \left[m_0 - 4t_\mu \sin^2 \left(\frac{k_\mu a_\mu}{2} \right) \right] \beta, \tag{1.142}$$

with $\mu = 1, 2, \cdots, d$ as spatial dimension, which provides corresponding film Hamiltonian

$$\begin{aligned} \mathscr{H}_{\text{Film}} = & \sum_{i_d, \boldsymbol{k}} \left(\Psi^\dagger_{i_d, \boldsymbol{k}} \mathscr{M}_0(\boldsymbol{k}) \Psi_{i_d, \boldsymbol{k}} + \Psi^\dagger_{i_d, \boldsymbol{k}} \mathscr{T}_d \Psi_{i_d+1, \boldsymbol{k}} + \Psi^\dagger_{i_d+1, \boldsymbol{k}} \mathscr{T}_d^\dagger \Psi_{i_d, \boldsymbol{k}} \right) \\ & + \sum_{i_d, \boldsymbol{k}} \Psi^\dagger_{i_d, \boldsymbol{k}} H_\parallel \Psi_{i_d, \boldsymbol{k}}, \end{aligned} \tag{1.143}$$

where we assume the film is perpendicular to d-direction, and $\boldsymbol{k}$ in $(d-1)$ super-plane reads $\boldsymbol{k} = (k_1, k_2, \cdots, k_{d-1})$. Then following procedures above, we solve one-dimensional problem along d-direction and projecting the Hamiltonian with the solved eigenstates, which equals a k-dependent unitary transformation, we would get the equivalent Hamiltonian with Dirac fermions. Notice that when reduced to the one-dimensional problem, by counting number of Gamma matrices, we find that there are always enough choice for us to reduce the equation into 2×2 sub-block-wised form, which then serves as the to be solved sub-block labelled by a series of indices.

1.3 Semi-infinite Geometry

Half-quantized Hall conductance was proposed earlier on the surface of strong topological insulator[**?**], i.e. considering semi-infinite geometry along z-direction, where single

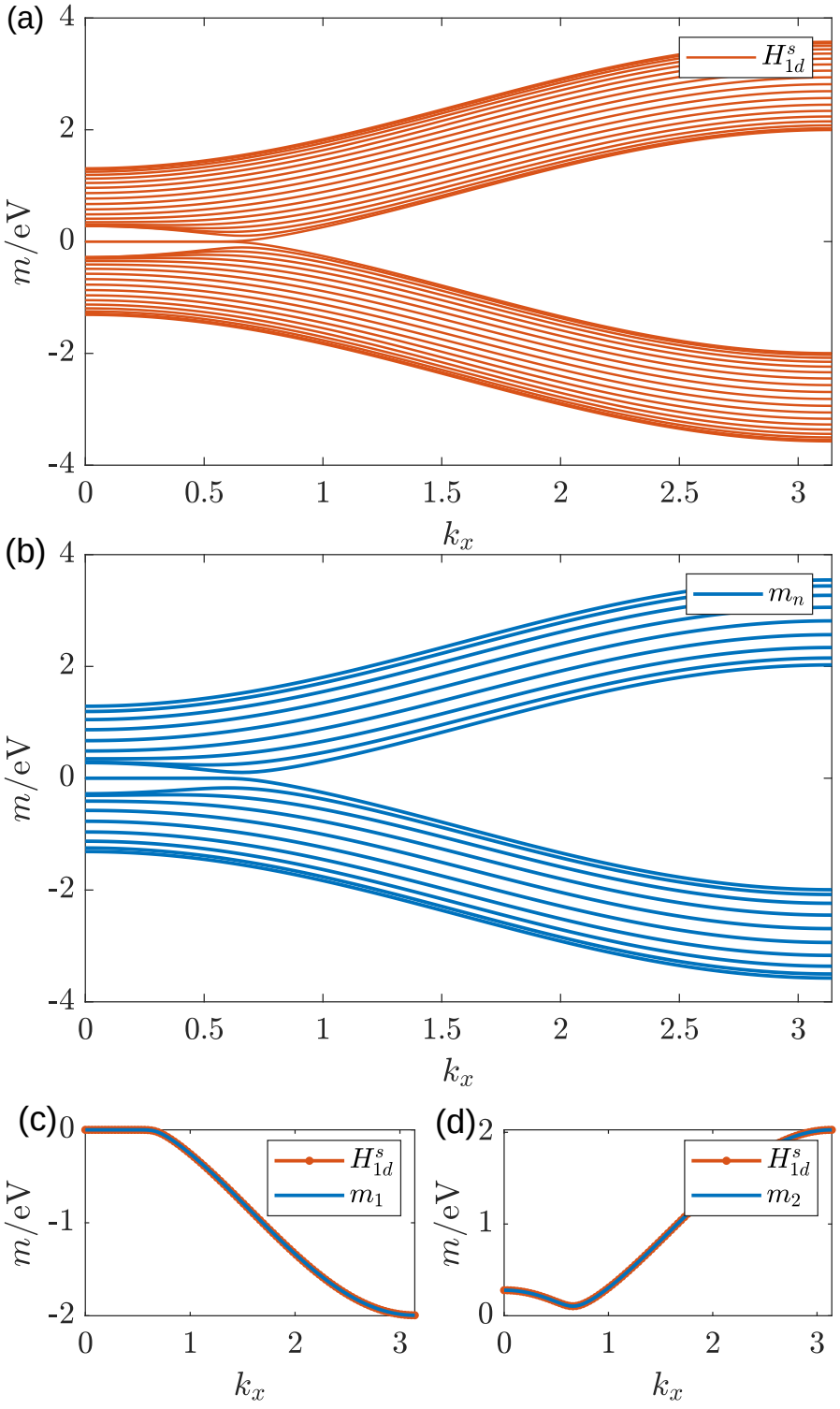


Figure 1.3: (a) Eigenvalues of H^s_{1d}. (b) m_n calculated from our equations. (c), (d) Compare between $m_{1,2}$ and corresponding eigenvalues of H^s_{1d}.

Dirac cone with half quantum conductance is expected resulted from so-called quantum anomaly in continuum model. Usually, quantum anomaly appears in quantum field theory as the failure of a symmetry of a theory's classical action to be a symmetry of any regularization of the full quantum theory[?], bounded to loop diagrams where ultraviolet divergence emerges where physics is unknown. In condensed matter, however, physics we care defined on lattice model as interaction among bounded states from the point-view of tight-binding naturally owns high-energy cut-off given by finite lattice parameter, and whether this term-analogy hosts may depend.

Given finite geometry with exact solution, we would also like to proceed and bring out the single Dirac cone together with its high-energy regularization in a somehow strict manner. The way to make it, however, is not that straightforward.

Before solving the problem, we note that general $E-\xi$ relation is inherited from above discussion as

$$\begin{aligned} E^2 &= (M+2t_\perp\cos\xi)^2+\lambda_\perp^2\sin^2\xi, \\ \cos\xi_\alpha^p &= \frac{-Mt_\perp+(-1)^{\alpha-1}\sqrt{M^2t_\perp^2-(t_\perp^2-\lambda_\perp^2/4)(M^2+\lambda_\perp^2-E^2)}}{2(t_\perp^2-\lambda_\perp^2/4)}, \\ \sin\xi_\alpha^p &= p\sqrt{1-\cos^2\xi_\alpha},\ p=\pm,\alpha=1,2. \end{aligned} \tag{1.144}$$

Then to our consideration, and to make life easier, we take our lattice to start from $i_z=-1$ to $-\infty$, and boundary condition stands as wavefunctions vanish at $i_z=0$ and $-\infty$.

The question firstly looks simpler since one would simply think that boundary condition at infinity may enormously save us from counting all possible function forms and help to reduce the problem into Jackie-Rebbi' zero-mode frame. However, that methodology works due to the explicit existence of 1-D chiral symmetry inside surface-state zone; the existence of high energy part, where 1-D solution should have evolved into bulk, on the other hand, would exactly break chiral symmetry. And if one insists solving the problem in this manner, one would get determinant from $\Phi(i_z=0)=0$ as

$$\lambda_1 m_2+\lambda_2 m_1=0,$$

where $\lambda_\alpha=\lambda_\perp\sin\xi_\alpha,\ m_\alpha=E-M-2t_\perp\cos\xi_\alpha$. One may deal this with low-energy zone by assuming $E=0$, but is less likely to obtain similar sight when k becomes large.

On the other hand, we may treat the problem in a limit-taken manner: firstly we assume finite geometry, where the top surface is put at $i_z=1$, equivalent to shifting our solution above down by $l/2$; then we take the limit $l=L_z+1\to+\infty$. For the first step, we need to shift all appeared functions as

$$f(i_z)\to f'(i_z)=f(i_z+l/2). \tag{1.145}$$

This operation leads to the re-definition of our even/odd element:

$$\begin{cases} E(\beta,i_z)=\dfrac{\cos[\xi(i_z+l/2)]}{\cos(\xi l/2)} \\ O(\beta,i_z)=\dfrac{\sin[\xi(i_z+l/2)]}{\sin(\xi l/2)} \end{cases}, \tag{1.146}$$

remembering that $\beta=e^{i\xi}$ as defined from Euler's formula. Similarly, we have basic

functions

$$\begin{cases} f_+(i_z) = \dfrac{\cos[\xi_1(i_z+l/2)]}{\cos(\xi_1 l/2)} - \dfrac{\cos[\xi_2(i_z+l/2)]}{\cos(\xi_2 l/2)} \\ f_-(i_z) = \dfrac{\sin[\xi_2(i_z+l/2)]}{\sin(\xi_2 l/2)} - \dfrac{\sin[\xi_2(i_z+l/2)]}{\sin_2(\xi l/2)} \end{cases}, \tag{1.147}$$

which manifestly satisfy boundary condition as

$$f_\pm(i_z = 0, -l) = 0. \tag{1.148}$$

Then accordingly, basis solutions continue to exist as

$$\begin{cases} \varphi(s, i_z) = C \begin{pmatrix} -is\lambda_\perp f_+(i_z) \\ t_\perp \eta f_-(i_z) \end{pmatrix}, \ E = m \\ \chi(s, i_z) = C \begin{pmatrix} t_\perp \eta f_-(i_z) \\ is\lambda_\perp f_+(i_z) \end{pmatrix}, \ E = -m \end{cases}, \ s = \pm,$$

with the same m defined above. Then we choose basis as

$$\Phi_1 = \begin{pmatrix} \varphi(+) \\ 0 \end{pmatrix}, \ \Phi_2 = \begin{pmatrix} 0 \\ \chi(-) \end{pmatrix},$$
$$\Phi_3 = \begin{pmatrix} \chi(+) \\ 0 \end{pmatrix}, \ \Phi_4 = \begin{pmatrix} 0 \\ \varphi(-) \end{pmatrix},$$

and project the Hamiltonian into

$$H_{\text{EFF}} = \lambda_\parallel(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + m\tau_z\sigma_z.$$

For the second step, we take the limit $l \to +\infty$. At this time, we may have to borrow some gained picture like Fig. **??** to provide some qualitative information. Notice that for ξ_1, $\operatorname{Im}\xi_1$ changes its sign from minus to positive after k_c, while for ξ_2, it is complex-conjugate to ξ_1 while becomes nearly π/l after k_c as shown in the figure. This feature hints that we have to separate cases for discussion.

When $k < k_c$, $\xi_1 = \xi_2^*$ with $\operatorname{Im}\xi_1 < 0 < \operatorname{Im}\xi_2$, this directly gives us the limit (lim sign omitted)

$$\begin{cases} E(\beta_1, i_z) = E(\beta_1, i_z) = e^{i\xi_1 i_z} \\ E(\beta_2, i_z) = E(\beta_2, i_z) = e^{-i\xi_2 i_z} \end{cases}, \tag{1.149}$$

considering that

$$E(\beta, i_z) = \frac{e^{i\xi(i_z+l/2)} + e^{-i\xi(i_z+l/2)}}{e^{i\xi l/2} + e^{-i\xi l/2}}, \ O(\beta, i_z) = \frac{e^{i\xi(i_z+l/2)} - e^{-i\xi(i_z+l/2)}}{e^{i\xi l/2} - e^{-i\xi l/2}}.$$

Note that the limiting form naturally conserves convergence when $i_z \to -\infty$. Then our basic functions become

$$f_\pm(i_z) = e^{i\xi_1 i_z} - e^{-i\xi_2 i_z}, \tag{1.150}$$

respecting boundary condition inherited. As for η, notice that

$$\cot \xi l/2 = i\frac{e^{i\xi l/2} + e^{-i\xi l/2}}{e^{i\xi l/2} - e^{-i\xi l/2}},$$

and we have

$$\cot \xi_1 l/2 = i = -\cot \xi_2 l/2, \tag{1.151}$$

which gives

$$\eta = \frac{-2(\cos\xi_1 - \cos\xi_2)}{i(\sin\xi_1 + \sin\xi_2)} = -2i \tan\frac{\xi_1 - \xi_2}{2}.$$

These reductions turn our basis wavefunctions into

$$\begin{cases} \varphi(s, i_z) = C \begin{pmatrix} s\lambda_\perp \\ 2t_\perp \tan\frac{\xi_1-\xi_2}{2} \end{pmatrix} \left(e^{i\xi_1 i_z} - e^{-i\xi_2 i_z}\right), \\ \chi(s, i_z) = C \begin{pmatrix} 2t_\perp \tan\frac{\xi_1-\xi_2}{2} \\ -s\lambda_\perp \end{pmatrix} \left(e^{i\xi_1 i_z} - e^{-i\xi_2 i_z}\right), \end{cases}, \ m(k < k_c) = 0. \tag{1.152}$$

And we could see that, from result in 'thick limit',

$$\operatorname{Im}\xi_1(k < k_c) = -\frac{\lambda_\perp}{2t_\perp}, \tag{1.153}$$

which makes it possible to treat

$$\tan\frac{\xi_1 - \xi_2}{2} = i \tanh\frac{\lambda_\perp}{2t_\perp} \approx i\frac{\lambda_\perp}{2t_\perp},$$

which gives an approximate basis

$$\begin{cases} \varphi(s, i_z) = C \begin{pmatrix} s \\ i \end{pmatrix} \left(e^{i\xi_1 i_z} - e^{-i\xi_2 i_z}\right), \\ \chi(s, i_z) = C \begin{pmatrix} i \\ -s \end{pmatrix} \left(e^{i\xi_1 i_z} - e^{-i\xi_2 i_z}\right), \end{cases}, \ m(k < k_c) = 0,$$

which could also be obtained solving continuum model by assuming zero-energy.

When $k > k_c$, we see that $\operatorname{Im}\xi_1 > 0$, $\operatorname{Re}\xi_1 = 0 = \operatorname{Im}\xi_2$, while approximately $\xi_2 \approx \pi/l$. Note that if we directly bring π/l into functions above, we would see divergence, and to treat it properly, we assume that $\xi_2 = \pi/(l + \Delta)$ which describes asymptotic behavior but a lower value. Note that Δ represents order comparable with l but suppressed, say $O(\ln l)$ like. ξ_1 part is still easy to handle but with a jumping choice

$$E(\xi_1, i_z) = O(\xi_1, i_z) = e^{-i\xi_1 i_z}, \tag{1.154}$$

for compensation of sign jump of $\operatorname{Im}\xi_1$. For ξ_2, it is also easy to verify that

$$O(\xi_2, i_z) = \cos\frac{\pi}{l} z = 1, \tag{1.155}$$

since we are considering $l \to +\infty$. And for its even component, we may have to let

$$\begin{aligned} E(\xi_2, i_z) &= \frac{e^{i\frac{\pi}{l+\Delta} i_z} e^{i\frac{\pi}{2}\frac{l}{l+\Delta}} + e^{-i\frac{\pi}{l+\Delta} i_z} e^{-i\frac{\pi}{2}\frac{l}{l+\Delta}}}{e^{i\frac{\pi}{2}\frac{l}{l+\Delta}} + e^{-i\frac{\pi}{2}\frac{l}{l+\Delta}}} \\ &= \frac{e^{i\frac{\pi}{l+\Delta} i_z} e^{i\frac{\pi}{2}\left(1-\frac{\Delta}{l}\right)} + e^{-i\frac{\pi}{l+\Delta} i_z} e^{-i\frac{\pi}{2}\left(1-\frac{\Delta}{l}\right)}}{e^{i\frac{\pi}{2}\left(1-\frac{\Delta}{l}\right)} + e^{-i\frac{\pi}{2}\left(1-\frac{\Delta}{l}\right)}} \end{aligned}$$

$$= \frac{ie^{i\frac{\pi}{l+\Delta}i_z}e^{-i\frac{\pi}{2}\frac{\Delta}{l}} - ie^{-i\frac{\pi}{l+\Delta}i_z}e^{i\frac{\pi}{2}\frac{\Delta}{l}}}{ie^{-i\frac{\pi}{2}\frac{\Delta}{l}} - ie^{i\frac{\pi}{2}\frac{\Delta}{l}}}$$
$$= \frac{-2\sin\left(\frac{\pi}{l+\Delta}i_z - \frac{\pi}{2}\frac{\Delta}{l}\right)}{2\sin\left(\frac{\pi}{2}\frac{\Delta}{l}\right)} = 1,$$

by recognizing that physics we care satisfies $i_z \ll \Delta$. After these, we again obtain

$$f_\pm(i_z) = e^{-i\xi_1 i_z} - 1. \tag{1.156}$$

And for η, we see

$$\cot \xi_1 l/2 = -i, \;\; \cot \xi_2 l/2 = 0,$$

and

$$\eta = \frac{-2(\cos\xi_1 - 1)}{-i\sin\xi_1} = -2i\tan\frac{\xi_1}{2}, \tag{1.157}$$

which maps result above by $\xi_2 \to 0$. Then for $k > k_c$ we could also write down basis wavefunctions as

$$\begin{cases} \varphi(s, i_z) = C \begin{pmatrix} s\lambda_\perp \\ 2t_\perp \tan\frac{\xi_1}{2} \end{pmatrix} \left(e^{-i\xi_1 i_z} - 1\right), \; E = m \\ \chi(s, i_z) = C \begin{pmatrix} 2t_\perp \tan\frac{\xi_1}{2} \\ -s\lambda_\perp \end{pmatrix} \left(e^{-i\xi_1 i_z} - 1\right), \; E = -m \end{cases}, \; m(k > k_c) = m_0(k). \tag{1.158}$$

Now by choosing basis for surface states as

$$\Phi_1^s = \begin{pmatrix} \varphi(+) \\ 0 \end{pmatrix}, \; \Phi_2^s = \begin{pmatrix} 0 \\ \chi(-) \end{pmatrix}, \tag{1.159}$$

with unified form

$$\begin{cases} \varphi(s, i_z) = C \begin{pmatrix} s\lambda_\perp \\ 2t_\perp \tan\frac{\xi_1 - \xi_2}{2} \end{pmatrix} \left(e^{i\,\mathrm{Re}\,\xi_1 i_z} e^{|\,\mathrm{Im}\,\xi_1| i_z} - e^{-i\xi_2 i_z}\right), \; E = m \\ \chi(s, i_z) = C \begin{pmatrix} 2t_\perp \tan\frac{\xi_1 - \xi_2}{2} \\ -s\lambda_\perp \end{pmatrix} \left(e^{i\,\mathrm{Re}\,\xi_1 i_z} e^{|\,\mathrm{Im}\,\xi_1| i_z} - e^{-i\xi_2 i_z}\right), E = -m \end{cases}, \; m(k) = \Theta(-m_0(k))m_0(k), \tag{1.160}$$

we may recover basis wavefunctions above, and this projects into two-band Hamiltonian for surface states as

$$H^s_{\mathrm{EFF}} = \lambda_\parallel(\sin k_x \sigma_x + \sin k_y \sigma_y) + m\sigma_z. \tag{1.161}$$

1.4 Critical Parameters

Someone being careful may have noticed the singularity in Eq (1.114), if one lets $\lambda_\perp = 2t_\perp$. Under such setting, one could still find surface states under suitable parameters, which is easy to verify by just going back and solving the thin film Hamiltonian numerically. Then, there seems to be something wrong with our analytic solution, as we would see.

At first glance, one may say that at this time, from Eq (1.113), the quadratic form of $\cos\xi$ is just eliminated by the following $\sin\xi$ term as

$$E^2 = M^2 + 4Mt_\perp\cos\xi + 4t_\perp^2\cos^2\xi + \lambda_\perp^2\sin^2\xi = M^2 + 4Mt_\perp\cos\xi + 4t_\perp^2,$$

from which one yields a single solution

$$\cos\xi = \frac{E^2 - 4t_\perp^2 - M^2}{4Mt_\perp}.$$

This means that we have kicked out α index above, and the only uncertainty for the solution is the odevity of ξ labelled by p, as

$$\sin\xi^p = p\sqrt{1-\cos^2\xi},\ p = \pm.$$

And correspondingly, we only have two basic solutions as

$$\phi_p^s = \begin{pmatrix} s\lambda_\perp p\sin\xi \\ E - M - 2t_\perp\cos\xi \end{pmatrix} \equiv \begin{pmatrix} p\lambda \\ m \end{pmatrix},\ \lambda = s\lambda_\perp\sin\xi,\ m = E - M - 2t_\perp\cos\xi.$$

The general solution should be the linear combination as

$$\Phi^s = \sum_p C_p\phi_p^s e^{i\xi^p i_z},$$

which is somehow similar to the trivial case without SOC discussed earlier. At this time, however, requiring the constraints by boundary condition

$$\Phi^s(\pm l/2) = 0, l = n_z + 1$$

would give the so-called 'overdetermined set of equations' as

$$\begin{pmatrix} \lambda e^{i\xi l/2} & -\lambda e^{-i\xi l/2} \\ \lambda e^{-i\xi l/2} & -\lambda e^{i\xi l/2} \\ m e^{i\xi l/2} & m e^{-i\xi l/2} \\ m e^{-i\xi l/2} & m e^{i\xi l/2} \end{pmatrix} \begin{pmatrix} C_+ \\ C_- \end{pmatrix} = 0,$$

requiring which to have both exact and non-zero solutions would lead to

$$\lambda m = 0,\ \sin\xi l = 0.$$

From the latter one has $\xi = n\pi/l = n\pi/(n_z+1)$, where n is an integer, and bring this into $\lambda m = 0$ would give $\sin(n\pi/l) = 0$ or $E = M + 2t_\perp\cos(n\pi/l)$. If the first one stands, one concludes that $n = ml$ where m is another integer and $\sin\xi = 0$, $\cos\xi = \pm 1$, however, this is indeed also the situation in which the second one holds, as one could solve the coupled equations directly as

$$\begin{cases} \cos\xi = \dfrac{E^2 - 4t_\perp^2 - M^2}{4Mt_\perp} \\ E = M + 2t_\perp\cos(\xi) \end{cases} \Longrightarrow E = M \pm 2t_\perp,$$

i.e., $\cos\xi = \pm 1, \xi = m\pi$. Nevertheless, this solution could not give general zero modes, compared with Eq (1.122). These formal solutions are more likely to be trivial quantum-well states.

Then what is wrong? Clearly, by fixing suitable other parameters and only varying these two as $\mathscr{R} = \dfrac{\lambda_\perp}{2t_\perp}$, one could easily find non-trivial phase before and after $\mathscr{R} = 1$ with

the same topological index (generally, as long as $m_0 > t_\perp$ near this 'critical' point, see (**??**), or loosely $m_0 > 0$ since $t_\perp > 0$ has been assumed), then we could try to approach this 'limit'.

Let us define

$$\epsilon = 1 - \mathscr{R}^2 = 1 - \frac{\lambda_\perp^2}{4t_\perp^2}, \tag{1.162}$$

which is a small number. Now we would like to concern about the limit $\epsilon \to 0^+$, and the other side is similar. Notice that

$$M = M_0(\boldsymbol{k}) = m_0(\boldsymbol{k}) - 2t_\perp = m_0 - 4t_\parallel \left(\sin^2\frac{k_x a}{2} + \sin^2\frac{k_y b}{2}\right) - 2t_\perp,$$

and we would like to care about the zone $0 < m_0 < 2t_\perp$ so that $M < 0$ for all $\boldsymbol{k}$, which falls in the area with non-trivial topology for our thin film away from the singularity with enough thickness while giving us a benefit when judging its sign. A fine example is s shown in the calculation above.

Now,

$$\begin{aligned}
\lim_{\epsilon\to 0^+} \cos\xi_\alpha^p &= \lim_{\epsilon\to 0^+} \frac{-Mt_\perp + (-1)^{\alpha-1}\sqrt{M^2t_\perp^2 - (t_\perp^2 - \lambda_\perp^2/4)(M^2 + \lambda_\perp^2 - E^2)}}{2(t_\perp^2 - \lambda_\perp^2/4)} \\
&= \lim_{\epsilon\to 0^+} \frac{-Mt_\perp + (-1)^{\alpha-1}|M|t_\perp\sqrt{-(\lambda_\perp^2 - E^2)/M^2 + \lambda_\perp^2/4t_\perp^2[1 + (\lambda_\perp^2 - E^2)/M^2]}}{2t_\perp^2\epsilon} \\
&= \lim_{\epsilon\to 0^+} \frac{M}{2t_\perp}\frac{-1 - (-1)^{\alpha-1}\sqrt{1 - \epsilon(1+O)}}{\epsilon},\ O = \frac{\lambda_\perp^2 - E^2}{M^2}, \\
&= \lim_{\epsilon\to 0^+} \frac{M}{2t_\perp}\frac{-1 - (-1)^{\alpha-1}[1 - \epsilon(1+O)/2]}{\epsilon},
\end{aligned}$$

and one has

$$\lim_{\epsilon\to 0^+} \cos\xi_\alpha^p = \begin{cases} -M/t_\perp\epsilon \to +\infty, & \alpha = 1 \\ (E^2 - \lambda_\perp^2 - M^2)/4Mt_\perp, & \alpha = 2 \end{cases}, \tag{1.163}$$

where the case $\alpha = 2$ gives finite $\cos\xi$ and is just the one we discussed above. To show that these solutions stand, we could take the same limit for the energy in Eq (1.122), and if they are still finite, we say that our solution holds in the singular limit, and clearly, we only need to deal with terms labelled by $\alpha = 1$.

Firstly, we take out the limit

$$\begin{aligned}
\lim_{\epsilon\to 0^+} \tan\xi_1^p &= p \lim_{\epsilon\to 0^+} \frac{\sqrt{1 - \cos^2\xi_1}}{\cos\xi_1} = p \lim_{\epsilon\to 0^+} -\frac{t_\perp\epsilon}{M}\sqrt{1 - \frac{M^2}{t_\perp^2\epsilon^2}} = p \lim_{\epsilon\to 0^+} \sqrt{\frac{t_\perp^2\epsilon^2 - M^2}{M^2}} \\
&= ip \lim_{\epsilon\to 0^+} \sqrt{1 - \frac{t_\perp^2}{M^2}\epsilon^2} = ip.
\end{aligned}$$

Then, we evaluate

$$\begin{aligned}
\lim_{\epsilon\to 0^+}\frac{\cos(\xi_1^p l/2)}{\cos\xi_1^p} &= \lim_{\epsilon\to 0^+}\frac{(\cos\xi_1+ip\sin\xi_1)^{l/2}+(\cos\xi_1-ip\sin\xi_1)^{l/2}}{2\cos\xi_1},\ x=\cos\xi_1,\\
&= \lim_{x\to+\infty}\frac{(x+ip\sqrt{1-x^2})^{l/2}+(x-ip\sqrt{1-x^2})^{l/2}}{2x},\ i\sqrt{1-x^2}=-x\sqrt{1-\frac{1}{x^2}}\\
&= \lim_{x\to+\infty}\frac{1}{2}x^{l/2-1}\left[\left(1-p\sqrt{1-\frac{1}{x^2}}\right)^{l/2}+\left(1+p\sqrt{1-\frac{1}{x^2}}\right)^{l/2}\right]\\
&= \lim_{x\to+\infty}\frac{1}{2}x^{l/2-1}\left[\left(\frac{1}{2x^2}\right)^{l/2}+\left(2-\frac{1}{2x^2}\right)^{l/2}\right]\\
&= \lim_{x\to+\infty}(2x)^{l/2-1}\geq 1,
\end{aligned}$$

where in the last line we used the fact that $l=n_z+1\geq 2$, and for our thin film, $l>2$ in fact. Then $\cos(\xi_1^p l/2)$ is at least at the same order of infinity compared with $\cos\xi_1^p$, and one also has

$$\lim_{\epsilon\to 0^+}\tan(\xi_1^p l/2)=ip.$$

Now, for energy concerned, one has

$$\begin{aligned}
\lim_{\epsilon\to 0^+}E_+ &= M+2t_\perp\lim_{\epsilon\to 0^+}\frac{\cos\xi_1 g^+(\xi_1)-\cos\xi_2 g^+(\xi_2)}{g^+(\xi_1)-g^+(\xi_2)},\ g^+(x)=\frac{\tan(xl/2)}{\sin x},\\
&= M+2t_\perp\lim_{\epsilon\to 0^+}\frac{\tan(\xi_1 l/2)/\tan\xi_1-\cos\xi_2 g^+(\xi_2)}{\tan(\xi_1 l/2)/(\cos\xi_1\tan\xi_1)-g^+(\xi_2)}\\
&= M+2t_\perp\frac{1-\cos\xi_2 g^+(\xi_2)}{0-g^+(\xi_2)}=M+2t_\perp\sin\xi_2\left(\frac{1}{\tan\xi_2}-\frac{1}{\tan(\xi_2 l/2)}\right),
\end{aligned}$$

which is finite even if in the limit $\sin\xi_2\to 0$ as

$$E_+\to M+2t_\perp(1-2/l).$$

And for the other one,

$$\begin{aligned}
\lim_{\epsilon\to 0^+}E_- &= M+2t_\perp\lim_{\epsilon\to 0^+}\frac{\cos\xi_1 g^-(\xi_1)-\cos\xi_2 g^-(\xi_2)}{g^-(\xi_1)-g^-(\xi_2)},\ g^+(x)=\frac{1}{\sin x\tan(xl/2)},\\
&= M+2t_\perp\lim_{\epsilon\to 0^+}\frac{1/[\tan(\xi_1 l/2)\tan\xi_1]-\cos\xi_2 g^-(\xi_2)}{1/[\tan(\xi_1 l/2)\sin\xi_1]-g^-(\xi_2)}\\
&= M+2t_\perp\frac{\cos\xi_2 g^-(\xi_2)+1}{g^-(\xi_2)}=M+2t_\perp\cos\xi_2[1+\tan(\xi_2 l/2)\tan\xi_2].
\end{aligned}$$

By discussion above $E_+=-E_-$, which further gives constraint as

$$M+t_\perp\left[2\cos\xi_2+\sin\xi_2\left(\tan(\xi_2 l/2)-\frac{1}{\tan(\xi_2 l/2)}\right)\right]=0.$$

Now we are able to discuss some physical meaning. Since

$$\begin{aligned}
\lim_{\epsilon\to 0^+}\beta_\alpha^p &= \lim_{\epsilon\to 0^+}(\cos\xi_\alpha+ip\sin\xi_\alpha)\propto\lim_{\epsilon\to 0^+}\left(\frac{1}{\epsilon}+ip\sqrt{1-\frac{1}{\epsilon^2}}\right)\\
&= \lim_{\epsilon\to 0^+}\frac{1-p\sqrt{1-\epsilon^2}}{\epsilon}=\lim_{\epsilon\to 0^+}\frac{1-p(1-\epsilon^2/2)}{\epsilon},
\end{aligned}$$

one finds that

$$\lim_{\epsilon\to 0^+} \beta_\alpha^p \propto \begin{cases} \epsilon/2 \to 0, & p = + \\ 2/\epsilon \to +\infty, & p = - \end{cases}, \tag{1.164}$$

and none of them represents physical situation for surface states, as one leads to zero while the other gives un-normalizable wavefunctions. And in this way we argue that the singularity $\mathscr{R} = 1$ is in fact an unphysical point. Then why direct numerical result would show the surface state? Here we argue by claiming that any numerical solution has an accuracy limit, and if one treat the numerical solution as a strict one, the corresponding model Hamiltonian would not have $\mathscr{R} = 1$ but a small shift. On the other hand, one could also argue that since we are dealing with a thin film, then it is neither purely 2-D nor 3-D but lies between that, and the finite thickness forces fluctuation on these parameters and effect for reduced dimensions appears. Or from the fundamental point of view of the energy-time duality together with their uncertainty relation but a more practical way, any exact parameter is impossible within finite time after preparation of system.

Chapter 2

Quantum Spin Hall Family

2.1 Conventional Quantum Spin Hall Effect

Quantum spin Hall effect (QSHE)[27], or generally, 2-D topological insulator (TI) state[17], is a long-known phenomenon originated from spin Hall effect[38, 54] driven by transverse force on spin current, like Lorentz force on electric current that generates Hall effect. However, different from its 3-D partner, intrinsic 2-D TI is rather rare, which is mainly found in quantum-well structures[3] and predicted to be realizable in transition metal dichalcogenides (TMD)[46]. On the other hand, topological field theory[43] and general classification scheme[**?**, 49, 10] suggests a dimensional reduction procedure, which claims that both 2-D and 3-D TI root originally in $(4+1)$-D 'Chern insulator' described by second Chern class. In this way, a natural thought emerges that we may reduce one dimension of 3-D TI — say to make it a film — and obtain a quasi-2D system that exhibits QSHE.

This thought has been proposed decade more ago, first theoretically obtained for 2-D TI[68] and later 3-D one[32, 51], people found both exponentially decaying and oscillating small gap[65] for helical edge state[11, 34] or surface Dirac cone — characteristic states of 2-D and 3-D TI, respectively. Under such consideration, physics for the proposed quasi-2D QSHE becomes rather clear. Now consider our effective model, since every sub-block $h_{n,\chi}$ is decoupled from each other, and clearly we could see from mass term that $m_{n\geq 2}$ is trivial given by the formula

$$\sigma_H^n/(e^2/h) = \frac{\mathrm{sgn}(m_n(\infty)) - \mathrm{sgn}(m_n(0))}{2} = \mathrm{sgn}(m_n(\infty))\Theta(-m_n(\infty)m_n(0)), \qquad (2.1)$$

where the two-part contribution in the second term comes from anomaly and Dirac mass, respectively. Then the only possible non-trivial bands come from $h_{1,\chi}$:

$$h_{1,\chi} = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \chi m_1 \sigma_z,\ \chi = \pm.$$

Notice that as we have shown, for our system with single Dirac cone at Γ point, effective m_1 is always minus at high energy/large k, then it would suffice that when the finite-size gap $m_1(k=0) > 0$, the system is non-trivial and exhibit 2-D QSHE, since χ serves as a good 'spin' number. As a practical example, we have tuned layer number L_z under previous parameters and show both oscillating mass term, oscillating gap and oscillating $\mathbb{Z}_2$ index describing the system, as shown in Fig. 2.1(a) (b). In the calculation, we have employed symmetry indicator method[18] to simplify the index calculation. In this

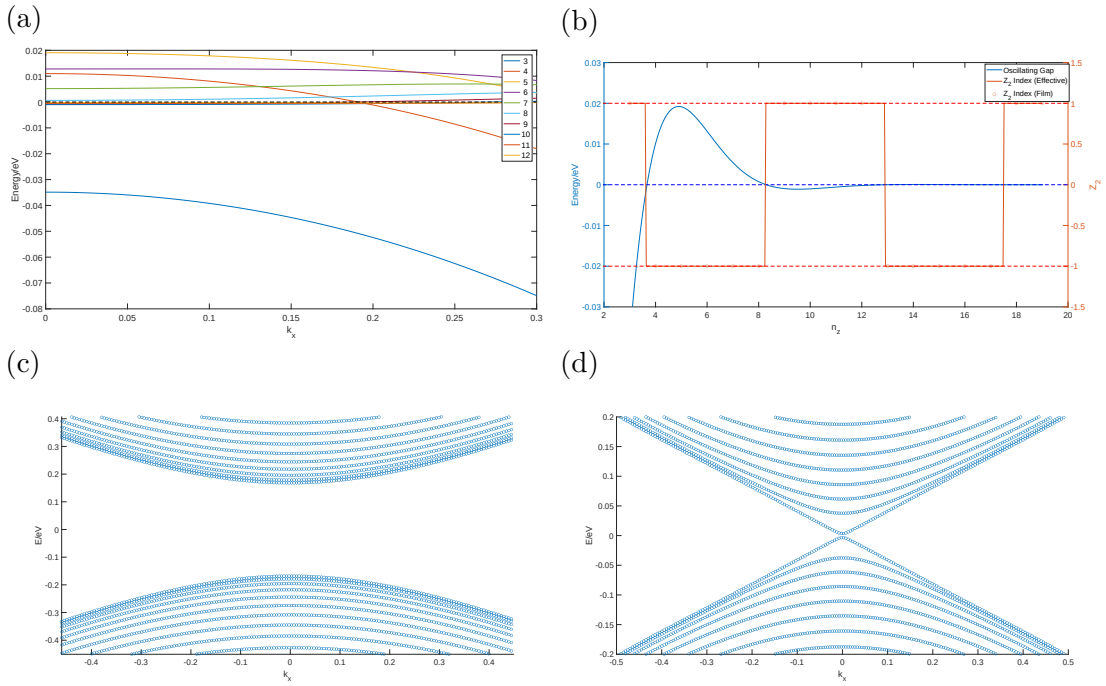


Figure 2.1: (a) Oscillating mass term $m_1(k)$. (b) Oscillating gap $m_1(0)$ and jumping $\mathbb{Z}_2$ index. (c) Band structure of trivial insulator with $L_z = 2$. (d) Band structure of topological insulator with $L_z = 5$. In (c) and (d) we set $L_y = 50$.

method, inversion symmetry (for the film, it is 3-D parity symmetry) is high-lighted whose basic action is

$$\mathscr{P} \left| i_z \right\rangle = \left| -i_z \right\rangle , \tag{2.2}$$

while considering inner DoFs of Dirac equation would lead to the back-diagonal representation

$$P = \begin{pmatrix} & & P_0 \\ & \iddots & \\ P_0 & & \end{pmatrix} , \ P_0 = \sigma_0 \tau_z . \tag{2.3}$$

Then it becomes clear that

$$\langle i_z | \mathscr{P} | \Phi \rangle = P_0 \Phi(-i_z). \tag{2.4}$$

For instance,

$$P_0 \Phi_1(-i_z) = \sigma_0 \tau_z C \begin{pmatrix} -i\lambda_\perp f_+(-i_z) \\ t_\perp \eta f_-(-i_z) \\ 0 \\ 0 \end{pmatrix} = C \begin{pmatrix} -i\lambda_\perp f_+(i_z) \\ t_\perp \eta f_-(i_z) \\ 0 \\ 0 \end{pmatrix} = \Phi_1(i_z),$$

and similarly, one may conclude that

$$\mathscr{P} \left| \Phi_i \right\rangle = (-1)^{i-1} \left| \Phi_i \right\rangle , i = 1, 2, 3, 4. \tag{2.5}$$

Then after projection, it becomes

$$P_{\text{EFF}} = \tau_0 \sigma_z \tag{2.6}$$

under new representation, for one block. Now for our $\mathbb{Z}_2$ index, with the existence of inversion symmetry, such an index could be indicated by

$$\delta_a = \prod_m \xi_m(\Lambda_a), \tag{2.7}$$

where the product is over the Kramers pairs of occupied bands and ξ_m represents parity eigenvalue, compared with original formula

$$(-1)^\nu = \prod_a \delta_a,\ \delta_a \equiv \frac{\mathrm{Pf}[w(\Lambda_a)]}{\sqrt{\det[w(\Lambda_a)]}} = \pm, \tag{2.8}$$

where $\nu = 1$ or multiplied -1 result is recognized as non-trivial, and a unitary matrix $w_{mn}(k) = \langle u_m(k)|\mathscr{T}|u_n(-k)\rangle$ with periodic Bloch wavefunctions and time-reversal operator $\mathscr{T}^2 = -\mathbb{I}$, which is antisymmetric at high-symmetry point Λ_a in Brillouin zone is introduced. To give concrete example, we have calculated open-y thin film band structure with $L_z = 2, 5$, respectively, as shown in Fig. 2.1(c) (d). While the former is $\mathbb{Z}_2$ trivial, the latter is not with helical edge states in the bulk gap. Notice that it also contains finite-size gap with exponential decay to L_y.

2.2 Half Quantum Mirror Hall Effect

The appearance of finite-size gap is a manifestation of 2-D surface Dirac cone of a 3-D TI. With enough thickness, such a gap could on one hand, be ignored, and on the other hand, be disabled by tuning the chemical potential away from the gap. In the second scheme which is more reasonable (though it may happen that the first one is also efficient realistically), we have indeed stepped into metallic phase, and one natural question arises as whether we have some quantity protected topologically. Luckily, the answer to that is positive.

Since we have been deviated from the gap, the conductance formula should be decorated correspondingly as (still consider single Dirac cone at Γ point)

$$\begin{aligned}\sigma_H^n/(e^2/h) &= \frac{1}{2\pi}\int_{E_n(k)<\mu_F} \mathrm{d}^2k\Omega_z^n(k) \\ &= \frac{1}{2}\left[\mathrm{sgn}(m_n(\infty)) - \cos\varphi_F^n\right],\ \cos\varphi_F^n \equiv \frac{m_n(k_F^n)}{E_n(k_F^n)},\end{aligned} \tag{2.9}$$

where k_F^n is nth Fermi vector for nth band which satisfies that

$$E_n(k_F^n) = \mu_F.$$

This holds when μ_F lies within nth band, and when it is not, we just pick $k_F^n = 0$. Now we may consider the case when chemical potential has escaped from the first gap (exponential surface state gap), while confined within the second gap, i.e., $|\mu_F| < |m_2(0)|$. At this time, as discussed before, all $m_{n\geq 2}$ are trivial and contribute no Hall conductance. However, for $n = 1$, since $m_1(k < k_c)$ (for simplicity we do not distinguish k_c and k_{c1} here) is all exponentially small (compared to lattice hopping parameters), one could treat

$$\cos\varphi_F^1 \approx 0. \tag{2.10}$$

Or if one just adopt the approximate analytic formula, one yields $\cos\varphi_F^1 = 0$. And one gets

$$\sigma_H^{1,\chi} = \frac{1}{2}\frac{e^2}{h}\,\mathrm{sgn}(\chi m_1(\infty)), \tag{2.11}$$

which provides a half-quantized electric Hall conductance with $\chi = \pm$ as good quantum number. Then combination of two sub-blocks inside $n = 1$ gives us a half-quantized version of spin Hall effect, named as *half quantum mirror Hall effect*, where mirror comes from 3-D mirror symmetry $\mathscr{M}_z$, as we shall see below. And here, we may firstly give

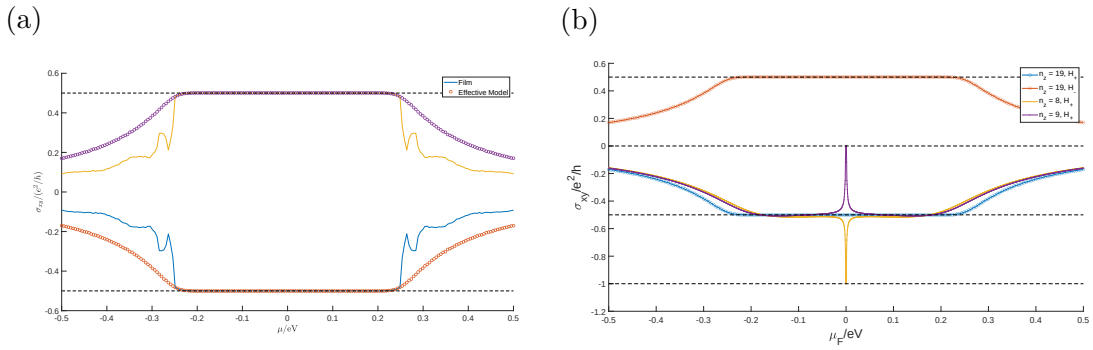


Figure 2.2: (a) HQMHE with $L_z = 19$, calculated by thin film and effective model, respectively. (b) HQMHE with $L_z = 8, 9, 19$, in former two case only $\chi = +$ part is shown.

a convincing picture to the proposal above, as shown in Fig. 2.2, where half-quantized nature is shown without doubt.

2.2.1 Parity Invariant Regime and Parity Anomaly

Most important thing to notice is the half-quantized nature comes from high energy contribution, i.e., part of integral of curvature form on closed manifold. This 'part' is chosen carefully so that it contains all Brillouin zone but

- stays away from low energy/Dirac point;
- stops inside parity invariant regime.

By parity invariant regime (PIR)[69] we refer to the k-space zone inside which 2-D surface Dirac cone owns 2-D parity symmetry (which is mirror symmetry with arbitrary in plane axis as mirror invariant line, or plane if we extend it to virtual 3D), e.g., choose

$$\mathscr{M}_x : \boldsymbol{k} \to (k_x, -k_y),$$

then by parity invariant one requires $h = \boldsymbol{d}(\boldsymbol{k}) \cdot \boldsymbol{\sigma}$ to satisfy that

$$U_{M_x}^\dagger d(k_x, -k_y) U_{M_x} = d(k_x, k_y),$$

where U_{M_x} is unitary matrix defined by

$$\mathscr{M}_x c_{\boldsymbol{k}}^\dagger \mathscr{M}_x = c_{k_x,-k_y}^\dagger U_{M_x}^\dagger,$$

and is to be determined by specific inner DoFs. In our case, since

$$h_{1,\chi} = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \chi m_1 \sigma_z,$$

with σ projected spin DoF preserved, we need to specify the action of $\mathscr{M}_x$ on spin, which could be done by analysing usual angular momentum due to isomorphism of Lie algebra between $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$. By comparing components of $L \sim r \times \mathrm{d}_t r$ with and without mirror operation, one is easy to get that for spin

$$\mathscr{M}_x : \boldsymbol{s} \to (s_x, -s_y, -s_z),$$

then it is proper to choose

$$U_{M_x} \propto \sigma_x$$

under such representation. Then the requirement above yields that

$$\begin{cases} \sin(k_x a) = \sin(k_x a) \\ -\sin(-k_y b) = \sin(k_y b) \\ -m_1(k_x, -k_y) = m_1(k_x, k_y) \end{cases}.$$

While the first two requirements are naturally satisfied, the third one hold only when

$$m_1(k) = 0,$$

i.e., $k < k_c$ since $m_1(k)$ depended on $m_0(k)$ thus on quadratic (k_x^2, k_y^2) instead of linear momentum. Similar requirement applies to arbitrary n-axis and would lead to the same result. We are then able to define PIR as zone where $k < k_c$ if the exponentially small gap is ignored by enough thickness.

One comment is mirror symmetry (or 2-D parity symmetry) here is defined for a 2-D system; as we would see later, mirror symmetry operator defined for a 3-D system would be much different from the one defined here. And the virtual 3D we have adopted here refers to the inner spin DoF rather than a true spatial one.

Another comment here is about the possible worrying finite-size gap, which seems to contain the tendency to destroy the effect. Admittedly, when chemical potential lies in that gap, the Hall conductance must be quantized — either zero or one unit counting on triviality of the gap — protected by Thouless, Kohmoto, Nightingale and den Nijs (TKNN) theorem[55], since integral now is curvature form over full closed manifold; however, when away from Dirac point, since the mass term is exponentially small, it is much less comparable with linear part thus could be viewed approximately to satisfy PIR condition, then our claim enforces the Hall conductance to be half-quantized. To show this, we have calculated corresponding conductance varying with chemical potential with relatively thin layer number as $L_z = 8, 9$, which lies within $\mathbb{Z}_2$ non-trivial and trivial zone as a 2-D system, respectively. As presented in Fig. 2.2(b), upon two situations the Hall conductance shows dip to one or ascend to zero behavior when within the gap over the base minus-half line when within far-large PIR, correspondingly. This conforms our picture above.

Now one very question arises as why high energy part contributes, say exactly half-quantized conductance. Theme here is deeply related to quantum anomaly originated from high-energy physics, and is known as chiral anomaly[1, 40] for even spacetime or odd space, together with parity-anomaly[48, 25, 50, 13, 29] for odd spacetime. Simply

speaking, yet unknown high-energy physics with non-trivial topological structure manifests itself in effective low-energy theory through quantum fluctuation (loops), ultraviolet (UV) regularization and renormalization. When it comes to condensed matter, one in fact finds that due to the existence of finite lattice parameter served as mutual high-energy cutoff, as long as we view physics inside as bound states hopping from tight-binding, we would actually in principle know 'high-energy' physics, denoted as physics away from emergent Dirac point, then no anomaly exists. Nevertheless, let us borrow the name here. Following the no-go Nielsen-Ninomiya theorem, a quadratic lattice Hamiltonian must hold even number of chiral Fermions if requirements translational invariance, locality, Hermiticity, local charge quantization and conservation are fulfilled. Just like one possible stable topological semimetal phase in 3-D space known as Weyl semimetal[26, 2, 58], where one massless Dirac pair is split into two Weyl Fermions connected by Fermi arc in momentum space. On the other hand, the holographic violation[45] on the boundary as odd number of chiral edge states/Dirac cones stands exactly as a consequence of anomaly. Somehow similar to spontaneous-symmetry-breaking, 'anomaly' refers to the phenomenon that a classical symmetry fails at quantum (loop) level, like Ward identity for axial current and Adler-Bell-Jackiw (ABJ) anomaly, i.e., it is the unknown high-energy part that breaks the desired low-energy symmetry. Back to edge state, in the low-energy zone, such state behaves as Weyl type massless Dirac Fermion thus conserves the symmetry (for concrete examples, chiral symmetry for 2-D Chern insulator and parity symmetry for 3-D TI), while for high energy, the non-zero mass term forces symmetry breaking as 'anomaly', such anomalous part is topologically nontrivial and contributes half quantum Hall conductance[69].

Simply speaking, it is the domain-wall structure of mass term in k-space that makes half-quantization (also integer quantization) possible; at the domain where mass is zero, two parts separated (always possible, protected by *Jordan curve theorem*, fails for multi-Dirac points) contribute half-integer curvature integral. Physical objects to imagine might be merons (half-skyrmions) with spin texture covering semi-sphere.

2.2.2 Mirror Symmetry in TI Film

Mirror symmetry for TI film is a 3-D one, which emerges as a combination of 3-D parity/inversion symmetry and C_{2z} as for z-mirror. With spin-orbital inner DoFs, one has

$$\mathscr{M}_z \Psi_{i_z,\boldsymbol{k}} \mathscr{M}_z^{-1} = U_z \Psi_{-i_z,\boldsymbol{k}},\ \mathscr{M}_z \Psi^\dagger_{i_z,\boldsymbol{k}} \mathscr{M}_z^{-1} = \Psi^\dagger_{-i_z,\boldsymbol{k}} U_z^\dagger, \tag{2.12}$$

where under our representation $U_z = -i\sigma_z \otimes \tau_z$ is a unitary matrix with imaginary eigenvalue $-i\chi,\ \chi = \pm$. Notice that the transformation is independent of $\boldsymbol{k}$ and we shall omit that for simplicity. Furthermore, one sees that

$$U_z^\dagger = -U_z,\ U_z^2 = -1.$$

Under $\{\Psi_{i_z,\boldsymbol{k}}\}_{i_z}$ representation, mirror operator is represented as

$$M_z = \begin{pmatrix} & & U_z \\ & \iddots & \\ U_z & & \end{pmatrix},\ U_z = -i\sigma_z\tau_z, \tag{2.13}$$

which switches elements upon mirror surface. And one sees apparently that this is a combination of P and $C_{2z} \equiv e^{-is_z\pi} = -i\sigma_z$, where the minus sign is chosen manually to

correspond with projected χ. Speaking of this, clearly that

$$\langle i_z|\mathscr{M}_z|\Phi\rangle = U_z\Phi(-i_z), \tag{2.14}$$

and again for instance,

$$U_z\Phi_1(-i_z) = -i\sigma_z\tau_z C\begin{pmatrix}-i\lambda_\perp f_+(-i_z)\\ t_\perp\eta f_-(-i_z)\\ 0\\ 0\end{pmatrix} = -iC\begin{pmatrix}-i\lambda_\perp f_+(i_z)\\ t_\perp\eta f_-(i_z)\\ 0\\ 0\end{pmatrix} = -i\Phi_1(i_z),$$

then similarly, one may conclude that

$$\mathscr{M}_z\left|\Phi_i\right\rangle = -i(-1)^{\Theta(i-2.5)}\left|\Phi_i\right\rangle, i = 1,2,3,4. \tag{2.15}$$

Then after projection, it becomes

$$M_{\mathrm{EFF}} = -i\tau_z\sigma_0 \tag{2.16}$$

under new representation, for one block.

We firstly examine the mirror symmetry by noticing that

$$\mathscr{M}_z\mathscr{H}\mathscr{M}_z^{-1} = \sum_{i_z,i_z'}\Psi_{-i_z}^\dagger U_z^\dagger H(i_z,i_z')U_z\Psi_{-i_z'},$$

which would require

$$U_z^\dagger H(i_z,i_z')U_z = H(-i_z,-i_z').$$

This is easy to check by noticing that

$$U_z^\dagger H_{2D}U_z = H_{2D},\ U_z^\dagger\mathscr{T}_zU_z = \mathscr{T}_z^\dagger,$$

where by H_{2D} I mean both linear part $H_\parallel$ and mass term $\mathscr{M}_0$.

Then in spirit of commuting $[\mathscr{H},\mathscr{M}_z] = 0$, we could simultaneously diagonalize both and in principle block-diagonalize $\mathscr{H}$ in $\mathscr{M}_z$ representation, which is done by mirror transformation

$$\phi_{i_z,\chi} = \frac{1}{\sqrt{2}}(\Psi_{i_z} + i\chi U_z\Psi_{-i_z}),\ i_z > 0, \tag{2.17}$$

as a combination of mirrored operators, and it is easy to see

$$\mathscr{M}_z\phi_{i_z,\chi}\mathscr{M}_z^\dagger = \frac{1}{\sqrt{2}}(U_z\Psi_{-i_z} + i\chi U_z^2\Psi_{i_z}) = -i\chi\phi_{i_z,\chi}.$$

Our choice of $\chi = \pm$ here directly corresponds to projected effective $\tau_z = \pm$ as indicated above. Here, one should be aware about the domain $i_z > 0$, utilized to avoid double counting. On the other hand, $i_z = 0$ case, which appears when z-layer number L_z is odd, is a bit tricky, but one could intuitively relate coherent superposition image of wave to the transformation and treat i_z as frequency, then one concludes that instead of divided by $\sqrt{2}$, it should be

$$\phi_{0,\chi} = \frac{1}{2}(\Psi_0 + i\chi U_z\Psi_0) = \Psi_0\frac{1+\chi\sigma_z\tau_z}{2}, \tag{2.18}$$

and one immediately reads that each χ contains two Fermionic operators as two states split from four. More explicitly, let

$$\Psi_0 = (c_{1\uparrow}, c_{2\uparrow}, c_{1\downarrow}, c_{2\downarrow})^T,$$

and one has

$$\phi_{0,+} = (c_{1\uparrow}, 0, 0, c_{2\downarrow})^T,\ \phi_{0,-} = (0, c_{2\uparrow}, c_{1\downarrow}, 0)^T.$$

One special property for $i_z = 0$ is

$$U_z \phi_{0,\chi} = -i\chi\phi_{0,\chi}. \tag{2.19}$$

It is also important before procedure verifying Fermionic commutation relation. For $i_z > 0$, we have

$$\begin{aligned}\{\phi_{i_z,\chi}, \phi^\dagger_{i'_z,\chi'}\} &= \frac{1}{2}\{\Psi_{i_z} + i\chi U_z \Psi_{-i_z}, \Psi^\dagger_{i'_z} - i\chi'\Psi^\dagger_{-i'_z}U_z^\dagger\} \\ &= \frac{1}{2}\left(\{\Psi_{i_z}, \Psi^\dagger_{i'_z}\} + \chi\chi' U_z\{\Psi_{-i_z}, \Psi^\dagger_{-i'_z}\}U_z^\dagger\right) \\ &= \frac{1}{2}\delta_{i_z,i'_z}(1 + \chi\chi') = \delta_{i_z,i'_z}\delta_{\chi,\chi'},\end{aligned}$$

and when $i_z = i'_z = 0$,

$$\begin{aligned}\{\phi_{0,\chi}, \phi^\dagger_{0,\chi'}\} &= \{\Psi_0, \Psi_0^\dagger\}\frac{1 + \chi\sigma_z\tau_z}{2}\frac{1 + \chi'\sigma_z\tau_z}{2} \\ &= \frac{1 + \chi\chi' + (\chi + \chi')\sigma_z\tau_z}{4} = \delta_{\chi,\chi'}\frac{1 + \chi\sigma_z\tau_z}{2},\end{aligned}$$

which corresponds to our two-state operator. However, since this is not canonical commutation relation, we may have to replace ϕ_0 back with Ψ_0 in the end.

One last step before starting is finding inverse transformation, which turns out to be

$$\begin{cases}\Psi_{i_z} = \dfrac{1}{\sqrt{2}}(\phi_{i_z,+} + \phi_{i_z,-}) \\ \Psi_{-i_z} = \dfrac{i}{\sqrt{2}}U_z(\phi_{i_z,+} - \phi_{i_z,-})\end{cases},\ i_z > 0, \tag{2.20}$$

and

$$\Psi_0 = \phi_{0,+} + \phi_{0,-} = iU_z(\phi_{0,+} - \phi_{0,-}). \tag{2.21}$$

Now we are ready to perform mirror transformation on Hamiltonian. Still by ignoring $\boldsymbol{k}$, we have three parts to be transformed. $i_z > 0$ and $i_z < 0$ parts are trivial as Hamiltonian is fully preserved, and we get

$$\sum_{i_z>0,\chi}\left(\phi^\dagger_{i_z,\chi}H_{2D}\phi_{i_z,\chi} + \phi^\dagger_{i_z,\chi}\mathscr{T}_z\phi_{i_z+1,\chi} + \text{h.c.}\right).$$

For the third term, in even layer number geometry, we need to evaluate z-hopping between $i_z = \pm 1/2$, which gives us

$$\begin{aligned}\Psi^\dagger_{-1/2}\mathscr{T}_z\Psi_{1/2} + \text{h.c.} &= -\frac{i}{2}(\phi^\dagger_{1/2,+} - \phi^\dagger_{i_z,-})U_z^\dagger\mathscr{T}_z(\phi_{1/2,+} + \phi_{1/2,-}) + \text{h.c.} \\ &= -\frac{1}{2}\sum_\chi i\chi\left[\phi^\dagger_{1/2,\chi}\left(U_z^\dagger\mathscr{T}_z - \mathscr{T}_z^\dagger U_z\right)\phi_{1/2,\chi} + \phi^\dagger_{1/2,\chi}\left(U_z^\dagger\mathscr{T}_z + \mathscr{T}_z^\dagger U_z\right)\phi_{1/2,-\chi}\right] \\ &= \sum_\chi i\chi\phi^\dagger_{1/2,\chi}\mathscr{T}_z^\dagger U_z\phi_{1/2,\chi},\end{aligned}$$

where we have noticed that

$$\begin{aligned} U_z^\dagger \mathscr{T}_z \pm \mathscr{T}_z^\dagger U_z &= (U_z^\dagger \mathscr{T}_z U_z^\dagger \pm \mathscr{T}_z^\dagger) U_z = (-U_z \mathscr{T}_z U_z^\dagger \pm \mathscr{T}_z^\dagger) U_z \\ &= (-\mathscr{T}_z^\dagger \pm \mathscr{T}_z^\dagger) U_z. \end{aligned}$$

And totally this serves us two decorated Hamiltonians describing semi-magnetic TI thin film with half-thickness

$$\mathscr{H}_\chi = \sum_{i_z=1/2}^{(L_z-1)/2} \left(\phi_{i_z,\chi}^\dagger H_{2D} \phi_{i_z,\chi} + \phi_{i_z,\chi}^\dagger \mathscr{T}_z \phi_{i_z+1,\chi} + \text{h.c.} \right) + i\chi \phi_{1/2,\chi}^\dagger \mathscr{T}_z^\dagger U_z \phi_{1/2,\chi}. \tag{2.22}$$

The Hamiltonian is Hermitian thus well-defined since

$$(i\chi \mathscr{T}_z^\dagger U_z)^\dagger = -i\chi U_z^\dagger \mathscr{T}_z = i\chi U_z \mathscr{T}_z = i\chi \mathscr{T}_z^\dagger U_z.$$

This Hamiltonian describes a TI film with half-thickness, decorated by one magnetic layer *on* the end of the chain.

For L_z odd, story becomes a bit tedious, but basically, $i_z \neq 0$ parts are also trivial under transformation. Accounting for $i_z = 0$ one needs its on-site 2-D part and hopping to $i_z = \pm 1$, and for the former,

$$\begin{aligned} \Psi_0^\dagger H_{2D} \Psi_0 &= (\phi_{0,+}^\dagger + \phi_{0,-}^\dagger) H_{2D} (\phi_{0,+} + \phi_{0,-}) \\ &= \sum_\chi \phi_{0,\chi}^\dagger H_{2D} \phi_{0,\chi} + \phi_{0,\chi}^\dagger H_{2D} \phi_{0,-\chi} + \text{h.c.}, \end{aligned}$$

while the last two terms vanishes due to anti-commutation relation, and this continues to serve us as a trivial transformation. For the latter, one could show that

$$\begin{aligned} (0 \leftrightarrow \pm 1) &= \frac{1}{\sqrt{2}} \sum_\chi \left[\phi_{0,\chi}^\dagger (\mathscr{T}_z + i\chi \mathscr{T}_z^\dagger U_z) \phi_{1,\chi} + \phi_{0,\chi}^\dagger (\mathscr{T}_z - i\chi \mathscr{T}_z^\dagger U_z) \phi_{1,-\chi} + \text{h.c.} \right] \\ &= \sqrt{2} \sum_\chi \left(\phi_{0,\chi}^\dagger \mathscr{T}_z \phi_{1,-\chi} + \text{h.c.} \right), \end{aligned}$$

where we have used

$$\begin{aligned} \phi_{0,\chi}^\dagger (\mathscr{T}_z \pm i\chi \mathscr{T}_z^\dagger U_z) &= \phi_{0,\chi}^\dagger (\mathscr{T}_z \pm i\chi U_z \mathscr{T}_z) = \phi_{0,\chi}^\dagger (\mathscr{T}_z \pm U_z^\dagger U_z \mathscr{T}_z) \\ &= \phi_{0,\chi}^\dagger \mathscr{T}_z (1 \pm 1). \end{aligned}$$

Discussion above leads to mirror separated Hamiltonian for odd layer number as

$$\begin{aligned} \mathscr{H}_\chi = &\sum_{i_z=0}^{(L_z-1)/2} \phi_{i_z,\chi}^\dagger H_{2D} \phi_{i_z,\chi} + \sum_{i_z=1}^{(L_z-1)/2} \left(\phi_{i_z,\chi}^\dagger \mathscr{T}_z \phi_{i_z+1,\chi} + \text{h.c.} \right) \\ &+ \sqrt{2} (\phi_{0,\chi}^\dagger \mathscr{T}_z \phi_{1,\chi} + \text{h.c.}). \end{aligned} \tag{2.23}$$

While at first glance the Hamiltonian seems to lose χ dependence, that side of information is indeed embedded in $\phi_{0,\chi}$, which is not a canonical Fermionic operator itself. Notice that under spin-orbital representation, DoFs yields totally $4L_z$ bands or states, and correspondingly, each χ is expected to contain half of that which is $2L_z$. It is rather easy to

see that in even case, however, for odd L_z mirror Hamiltonian as we have written, each χ part contains $2(L_z+1)$ bands, which originates from

$$\phi_{0,\chi} = \Psi_0 \frac{1+\chi\sigma_z\tau_z}{2},$$

where only two states are projected into each χ as shown above. Bring this back and one gets

$$\begin{aligned}
\mathscr{H}_\chi &= \sum_{i_z=1}^{(L_z-1)/2} \left(\phi^\dagger_{i_z,\chi} H_{2D} \phi_{i_z,\chi} + \phi^\dagger_{i_z,\chi} \mathscr{T}_z \phi_{i_z+1,\chi} + \text{h.c.}\right) \\
&+ \Psi_0^\dagger \frac{1+\chi\sigma_z\tau_z}{2} H_{2D} \frac{1+\chi\sigma_z\tau_z}{2} \Psi_0 + \sqrt{2}\Psi_0^\dagger \frac{1+\chi\sigma_z\tau_z}{2} \mathscr{T}_z \phi_{1,\chi} + \text{h.c.} \\
&= \sum_{i_z=1}^{(L_z-1)/2} \left(\phi^\dagger_{i_z,\chi} H_{2D} \phi_{i_z,\chi} + \phi^\dagger_{i_z,\chi} \mathscr{T}_z \phi_{i_z+1,\chi} + \text{h.c.}\right) \\
&+ \Psi_0^\dagger \frac{1+\chi\sigma_z\tau_z}{2} H_{2D} \Psi_0 + \Psi_0^\dagger \tilde{\mathscr{T}}_{0,\chi} \phi_{1,\chi} + \text{h.c.},
\end{aligned} \tag{2.24}$$

with a newly defined

$$\tilde{\mathscr{T}}_{0,\chi} = \sqrt{2}\frac{1+\chi\sigma_z\tau_z}{2}\mathscr{T}_z = \frac{\sqrt{2}t_\perp}{2}(\tau_z + \chi\sigma_z) - \frac{i\sqrt{2}\lambda_\perp}{4}(\sigma_z\tau_z + i\chi\tau_y). \tag{2.25}$$

Because of the extra factor brought by $\phi_{0,\chi}$, we have manually introduced two fake trivial flat bands in each mirror Hamiltonian, which are unphysical and should be abandoned. Also notice that set $\{\Psi_0, \phi_{i_z\geq 1,\chi}\}$ make up of full Fermionic operators in our system. This Hamiltonian also describes a TI film with slightly less than half-thickness, and coupled with an *additional* half-layer with magnetism.

Though with slightly deviated physical picture describing even and odd layer number film, physics here for HQMHE is the same. For the new half-film of each χ, there ought to be two Dirac surface states within topologically non-trivial regime, then totally we should have four Dirac cones (non-degenerate); now however, because of the extra magnetic layer only on one side, the nearby-surface Dirac cone is gapped out while the opposite-surface one remains, for each χ. Then totally, four cones are left with two, which contribute inverse half-quantized Hall conductance separately. This composes the complete picture. For one more comment, the additional magnetism appears at $i_z = 0$, i.e., middle of original film, where mirror current reaches its maximum, as the anomaly contribution from bulk states continued from surface Dirac cone; for each χ, time reversal is broken explicitly by this magnetism, but as a total, TI film does not. For the physics of TI film plus one surface magnetism, we refer that to quantum Hall family, which gives us a solely $1/2$; and here, we may call it as $1/2 - 1/2$.

2.2.3 Mirror Layer Chern Number

Due to the film geometry, it is natural to introduce the so-call layer Hall conductance[**?**] by splitting eigenstates into summation of layer-dependent one as

$$\sigma_H(l) = \frac{e^2}{h}\frac{1}{\pi} \sum_{E_n<\mu_F<E_m} \sum_{l'} \int \mathrm{d}^2k\, \mathrm{Im}\, \frac{\bar{v}_x^{nm}(l)\bar{v}_y^{mn}(l')}{(E_n - E_m)^2}, \tag{2.26}$$

where expectation value of velocity operator is

$$\bar{v}_i^{mn}(l) = \langle m(l)|\partial_{k_i} H|n(l)\rangle\,,$$

as the assumed only existing diagonal element. Basically there are two quantities to calculate as layer-resolved and cumulative layer Hall conductance.

On the other hand, because of the existence of mirror symmetry, one may separate and define mirror-layer-Chern number. Again, mirror symmetry under film geometry about z is defined by a combination of inversion and C_{2z} rotation, i.e., $M_z = C_{2z}P$, and is represented as

$$M_z = \begin{pmatrix} & & U_z \\ & \iddots & \\ U_z & & \end{pmatrix},\ U_z = -i\sigma_z\tau_z, \tag{2.27}$$

which switches elements upon mirror surface. There are clearly two eigenvalues of this operator $-i\chi,\ \chi = \pm$, with respect to even or odd configuration transformed under mirror operator, based on which corresponding mirror projection operator could be defined as

$$P_\chi^{M_z} = \frac{1 + i\chi M_z}{2}, \tag{2.28}$$

and our Hamiltonian is split into two parts with respect to mirror symmetry

$$H_\chi^{\text{Proj}} = P_\chi^{M_z} H, \tag{2.29}$$

where by H we denote it to be matrix form of $\mathscr{H}_{\text{Film}}$. In Fig. 2.2(a), we have also calculated Hall conductance labelled by mirror eigenvalue for film. This could be done either by using $\mathscr{H}_\chi$ we defined above, or just use projected Hamiltonian H_χ^{Proj}.

For eigenstates, notice that since $[M_z, H] = 0$,

$$\begin{cases} H\,|n\rangle = E_n\,|n\rangle \\ H(M_z\,|n\rangle) = M_z H\,|n\rangle = E_n(M_z\,|n\rangle) \end{cases}, \tag{2.30}$$

while generally,

$$M_z\,|n\rangle \neq -i\chi\,|n\rangle\,,$$

i.e., one may write

$$|n\rangle = c^+\,|n^+\rangle + c^-\,|n^-\rangle\,,$$

where

$$M_z\,|n^\chi\rangle = -i\chi\,|n^\chi\rangle\,,\ H\,|n^\chi\rangle = E_n\,|n^\chi\rangle\,,$$

then it is easy to check this makes Eq. (2.30) accurate. What is more, one has

$$H^\chi\,|n^{\chi'}\rangle = P_\chi^{M_z} H\,|n^{\chi'}\rangle = E_n\frac{1 + i\chi M_z}{2}\,|n^{\chi'}\rangle = E_n\frac{1 + \chi\chi'}{2}\,|n^{\chi'}\rangle = \delta_{\chi\chi'}E_n\,|n^{\chi'}\rangle\,,$$

or

$$H^\chi\,|n^{\chi'}\rangle = E_n^{\chi'}\,|n^{\chi'}\rangle\,,\ E_n^{\chi'} = \begin{cases} E_n, & \chi' = \chi \\ 0, & \chi' = \bar{\chi} \end{cases}, \tag{2.31}$$

as expected. Then we are able to define mirror Chern number merely based on H^χ, as

$$\sigma_H^\chi = \frac{e^2}{h}\frac{\text{Im}}{\pi}\left[\sum_{E_n^\chi < \mu < E_m^\chi}\int \mathrm{d}^2k \frac{\langle n^\chi|\partial_{k_x}H^\chi|m^\chi\rangle\,\langle m^\chi|\partial_{k_y}H^\chi|n^\chi\rangle}{(E_n^\chi - E_m^\chi)^2}\right], \tag{2.32}$$

and to obtain layer Chern number, one may decouple

$$\langle n^\chi|\partial_{k_x}H^\chi|m^\chi\rangle\langle m^\chi|\partial_{k_y}H^\chi|n^\chi\rangle=\sum_{l,l',i,i'}\langle n^\chi(l)|\partial_{k_x}H^\chi|m^\chi(l')\rangle\langle m^\chi(i)|\partial_{k_y}H^\chi|n^\chi(i')\rangle\,,$$

with $|n(l)\rangle=P_l|n\rangle$, where P_l is z-position projection operator; or one could squeeze the zero-part of each $|n(l)\rangle$ to $|n\rangle=\oplus_l|n_S(l)\rangle$. Thing different from the above is now

$$v^\chi_{k_i}=\partial_{k_i}H^\chi=P^{M_z}_\chi\partial_{k_i}H=P^{M_z}_\chi v_{k_i}$$

would contain off-diagonal component induced by

$$P^{M_z}_\chi=\frac{1}{2}\begin{pmatrix}1&&&&i\chi U_z\\&\ddots&&\iddots&\\&&1+i\chi U_z&&\\&\iddots&&\ddots&\\i\chi U_z&&&&1\end{pmatrix},\tag{2.33}$$

i.e., by denoting

$$v_{k_i}=\bigoplus_l v_{k_i}(l),\tag{2.34}$$

one finds that aside for diagonal hopping as

$$\langle n_S(l)|v_{k_i}(l)|m_S(l)\rangle\,,\tag{2.35}$$

one also has

$$\begin{aligned}\langle n(l)|M_zv_{k_i}(l)|m(l')\rangle&=\langle n(l)|\begin{pmatrix}&&U_zv_{k_i}(L_z)\\&\iddots&\\U_zv_{k_i}(1)&&\end{pmatrix}|m(l')\rangle\\&=\langle n_S(l)|\,U_zv_{k_i}(L_z+1-l)\,|m_S(l')\rangle\,\delta_{L_z+1-l,l'},\end{aligned}$$

and if we assume that v_{k_i} does not vary with z denoted as $v^S_{k_i}$, above formula is reduced into

$$\langle n(l)|M_zv_{k_i}(l)|m(l')\rangle=\langle n_S(l)|\,U_zv^S_{k_i}\,|m_S(l')\rangle\,\delta_{l+l',L_z+1},\tag{2.36}$$

i.e., mirror symmetric sites contribute to current under mirror symmetry. Then totally one write

$$\langle n^\chi(l)|\partial_{k_x}H^\chi|m^\chi(l')\rangle=\frac{1}{2}\langle n^\chi_S(l)|\left(v^S_{k_i}\,|m^\chi_S(l')\rangle\,\delta_{l,l'}+U_zv^S_{k_i}\,|m^\chi_S(l')\rangle\,\delta_{l+l',L_z+1}\right).\tag{2.37}$$

Bring this back and totally we get mirror layer Chern number

$$\sigma^\chi_H(l)=\frac{e^2}{h}\frac{\mathrm{Im}}{\pi}\left[\sum_{E^\chi_n<\mu<E^\chi_m}\sum_{l'}\int\mathrm{d}^2k\frac{\bar v^{nm}_{\chi,k_x}(l)\bar v^{mn}_{\chi,k_y}(l')}{(E^\chi_n-E^\chi_m)^2}\right],\tag{2.38}$$

with

$$\bar v^{nm}_{\chi,k_i}(l)=\frac{1}{2}\langle n^\chi_S(l)|\left(v^S_{k_i}\,|m^\chi_S(l)\rangle+U_zv^S_{k_i}\,|m^\chi_S(L_z+1-l)\rangle\right).\tag{2.39}$$

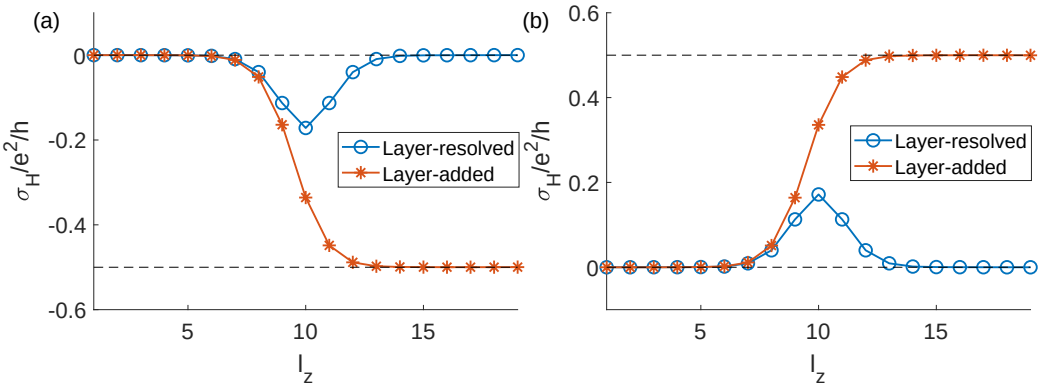


Figure 2.3: Still $L_z = 19$, and $\mu \approx 0$. Mirror layer-resolved and layer-cumulated Chern number for (a) $\chi = +$ and (b) $\chi = -$ part, respectively.

To show the essence, we have calculated mirror layer-resolved and layer-cumulated Chern number to show how Hall conductance changes with layer number for each H_χ^{Proj}, as shown in Fig. 2.3. We choose chemical potential to lie near zero so that only contribution from two gapless Dirac cones are effectively accounted. As within expectation, for $\chi = \pm$ parts, the cumulated Hall conductance approaches $\pm 1/2$ quantum unit, which again shows half quantum mirror Hall effect undoubtedly. What is more, by investigating into the resolved Hall conductance by layer labelled by mirror eigenvalue, one could see that each $1/2$ is contributed from the bulk near mirror surface instead of the surface. This could also

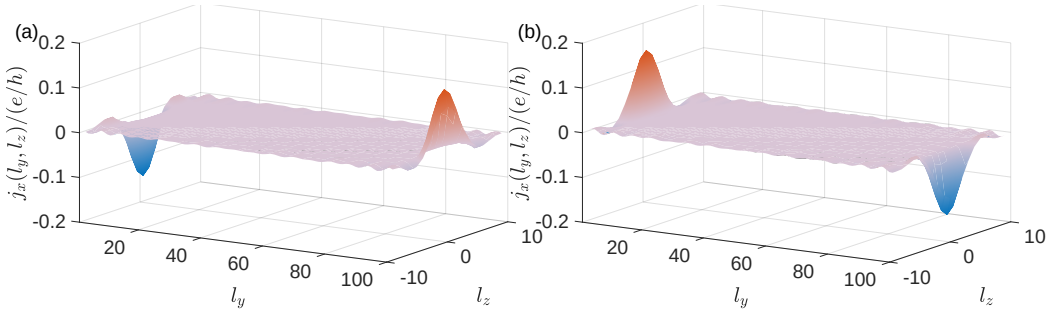


Figure 2.4: Still $L_z = 19$, and $\mu = 0.2$eV. Mirror local current density for (a) $\chi = +$ and (b) $\chi = -$ part, respectively.

be illustrated by local current density for each mirror part, shown in Fig. 2.4, where inverse electric current appears for each part, which would cancel each other totally. Such results show unambiguously that HQMHE comes from 'anomaly', i.e., high energy part contribution.

As a last comment, to detect this, one has to be reminded by the metallic nature of our system, then common two-terminal or even six-terminal measurements may fail to detect solely this effect, caused by non-zero longitudinal conductance/resistance. However, notice that Hall conductance (under Kubo's formalism we calculated Hall conductivity, such bulk quantity is mapped to boundary as dissipationless current which is thus quantized with conductance the same unit as Hall conductivity, and dimension here happens to be the same due to 2-D expression) directly generate current proportional to quantum unit, while longitudinal conductance is proportional to both the width of quasi-2D system and longitudinal conductivity which could be view as a constant (minimal conductivity for

Dirac cone for instance). Then by detecting merely two-terminal conductance with different sample width, one would see a linear changing behavior with conductance-intercept at zero width by real continuation to be quantum unit — similar and standard result as in conventional QSHE case where two-unit conductance emerges. On the other hand, by manually setting mirror-gauge field to the effect, there should be intertextual behavior; say now a six-terminal measurement with $1 \sim 6$ terminals clockwise from the left, then by $6 \to 2$ electric current, there should be $1 \to 4$ mirror current (no net electric current) which by the gauge field, again induces electric current $5 \to 3$, which is detectable.

2.2.4 Break-Down of Mirror Symmetry

It is not that easy to find a strong TI with mirror symmetry easily growable. It is ideal that a crystal is born with nice mirror symmetry or manually given that by a twin boundary, however, such a strict restriction may break down in real experiments. For instance, glide symmetry instead of a mirror one may appear in real materials. When mirror symmetry is not broken severely, we may still have a chance to observe the enhanced current or width-dependent conductance.

Chapter 3

On-Site Field

By on-site field I mean field like $V(\boldsymbol{k}, l_z)$, i.e., it could have sites-related part, but only in $x-y$ plane and could be transformed into single-wavevector dependent form, while it could only contain single-site-dependent form along z. Such a field thus still lies in category of single-particle physics as a processible one in this frame. Field here could be but not limited to external/internal field/order like Zeeman splitting field, magnetic field (rational flux and magnetic Brillouin zone might be required), magnetic order and superconducting order. Every term would play its own important role once form established, through our projection, or equivalently, unitary transformation. In fact, linear term in Dirac equation could be viewed as a special on-site field generated by, say spin-orbital coupling. And its discussion has already been done in deriving the projected model.

3.1 Zeeman Field

Recognize σ as spin DoF, general Zeeman splitting term (homogeneous in $x-y$ plane) would be

$$\mathscr{V}_Z(\boldsymbol{k}) = \sum_{l_z,\boldsymbol{k}} \Psi^\dagger_{l_z,\boldsymbol{k}} V^z_Z(l_z) \Psi_{l_z,\boldsymbol{k}}, \tag{3.1}$$

where

$$V^z_Z(l_z) = \boldsymbol{V}_Z(l_z) \cdot \boldsymbol{\sigma}\tau_0, \tag{3.2}$$

where $\boldsymbol{V}_Z = (V_x, V_y, V^z_Z)$ represents vectorized Zeeman field. Its effective action could also be described using projection $\langle \Phi_m | V^z_Z | \Phi_{m'} \rangle$ $(m, m' = 1, 2, 3, 4)$.

z-Zeeman

Now since $V^z_Z = V^z_Z \sigma_z \tau_0$ is diagonal in its original form, it is simple to see

$$\langle \Phi_{1,3} | V^z_Z | \Phi_{2,4} \rangle = 0,$$

which holds for all possible index combination. For the remaining part we could define two integrals counting different n-blocks as

$$\begin{cases} I^{z,nn'}_S & = \langle \Phi^n_1 | V^z_Z | \Phi^{n'}_1 \rangle \\ iI^{z,nn'}_A & = \langle \Phi^n_1 | V^z_Z | \Phi^{n'}_3 \rangle \end{cases}, \tag{3.3}$$

which varies with wavevector $\boldsymbol{k}$, then by utilizing basis solution above we have

$$\begin{aligned} I_S^{z,nn'}(\boldsymbol{k}) &= \sum_{l_z} \varphi(+,\boldsymbol{k},l_z)^\dagger \tau_0 \varphi(-,\boldsymbol{k},l_z) \\ &= |C_n C_{n'}| \sum_{l_z} V_Z^z(l_z)[\lambda_\perp^2 (f_+^n)^* f_+^{n'} + t_\perp^2 \eta_n \eta_{n'} (f_-^n)^* f_-^{n'}](\boldsymbol{k}) = (I_S^{z,n'n}(\boldsymbol{k}))^*, \quad (3.4a) \\ iI_A^{z,nn'}(\boldsymbol{k}) &= \sum_{l_z} \varphi(+,\boldsymbol{k},l_z)^\dagger \tau_0 \chi(+,\boldsymbol{k},l_z) \\ &= i|C_n C_{n'}| \sum_{l_z} V_Z^z(l_z) \lambda_\perp t_\perp [\eta^{n'} (f_+^n)^* f_-^{n'} + \eta^n (f_-^n)^* f_+^{n'}] = -i(I_A^{z,n'n}(\boldsymbol{k}))^*, \quad (3.4b) \end{aligned}$$

respecting (anti-)symmetric part projection of Zeeman field to z, i.e., by defining

$$\begin{cases} V_S^z(l_z) = [V_Z^z(l_z) + V_Z^z(-l_z)]/2 \\ V_A^z(l_z) = [V_Z^z(l_z) - V_Z^z(-l_z)]/2 \end{cases}, \tag{3.5}$$

two integrals above are equivalent to

$$\begin{aligned} I_S^{z,nn'} &= |C_n C_{n'}| \sum_{l_z} V_S^z(l_z)[\lambda_\perp^2 (f_+^n)^* f_+^{n'} + t_\perp^2 \eta_n \eta_{n'} (f_-^n)^* f_-^{n'}], \\ iI_A^{z,nn'} &= i|C_n C_{n'}| \sum_{l_z} V_A^z(l_z) \lambda_\perp t_\perp [\eta^{n'} (f_+^n)^* f_-^{n'} + \eta^n (f_-^n)^* f_+^{n'}]. \end{aligned} \tag{3.6}$$

Note that $I_{S/A}^{z,nn}$ are real since η is real while off-block-diagonal crossing integrals are essentially not in general. Meanwhile, one finds

$$\begin{cases} \langle \Phi_1^n | V_Z^z | \Phi_1^{n'} \rangle = \langle \Phi_3^n | V_Z^z | \Phi_3^{n'} \rangle = -\langle \Phi_2^n | V_Z^z | \Phi_2^{n'} \rangle = -\langle \Phi_4^n | V_Z^z | \Phi_4^{n'} \rangle \\ \langle \Phi_1^n | V_Z^z | \Phi_3^{n'} \rangle = -\langle \Phi_2^n | V_Z^z | \Phi_4^{n'} \rangle \end{cases}, \tag{3.7}$$

by direct calculation, then by defining two projected z-Zeeman matrices

$$I_{S/A}^z \equiv \begin{pmatrix} I_{S/A}^{z,11} & I_{S/A}^{z,12} & I_{S/A}^{z,13} & \cdots \\ I_{S/A}^{z,21} & I_{S/A}^{z,22} & \ddots & \\ I_{S/A}^{z,31} & \ddots & \ddots & \\ \vdots & & & \end{pmatrix}_{L_z \times L_z}, \tag{3.8}$$

we arrive at final effective Hamiltonian for z-Zeeman term

$$V_{\mathrm{EFF}}^z(\boldsymbol{k}) = (\mathbf{I}_S^z(\boldsymbol{k})\tau_0 - \mathbf{I}_A^z(\boldsymbol{k})\tau_y)\,\sigma_z. \tag{3.9}$$

The attached Dirac matrices are found by considering any example, say for arbitrary blocks n, n' and by ignoring block index, one writes

$$\begin{pmatrix} I_S & & iI_A & \\ & -I_S & & -iI_A \\ -iI_A & & I_S & \\ & iI_A & & -I_S \end{pmatrix} = I_S \tau_0 \sigma_z - I_A \tau_y \sigma_z.$$

x & y-Zeeman

The projected Hamiltonian could be found similarly, and here we make it explicit. Also notice that there is only imaginary unit difference of x and y due to the similar form of $\sigma_x\tau_0$ and $\sigma_y\tau_0$, then it would suffice that we only consider one of them, say x.

Now $V_Z^z = V_x\sigma_x\tau_0$ and due to its back-diagonal form, we have

$$\langle\Phi_{1,3}|V_Z^x|\Phi_{1,3}\rangle = 0 = \langle\Phi_{2,4}|V_Z^x|\Phi_{2,4}\rangle\,.$$

For the non-zero part, similarly two integrals could be defined as

$$\begin{cases} I_S^{x,nn'} &= \langle\Phi_1^n|V_Z^x|\Phi_4^{n'}\rangle \\ I_A^{x,nn'} &= -\langle\Phi_1^n|V_Z^x|\Phi_2^{n'}\rangle \end{cases}, \tag{3.10}$$

with explicit form

$$\begin{aligned} I_S^{x,nn'} &= \sum_{l_z}\varphi(+)^\dagger\tau_0\varphi(-) = |C_nC_{n'}|\sum_{l_z}V_x(l_z)[-\lambda_\perp^2(f_+^n)^*f_+^{n'} + t_\perp^2\eta_n\eta_{n'}(f_-^n)^*f_-^{n'}] \qquad (3.11a)\\ &= (I_S^{x,n'n})^*, \\ I_A^{x,nn'} &= -\sum_{l_z}\varphi(+)^\dagger\tau_0\chi(-) = -i|C_nC_{n'}|\sum_{l_z}V_x(l_z)\lambda_\perp t_\perp[\eta^{n'}(f_+^n)^*f_-^{n'} - \eta^n(f_-^n)^*f_+^{n'}] \qquad (3.11b)\\ &= (I_A^{x,n'n})^*, \end{aligned}$$

and also $I_{S/A}^{x,nn}$ are real, where the anti-symmetric part is seen by

$$-i[\eta^n(f_+^n)^*f_-^n - \eta^n(f_-^n)^*f_+^n] = 2\eta^n\,\mathrm{Im}((f_+^n)^*f_-^n).$$

And one also finds

$$\begin{cases} \langle\Phi_1^n|V_Z^x|\Phi_4^{n'}\rangle = \langle\Phi_3^n|V_Z^x|\Phi_2^{n'}\rangle\,, \\ \langle\Phi_1^n|V_Z^x|\Phi_2^{n'}\rangle = -\langle\Phi_3^n|V_Z^x|\Phi_4^{n'}\rangle \end{cases} \tag{3.12}$$

To find Dirac matrices attached, still chose arbitrary example and see

$$\begin{pmatrix} & -I_A & & I_S \\ -I_A & & I_S & \\ & I_S & & I_A \\ I_S & & I_A & \end{pmatrix} = I_S\tau_x\sigma_x - I_A\tau_z\sigma_x.$$

Then similarly we arrive at effective Hamiltonian for x-Zeeman term

$$V_{\mathrm{EFF}}^x(\boldsymbol{k}) = (\mathbf{I}_S^x(\boldsymbol{k})\tau_x - \mathbf{I}_A^x(\boldsymbol{k})\tau_z)\,\sigma_x. \tag{3.13}$$

Similarly, we could arrive at y-Zeeman projection as

$$V_{\mathrm{EFF}}^x(\boldsymbol{k}) = (\mathbf{I}_S^y(\boldsymbol{k})\tau_x - \mathbf{I}_A^y(\boldsymbol{k})\tau_z)\,\sigma_y, \tag{3.14}$$

by realizing that under the same field-strength configuration,

$$\langle\Phi_{1,3}^n|V_Z^y|\Phi_{2,4}^{n'}\rangle = -i\,\langle\Phi_{1,3}^n|V_Z^x|\Phi_{2,4}^{n'}\rangle\,, \tag{3.15}$$

i.e.,

$$\begin{pmatrix} & iI_A & & -iI_S \\ -iI_A & & iI_S & \\ & -iI_S & & -iI_A \\ iI_S & & iI_A & \end{pmatrix} = I_S\tau_x\sigma_y - I_A\tau_z\sigma_y.$$

Notice that matrix $\mathbf{I}^{x,y}_{S/A}$ has the same form as with Eq. (3.8), but with z replaced with x, y. Also notice that $\sigma_{x,y}$ are preserved, since they still represent spin degrees of freedom, and Zeeman nature is thus unchanged.

In sum, general Zeeman term homogeneous in $x-y$ plane while varies with z would be projected into the form

$$V_Z^{\text{Proj}} = \boldsymbol{V}_Z^{\text{EFF}} \cdot \boldsymbol{\sigma}, \tag{3.16}$$

where effective Zeeman field reads

$$\begin{aligned} \boldsymbol{V}_Z^{\text{EFF}} &= (\mathbf{I}_S^x(\boldsymbol{k})\tau_x - \mathbf{I}_A^x(\boldsymbol{k})\tau_z, \mathbf{I}_S^y(\boldsymbol{k})\tau_x - \mathbf{I}_A^y(\boldsymbol{k})\tau_z, \mathbf{I}_S^z(\boldsymbol{k})\tau_0 - \mathbf{I}_A^z(\boldsymbol{k})\tau_y) \\ &= \vec{\mathbf{I}}_S\tau_x - \vec{\mathbf{I}}_A\tau_z, \end{aligned} \tag{3.17}$$

where $\vec{\mathbf{I}}_P = (\mathbf{I}_P^x, \mathbf{I}_P^y, i^P\mathbf{I}_P^z\tau_x)$, with

$$i^P \equiv \begin{cases} 1, & P = S \\ i, & P = A \end{cases},$$

i.e., P counts for parity.

3.2 Gate-Voltage

This term is used mainly to counts for the asymmetric bias over the film, either by manual set or substrate drag.

Now, gate-voltage as on-site electrical potential reads

$$\mathscr{V}_G(\boldsymbol{k}) = \sum_{l_z,\boldsymbol{k}} \Psi^\dagger_{l_z,\boldsymbol{k}} V_G(l_z) \Psi_{l_z,\boldsymbol{k}}, \tag{3.18}$$

where

$$V_G(l_z) = V(l_z)\sigma_0\tau_0, \tag{3.19}$$

and similarly, due to its diagonal form, we still have

$$\langle \Phi_{1,3} | V_G | \Phi_{2,4} \rangle = 0,$$

for all index combination. And again, two integrals are defined as

$$\begin{cases} V_S^{nn'} = \langle \Phi_i^n | V_Z^z | \Phi_i^{n'} \rangle \\ iV_A^{nn'} = \langle \Phi_1^n | V_Z^z | \Phi_3^{n'} \rangle \end{cases}, \tag{3.20}$$

with again $V^{nn}_{S/A}$ terms real; integrals equal

$$V_S^{nn'} = |C_n C_{n'}| \sum_{l_z} V(l_z) \left[\lambda_\perp^2 (f_+^n)^* f_+^{n'} + t_\perp^2 \eta^n \eta^{n'} (f_-^n)^* f_-^{n'} \right] = (V_S^{n'n})^*, \tag{3.21a}$$

$$iV_A^{nn'} = |C_n C_{n'}| \sum_{l_z} V(l_z) i\lambda_\perp t_\perp \left[\eta^{n'}(f_+^n)^* f_-^{n'} + \eta^n (f_-^n)^* f_+^{n'}\right] = -i(V_A^{n'n})^*, \tag{3.21b}$$

with

$$\langle \Phi_1^n | V_Z^z | \Phi_3^{n'} \rangle = \langle \Phi_2^n | V_Z^z | \Phi_4^{n'} \rangle . \tag{3.22}$$

By finding attached Dirac matrices as

$$\begin{pmatrix} V_S & & iV_A & \\ & V_S & & iV_A \\ -iV_A & & V_S & \\ & -iV_A & & V_S \end{pmatrix} = V_S \tau_0 \sigma_0 - V_A \tau_y \sigma_0,$$

we arrive at projected gate-voltage potential

$$V_G^{\text{Proj}}(\boldsymbol{k}) = \mathbf{V}_S(\boldsymbol{k}) \tau_0 \sigma_0 - \mathbf{V}_A(\boldsymbol{k}) \tau_y \sigma_0, \tag{3.23}$$

with two $L_z \times L_z$ matrices defined similarly. Again one sees that spin-related DoF as σ part has been totally conserved. And what is more important is apart from a somehow trivial shift represented by $\mathbf{V}_S$, the other term $\mathbf{V}_A$ contains effective-orbital action upon τ_y. Such a term should be responsible for Rashba type spin-orbital coupling, since asymmetric field configuration reveals existence of gradient electric potential which generates coupling by

$$\nabla V \cdot (\boldsymbol{\sigma} \times \boldsymbol{p}),$$

we shall discuss the effect in detail later.

3.3 Orbital Order

Motivation for such a term is non-trivial. Notice that when explaining HQMHE using mirror symmetry, we decouple even layer-number Hamiltonian into its mirror-eigen-form explained as a half-length chain with magnetic field at one end, which is

$$i\chi \phi_{1/2,\chi}^\dagger \mathscr{T}_z^\dagger U_z \phi_{1/2,\chi},$$

where

$$\mathscr{T}_z^\dagger U_z = (t_\perp \beta + i\frac{\lambda_\perp}{2}\alpha_z)(-i\sigma_z\tau_z) = -i(t_\perp \sigma_z + \frac{\lambda_\perp}{2}\tau_y), \tag{3.24}$$

is effective magnetic field under mirror representation. Here again σ is preserved as real spin, and the first term represents a normal Zeeman coupling, which locates at one side of a TI film with half thickness, effectively.

Now as a forward-looking, we claim that such a 'TI film with one-side magnetism' would reveal half-quantized Hall conductance as a metallic topological phase, whose existence depends highly on the single surface Dirac-cone which parity symmetry at low-energy. Now since σ_z plays such a role, one may expect τ_y to step in with also a non-trivial manner, and it indeed does.

Given

$$\mathscr{V}_O^y(\boldsymbol{k}) = \sum_{l_z, \boldsymbol{k}} \Psi_{l_z,\boldsymbol{k}}^\dagger V_O^y(l_z) \Psi_{l_z,\boldsymbol{k}}, \tag{3.25}$$

where

$$V_O^y(l_z) = V^y(l_z) \sigma_0 \tau_y, \tag{3.26}$$

and due to its block-diagonal form, we still have

$$\langle\Phi_{1,3}|V_O^y|\Phi_{2,4}\rangle = 0,$$

with two integrals with real $n-n$ components written as

$$\begin{cases} V_A^{y,nn'} = \langle\Phi_1^n|V_O^y|\Phi_1^{n'}\rangle \\ iV_S^{y,nn'} = \langle\Phi_1^n|V_O^y|\Phi_3^{n'}\rangle \end{cases}, \tag{3.27}$$

and they equal

$$\begin{aligned} V_A^{y,nn'} &= \sum_{l_z} V^y(l_z)\varphi^n(+)^\dagger\tau_y\varphi^{n'}(+) \\ &= |C_nC_{n'}|\sum_{l_z} V^y(l_z)\lambda_\perp t_\perp\left[\eta^{n'}(f_+^n)^*f_-^{n'} + \eta^n(f_-^n)^*f_+^{n'}\right] = (V_A^{y,n'n})^*, \end{aligned} \tag{3.28a}$$

$$\begin{aligned} iV_S^{y,nn'} &= \sum_{l_z} V^y(l_z)\varphi^n(+)^\dagger\tau_y\chi^{n'}(+) \\ &= i|C_nC_{n'}|\sum_{l_z} V^y(l_z)\left[\lambda_\perp^2(f_+^n)^*f_+^{n'} + t_\perp^2\eta^n\eta^{n'}(f_-^n)^*f_-^{n'}\right] = -i(V_S^{y,n'n})^*, \end{aligned} \tag{3.28b}$$

with relation

$$\begin{cases} \langle\Phi_1^n|V_O^y|\Phi_1^{n'}\rangle = -\langle\Phi_2^n|V_O^y|\Phi_2^{n'}\rangle = \langle\Phi_3^n|V_O^y|\Phi_3^{n'}\rangle = -\langle\Phi_4^n|V_O^y|\Phi_4^{n'}\rangle \\ \langle\Phi_1^n|V_O^y|\Phi_3^{n'}\rangle = -\langle\Phi_2^n|V_O^y|\Phi_4^{n'}\rangle \end{cases}. \tag{3.29}$$

Then attached Dirac matrices are found as

$$\begin{pmatrix} V_A & & iV_S & \\ & -V_A & & -iV_S \\ -iV_S & & V_A & \\ & iV_S & & -V_A \end{pmatrix} = V_A\tau_0\sigma_z - V_S\tau_y\sigma_z,$$

and projected form is

$$V_O^{y,\mathrm{Proj}}(\boldsymbol{k}) = \mathbf{V}_A^y(\boldsymbol{k})\tau_0\sigma_z - \mathbf{V}_S^y(\boldsymbol{k})\tau_y\sigma_z. \tag{3.30}$$

Notice that spin DoF here is not preserved due to explicit influence of orbital order, which would interact with spin order through spin-orbital coupling. Also, we see here that antisymmetric part is attached with $\tau_0\sigma_z$ matrix that acts as if a Zeeman field. In this way we may expect similar physics as half-quantized QAHE but with different physical origins.

Again similarly, τ_x has parallel behavior. Now

$$\begin{cases} V_A^{x,nn'} = \langle\Phi_1^n|V_O^x|\Phi_1^{n'}\rangle \\ V_S^{x,nn'} = -\langle\Phi_1^n|V_O^x|\Phi_3^{n'}\rangle \end{cases}, \tag{3.31}$$

with $V_O^x(l_z) = V^x(l_z)\sigma_0\tau_x$, and they equal

$$V_A^{x,nn'} = \sum_{l_z} V^x(l_z)\varphi^n(+)^\dagger\tau_x\varphi^{n'}(+)$$

$$= i|C_n C_{n'}| \sum_{l_z} V^x(l_z)\lambda_\perp t_\perp \left[\eta^{n'}(f_+^n)^* f_-^{n'} - \eta^n (f_-^n)^* f_+^{n'}\right] = (V_A^{x,n'n})^*, \tag{3.32a}$$

$$V_S^{x,nn'} = -\sum_{l_z} V^x(l_z)\varphi^n(+)^\dagger \tau_x \chi^{n'}(+)$$

$$= |C_n C_{n'}| \sum_{l_z} V^x(l_z) \left[\lambda_\perp^2 (f_+^n)^* f_+^{n'} - t_\perp^2 \eta^n \eta^{n'} (f_-^n)^* f_-^{n'}\right] = (V_S^{x,n'n})^*, \tag{3.32b}$$

with relation

$$\begin{cases} \langle \Phi_1^n | V_O^x | \Phi_1^{n'} \rangle = \langle \Phi_2^n | V_O^x | \Phi_2^{n'} \rangle = -\langle \Phi_3^n | V_O^x | \Phi_3^{n'} \rangle = -\langle \Phi_4^n | V_O^x | \Phi_4^{n'} \rangle \\ \langle \Phi_1^n | V_O^x | \Phi_3^{n'} \rangle = \langle \Phi_2^n | V_O^x | \Phi_4^{n'} \rangle \end{cases}, \tag{3.33}$$

and attached Dirac matrices

$$\begin{pmatrix} V_A & & -V_S & \\ & V_A & & -V_S \\ -V_S & & -V_A & \\ & -V_S & & -V_A \end{pmatrix} = V_A \tau_z \sigma_0 - V_S \tau_x \sigma_0,$$

and projected form is

$$V_O^{x,\mathrm{Proj}}(\boldsymbol{k}) = \mathbf{V}_A^x(\boldsymbol{k})\tau_z\sigma_0 - \mathbf{V}_S^x(\boldsymbol{k})\tau_x\sigma_0. \tag{3.34}$$

3.4 Superconducting Order

3.5 Effective Mass

Here, we point it out that when projected on-site field contains σ_z part, it could be combined with mass term and further diagonalized at least formally, which then maps mass into effective mass, i.e., generally,

$$\left[\bigoplus_{n=1}^{L_z} m_n(\boldsymbol{k})\tau_z + \mathbf{I}_P^\mu(\boldsymbol{k})\tau_\mu\right] \longmapsto \bigoplus_{p=1}^{2L_z} \tilde{m}_p(\boldsymbol{k}), \tag{3.35}$$

where $\tilde{m}_p(\boldsymbol{k})$ is the effective mass labelled by p. Such an index may be further separated by additional symmetry like mirror.

Chapter 4

Quantum Hall Family

The famous story of Hall effect[21], quantum Hall effect[28] and TKNN number[55] is not repeated here but put in the appendix.

4.0 Mirror Symmetry for Zeeman Field

Quantized Hall effect in this chapter mainly refers to quantum anomalous Hall effect, where term 'anomalous' means non-trivial Hall conductance appears in realized lattice system without help from external magnetic field. Usually one would expect intrinsic magnetic order inside the material with non-zero average value. On the other hand, since in principle, non-zero Hall conductance only needs breaking of time-reversal, one may expect more kinds of orders that would work. Here we only emphasize on Zeeman order here and find out what mirror symmetry could bring to it.

As a general requirement, one needs

$$U_z^\dagger V_Z(l_z) U_z = V_Z(l_z) \tag{4.1}$$

for standing of mirror symmetry. And when specified to z-Zeeman, on needs

$$V_Z(l_z) = V_Z(-l_z), \tag{4.2}$$

as symmetric z-Zeeman distribution, which would gain the form after mirror transformation,

$$\mathscr{V}_\chi = \sum_{l_z=1/2}^{(L_z-1)/2} \phi_{l_z,\chi}^\dagger V_S(l_z) \phi_{l_z,\chi}, \qquad L_z \text{ even} \tag{4.3a}$$

$$\mathscr{V}_\chi = \sum_{l_z=1}^{(L_z-1)/2} \phi_{l_z,\chi}^\dagger V_S(l_z) \phi_{l_z,\chi} + \Psi_0^\dagger P_\chi^{M_z} V_S(0) \Psi_0, \qquad L_z \text{ odd} \tag{4.3b}$$

where $\boldsymbol{k}$ is hidden while

$$V_S(l_z) = \frac{V_Z(l_z) + V_Z(-l_z)}{2} = V_Z(l_z) \tag{4.4}$$

as symmetrized Zeeman field. Such a form is similar to 2-D part under mirror transformation.

4.1 Conventional Quantum Anomalous Hall Effect

As a conceptional extension from QHE to lattice system in condensed matter, quantum anomalous Hall effect (QAHE), aka Chern insulator has been a long desired phenomenon since its proposal[20]. However, its actual realization[7, 5, 8] is even much later than younger family members as 2-D and 3-D TIs. The reason is just the same as scarcity of intrinsic 2-D QSH system so does its nowadays solution.

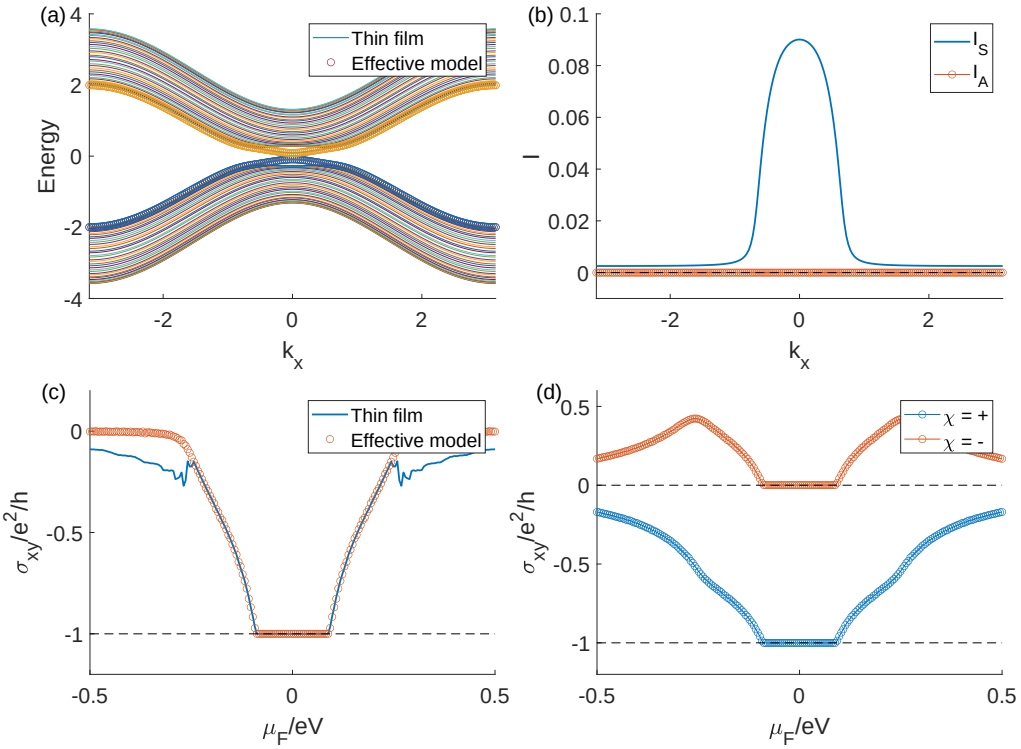


Figure 4.1: Conventional QAHE. Here top and bottom two layers with Zeeman field immersed are chose upon total 19 layers. z-Zeeman field were chosen to be parallel with the same strength $V_z = 0.1$eV. (a) Band structure calculated from thin film Hamiltonian and four-band effective model. (b) Lowest block integrals. (c) Hall conductance calculated from thin film Hamiltonian and four-band effective model. (d) Hall conductance calculated from two sub-blocks.

Recall the HQMHE introduced before, where two massless Dirac cones with inverse half-quantized Hall conductance were show explicitly. Now imagine that we add Zeeman field at both top and bottom several layers so that surface states are dramatically influenced. In fact, Zeeman field here acts like an emergent Higgs field that generates mass to both originally massless Dirac cones, with the same mass sign. Nevertheless, recall that HQMHE tells us that two cones own inverse high-energy mass sign, thus by the simple formula

$$c_n^\chi = \frac{1}{2}\left[\tilde{m}_{n,\chi}(\infty) - \tilde{m}_{n,\chi}(0)\right], \tag{4.5}$$

where $n \in [1, L_z]$ denotes different bands (in this discussion limited to $n = 1$) and $\chi = \pm$ for mirror index under separable case, we see that one band contributes $C = 1$ while the other is trivial. Notice that one trick is hided here as in HQMHE, two Dirac cones are separated by mirror symmetry, which shares living on two-side nature like parity

symmetry, then two Dirac bands trivial or non-trivial are also separated by mirror plane, as we would see later.

As the simplest explanation, we may directly preserve $n = 1$ block as weak Zeeman approximation

$$\begin{aligned} H_{\mathrm{EFF}} &= \lambda_{\parallel}\tau_0(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \begin{pmatrix} m+I_S & 0 & iI_A & 0 \\ 0 & -m-I_S & 0 & -iI_A \\ -iI_A & 0 & -m+I_S & 0 \\ 0 & iI_A & 0 & m-I_S \end{pmatrix} \\ &= \lambda_{\parallel}\tau_0(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + m\tau_z\sigma_z + I_S\tau_0\sigma_z - I_A\tau_y\sigma_z, \end{aligned} \tag{4.6}$$

where we have neglected index $n = 1$. By numerically calculating $m, I_{S/A}$, we could reproduce corresponding lowest-four-band structure as shown in Fig. 4.1(a) compared with band structure calculated directly from H_{Film} with Zeeman field, and the consistency is shown clearly. Compare of Hall conductance between the four-band model and film Hamiltonian is also shown in Fig. 4.1(c), where both cases show $C = -1$ plateau inside Zeeman gap. What is more, under our symmetric Zeeman field configuration, I_A vanishes as shown in Fig. 4.1(b), then our block Hamiltonian could be further split into two sub-blocks as

$$H_{\mathrm{EFF},\chi} = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + (\chi m(\boldsymbol{k}) + I_S(\boldsymbol{k}))\sigma_z, \tag{4.7}$$

whose Hall conductance is shown in Fig. 4.1(d), from which one sees clearly that $\chi = +$ sub-block contributes unit Hall plateau while $\chi = -$ sub-block is trivial. This is essentially because, $\chi = \pm$ determines sign of high-energy mass of χm, then together with I_S as equivalent low-energy mass, whether the global gap is opened in a (non-)trivial way is given. The Chern numbers of the bands are given by

$$c^{\chi} = \frac{1}{2}\left\{\mathrm{sgn}[\chi m(\pi,\pi) + I_S(\pi,\pi)] - \mathrm{sgn}[\chi m(0) + I_S(0)]\right\}. \tag{4.8}$$

For a TI film with sufficient thickness, $m(0) \to 0$ and $|I_S(\pi,\pi)| < |m(\pi,\pi)|$, thus there is always one trivial band and one nontrivial band. This explains why one sees chiral edge state in this system, upon the boundary of $\chi = +$ sub-block and trivial vacuum. This is shown unambiguously in local density of states by Fig. 4.2.

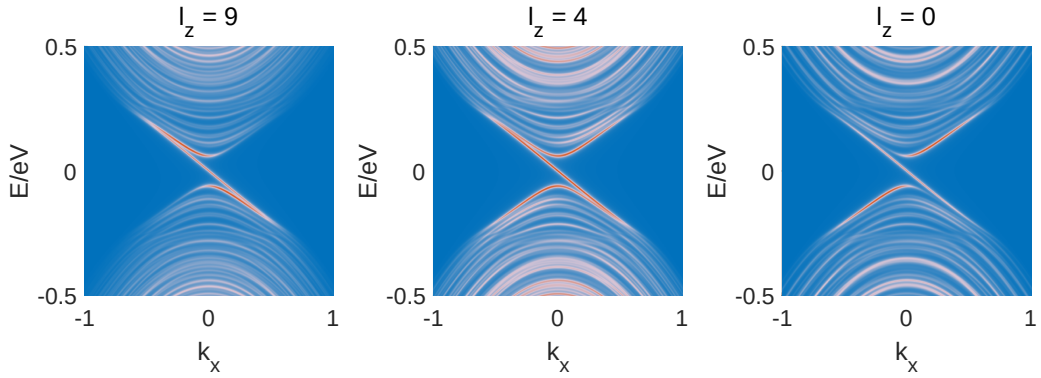


Figure 4.2: Local density of states of conventional quantum anomalous Hall system with open-y geometry by $L_y = 20$ at y-front surface, several representative heights have been chosen to show its l_z-dependent behavior. (Notice that $L_z = 19$ is put z-symmetrically)

Especially notice that our Zeeman field configuration is a symmetric one, i.e., mirror symmetry is respected. This is revealed since the projected mirror operator is represented

to be proportional to τ_z, i.e., each χ sub-block we have defined above is a mirror-projected one under weak Zeeman approximation, and what we have shown is mirror Chern number. What is more, since mirror or parity symmetry is included in our basis $|\Phi\rangle$ naturally, we

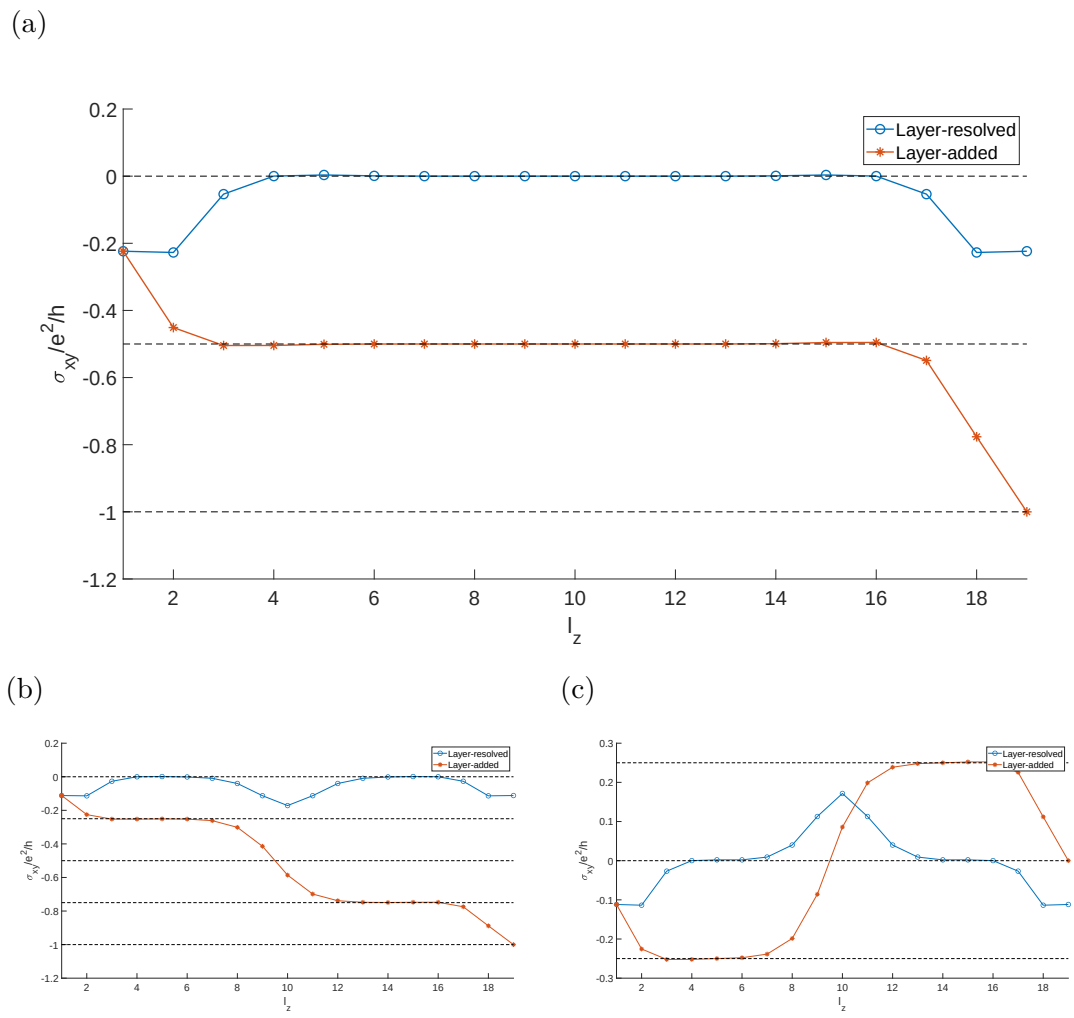


Figure 4.3: (a) Layer resolved and added Chern number. (b) Mirror layer resolved and added Chern number with $\chi = +$. (c) Mirror layer resolved and added Chern number with $\chi = -$.

should be aware that both $C = 1$ and $C = 0$ would have a distribution symmetric about the film, and this is revealed from mirror layer Chern number. As shown in Fig. 4.3, for $\chi = +$ part in (b), Chern number shows

$$C^{\chi=+} = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1, \tag{4.9}$$

while for $\chi = -$ in (c), we see

$$C^{\chi=-} = \frac{a}{4} - \frac{a}{2} + \frac{a}{4} = 0, \ a \neq 1. \tag{4.10}$$

By $1/4$ or $a/4$, we mean surface contribution, while $1/2$ or $a/2$ means bulk contribution from anomaly. In non-trivial case by $\chi = +$, $1/4$ is accurate due to the existence of mass domain-wall, aka point-wise parity invariant regime; however, for trivial part by $\chi = -$,

no such property is guaranteed, and the best we could expect is $a/4$. As a result, no half-quantization exists at surface of QAH film. One may argue for non-perfectness of $1/4$ for $\chi = +$, and our explanation for that is lattice constant limitation. One could then expect perfect 1/4-tization in continuum model.

4.1.1 Higher Chern Numbers

Here general picture for origin of high Chern systems under z-Zeeman field is given as a special application. Notice that given homogenous Zeeman field whose strength is denoted as V_z,

$$\begin{aligned}\langle\Phi_i^n|V_Z|\Phi_j^{n'}\rangle &= V_z\sum_{l_z}\Phi_i^{n\dagger}(l_z)\sigma_z\tau_0\Phi_j^{n'}(l_z)\\ &= V_z(-1)^{i+1}\sum_{l_z}\Phi_i^{n\dagger}(l_z)\Phi_j^{n'}(l_z) = V_z(-1)^{i+1}\delta_{i,j}\delta_{n,n'},\end{aligned} \tag{4.11}$$

then clearly, only diagonal part of $\mathbf{I}_S^z$ is preserved, i.e.,

$$\begin{cases}\mathbf{I}_A^z = 0\\ \mathbf{I}_S^z = V_z\mathbb{I}_{L_z}\end{cases}. \tag{4.12}$$

Now the projected Hamiltonian becomes

$$H_{\text{EFF}} = \bigoplus_{\chi=\pm,n=1}^{L_z} H_{n,\chi},\ H_{n,\chi} = \lambda_\parallel(\sin(k_xa)\sigma_x + \sin(k_yb)\sigma_y) + (V_z + \chi m_n)\sigma_z, \tag{4.13}$$

as a sub-block direct-sum form over L_z masses and mirror index, which provides us a natural platform to study Chern number for each sub-block, given by

$$c_n^\chi = \frac{1}{2}\left\{\text{sgn}[\chi m_n(\pi,\pi) + V_z] - \text{sgn}[\chi m_n(0) + V_z]\right\}. \tag{4.14}$$

To make the picture further clearer, let us consider the case V_z increases from zero. When V_z at first time varies beyond $m_1(0)$ (either gapless or finite-size gap one for STI film, and in the former case arbitrarily small V_z works), since $m_1(\pi,\pi) < 0$, we see that

$$\begin{aligned}c_1^+ &= \frac{1}{2}\left\{\text{sgn}[m_1(\pi,\pi) + V_z] - \text{sgn}[m_1(0) + V_z]\right\} = \frac{1}{2}(-1-1) = -1,\\ c_1^- &= \frac{1}{2}\left\{\text{sgn}[-m_1(\pi,\pi) + V_z] - \text{sgn}[-m_1(0) + V_z]\right\} = \frac{1}{2}(1-1) = 0,\end{aligned}$$

while all other bands remain trivial. Similar discussion could be pushed forward to the case when V_z increases further. And one could easily obtain the image that as long as $|m_n(0)| < V_z < |m_{n+1}(0)|$, we would obtain total Chern number $C = n$. Chern number for each non-trivial band adds up together instead of cancelling each other, because sgn function at $k = 0$ is fixed by V_z, and speaking of this, due to the sign-jumping behavior of m_n varying with n, we see that the mirror index of newly obtained non-trivial band also jumps between $\chi = \pm$. Such a picture holds as long as $V_z < |m_1(\pi,\pi)|$ shown in Fig. 4.4. The absence of $C = -2 \sim -4$ is because $|m_{2\sim4}(0)|$ have rather close values, and they are exceeded at the same time. Also notice that maximum $|C|_{\max} = 19 = L_z$, which is our layer number.

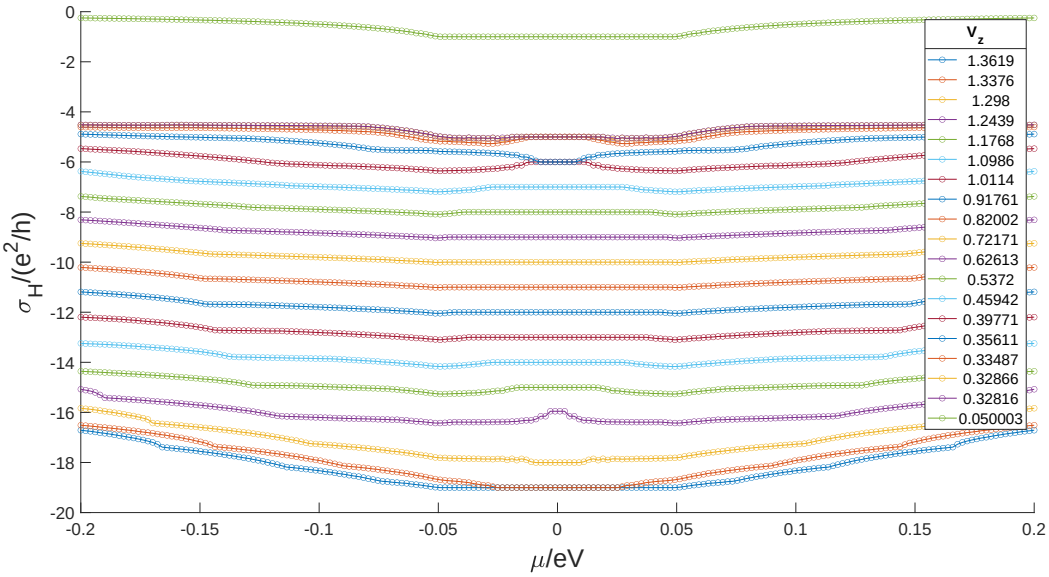


Figure 4.4: High Chern number varies with V_z.

Picture above needs modification when V_z increases further, exceeding high-energy part of $|m_n(\pi,\pi)|$. As one could easily conclude, $V_z > |m_n(\pi,\pi)|$ means loss of n non-trivial bands upon $C = -19$ background, and n-bands here are indeed trivialized and driven into trivial flat bands, or Zeeman level. Finally, when V_z is beyond bandwidth, all bands are trivialized and driven into Zeeman splitting energy level labelled by spin index.

We emphasize here this is the general picture that gives rise to higher Chern number using Zeeman field, and it is quite easily understandable. All other Zeeman configurations that lead to higher Chern number origin from discussion and picture above.

4.1.2 General Symmetry

Cases we have discussed above are all mirror-symmetric, based on which χ-labelled sub-blocks could be separated. On the other hand, when $\mathbf{I}_A \neq 0$, mirror symmetry is broken, then question emerges as how could we distinguish non-trivial bands. The general simple answer is to utilize effective mass picture mentioned before.

As an example, we may now consider general weak Zeeman field and preserve only $n = 1$ block for the investigation. The projected four-band effective Hamiltonian is written as

$$\begin{aligned} H_{\text{EFF}} &= \lambda_{\parallel}\tau_0(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + m\tau_z\sigma_z + I_S\tau_0\sigma_z - I_A\tau_y\sigma_z \\ &= \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + (m\tau_z + I_S\tau_0 - I_A\tau_y)\sigma_z. \end{aligned} \tag{4.15}$$

In the second line, we have explicitly gathered terms that contribute to the mass, i.e., σ_z term, and diagonalizing this leads to effective mass

$$\tilde{m}_{\pm} = I_S \pm \sqrt{m^2 + I_A^2} \tag{4.16}$$

with k-dependence. Then the effective Hamiltonian is again separable as

$$\tilde{H}_{\text{EFF},\pm} = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \tilde{m}_{\pm}(\boldsymbol{k})\sigma_z. \tag{4.17}$$

Based on this, upon weak Zeeman and assume $|m(\pi,\pi)| > I_S(\pi,\pi),\ I_S(0) > 0$, we see

• $I_S(0) > |I_A(0)|$, then $\tilde{m}_-$ is non-trivial and contributes Chern number one, while $\tilde{m}_+$ is trivial. This case corresponds to the case symmetric order overwhelms antisymmetric order, like top-bottom parallel Zeeman setup. Also notice that since $\tilde{m}_-(0) \leq \tilde{m}_+(0)$, it is always the band with smaller gap that is non-trivial;
• $I_S(0) = |I_A(0)|$, in such case, we see that $\tilde{m}_-(0) = 0$ is non-trivial and contributes half-quantized anomalous Hall effect as we shall discuss later, while $\tilde{m}_+$ is still trivial. This case corresponds to the case symmetric order has a fine match with antisymmetric order, for instance we may only add Zeeman field at top of the film;
• $I_S(0) < |I_A(0)|$, now $\tilde{m}_\pm > 0$ always stand, and both bands are trivial. This case corresponds to the case symmetric order is overwhelmed by antisymmetric order, like the so-called axion insulator realization where people add Zeeman field at top and bottom but with inverse polarization.

In the discussion above, still we obtain two separable sub-blocks by considering effective mass. Such a separable feature should correspond to some symmetry, and let us try to figure it out. Re-writing of effective mass leads to

$$\begin{pmatrix} \tilde{m}_+ & \\ & \tilde{m}_- \end{pmatrix} = I_S\tau_0 + \sqrt{m^2 + I_A^2}\tau_z,$$

given by unitary transformation as

$$U_M^\dagger(m\tau_z + I_S\tau_0 - I_A\tau_y)U_M = \begin{pmatrix} \tilde{m}_+ & \\ & \tilde{m}_- \end{pmatrix}, \tag{4.18}$$

where $U_M = (|+\rangle, |-\rangle)$ as unitary matrix formed by eigenstates of $(m\tau_z + I_S\tau_0 - I_A\tau_y)$, which is a k-dependent one diagonalizing mass. Explicitly, it is given by

$$|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} i\mathrm{sgn}(I_A)\sqrt{1+m/M} \\ \sqrt{1-m/M} \end{pmatrix}, \ |-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -i\mathrm{sgn}(I_A)\sqrt{1-m/M} \\ \sqrt{1+m/M} \end{pmatrix},$$

where $M \equiv \sqrt{m^2 + I_A^2}$. Then notice that τ_z offers symmetry index for effective mass, we could transform it back and get original symmetry operator (in the projected representation) which separates two blocks as

$$\begin{aligned} S(\boldsymbol{k}) &= U_M(\boldsymbol{k})\tau_z U_M^\dagger(\boldsymbol{k}) = |+\rangle\langle+| - |-\rangle\langle-| \\ &= \begin{pmatrix} m/M & i\,\mathrm{sgn}(I_A)\sqrt{1-m^2/M^2} \\ -i\,\mathrm{sgn}(I_A)\sqrt{1-m^2/M^2} & -m/M \end{pmatrix}(\boldsymbol{k}) \\ &= \cos\varphi(\boldsymbol{k})\tau_z - \mathrm{sgn}[I_A(\boldsymbol{k})]\sin\varphi(\boldsymbol{k})\tau_y, \end{aligned} \tag{4.19}$$

where $\cos\varphi(\boldsymbol{k}) \equiv m/M$. Eigenvalues of $S(\boldsymbol{k})$ (which is clearly ± 1) label separable sub-blocks while eigenstates ($|\pm\rangle$ in this case) offer a concrete way to construct the separation (inverse transformation, representation basis). On the other hand, essentially, from the form we see that $S(\boldsymbol{k})$ is just the normalized mass term.

Generally, processable separation does not need k-global symmetry, effective mass has already made this. Now given generally

$$\left[\bigoplus_{n=1}^{L_z} m_n(\boldsymbol{k})\tau_z + \mathbf{I}^\mu(\boldsymbol{k})\tau_\mu\right]\sigma_z, \ \mu = 0,1,2,3, \tag{4.20}$$

where $\mathbf{I}$ as $L_z \times L_z$ matrices include all possible on-site field action. Diagonalizing such a mass term reads

$$U_M^\dagger(\boldsymbol{k}) \left[\bigoplus_{n=1}^{L_z} m_n(\boldsymbol{k})\tau_z + \mathbf{I}^\mu(\boldsymbol{k})\tau_\mu \right] U_M(\boldsymbol{k}) = \bigoplus_{n=1}^{2L_z} \tilde{m}_n(\boldsymbol{k}) \tag{4.21}$$

as effective masses, corresponding to $2L_z$ 2×2 sub-blocks.

Claim 1. *$\boldsymbol{k}$-dependent symmetry operator is written (guessed) as*

$$S(\boldsymbol{k}) = \mathbb{I} - 2P(\boldsymbol{k}), \tag{4.22}$$

where $P(\boldsymbol{k})$ is mass projection operator defined from lowest L_z eigenstates of mass term.

Proof. By definition

$$P(\boldsymbol{k}) = \sum_n |-\rangle_n \langle -|_n \,, \; n \in [1, L_z], \tag{4.23}$$

where

$$\left[\bigoplus_{n=1}^{L_z} m_n(\boldsymbol{k})\tau_z + \mathbf{I}^\mu(\boldsymbol{k})\tau_\mu \right] |-\rangle_n = \tilde{m}_n(\boldsymbol{k}) \, |-\rangle_n \,. \tag{4.24}$$

Then to show $S(\boldsymbol{k})$ is a mass symmetry operator, we only need to show

$$[M_P, P] = M_P \sum_n |-\rangle_n \langle -|_n - \sum_n |-\rangle_n \langle -|_n M_P = 0,$$

which is obvious. Notice that we have re-defined

$$M_P(\boldsymbol{k}) = \left[\bigoplus_{n=1}^{L_z} m_n(\boldsymbol{k})\tau_z + \mathbf{I}^\mu(\boldsymbol{k})\tau_\mu \right]$$

as projected (using our 1-D solution basis) mass term ($P(\boldsymbol{k})$ insists of lowest L_z eigenstates of $M_P(\boldsymbol{k})$). □

Such a symmetry operator is both Hermitian and unitary, belonging to $GL(2L_z, \mathbb{C})$, and could be recognized to afford special transport path upon local trivialization $U_{\mathrm{BZ}}(\boldsymbol{k}) \times GL(2L_z, \mathbb{C})$.

However, the symmetry is too loose, new aspect needed ...

As a symmetry operator formed by projection operator, it might be just too loose. Now denote

$$M_P \, |n\rangle = \tilde{m}_n \, |n\rangle \,, \tag{4.25}$$

then since

$$P_n = |n\rangle\langle n| \tag{4.26}$$

as the nth projector also satisfying that $[P_n, M_P] = 0$, while spin degrees of freedom is decoupled in effective mass picture, we have

$$[P_n, H_{\mathrm{EFF}}] = 0, \tag{4.27}$$

as k-dependent mass-projection symmetry. This operator helps us to extract information related to each Dirac sub-band. Such a fact is easily seen since

$$M_P P_n = \tilde{m}_n P_n, \tag{4.28}$$

and

$$H_{\mathrm{EFF}} P_n = P_n \otimes [\lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \tilde{m}_n(\boldsymbol{k})\sigma_z]. \tag{4.29}$$

4.2 Half-Quantized Anomalous Hall Effect

In writer's opinion, proposal for such an effect origins from two sides: one is the half-quantization at gapped surface of 3-D TI[12], and the other is physical geometry in realization of QAHE and the so-called axion insulator (AI) in TI film[7, 36]. The first proposal attributes half-quantization to surface Hall conductance while the latter gives two schematic pictures with different magnetic configurations, and their combination gives rise to magnetism only within one side of TI film[37, 69], as the simplest situation where half-quantized anomalous Hall effect emerges, together with first stable realization of quantum anomalous semimetals[15] in $(2+\epsilon)$ spatial dimensions. In the discussion below, we shall ignore finite-size effect, which opens an exponentially small gap for surface states and causes exponentially small energy shift between parity states.

As we have discussed above, adding Zeeman field upon one side of TI film would lead to the case $|I_S(k < k_c)| = |I_A(k < k_c)|$ within $n = 1$ block, then further mass-diagonalization leads to one trivial effective mass and the other one which is massless at low energy while becoming massive at high energy. Here $k < k_c$ is required to give an area instead of a point where effective mass is zero, i.e., the parity invariant regime. Such a property is satisfied by the existence of surface states of 3-D TI, which localize near the top and bottom surfaces with the same decaying behavior, labelled by parity or mirror index thus distribute equally at two surfaces. Then the one-side Zeeman field would give rise to $|V_S(l_z)| = |V_A(l_z)|$ as symmetric or antisymmetric Zeeman distribution with also the same strength, and it becomes clear that the expectation value of $V_{S/A}$ would have the same amplitude under given surface states, i.e., $|I_S| = |I_A|$, at least at low energy where localization of surface states is rather good. This is verified by numerical calculation shown in Fig. 4.5(b), where one recognizes $I_S = -I_A$ at low energy.

Such a result naturally leads to HQAHE, but firstly, one may check for correct band structure re-produced by the effective four-band model, compared with the lowest-four bands result from direct diagonalization of thin film Hamiltonian. As presented in Fig. 4.5(a), they show almost prefect correspondence, although small derivation is expected since we only conserve $n = 1$ block. Meanwhile, compare of Hall conductance is also showed in (c), where both results show half-quantization explicitly.

Now it is time for discussion of physics behind. The effective mass picture leads to further splitting of Hamiltonian as

$$\tilde{H}_- = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \tilde{m}_-(\boldsymbol{k})\sigma_z, \tag{4.30a}$$

$$\tilde{H}_+ = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \tilde{m}_+(\boldsymbol{k})\sigma_z, \tag{4.30b}$$

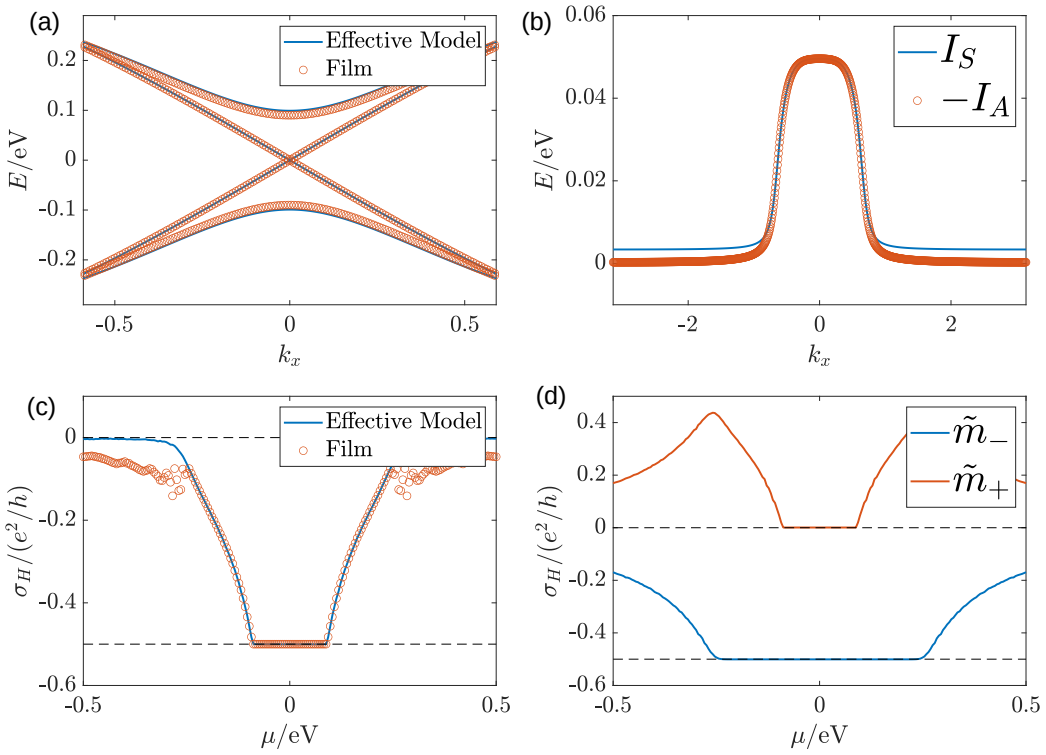


Figure 4.5: Half-quantized anomalous Hall effect. (a) Band structure from four-band effective model and thin film with lowest-four bands. (b) Two integrals I_S and $-I_A$. (c) Hall conductance from four-band effective model and thin film. (d) Hall conductance for two separable sub-blocks with effective masses $\tilde{m}_-$ and $\tilde{m}_+$, respectively.

with

$$\tilde{m}_\chi = I_S + \chi\sqrt{m^2 + I_A^2}.$$

As discussed before, m here as $m_1(\boldsymbol{k})$ has an approximate analytic expression when L_z is thick as

$$m(\boldsymbol{k}) \approx \Theta(k - k_c)m_0(\boldsymbol{k}), \tag{4.31}$$

where k_c is determined by $m_0(k_c) = 0$ or more strictly $m_0(k_c) = \lambda_\perp/4t_\perp^2$. However, for a thick-enough film, the former one is already enough to suppress mass to zero. Then by approximately recognizing k_c as also the critical k-value inside which $I_S = -I_A$, we obtain

$$\begin{cases} \tilde{m}_-(k < k_c) = I_S - \sqrt{0 + I_A^2} = 0 \\ \tilde{m}_+(k < k_c) = I_S + \sqrt{0 + I_A^2} = 2I_S > 0 \end{cases}. \tag{4.32}$$

Again by weak Zeeman, high-energy behavior is determined by χ here, then by setting Fermi surface μ_F inside parity invariant regime $k_F < k_c$, we have

$$\sigma_H^-/(e^2/h) = \frac{1}{2}\{\mathrm{sgn}(\chi = -) - \mathrm{sgn}[\tilde{m}_-(k_F)]\} = \frac{1}{2}(-1 - 0) = -\frac{1}{2}, \tag{4.33a}$$

$$\sigma_H^+/(e^2/h) = \frac{1}{2}\{\mathrm{sgn}(\chi = +) - \mathrm{sgn}[\tilde{m}_+(k_F)]\} = \frac{1}{2}(1 - 1) = 0. \tag{4.33b}$$

Nevertheless, merely half-quantization could not offer more information, especially when one wants to relate this to film geometry. Physically thinking, the single Dirac cone on boundary of 3-D TI splits into two with film geometry, separated by infinite $x-y$ planes

(periodic boundary condition), living at top and bottom surfaces, respectively (notice that we have ignored finite-size so that we do not need to worry about energy splitting between even and odd parity states, and top or bottom should be able to serve as good quantum number). Then, adding Zeeman field at one side will drive related massless Dirac cone into a massive one, while the other living at the contrary surface should remain gapless (and the finite-size gap would be further suppressed if considering). Now one massless Dirac cone plus one massive Dirac fermion serve us half-quantization, and the only question here is who actually contributes to that.

Before our strict derivation, people have guessed for a model[69] focusing on hybridization between two surface states, and we may call such a Hamiltonian written under 'surface-state-representation' (SSR). In fact, by a two-step unitary transformation, our four-band effective model, which may be called to stand under 'parity/mirror-representation' (PMR), could be transformed into the model proposed before. Now define

$$U_1 = e^{i\pi\tau_y\sigma_z/4},\ U_2 = e^{i\pi\tau_x\sigma_0/4}, \tag{4.34}$$

then their combination $U \equiv U_1U_2$ would transform effective four-band Hamiltonian under PMR into SSR as (with recognition $H_{PMR} = H_{\text{EFF}}$ written above)

$$\begin{aligned} H_{SSR} &= U^\dagger H_{\text{PMR}} U = \lambda_\parallel\tau_z(\sin(k_xa)\sigma_y - \sin(k_yb)\sigma_x) + I_S\tau_0\sigma_z - I_A\tau_z\sigma_z + m(\boldsymbol{k})\tau_x\sigma_0 \\ &= \begin{pmatrix} \lambda_\parallel(\sin(k_xa)\sigma_y - \sin(k_yb)\sigma_x) + I_-\sigma_z & m(\boldsymbol{k})\sigma_0 \\ m(\boldsymbol{k})\sigma_0 & -\lambda_\parallel(\sin(k_xa)\sigma_y - \sin(k_yb)\sigma_x) + I_+\sigma_z \end{pmatrix}, \end{aligned} \tag{4.35}$$

where we have defined $I_\pm = I_S \pm I_A$. Then, by relation $I_S = -I_A$ as shown above, we have

$$I_- = 2I_S,\ I_+ = 0,$$

inside parity invariant regime, and we obtain a gapped Dirac fermion plus a massless Dirac cone by considering $m(k < k_c) = 0$ at the same time. However, this low-energy picture of H_{SSR} near $k = 0$ is incomplete. At first glance, there are two effective low-energy continuum models as

$$\begin{cases} \lambda_\parallel(\boldsymbol{k}\times\boldsymbol{\sigma})_z + 2I_S(0)\sigma_z \\ -\lambda_\parallel(\boldsymbol{k}\times\boldsymbol{\sigma})_z \end{cases},$$

and direct calculation of Berry curvature seems to hint that massive band is non-trivial. But to obtain the desired half-quantization, one has to do the curvature integral to infinity, in which case Fermi surface is physically unobservable. What is more, and to make the picture clear, we may come back to 'effective-mass picture' (EMP), and utilize Eq. (4.33) to give us proper Hall conductance quantization. Then immediately we see that gapped one is trivial, and it is the high-energy part of the gapless Dirac fermion that provides half-quantized Hall conductance. As discussed above, trivially gapped surface state gives at most nearly half-quantized surface conductance.

4.2.1 Surface State Representation

There is still a question as what representation does EMP stays in? Re-considering effective mass diagonalization, at low energy, one has

$$U_M(k < k_c) = \frac{1}{\sqrt{2}}\begin{pmatrix} i\,\text{sgn}(I_A) & -i\,\text{sgn}(I_A) \\ 1 & 1 \end{pmatrix},$$

then by counting spin DoF as σ_0 here, we have basis transformation

$$\begin{aligned}&(|\Phi_1\rangle, |\Phi_2\rangle, |\Phi_3\rangle, |\Phi_4\rangle)(U_M \otimes \sigma_0)\\ =&\frac{1}{\sqrt{2}}\left(i\,\mathrm{sgn}(I_A)\,|\Phi_1\rangle + |\Phi_3\rangle, i\,\mathrm{sgn}(I_A)\,|\Phi_2\rangle + |\Phi_4\rangle, -i\,\mathrm{sgn}(I_A)\,|\Phi_1\rangle + |\Phi_3\rangle, -i\,\mathrm{sgn}(I_A)\,|\Phi_2\rangle + |\Phi_4\rangle\right).\end{aligned} \tag{4.36}$$

Especially, for example,

$$\frac{1}{\sqrt{2}}\left[i\,\mathrm{sgn}(I_A)\Phi_1(l_z) + \Phi_3(l_z)\right] = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\,\mathrm{sgn}(I_A)\\ 0\\ 0\end{pmatrix}\left[\mathrm{sgn}(I_A)\lambda_\perp f_+(l_z) + t_\perp \eta f_-(l_z)\right],$$

which should localize at one surface. To make this clearer, we directly calculate the basis

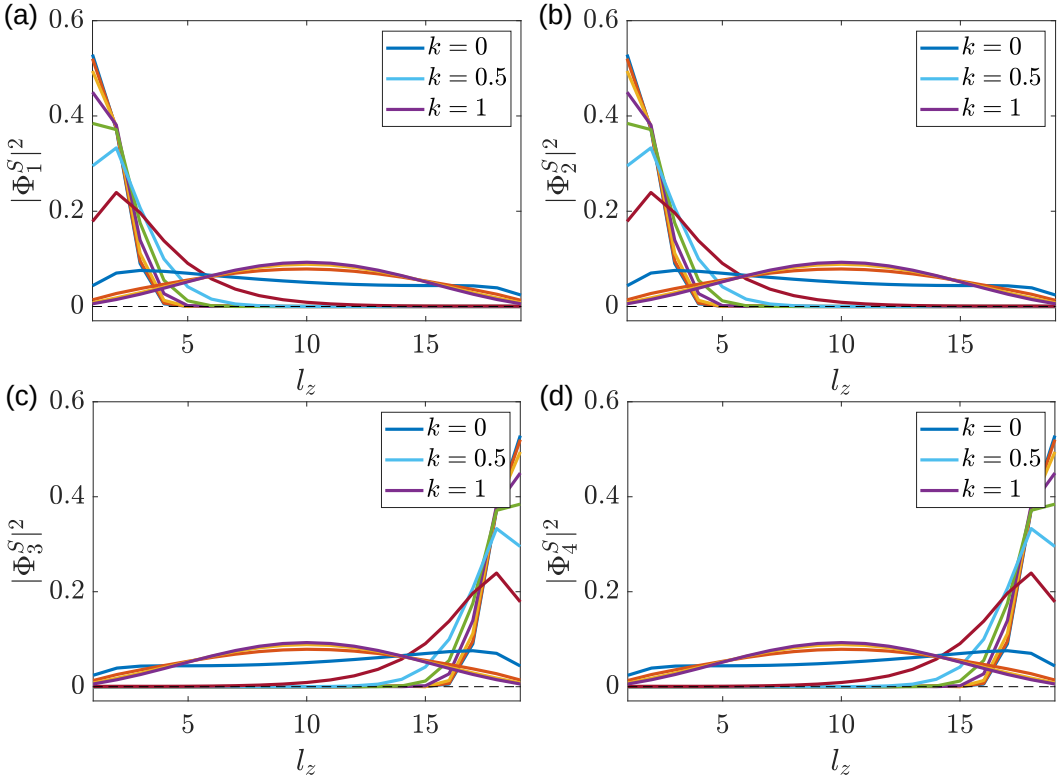


Figure 4.6: Basis wavefunction distribution under surface-state representation with different k.

wavefunction distribution under surface-state representation with different k as shown in Fig. 4.6, and one indeed sees surface states localized at top or bottom surface for $k < k_c$. Our label here follows that by defining basis states as $|\Phi^S\rangle$ under SSR,

$$\begin{aligned}&(|\Phi_1^S\rangle, |\Phi_2^S\rangle, |\Phi_3^S\rangle, |\Phi_4^S\rangle) = (|\Phi_1\rangle, |\Phi_2\rangle, |\Phi_3\rangle, |\Phi_4\rangle)(U_M \otimes \sigma_0)\\ =&\frac{1}{\sqrt{2}}\left(i\,\mathrm{sgn}(I_A)\sqrt{1+\frac{m}{M}}\,|\Phi_1\rangle + \sqrt{1-\frac{m}{M}}\,|\Phi_3\rangle, i\,\mathrm{sgn}(I_A)\sqrt{1+\frac{m}{M}}\,|\Phi_2\rangle + \sqrt{1-\frac{m}{M}}\,|\Phi_4\rangle\right.\\ &\left.-i\,\mathrm{sgn}(I_A)\sqrt{1-\frac{m}{M}}\,|\Phi_1\rangle + \sqrt{1+\frac{m}{M}}\,|\Phi_3\rangle, -i\,\mathrm{sgn}(I_A)\sqrt{1-\frac{m}{M}}\,|\Phi_2\rangle + \sqrt{1+\frac{m}{M}}\,|\Phi_4\rangle\right)\end{aligned} \tag{4.37}$$

as a unitary transformation. Under our signal, $(|\Phi_1^S\rangle, |\Phi_2^S\rangle)$ compose basis with $\tilde{m}_+$, while $(|\Phi_3^S\rangle, |\Phi_4^S\rangle)$ for $\tilde{m}_-$.

4.2.2 General Symmetry Indicates Layer Information

Inspired by idea above, we may have a chance to define '*general symmetry induced layer Chern number*' and calculate it, similar to mirror layer Chern number we defined before. Notice that for effective lowest four-band under SSR basis, we have

$$\langle\Phi^S|H_{\text{Film}}|\Phi^S\rangle = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \begin{pmatrix}\tilde{m}_+ & \\ & \tilde{m}_-\end{pmatrix}\sigma_z, \tag{4.38}$$

with

$$|\Phi^S\rangle = |\Phi\rangle\,(U_M\sigma_0),$$

where U_M diagonalizes mass term and is defined in Eq. (4.21). Furthermore, we notice that SSR in this case is just a special representation for the *effective mass representation* (EMR), i.e.,

$$|\Phi^S\rangle \xrightarrow{\text{general}} |\Phi^M\rangle\,, \tag{4.39}$$

which could be further fully diagonalized into energy representation with sub-block transformation. To make it clearer, we see that in the case under studying, we may define another unitary matrix that diagonalizes Hamiltonian in EMR (SSR in the case)

$$U_E = \bigoplus_n U_E^n, \tag{4.40}$$

with

$$(U_E^n)^\dagger(\lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \tilde{m}_n)U_E^n = E_n\sigma_z, \tag{4.41}$$

where energy

$$E_n = \sqrt{\lambda_{\parallel}^2[\sin^2(k_x a) + \sin^2(k_y b)] + \tilde{m}_n^2}. \tag{4.42}$$

Accordingly, we have

$$U_E^n = (|E_n+\rangle\,, |E_n-\rangle), \tag{4.43}$$

which is composed of eigenstates of each sub-block in EMR. Now, by denoting

$$|\Phi^E\rangle = |\Phi^M\rangle\,U_E = |\Phi\rangle\,(U_M\sigma_0)U_E, \tag{4.44}$$

we have

$$\langle\Phi^E|H_{\text{Film}}|\Phi^E\rangle = \bigoplus_n E_n\sigma_z, \tag{4.45}$$

and since $|\Phi^E\rangle$ still contains position information along l_z, we could measure layer-dependent observable with such states as probes accordingly, say the effective-mass labelled layer Chern number (notice that mirror layer Chern number is a special case under such a picture). To be clearer, we have formally

$$\begin{array}{ccccccc} \text{Original} & \longrightarrow & \text{PMR} & \xrightarrow{U_M} & \text{EMR} & \xrightarrow{U_E} & \text{Energy.} \\ H_{\text{Film}} & & |\Phi\rangle & & |\Phi^M\rangle & & |\Phi^E\rangle \end{array}$$

However, one may have to be aware that U_M and U_E may could not be well-defined with a single chart.

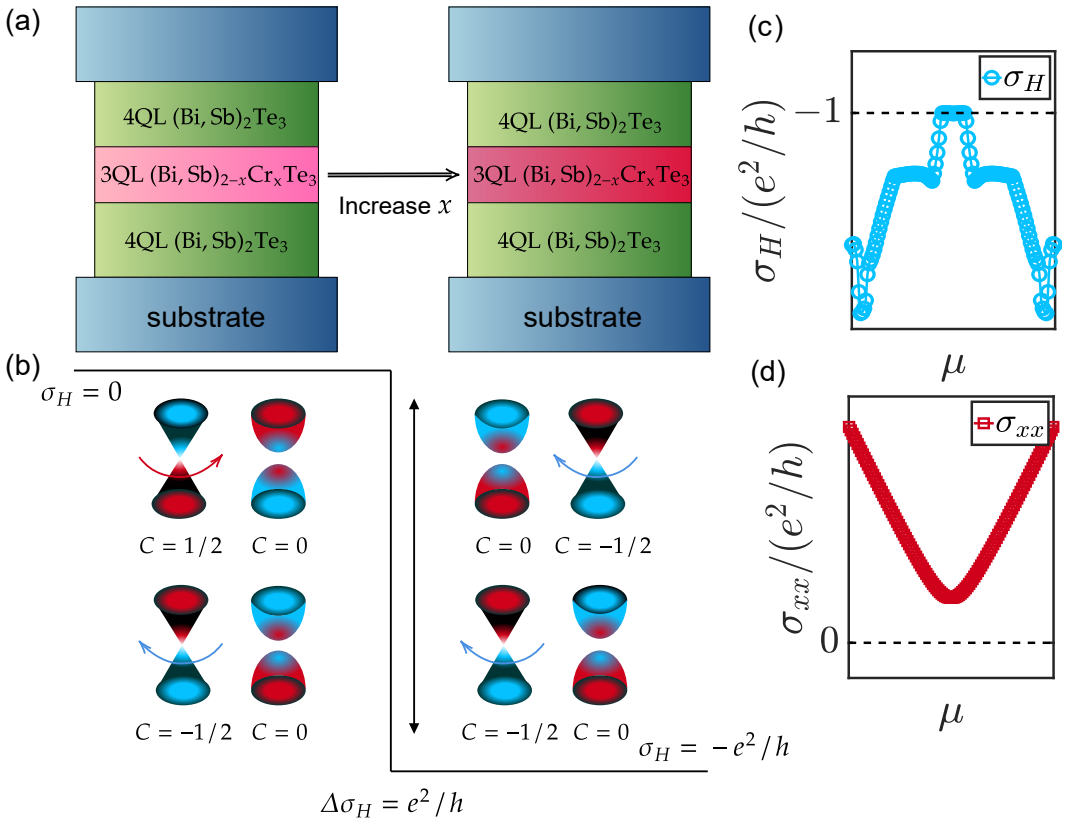


Figure 4.7: (a) Schematic of the magnetic sandwich heterosturcture of a $(Bi, Sb)_2Se_3$ TI film with the concentration x of magnetically doped Cr atoms. (b) A transition from two pairs of massless and massive Dirac fermions with no net Hall conductance $\sigma_H = 0$ at low concentration x to that with a quantized Hall conductance $\sigma_H = -\frac{e^2}{h}$ at higher concentration x (the sign depending on the direction of magnetization). C represents the Hall conductance in the unit of e^2/h, while color represents the sign-value of Berry curvature with blue for minus and red for positive. The masses of a pair of massless and massive Dirac fermions at lower energy exchange by increasing the concentration x while the higher energy part of the Dirac fermions remain almost unchanged. (c) Schematic of the quantized Hall conductance σ_{xy} and (d) the longitudinal conductivity σ_{xx} as function of the chemical potential μ at a higher doping concentration x.

4.3 Metallic Quantized Anomalous Hall Effect without Chiral Edge States

We consider a symmetric topological insulator thin film with a magnetic doped layer at the center $m\mathrm{QLX_2Te_3}/3\mathrm{QLX}_{2-x}\mathrm{Cr}_x\mathrm{Te_3}/m\mathrm{QLX_2Te_3}$ with X = (Bi, Sb) and $m = 4$ as shown in Fig. 1. A larger integer m does not change the main result in this proposal. Bi_2Te_3, Sb_2Te_3 and Bi_2Se_3 are prototypes of strong topological insulators with single Dirac cones of the surface electrons [63]. 1QL means a quintuple layer of X and Te atoms, and is about 1nm in Bi_2Te_3. The Dirac cone of surface states was observed explicitly by the ARPES [9, 66] and was also evidenced by a series of transport measurements. The exchange interaction between the p-orbital electron from Bi and Te and magnetic ions Cr may induce a finite magnetization in $\mathrm{X}_{2-x}\mathrm{Cr}_x\mathrm{Te_3}$[61, 57]. Tuning the concentration x of Cr can change the exchange interaction in the magnetic layer, and even makes it a ferromagnetic insulator [67]. The magnetic element Cr was modulation-doped only near the center layer. The non-doped layers are thick enough such that the top and bottom surface electrons do not open energy gap. The topological nature of the band structures of Bi_2Se_3 and Bi_2Te_3 can be well described by the tight-binding model for the electrons of $P_{z,\uparrow}$ and $P_{z,\downarrow}$ orbitals from (Bi and Te or Se atoms near the Fermi energy [63, 31],

$$H_{TI} = \sum_l \Psi_l^\dagger \mathscr{M} \Psi_l + \sum_{l,\alpha=x,y,z} \left(\Psi_l^\dagger \mathscr{T}_\alpha \Psi_{l+\alpha} + \Psi_{l+\alpha}^\dagger \mathscr{T}_\alpha^\dagger \Psi_l \right) \tag{4.46}$$

where $\mathscr{M} = (m_0 - 2\sum_\alpha t_\alpha)\sigma_0\tau_z$, $\mathscr{T}_\alpha = t_\alpha\sigma_0\tau_z - i\frac{\lambda_\alpha}{2}\sigma_\alpha\tau_x$, $\Psi_l^\dagger$ and Ψ_l are the four-component creation and annihilation operators at position $l = (l_x, l_y, l_z)$. The Pauli matrices σ_α and τ_α act on the spin and orbital indices, respectively. Adapting a model homogeneous in $x-y$ plane leads to $t_\parallel = t_x = t_y$, $t_\perp = t_z$, $\lambda_\parallel = \lambda_x = \lambda_y$, $\lambda_\perp = \lambda_z$, together with lattice constant $a = b$ in plane. The magnetic effect induced by the magnetic ions of Cr is modeled by introducing the Zeeman field along the z direction, $V_Z = \sum_l V_z(l_z)\Psi_l^\dagger\sigma_z\tau_0\Psi_l$. $V_z(l_z) = \alpha t_\perp$ in the magnetic doped layers (using $t_\perp$ as a unit), *i.e.*, $l_z = \pm 1/2, \cdots, \pm m_z/2$, and equals zero in the non-doped layers. Here we ignore the possible change of the bulk gap m_0 in $\mathrm{X}_{2-x}\mathrm{Cr}_x\mathrm{Te_3}$ caused by doping. Another potential material candidate is a thin film of $5\mathrm{QLX_2Te_3}/1\mathrm{SLMnBi_2Te_4}/5\mathrm{QLX_2Te_3}$, in which the layer of $MnBi_2Te_4$ is ferromagnetic [30, 42].

The band structure and the quantized Hall conductance- We consider the periodic boundary condition in the x and y direction and open boundary condition in the z direction with finite L_z layers. The band structure of the film is calculated numerically by means of the exact diagonalization method as shown in Fig. 2(a) in the absence of magnetic layers ($\alpha = 0$) and (b) in the presence of magnetic layers ($\alpha = 2$). It is observed that there exist a pair of massless Dirac fermions in both cases. The dispersions are doubly degenerated near the crossing point at $k = 0$. The presence of the Zeeman field does not open energy gap in the surface states while the Zeeman field α varies from 0 to 0.9. It is reasonable that the massless surface electrons are mainly located near the top and bottom surfaces which are far aways from the magnetic ions in the magnetic layers (see Fig. S4 in Ref. [**?**]). After having the numerical energy eigenvalues and eigenvectors, the Hall conductance can be calculated numerically by means of the Kubo formula for electric conductivity [35]. The Hall conductance becomes nonzero in the presence of α when the Fermi level crosses the conduction and valence bands. As shown in Fig. 2(c), a plateau

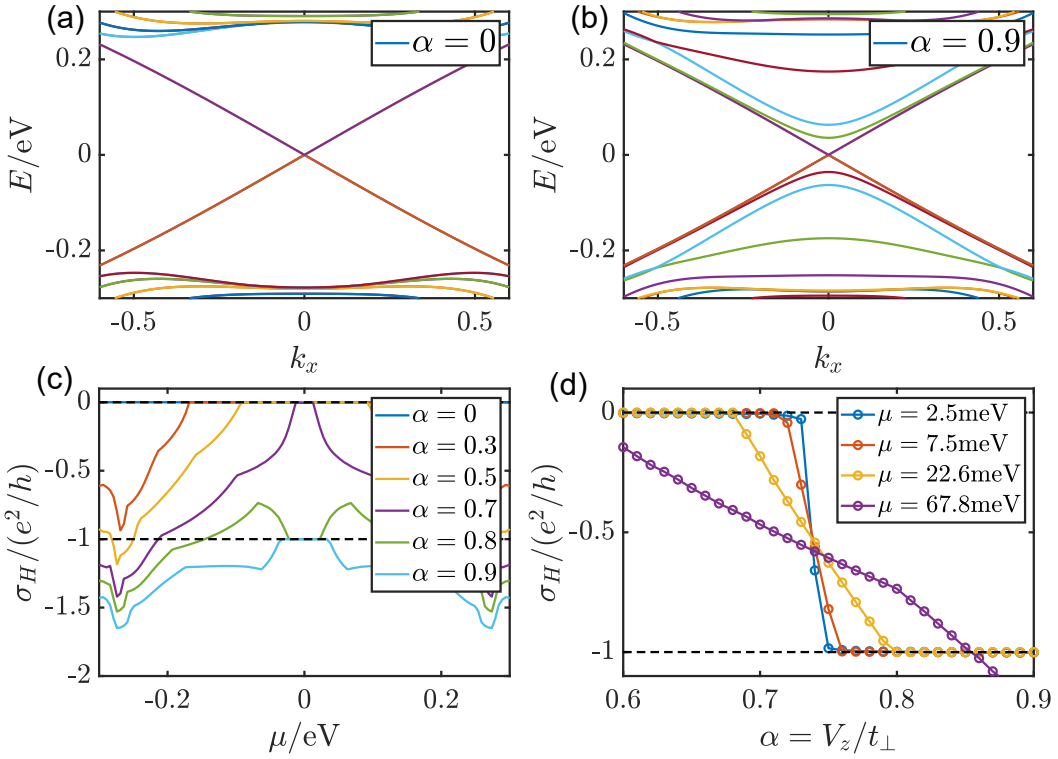


Figure 4.8: The band structure near the Γ point with $k_y = 0$ (a) in the absence of magnetic doping ($\alpha = 0$) and (b) in the the presence of magnetically doping ($\alpha = 2$). The gapless dispersions for the surface states in (a) and (b) are doubly degenerated. (c) The calculated Hall conductance as a function of chemical potential μ_F. (d) The Hall conductance as a function of α at different chemical potentials. We set the model parameters as $\lambda_\parallel = 0.41$ eV, $\lambda_\perp = 0.44$ eV, $t_\parallel = 0.566$ eV, $t_\perp = 0.4$ eV, $m_0 = 0.28$ eV, $a = b = 1$ nm and $c = 0.5$ nm if with no specific indication[63]. The thickness $L_z = 22$ and the magnetic layers $m_z = 6$.

of zero Hall conductance appears near $\mu_F = 0$ for a weak field, while for a strong Zeeman field α, a flat plateau of $\sigma_H = -\frac{e^2}{h}$ appears. Detailed calculation presented in Fig. 2(d) shows the Hall conductance changes from zero to $-\frac{e^2}{h}$ with increasing the Zeeman field α for fixed chemical potentials. Consider there is no band gap while α changes from 0 to 0.9. The longitudinal conductance must be finite such that the Hall resistivity is not quantized. Thus the appearance of the Hall conductance indicates that it differs from the conventional quantum anomalous Hall effect in an insulating phase.

Equivalent Dirac-like fermions- To explore the physical origin of the quantized Hall conductance, we study the band structure of of the thin film in the presence of the Zeeman field. First we adopt the Fourier transformation in the x and y direction $\Psi_{l_z,\mathbf{k}} = \sum_{l_x,l_y} \exp[il_x k_x + il_y k_y]\Psi_{l_x,l_y,l_z}$. The tight binding model in 4.46 with the Zeeman field $H_{tot} = H_{TI} + V_Z$ can be split into two parts $H_{tot} = H_\parallel + H_{1D}(\alpha)$. The in-plane spin-orbital coupling $H_\parallel = \sum_{l_z,\mathbf{k}} \Psi^\dagger_{l_z,\mathbf{k}} \lambda_\parallel (\sin k_x \sigma_x + \sin k_y \sigma_y)\tau_x \Psi_{l_z,\mathbf{k}}$. The part $H_{1D}(\alpha)$ for each $\mathbf{k}$ is equivalent to a one-dimensional topological insulator with the k-dependent band gap $m(\mathbf{k}) = m_0 - 4t_\parallel \left(\sin^2 \frac{k_x}{2} + \sin^2 \frac{k_y}{2}\right)$ in a Zeeman field. In the case, $[\sigma_z, H_{1D}] = 0$ such that H_{1D} can be diagonalized to have a series of energy eigenvalues $\tilde{m}_{n,\chi}(k_x, k_y)$ and eigenvectors $\tilde{\Phi}_{k,n,\chi} = \sum_{l_z} U_{n,\chi;l_z} \Psi_{l_z,k}$ with $n = 1, ..., L_z$ and $\chi = \pm$. The double degeneracy is caused by the time-reversal symmetry and one-dimensional inversion symmetry. Using

the eigenvectors as new basis, we find that H_{tot} is equivalently reduced to a series of two-dimensional Dirac-like models $H_{tot} \equiv \sum_{\mathbf{k},n,\chi=\pm1} \widetilde{\Phi}^{\dagger}_{\mathbf{k},n,\chi} h_{n,\chi}(\mathbf{k}) \tilde{\Phi}_{\mathbf{k},n,\chi}$ with

$$h_{n,\chi}(\mathbf{k}) = \lambda_{\parallel}(\sin k_x \sigma_x + \sin k_y \sigma_y) + \tilde{m}_{n,\chi}(\mathbf{k},\alpha)\sigma_z. \tag{4.47}$$

The energy dispersions are $E_{n,\chi,\pm} = \pm\sqrt{\lambda_{\parallel}^2(\sin^2 k_x + \sin^2 k_y) + \tilde{m}_{n,\chi}^2}$ in which $\tilde{m}_{n,\chi}$ plays a role of momentum-dependent mass term for the Dirac fermions.

In the absence of magnetic doping, i.e., $\alpha = 0$, the one-dimensional H_{1D} can be solved exactly. For detail, the solutions of the energy and wave function can be seen in Section SI in Ref. [**?**]. The masses have a relation $\tilde{m}_{n,+} = -\tilde{m}_{n,-} = m_n$. For $m(\mathbf{k}) > 0$, H_{1D} is topologically nontrivial, and has zero energy modes $m_1 = 0$ [53]; for $m(\mathbf{k}) < 0$, H_{1D} is topologically trivial, and the lowest energy modes $m_1 = m(\mathbf{k})$. Here the film is thick enough such that the finite size effect can be ignored [33]. This indicates that there exist a pair of the gapless Dirac cones of the surface electrons. The electron states of $m(\mathbf{k}) < 0$ are no longer confined near the surfaces, and evolve into the bulk. It is worth stressing that nonzero m_1 for $m(\mathbf{k}) < 0$ or a large $\mathbf{k}$ allows us defining the chirality of the gapless Dirac surface states χ even in two dimensions, one has right hardness ($\chi = +1$) and another has left handness ($\chi = -1$) even in this two-dimensional case. Here the complete band structure of the gapless Dirac fermions in the entire Brillouin zone consists of the surface electrons localized near the surface for $m(\mathbf{k}) > 0$ or small $\mathbf{k}$ and those extended in the z direction for $m(\mathbf{k}) < 0$ or large $\mathbf{k}$ (see in Fig. 3a). For $n \geq 2$, all $m_n(\mathbf{k})$ at $\mathbf{k} = 0$ are not equal to zero, which means the energy bands $E_{n,\chi}$ open an energy gap at the point. For a small $\mathbf{k}$, $h_{n,\chi}(\mathbf{k}) \simeq \lambda_{\parallel}(k_x\sigma_x + k_y\sigma_y) + \chi m_n(0)\sigma_z$. In other words, all the bands can be regarded as massive Dirac fermions. The spatial distribution of the wave function of the states of $m_{n=1} = 0$ are mainly distributed near the top and bottom surfaces as expected. The states of nonzero $m_{n=1}$ or at large k are spatially distributed in the bulk, which represents that the the surface states evolve into the bulk states with the variation of the wave vector $\mathbf{k}$.

In the presence of magnetic doping, the Zeeman field V_Z will change the band structures by altering effective mass $\tilde{m}$, while linear part vertical to z-direction remains unchanged due to degrees of freedom decoupling. In the basis of the energy eigenstates of $H_{1D}(\alpha)$ at $\alpha = 0$, the Zeeman term can be expressed as $\alpha \mathbf{I}_S(\mathbf{k})\tau_0\sigma_z$, where $\mathbf{I}_S(\mathbf{k})$ is a $L_z \times L_z$ matrix. Each element in $\mathbf{I}_S(\mathbf{k})$ can be calculated numerically. Thus H_{1D} is projected into the form $\left(\bigoplus_{n=1}^{L_z} m_n\tau_z + \alpha \mathbf{I}_S(\mathbf{k})\tau_0\right)\sigma_z$, and further diagonalizing this provides a bijection which maps the Hamiltonian into the mass term $\oplus_n \tilde{m}_{n,\chi}(\mathbf{k},\alpha)\sigma_z$. Confining to the subspace with $\sigma_z = +$, we could then track the evolution and interaction of the mass terms $\tilde{m}_{n,\chi}$ between $n = 1$ and $n = 2$ blocks with increasing α for given χ. What stands out in the process is an exotic grafting behavior signed in Fig. 3: viewing from left to right, while the masses $\tilde{m}_{n,+}(n = 1, 2)$ maintain their topological shapes, $\tilde{m}_{n,-}$, which represent one massless Dirac cone plus one massive Dirac cone, will fully exchange their low-energy parts with increasing α, i.e., *massless* $\longleftrightarrow$ *massive*. By increasing α $\tilde{m}_{n=1,-}$ and $\tilde{m}_{n=2,-}$ behave as if they cross around $\alpha \approx 0.74$ and then separate, during which detailed dynamic exchange reveals. On the other hand, what essentially remains unchanged is the high-energy part of each cone, then since $\tilde{m}_{n,\chi}$ of $n = 1, 2$ are naturally assigned with opposite sign for high-energy part. As a result, viewed from low-energy perspective as presented, the high-energy mass exchange between massless and massive cones. Such a Zeeman field induced mass exchange of the massless and massive Dirac fermions is closely associated with the change of the Hall conductance.

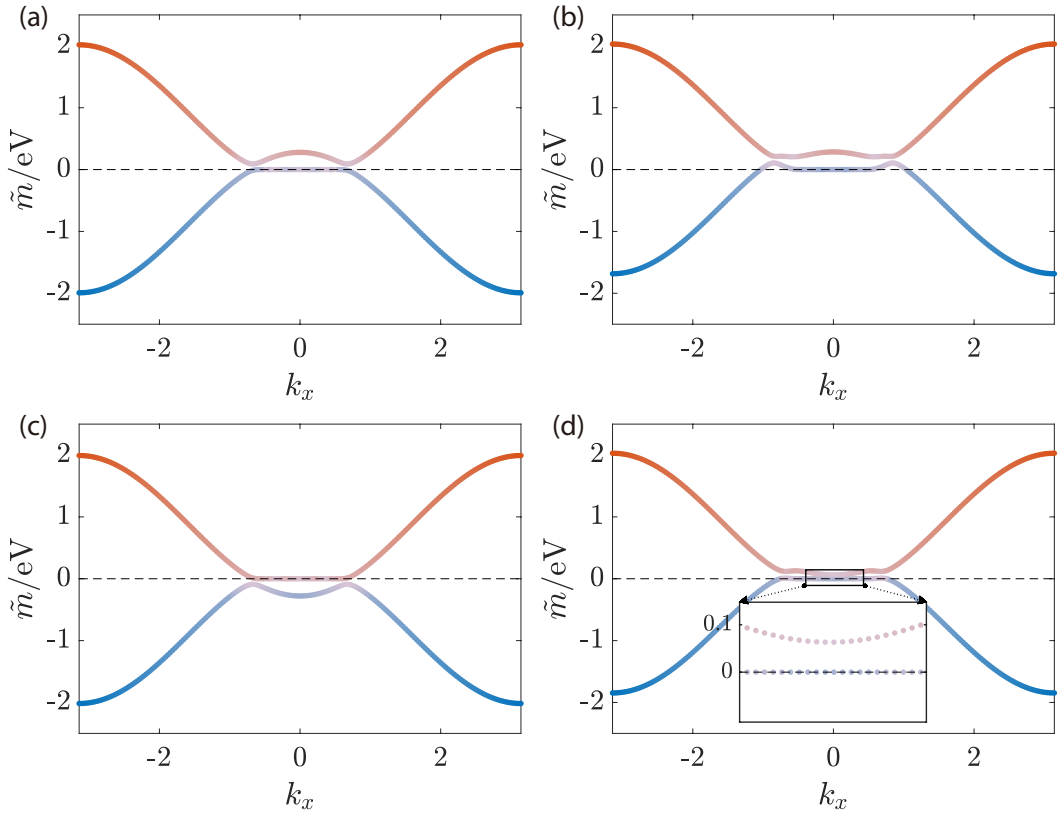


Figure 4.9: The evolution of the effective mass $\tilde{m}_{n,\chi}(k_x, k_y = 0)$ with the Zeeman field $V_Z = \alpha t_\perp$. (a), (c) represent $\tilde{m}_{1/2,\chi}$ at α= 0 for different $\chi = \pm$ separately while (b) and (d) stand for corresponding effective mass configurations when $\alpha = 2$.

Quantized Hall conductance- For the Dirac fermions in Eq. 4.47, the Hamiltonian can be expressed in terms of the Pauli matrices, $h_{n,\chi} = E_{n,+}\mathbf{d}(\mathbf{k}) \cdot \sigma$, with the spin texture $\mathbf{d} = (\lambda_\parallel \sin k_x, \lambda_\parallel \sin k_y, \tilde{m}_{n,\chi}(k_x, k_y))/E_{n,+}$. Using the Kubo formula for electric conductivity, the Hall conductance is given by

$$\sigma_H = -\frac{e^2}{h}\frac{1}{4\pi}\int \frac{dk_x dk_y}{4\pi}\mathbf{d}\cdot\left[\partial_{k_x}\mathbf{d}\times\partial_{k_y}\mathbf{d}\right](f_{k,+} - f_{k,-}) \tag{4.48}$$

where $f_{k,\pm} = \Theta(\mu_F - E_{n,\pm})$ the Heaviside step function at zero temperature and μ_F the chemical potential [44, 53]. For the massive Dirac fermions, the values of $\tilde{m}_{n,\chi}$ at $\mathbf{k} = (0, 0)$ and $\mathbf{k} = (\pi, .\pi)$ have the same sign, and there does not exist band inversion in the first Brillouin zone. The bands are always topologically trivial such that the fully filled bands, i.e., $\mu_F = 0$, always have no Hall conductance, $\sigma_H = 0$ which is consistent with the TKNN theorem [56]. For massless Dirac fermions, $\tilde{m}_{n,\chi} = 0$ near $\mathbf{k} = 0$. In the regime, $\mathbf{d}\cdot\left[\partial_{k_x}\mathbf{d}\times\partial_{k_y}\mathbf{d}\right] = 0$ which indicates that the Berry curvature of the band vanishes. Nonzero Berry curvature comes from the part of nonzero $\tilde{m}_{n,\chi}$ or the regime of large k. The Hall conductance is half-quantized for the chemical potential μ_F located within the regime, $\sigma_H = \frac{e^2}{2h}sgn[\tilde{m}_{n,\chi}(\pi, \pi)]$. This is consistent with the symmetry analysis and numerical calculation [70, 16].

Based on the mass-exchange picture of massless and massive Dirac fermions, we have a theoretical explanation of the change of the Hall conductance induced by the Zeeman field in Fig. 2(c), (d). The film hosts a series of massive and massless Dirac fermions. For our purpose, we focus on the bands of $n = 1$ and $n = 2$ as all other massive Dirac fermions ($n \geq 3$) have no contribution to the Hall conductance when they are fully filled for the chemical potential near $\mu = 0$. In the absence of the Zeeman field, the film hosts a pair of massless Dirac fermions, between which one has $+\frac{e^2}{2h}$ and another has $-\frac{e^2}{2h}$ due to the sign difference of the mass terms at large k. The total Hall conductance is zero as expected. The presence of a weak Zeeman field does not change this situation. Equipped with a holistic view, with increasing the Zeeman field further, one massless Dirac fermion and one massive Dirac fermion exchange their masses, whose higher energy parts have different sign, which remains unchanged. Equivalently, the massless Dirac fermion changes the sign of massive term at higher energy viewed from low-energy perspective. Consequently, its Hall conductance changes from $+\frac{e^2}{2h}$ from $-\frac{e^2}{2h}$ (the sign is determined by the sign of the Zeeman field). The addition of two massless Dirac fermions gives a quantized Hall conductance $-\frac{e^2}{2h} - \frac{e^2}{2h} = -\frac{e^2}{h}$.

Absence of chiral edge states There are no chiral edge states around the system boundary in these paired gapless Dirac fermions. The quantum Hall conductance is not governed by a TKNN number or the Chern number and does not satisfy the conventional bulk-edge correspondence [22]. We calculated the local density states at the y-front surface of a y-opened film in Fig. 4(a), where there is clearly no dispersion that connects the lateral surface valence and conduction bands, opposite to the conventional case. This illustrates explicitly that there do not exist the chiral edge states along the system boundary. The asymmetric local density of states between k_x and $-k_x$ reflects the fact that there exist chiral edge current for the filled bulk states. The states carrying chiral edge current gradually becomes prominent when immersing into middle of z from its top surface. Furthermore, it is found that there still exists a chiral edge current whose amplitude is proportional to the chemical potential due to the time-reversal symmetry breaking caused by the Zeeman coupling [71]. As the Zeeman field is parallel with the lateral surface, the

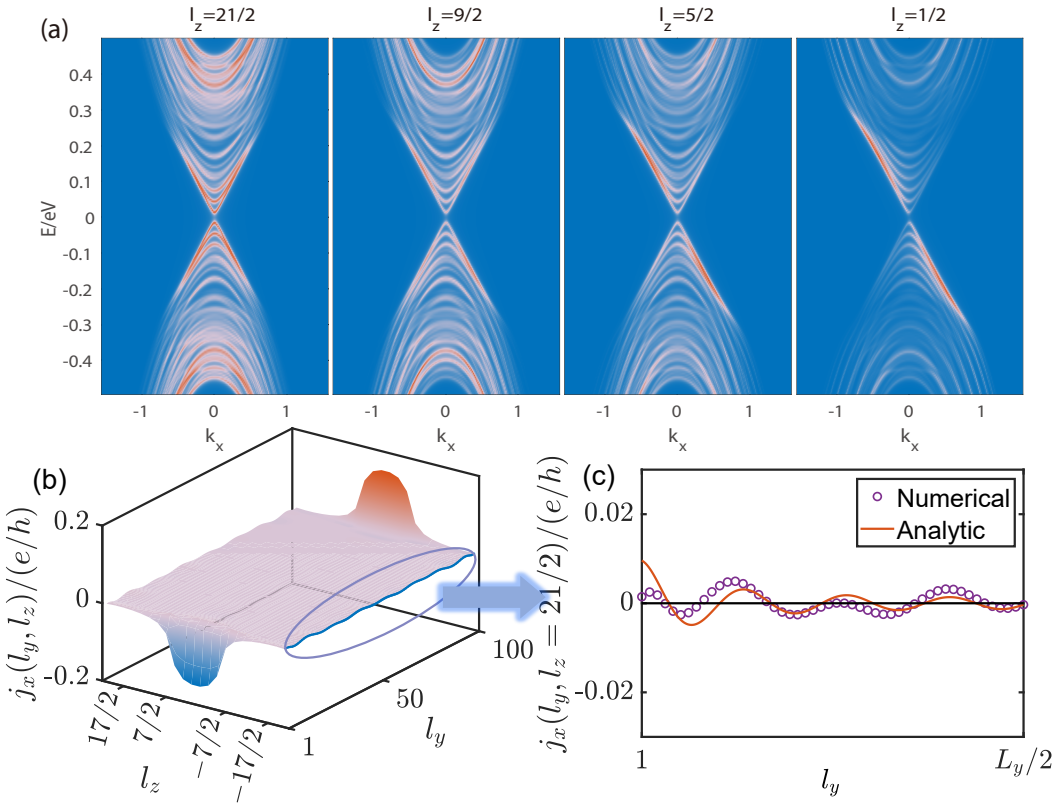


Figure 4.10: The local current distribution and local density of states of the thin film with $L_z = 22$, and $\alpha = 0.8$. (a) Local density of state $\rho(l_z, k_x, E)$ at the y-front of the film with width $L_y = 20$. Several representative l_z are picked out to show the absence of chiral edge states. (b) The local current density $j_x(l_y; l_z)$ for a film with width $L_y = 100$. The chemical potential $\mu_F = 0.1$eV. (c) The surface current density as a function of l_y at the surface $l_z = 11$ or -11, $j_x(l_y; l_z = 11)$, compared with the current formula $j_x(x) \propto J_1(2k_F x)/x$ [71].

lateral surface states remain gapless. We present the spatial distribution of the electric current density in Fig. 4(b). It shows that the current density is mainly distributed around the surface of the magnetic layers, and decays quickly into the bulk, which demonstrates that the electronic transport mainly occurs on the surface. The local current density on the surface in Fig. 4(c) hows that the current density on the surface decays slowly which obviously deviates the exponential law. We fit the numerical result by using the current formula $j_x(x) \propto J_1(2k_F x)/x$ in Ref. [71] under hard-wall boundary condition. $J_1(x)$ is the first Bessel function. Small deviation appears within expectation as α is finite, the overall shape which hints a power-law decay away from ends along y direction, however, is also clear, as indicated in Fig. 4(c). Also, it is worth of stressing that the current is induced by the Zeeman exchange interaction, and should be dispationless. Such behavior depends heavily on the metallic nature of surface Dirac cone.

Discussion and summary- In the field theory, the massless Dirac fermions possess the parity symmetry. When the Dirac fermions are coupled to electromagnetic field, its action fails to restore the symmetry in any regularization, and is characterized by a half-quantized Hall conductance. The discussion on parity anomaly in the condensed matter dated back to early 1980s [41, 47, 14]. It has attracted extensive interests since

the discovery of topological insulators as the massless Dirac fermions can exist on the surface of a three-dimensional insulator [64, 6, 59]. The thin film here provides a platform to explore the related physics of parity anomaly. The massless Dirac fermions on the surfaces accompany with presence of nonzero zero term $\tilde{m}_{n,\chi}$ at large k, which plays a role of the regulators of Dirac fermions in the field theory. Thus the nonzero Hall conductance is just determined by the sign of $\tilde{m}_{n,\chi}$ at $k = 0$ and large k, and independent of the specific form and amplitude of $\tilde{m}_{n,\chi}$. In this sense, this reflects the physics of quantum anomaly. However, we should keep in mind that the term has already broken the parity symmetry explicitly. The quantized Hall conductance is not a consequence of spontaneous symmetry breaking.

4.3.1 $\alpha = 0.9$ explained by only $n = 1, 2$

Here we focus on the case proposed as novel metallic quantum anomalous Hall effect (MQAHE) without chiral edge state (CES), i.e., TI film with $L_z = 22$ and uniform Zeeman field at middle six layers, where field strength is $V_z = \alpha t_\perp$, the same setting as in the main text. Essentially here we choose $\alpha = 0.9$ within the non-trivial regime, and our claim here is that physics of mass exchange could be well-presented for this case with only $n = 1, 2$ blocks. Choosing certain block or blocks in our effective Hamiltonian is equivalent to some cut-off approximation. This approximation would become useful inside certain parameter zone, like the long-known mean-field approximation, which may require more information near the critical point equivalent to increasing number of blocks in our model. To make symbols concise, we still let α to be included inside function $I_{S/A}$ here. Again here because of the symmetric distribution of Zeeman field, $I_A^{nn'} = 0$ for any component.

At this time, the projected effective 1-d Hamiltonian considering symmetric Zeeman coupling could be written as

$$\left[\left(\bigoplus_{n=1}^{2} m_n\right)\tau_z + \begin{pmatrix} I_S^{11} & I_S^{12} \\ I_S^{21} & I_S^{22} \end{pmatrix}\tau_0\right]\sigma_z = \begin{pmatrix} m_1\tau_z + I_S^{11}\tau_0 & I_S^{12}\tau_0 \\ I_S^{21}\tau_0 & m_2\tau_z + I_S^{22}\tau_0 \end{pmatrix}\sigma_z; \tag{4.49}$$

then by introducing

$$\tau_z \left|\chi\right\rangle = \chi \left|\chi\right\rangle, \tag{4.50}$$

we have

$$H_\chi^{1d,n=1,2} \equiv \begin{pmatrix} \chi m_1 + I_S^{11} & I_S^{12} \\ I_S^{21} & \chi m_2 + I_S^{22} \end{pmatrix}\sigma_z. \tag{4.51}$$

Now by adding linear term back we have effective Hamiltonian considering $n = 1, 2$ blocks together with their Zeeman hybridization

$$\begin{aligned} H_{\mathrm{EFF},\chi=\pm}^{n=1,2} &= \lambda_\parallel \rho_0(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \begin{pmatrix} \pm m_1(\boldsymbol{k}) + I_S^{11}(\boldsymbol{k}) & I_S^{12}(\boldsymbol{k}) \\ I_S^{21}(\boldsymbol{k}) & \pm m_2(\boldsymbol{k}) + I_S^{22}(\boldsymbol{k}) \end{pmatrix}\sigma_z \\ &= \lambda_\parallel \rho_0(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \mathrm{Re}(I_S^{12})(\boldsymbol{k})\rho_x\sigma_z - \mathrm{Im}(I_S^{12})(\boldsymbol{k})\rho_y\sigma_z \\ &\quad + E_\pm(\boldsymbol{k})\rho_0\sigma_z + \Delta_\pm(\boldsymbol{k})\rho_z\sigma_z, \end{aligned} \tag{4.52}$$

with σ representing spin DoF while ρ stands for DoF of two blocks, and

$$\begin{cases} E_\pm = (\pm m_1 + I_S^{11} \pm m_2 + I_S^{22})/2 \\ \Delta_\pm = (\pm m_1 + I_S^{11} \mp m_2 - I_S^{22})/2 \end{cases}. \tag{4.53}$$

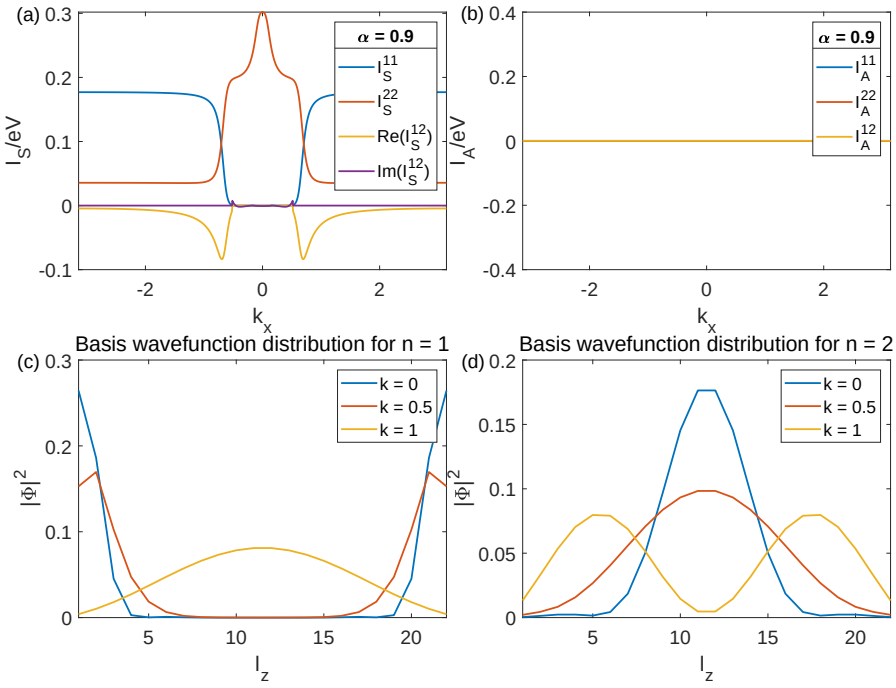


Figure 4.11: I_S, I_A and basis wavefunction distribution with $\alpha = 0.9$ for our 22-layer thin film with 3 layers of magnetic doping exactly lying in the middle. (a) I_S, especially notice that I_S^{12} is complex, and stands out only nearly outside low-energy zone with surface states. (b) I_A, which also vanishes. (c) Basis wavefunction distribution $|\Phi_i^{n=1}|^2$ near representative k. Especially at $k = 0$ there exists a strong sign of surface state. (d) Basis wavefunction distribution $|\Phi_i^{n=2}|^2$ near representative k.

Calculated values of $I_{S/A}$ together with basis wavefunction distribution of $n = 1, 2$ blocks at several k-points are shown in Fig. 4.11, from which one could see clearly that because of the characteristic surface state behavior of Φ^1 at small k, away from Zeeman field at middle and distinct from bulk state Φ^2, neither I_S^{11} nor I_S^{12} appears in the low-energy zone, then two gapless Dirac cones are preserved. What is more, one could directly see

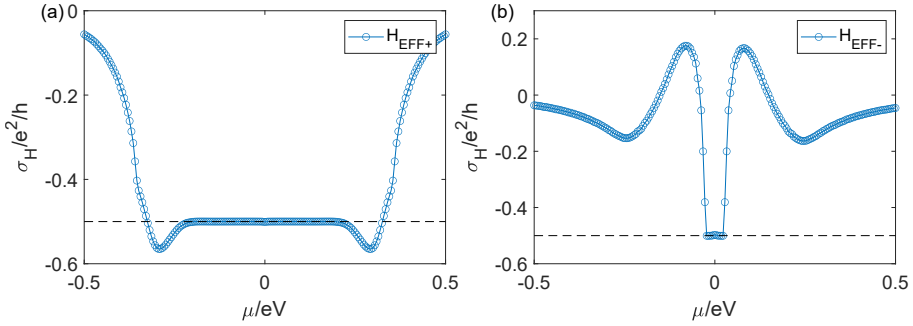


Figure 4.12: Hall conductance for (a) $H^{n=1,2}_{\mathrm{EFF}+}$ and (b) $H^{n=1,2}_{\mathrm{EFF}-}$.

half-quantization of Hall conductance for each $H^{n=1,2}_{\mathrm{EFF},\chi}$ as shown in Fig. 4.12. We have to emphasize here that since we only consider two lowest blocks, when Hall conductance ends up quantization upon varying chemical potential may be not accurate.

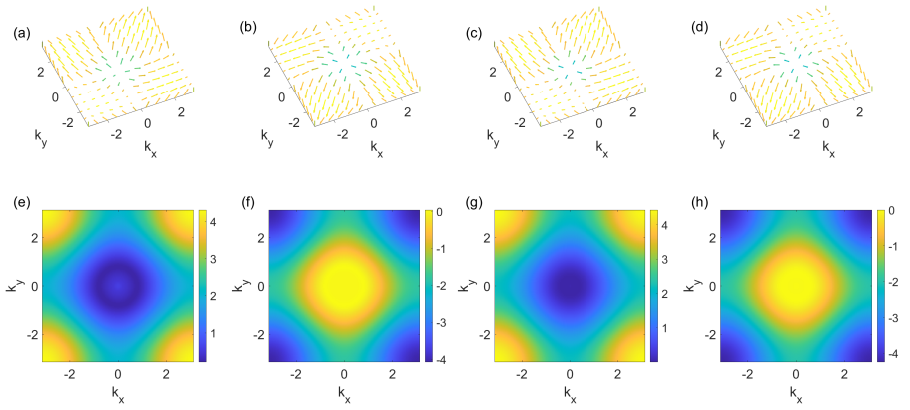


Figure 4.13: (a)∼(d) Spin configuration (borrow program[**?**]) (re-scaled) and (e)∼(h) spin-z configuration for $E_+ + \Lambda_+$, $E_+ - \Lambda_+$, $E_- + \Lambda_-$, $E_- - \Lambda_-$, respectively.

To make the picture further clearer, we notice that two gapless Dirac cones could be further separated by sub-block-diagonalization. Consider $H^{n=1,2}_{\mathrm{EFF}\pm}$ in Eq (4.52), we may introduce a two-step unitary transformation

$$U_\pm = U_2^\pm U_1^\pm,\ U_2^\pm = e^{i\rho_x\theta_2^\pm},\ U_1^\pm = e^{i\rho_y\theta_1^\pm}, \tag{4.54}$$

where the rotational angles are selected to be

$$\tan 2\theta_1^\pm = \frac{\mathrm{Re}(I_S^{12})}{\Delta_\pm},\ \tan 2\theta_2^\pm = \frac{\mathrm{Im}(I_S^{12})}{\delta_\pm},\ \delta_\pm = \sqrt{\mathrm{Re}(I_S^{12})^2 + \Delta_\pm^2}; \tag{4.55}$$

then by defining

$$\Lambda_\pm = \sqrt{\mathrm{Im}(I_S^{12})^2 + \delta_\pm^2} = \sqrt{\Delta_\pm^2 + |I_S^{12}|^2}, \tag{4.56}$$

it is ready to show that

$$U_{\pm}H^{n=1,2}_{\mathrm{EFF}\pm}U^{\dagger}_{\pm} = \lambda_{\parallel}\rho_0(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + (E_{\pm}\rho_0 + \Lambda_{\pm}\rho_z)\sigma_z, \tag{4.57}$$

which serves us again two separated sub-blocks. And the one with $|E_{\pm}| - \Lambda_{\pm}$ would contribute to gapless Dirac cones since

$$m_1(k=0) = 0,\ I_S^{12}(k=0) = 0,\ I_S^{11}(k=0) = 0,$$

as shown above, and this in turn gives us

$$(|E_{\pm}| - \Lambda_{\pm})(k=0) = \left|\frac{\pm m_2 + I_S^{22}}{2}\right|(k=0) - \sqrt{\frac{(\mp m_2 - I_S^{22})^2}{4}}(k=0) = 0. \tag{4.58}$$

Utilizing messages above, we are able to show both spin and spin-z configuration for four different diagonalized sub-blocks

$$\tilde{H}^{n=1,2}_{\mathrm{EFF},\chi\zeta} = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + (E_{\chi}(\boldsymbol{k}) + \zeta\Lambda_{\chi}(\boldsymbol{k}))\sigma_z,\ \chi,\zeta = \pm, \tag{4.59}$$

where ζ stands for eigenvalue for ρ_z. The result is shown in Fig. 4.13, and one sees two ((a), (c) or (e), (g)) trivial massive Dirac spin configurations together with two ((b), (d) or (f), (h)) non-trivial massless Dirac spin configurations that contribute two half-quantized Hall conductance with the same sign.

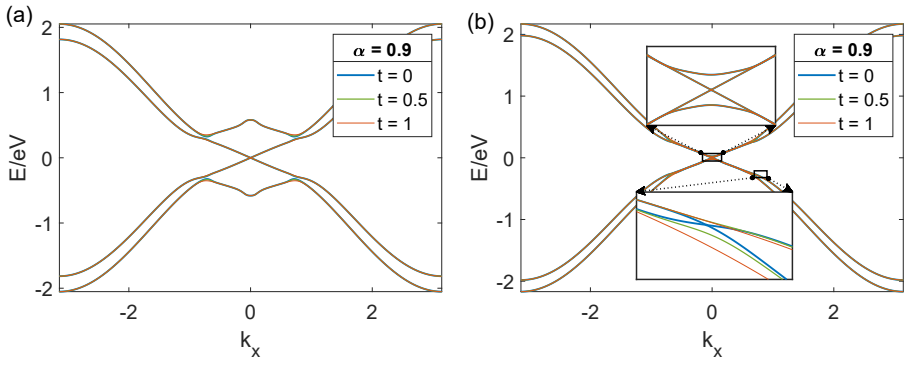


Figure 4.14: Calculated effective band structure with tuning $I_S^{12} \to tI_S^{12}$. (a) Band for $H^{n=1,2}_{\mathrm{EFF}+}$. (b) Band for $H^{n=1,2}_{\mathrm{EFF}-}$.

To gain physical picture from another side, and essentially to show the dynamical process of mass exchange under given α, we emphasize the role I_S^{12} plays here as an anti-crossing gap opening term. To observe this, we manually assign a coupling strength factor t to I_S^{12}, which is

$$H^{n=1,2}_{\mathrm{EFF}\pm}(t) = \lambda_{\parallel}\rho_0(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \begin{pmatrix} \pm m_1 + I_S^{11} & tI_S^{12} \\ tI_S^{21} & \pm m_2 + I_S^{22} \end{pmatrix}\sigma_z. \tag{4.60}$$

Diagonalizing this Hamiltonian with different t immediately tells us how I_S^{12} steps in under given situation $\alpha = 0.9$, and the result is shown in Fig. 4.14. Band structure (here we mainly refer to relative geometric arrangement of bands) for $H^{n=1,2}_{\mathrm{EFF}+}$ almost does not change, and what adding I_S^{12}, or equivalently increasing t offers us is a repulsive characteristic between gapless $n=1$ sub-block and gapped $n=2$ sub-block, which does not change topological shape. Nevertheless, things change dramatically for $H^{n=1,2}_{\mathrm{EFF}-}$, where

the repulsive characteristic makes the original ($t = 0$) crossing bands gain anti-crossing feature in high energy part of two sub-blocks with $\tau_z = -$. This behaviour becomes clearer by drawing fine band structure, as shown in the smaller graph inside Fig. 4.14(b). Then as claimed in the main text, this anti-crossing substantially reflects the mass exchange mechanism which brings us two massless Dirac cones with the same high energy sign.

The reason why we choose $\alpha = 0.9$ lies in the observation that when α becomes smaller, or equivalently approaching the critical point close to one, the simple picture above fails to continue to serve us MQAHE without CES. Explanation here is similar to why mean-field below higher critical dimension is only qualitatively right: one needs more information. Here by choosing merely $n = 1$ and $n = 2$, we have done a cut-off approximation ignoring other blocks and their interactions with massless Dirac cones through off-diagonal Zeeman integrals. The closer we approach critical zone of α, the larger number of blocks we need to explain the phenomenon.

4.3.2 Fine Process near Transition

To gain more physical insight, we calculate and plot more effective mass diagrams to show what has essentially happened during the transition. Setup here is the same as the main text except for changing Zeeman field strength. Under such investigation,

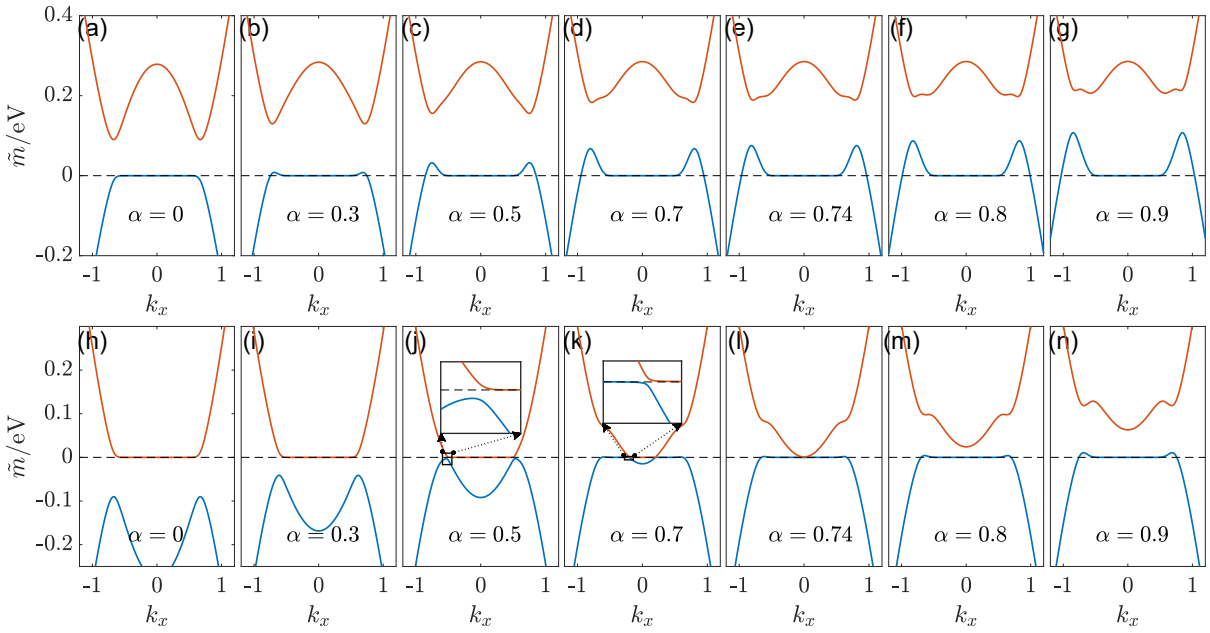


Figure 4.15: (a)∼(g) and (h)∼(n) show lowest-two **effective masses** varying with changing Zeeman field strength α belonging to set $[0, 0.3, 0.5, 0.7, 0.74, 0.8, 0.9]$ for $\chi = +$ and $\chi = -$, respectively. (a) (g) (h) (n) have already been shown as Fig. 3 in the main text but with a finer structure here.

the mass exchange process is expressed in detail. Notice that here we use high-energy part to label $n = 1, 2$ with a holistic view, although label change does not affect physical nature. As shown in Fig. 4.15(a)∼(g), $\chi = +$ part bears less change during the process. Nevertheless, for $\chi = -$ part, when α increases from 0 to 0.5 shown in (h)∼(j), minus high-energy branch as $\tilde{m}_{2,-}$ begins to touch zero energy; and by further increasing α, the zero mode part of positive high-energy branch as $\tilde{m}_{1,-}$ is gradually transferred to $\tilde{m}_{2,-}$

as shown in (j)∼(m), and the transfer rather than crossing nature is verified by the anti-crossing behavior between effective masses as shown in smaller graphs inside (j) and (k); finally after some critical value of $\alpha_c \in (0.7, 0.8)$, this grafting process is over, and new zero-energy surface state by $\tilde{m}_{2,-}$ now owns negative chirality when k is large enough, as shown in (m) and (n). Essentially, we may have

$$\alpha_c \approx 0.74 \tag{4.61}$$

to indicate $0 \to -1$ plateau transition after more detailed exploration (sign depends on Zeeman direction). After that, both gapless Dirac cones in the system would exhibit negative half-quantized Hall conductance, carried by $\tilde{m}_{1,+}$ and $\tilde{m}_{2,-}$ correspondingly, as indicated in (g) and (n). This closes the discussion by giving qualitative picture and quantitative transition description.

In the above discussion, to indicate unambiguously on effective mass, we have utilized mirror transformed Hamiltonian. As claimed above, physics as high energy exchange could be explained by merely 1-D part bands, and we kick out $H_{\parallel}$, restore $\boldsymbol{k}$, bring $\sigma_z \to s$ as its good quantum number for H_{1d} and get

$$\begin{aligned}\mathscr{H}^{1d}_{\chi,s} = \sum_{\boldsymbol{k}} \Bigg[& \sum_{i_z=1}^{(L_z-1)/2} \left(\phi^{\dagger}_{i_z,\boldsymbol{k}\chi}(M_0(\boldsymbol{k})\tau_z + sV_S(i_z))\phi_{i_z,\chi} + \phi^{\dagger}_{i_z,\chi}(t_{\perp}\tau_z - is\frac{\lambda_{\perp}}{2}\tau_x)\phi_{i_z+1,\chi} + \text{h.c.} \right) \\ & + \Psi^{\dagger}_{0,\boldsymbol{k}} \frac{1+\chi s\tau_z}{2}(M_0(\boldsymbol{k})\tau_z + sV(i_z=0))\Psi_{0,\boldsymbol{k}} + \Psi^{\dagger}_{0,\boldsymbol{k}}\tilde{\mathscr{T}}_{0,s,\chi}\phi_{1,\boldsymbol{k},\chi} + \text{h.c.} \Bigg],\end{aligned} \tag{4.62}$$

where

$$\tilde{\mathscr{T}}_{0,s,\chi} = \sqrt{2}\frac{1+\chi s\tau_z}{2}\mathscr{T}_z(\sigma_z \to s) = \frac{\sqrt{2}t_{\perp}}{2}(\tau_z + \chi s) - \frac{i\sqrt{2}\lambda_{\perp}}{4}(s\tau_z + i\chi\tau_y), \tag{4.63}$$

for L_z odd case; while for L_z even, we have

$$\mathscr{H}^{1d}_{\chi,s} = \sum_{i_z=1/2}^{(L_z-1)/2} \left(\phi^{\dagger}_{i_z,\chi}(M_0(\boldsymbol{k})\tau_z + V_S(i_z))\phi_{i_z,\chi} + \phi^{\dagger}_{i_z,\chi}\mathscr{T}_z\phi_{i_z+1,\chi} + \text{h.c.} \right) + i\chi\phi^{\dagger}_{1/2,\chi}\mathscr{T}_z^{\dagger}U_z\phi_{1/2,\chi}. \tag{4.64}$$

And effective masses $\tilde{m}_n(\boldsymbol{k})$ are just eigenvalues of $\mathscr{H}^{1d}_{\chi,s}$, with $\chi = \pm$ labelling mirror value.

Similarly, we could calculate the mirror layer Chern number, shown in Fig. 4.16. As is presented, both $\chi = \pm$ finally shows half-quantized Hall conductance, contributed from middle of the film rather than the surface, which is equivalent to saying that the conductance emerges from high-energy bulk rather than low-energy cone.

Despite the power of special mirror symmetry in our system, here we have to emphasize that the half plus half quantized nature is not protected by the mirror symmetry. Instead, the existence of mirror symmetry is a special case where we find two gapless surface states with opposite mirror label so that they are separable unambiguously. Nevertheless, focusing on the effective mass argument, one is capable to realize that the half-and-half nature relies only upon high-energy chirality. As one could perform at least numerically, either un-symmetric hopping or Zeeman field distribution does not affect the appearance of metallic quantized Hall conductance.

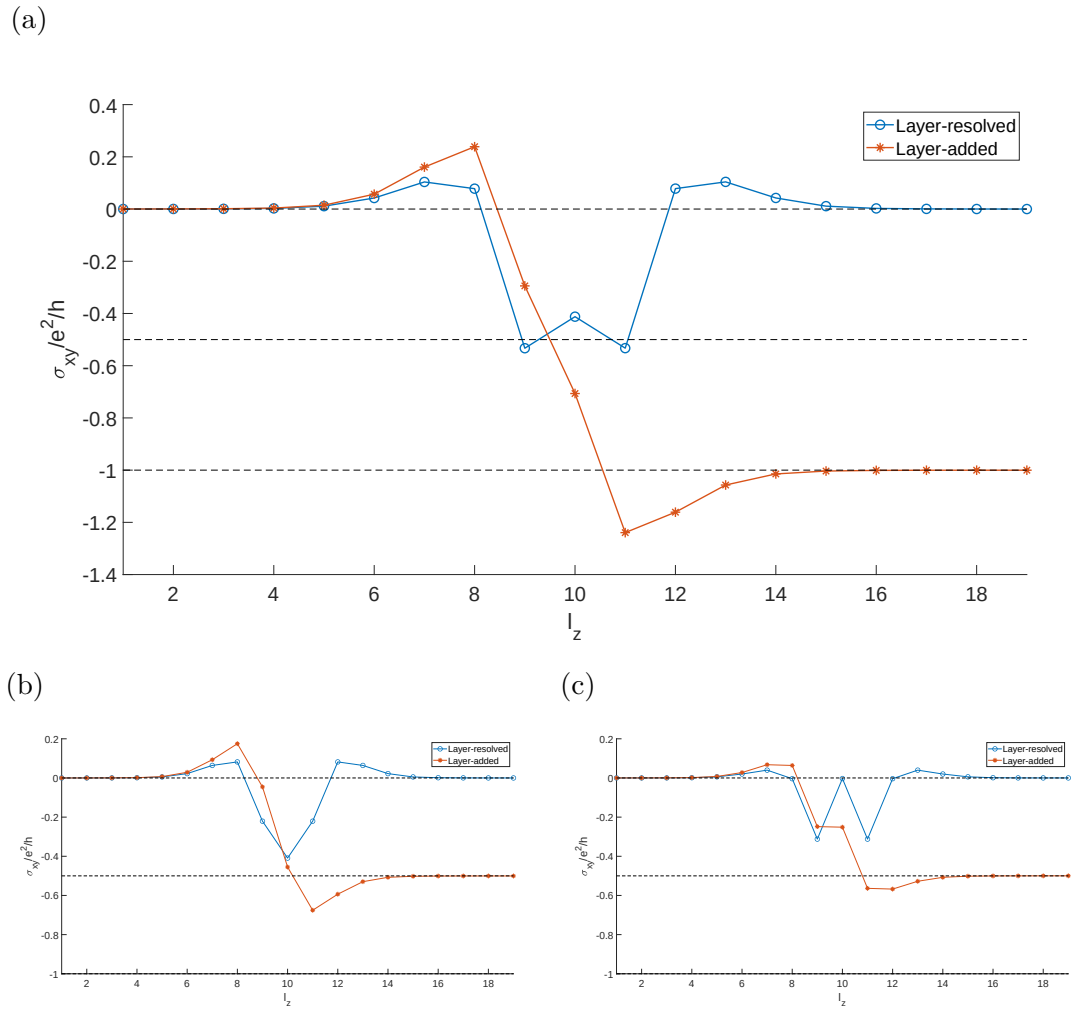


Figure 4.16: (a) Layer resolved and added Chern number. (b) Mirror layer resolved and added Chern number with $\chi = +$. (c) Mirror layer resolved and added Chern number with $\chi = -$.

4.4 QAHE with Anti-Ferromagnetic Order

This phenomenon originates from the observation that when L_z is even,

$$\begin{aligned} i\mathscr{T}_z^\dagger U_z &= i(t_\perp \beta + i\frac{\lambda_\perp}{2}\alpha_z)(-i\sigma_z\tau_z) \\ &= t_\perp \sigma_z\tau_0 + \frac{\lambda_\perp}{2}\sigma_0\tau_y, \end{aligned} \tag{4.65}$$

and one then may expect τ_y to play some role which has some similar or different impact like σ_z does. As derived before, such τ_y orbital order is projected effectively into

$$V_O^{y,\mathrm{Proj}}(\boldsymbol{k}) = \mathbf{V}_A^y(\boldsymbol{k})\tau_0\sigma_z - \mathbf{V}_S^y(\boldsymbol{k})\tau_y\sigma_z, \tag{4.66}$$

and especially under weak-field approximation, we only conserve $n = 1$ block component and write

$$V_y = V_A(\boldsymbol{k})\tau_0\sigma_z - V_S(\boldsymbol{k})\tau_y\sigma_z. \tag{4.67}$$

Then compared with z-Zeeman field case, we find that only $S \leftrightarrow A$ have been exchanged, and we are surely to say that

- τ_y at one side of TI: half-quantized anomalous Hall effect;
- Anti-symmetric order at two sides: quantum anomalous Hall effect;
- Symmetric order at two sides: so-called axion insulator phase.

Now the ultimate question is what physical meaning τ_y is. Clearly we see that τ_y breaks time reversal since $\mathscr{K}\tau_y = -\tau_y\mathscr{K}$, then this term is naturally related some magnetic-like order.

The role of τ_y still, dates back to the competition between it and σ_z.

4.5 Gauge Theory

Chapter 5

Topological Superconductor

5.1 Basics for Superconductor

Superconductor under Bardeen-Cooper-Schrieffer (BCS) theory originates from electron-phonon coupling induced Fermi surface instability in metal. The microscopic Hamiltonian with phenomenological parameters reads

$$H = H_e + H_p + H_{ep} \tag{5.1a}$$

$$H_e = \sum_k \xi_k c_k^\dagger c_k + \sum_{k,k',q} V_q c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k, \tag{5.1b}$$

$$H_p = \sum_q \hbar\omega_q \left(a_q^\dagger a_q + \frac{1}{2} \right), \tag{5.1c}$$

$$H_{ep} = i \sum_{k,q} A_q [c_{k+q}^\dagger a_q c_k + \text{h.c.}]. \tag{5.1d}$$

The effective Hamiltonian describing electrons comes from integrating phonons out, up to second order as shown in the second diagram in Fig. **??**, the effective electron-electron coupling becomes

$$\begin{aligned} \sim \langle H_{ep}^2 \rangle_p &= - \sum_{k,k',q,q'} \langle A_q A_{q'} [c_{k+q}^\dagger a_q c_k + \text{h.c.}][c_{k'+q'}^\dagger a_{q'} c_{k'} + \text{h.c.}] \rangle_p \\ &= -2 \sum_{k,k',q,q'} A_q A_{q'} c_{k+q}^\dagger c_k c_{k'}^\dagger c_{k'+q} \langle a_q a_{q'}^\dagger \rangle_p \\ &= - \sum_{k,k',q} \tilde{A}_q^2 c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k, \end{aligned} \tag{5.2}$$

which is an attractive interaction. Assume that this attraction overwhelms the Coulomb repulsion, which is usually possible in a metal with dense electron gas where Coulomb interaction is effectively screened, we can phenomenologically write down

$$H = \sum_k \xi_k c_k^\dagger c_k - \frac{g}{2} \sum_{k,k',q} c_{k+q}^\dagger c_{k'-q}^\dagger c_{k'} c_k \approx \sum_k \xi_k c_k^\dagger c_k - \frac{g}{2} \sum_{k,q} c_{k+q}^\dagger c_{-k-q}^\dagger c_{-k} c_k, \tag{5.3}$$

where we assume constant coupling and use maximal coupling volume principle. The wavevector q is bounded by phonon dispersion $|\omega_q| < \omega_D$ with ω_D usually Debye frequency.

Generally, the Fermionic operator $c_k = \{c_k^\alpha\}$ contains inner degrees of freedom, correspondingly

$$H = \sum_k \xi_k c_k^{\alpha,\dagger} c_k^\alpha - \frac{g}{2} \sum_{k,q} c_{k+q}^{\alpha,\dagger} c_{-k-q}^{\beta,\dagger} c_{-k}^\beta c_k^\alpha. \tag{5.4}$$

Introduce (self-consistent) mean-field order parameter, which also represents the superconducting gap, and the electronic correlation respecting Pauli principle

$$\Delta_{\alpha\beta}(k) = g \sum_q \langle c_{k+q}^\alpha c_{-k-q}^\beta \rangle = -\Delta_{\beta\alpha}(-k), \tag{5.5}$$

and by decomposing $\Delta_{\alpha\beta}(k) = \chi_{\alpha\beta}\Delta(k)$ with χ the spinor matrix for Fermionic inner degrees of freedom (spin, orbital, layer, $\cdots$), we recognize two cases

$$\begin{cases} \chi_{\alpha\beta} = -\chi_{\beta\alpha}, \ \Delta(k) = \Delta(-k), \ \text{Singlet} \\ \chi_{\alpha\beta} = \chi_{\beta\alpha}, \ \Delta(k) = -\Delta(-k), \ \text{Triplet} \end{cases}, \tag{5.6}$$

with the former for $s, d, \cdots$ wave pairing while the latter for $p, f, \cdots$ wave pairing. With introducing the pairing, the Hamiltonian becomes a non-interacting one (constant ignored)

$$H_M = \sum_k \xi_k^\alpha c_k^{\alpha,\dagger} c_k^\alpha + \frac{1}{2} \sum_k \Delta_{\alpha\beta}(k) c_k^{\alpha\dagger} c_{-k}^{\beta\dagger} + \text{h.c.} \tag{5.7}$$

Further introduce

$$\psi_k = \begin{pmatrix} c_k^{\alpha_1} \\ c_k^{\alpha_2} \\ \vdots \end{pmatrix}, \ \Psi_k = \begin{pmatrix} \psi_k \\ \psi_{-k}^\dagger \end{pmatrix}, \tag{5.8}$$

where $\psi^\dagger$ in second definition should be understood as a column vector, we have

$$H_M = \sum_k \psi_k^\dagger \xi_k \psi_k + \frac{1}{2}\psi_k^\dagger \Delta_k \psi_{-k}^\dagger - \frac{1}{2}\psi_k \Delta_k^* \psi_{-k} = \frac{1}{2} \sum_k \Psi_k^\dagger h_{\text{BdG}}(k) \Psi_k, \tag{5.9}$$

with $(\Delta_k)_{\alpha\beta} = \Delta_{\alpha\beta}(k)$, and

$$h_{\text{BdG}}(k) = \begin{pmatrix} \xi_k & \Delta_k \\ -\Delta_k^* & -\xi_k^T \end{pmatrix}. \tag{5.10}$$

On the other hand, when α includes spin degrees of freedom, represented by σ, a new basis known as Nambu spinor can be introduced as

$$\Phi_k = \begin{pmatrix} \psi_k \\ (i\sigma_y)\psi_{-k}^\dagger \end{pmatrix}, \tag{5.11}$$

under which the Hamiltonian equals

$$H_M = \sum_k \Phi_k^\dagger \begin{pmatrix} \xi_k & \Delta_k(-i\sigma_y) \\ -i\sigma_y \Delta_k^* & -\sigma_y \xi_{-k}^T \sigma_y \end{pmatrix} \Phi_k. \tag{5.12}$$

Notice that two electronic vectors in Nambu basis are time-reversal partners.

5.2 SMTI-SC Heterostructure

The Hamiltonian of 3-D **bulk** TI[**?**, **?**, **?**, 63, 9] is represented by the standard modified Dirac's equation[52] on lattice $\mathscr{H}_{TI} = \sum_{\boldsymbol{k}} \Psi^\dagger_{\boldsymbol{k}} H(\boldsymbol{k}) \Psi_{\boldsymbol{k}}$ where

$$H(\boldsymbol{k}) = \sum_\mu \lambda_\mu \sin(k_\mu a_\mu)\alpha_\mu + \left[m_0 - 4t_\mu \sin^2\left(\frac{k_\mu a_\mu}{2}\right)\right]\beta, \tag{5.13}$$

and given that $\lambda_\mu > 0$ and $t_\mu > 0$, the topology of the system is determined by the low-energy mass term m_0, which drives the system into trivial, weak TI or strong TI with four indices.

An anisotropic 3-D topological insulator **film** with limited number of sites along z direction, by Fourier transformation in $x-y$ plane, reads

$$\mathscr{H}_{\text{Film}} = \sum_{l_z,\boldsymbol{k}} \left(\psi^\dagger_{l_z,\boldsymbol{k}} \mathscr{M}_0(\boldsymbol{k}) \psi_{l_z,\boldsymbol{k}} + \psi^\dagger_{l_z,\boldsymbol{k}} \mathscr{T}_z \psi_{l_z+1,\boldsymbol{k}} + \text{h.c.}\right) + \sum_{l_z,\boldsymbol{k}} \psi^\dagger_{l_z,\boldsymbol{k}} H_\parallel \psi_{l_z,\boldsymbol{k}}, \tag{5.14}$$

with

$$H_\parallel = \lambda_\parallel[\sin(k_x a)\sigma_x\tau_x + \sin(k_y b)\sigma_y\tau_x], \tag{5.15}$$

and $\mathscr{M}_0(\boldsymbol{k}) = M_0(\boldsymbol{k})\sigma_0\tau_z = [m_0(\boldsymbol{k}) - 2t_\perp]\sigma_0\tau_z$, $\mathscr{T}_\mu = t_\mu\beta - i\frac{\lambda_\mu}{2}\alpha_\mu$, where

$$m_0(\boldsymbol{k}) = m_0 - 4t_\parallel\left(\sin^2\frac{k_x a}{2} + \sin^2\frac{k_y b}{2}\right). \tag{5.16}$$

Note that we have set $t_x = t_y = t_\parallel$, $t_z = t_\perp$, $\lambda_x = \lambda_y = \lambda_\parallel$, $\lambda_z = \lambda_\perp, a = b$. The Fermionic vector $\psi = (c_{1\uparrow}, c_{2\uparrow}, c_{1\downarrow}, c_{2\downarrow})^T$, corresponding to $\sigma\tau$ representing spin $\otimes$ orbital.

The **equivalently effective model**[**?**, **?**] is obtained by:
(i) Separate Hamiltonian at $\boldsymbol{k}$ as

$$\mathscr{H}_{\text{Film}}(\boldsymbol{k}) = \mathscr{H}_{1d}(\boldsymbol{k}) + \mathscr{H}_S(\boldsymbol{k}) \tag{5.17}$$

with

$$\mathscr{H}_{1d}(\boldsymbol{k}) = \sum_{l_z}\left(\psi^\dagger_{l_z,\boldsymbol{k}}\mathscr{M}_0(\boldsymbol{k})\psi_{l_z,\boldsymbol{k}} + \psi^\dagger_{l_z,\boldsymbol{k}}\mathscr{T}_z\psi_{l_z+1,\boldsymbol{k}} + \text{h.c.}\right), \tag{5.18a}$$

$$\mathscr{H}_S(\boldsymbol{k}) = \sum_{l_z}\psi^\dagger_{l_z,\boldsymbol{k}} H_\parallel \psi_{l_z,\boldsymbol{k}}. \tag{5.18b}$$

(ii) Solve $\mathscr{H}_{1d}$ with eigenstates ϕ^n_i whose energies are $(m_n, -m_n, -m_n, m_n)$ for $i = 1,2,3,4$ and $n = 1, \cdots, L_z$. Notice that the sequence of basis has been assigned so that $i = 1,2$ states and $i = 3,4$ states are separated by mirror symmetry. This procedure is general, depending only on the specific mirror or parity symmetry the system owns and not on specific parameter choices.
(iii) Project the film Hamiltonian and possible external field terms using Φ^n_i as

$$(H_{\text{equ}})^{nn'}_{ij} = \langle\phi^n_i|\, H_{\text{Film}} + V_{\text{ext}}\, |\phi^{n'}_j\rangle\,. \tag{5.19}$$

Basically, the projection of the original film Hamiltonian leads to

$$\tilde{H}_{\text{Film}}(\boldsymbol{k}) = \bigoplus_{n=1}^{L_z}\left[\lambda_\parallel(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + m_n(\boldsymbol{k})\tau_z\sigma_z\right], \tag{5.20}$$

where $2L_z$ Dirac fermions $\tilde{H}_{\text{Film}} = \oplus_{n,\chi} h_{n,\chi}(\boldsymbol{k})$ emerge as

$$h_{n,\chi}(\boldsymbol{k}) = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \chi m_n(\boldsymbol{k})\sigma_z, \tag{5.21}$$

with $\chi = \pm$ labelling the mirror eigenvalue.

Clearly, $\mathscr{M}_0(\boldsymbol{k})$ in $\mathscr{H}_{1d}(\boldsymbol{k})$ would be the main element dominating whether it has zero modes solution, under suitable boundary condition. In the numerical simulation, we set the model parameters to be fixed as $\lambda_{\parallel} = 0.41$eV, $\lambda_{\perp} = 0.44$eV, $t_{\parallel} = 0.566$eV, $t_{\perp} = 0.4$eV, $m_0 = 0.28$ eV, $a = b = 1$nm, $c = 0.5$nm[63].

For a semi-magnetic topological insulator attached with s-wave superconductor, the external field contains two parts:

• The Zeeman field

$$\mathscr{V}_Z(\boldsymbol{k}) = \sum_{l_z,\boldsymbol{k}} \psi^{\dagger}_{l_z,\boldsymbol{k}} V_Z(l_z)\psi_{l_z,\boldsymbol{k}},\ V_Z(l_z) = V_z(l_z)\sigma_z\tau_0; \tag{5.22}$$

• The proximate superconducting pairing (simplest form)

$$\begin{aligned}\mathscr{V}_{SC}(\boldsymbol{k}) &= \sum_{l_z,\boldsymbol{k}} \Delta_k(l_z)(c^{1\dagger}_{l_z,\boldsymbol{k}\uparrow}c^{1\dagger}_{l_z,-\boldsymbol{k}\downarrow} + c^{2\dagger}_{l_z,\boldsymbol{k}\uparrow}c^{2\dagger}_{l_z,-\boldsymbol{k}\downarrow}) + \text{h.c.} \\ &= \frac{1}{2}\sum_{l_z,\boldsymbol{k}} \psi^{\dagger}_{l_z,\boldsymbol{k}} \begin{pmatrix} & \Delta_k(l_z)\tau_0 \\ -\Delta_{-k}(l_z)\tau_0 & \end{pmatrix} \psi^{\dagger}_{l_z,-\boldsymbol{k}},\end{aligned} \tag{5.23}$$

which will become, by assuming s-wave pairing so that $\Delta_k = \Delta_{-k}$,

$$\mathscr{V}_{SC}(\boldsymbol{k}) = \frac{1}{2}\sum_{l_z,\boldsymbol{k}} \psi^{\dagger}_{l_z,\boldsymbol{k}}\Delta_k(l_z)(i\sigma_y)\tau_0\psi^{\dagger}_{l_z,-k}. \tag{5.24}$$

From here, we already see that two choices basis can be made: absorbing the $i\sigma_y$ matrix into pairing or into Fermionic operator. Since in the latter case, the chosen pairing will contain identity spinor matrix, we choose this convention.

Now we choose Nambu basis

$$\Phi_k = \begin{pmatrix} \psi_k \\ (i\sigma_y)\psi^{\dagger}_{-k} \end{pmatrix},\ \psi_k = \oplus_{l_z}\psi_{l_z,k}, \tag{5.25}$$

to write down the Bogoliubov-de-Gennes Hamiltonian as

$$\begin{aligned}\mathscr{H} &= \mathscr{H}_{\text{Film}} + \mathscr{V}_Z + \mathscr{V}_{SC} = \frac{1}{2}\sum_{\boldsymbol{k}} \Phi^{\dagger}_{\boldsymbol{k}} H_{\text{BdG}} \Phi_{\boldsymbol{k}} \\ &= \frac{1}{2}\sum_{\boldsymbol{k}} \begin{pmatrix} \psi^{\dagger}_{\boldsymbol{k}} & \psi_{-k}(-i\sigma_y) \end{pmatrix} \begin{pmatrix} H_{\text{Film}}(\boldsymbol{k}) + V_Z & \Delta_k\sigma_0\tau_0 \\ \Delta^*_k\sigma_0\tau_0 & -\sigma_y[H_{\text{Film}}(-\boldsymbol{k}) + V_Z]^T\sigma_y \end{pmatrix} \begin{pmatrix} \psi_k \\ (i\sigma_y)\psi^{\dagger}_{-k} \end{pmatrix},\end{aligned} \tag{5.26}$$

where quantities should be understood as

$$V_Z = \oplus_{l_z} V_Z(l_z),\ \Delta_k = \oplus_{l_z}\Delta_k(l_z), \tag{5.27}$$

and there should be an explicit chemical potential term $-\mu\sigma_0\tau_0$ added to the film Hamiltonian. Since it is easy to verify that

$$\sigma_y[H_{\text{Film}}(-\boldsymbol{k}) + V_Z]^T\sigma_y = H_{\text{Film}}(\boldsymbol{k}) - V_Z, \tag{5.28}$$

one concludes that

$$H_{\mathrm{BdG}}(\boldsymbol{k}) = \begin{pmatrix} H_{\mathrm{Film}}(\boldsymbol{k}) + V_Z & \Delta_k \sigma_0 \tau_0 \\ \Delta_k^* \sigma_0 \tau_0 & -H_{\mathrm{Film}}(\boldsymbol{k}) + V_Z \end{pmatrix}, \tag{5.29}$$

which serves as the starting point for analysis.

Before start, let us consider once more about symmetries own by the system. Without V_Z, the system owns time-reversal symmetry $\mathscr{T} = i\sigma_y \mathscr{K}$ if and only if Δ_k is real. Particle-hole symmetry switches particle and hole, together with inverting the momentum. Generally, we write

$$\mathscr{C}\psi_k \mathscr{C}^{-1} = (\psi_{-k}^\dagger U^\dagger)^T, \ \mathscr{C}\psi_k^\dagger \mathscr{C}^{-1} = (U\psi_{-k})^T, \tag{5.30}$$

where U represents inner degrees of freedom mixture, while ^{T} is transpose used to make sense of the equality. Then for general Hamiltonian $\mathscr{H} = \sum_k \psi_k^\dagger h(k) \psi_k$, particle-hold symmetry requires that

$$\begin{aligned} \mathscr{C}\mathscr{H}\mathscr{C}^{-1} &= \sum_k \mathscr{C}\psi_k^\dagger \mathscr{C}^{-1} \mathscr{C} h(k) \mathscr{C}^{-1} \mathscr{C} \psi_k \mathscr{C}^{-1} \\ &= \sum_k \psi_{-k}^T U^T h(k) U^* (\psi_{-k}^\dagger)^T = -\sum_k \psi_k^\dagger U^\dagger h^T(-k) U \psi_k + \mathrm{Tr}(H) = \mathscr{C}, \end{aligned}$$

which leads to requirement due to hermicity

$$U^\dagger h^T(-k) U = -h(k), \ \ \mathrm{Tr}(h) = 0. \tag{5.31}$$

Notice that we have chosen particle-hole operator to be unitary. Under BdG representation, it is natural to choose $U = \varsigma_y \sigma_y$ as Nambu degrees of freedom which switches creation and annihilation operators

$$(\Phi_{-k}^\dagger U^\dagger)^T = \left[\begin{pmatrix} \psi_{-k}^\dagger & \psi_k(-i\sigma_y) \end{pmatrix} \begin{pmatrix} & -i\sigma_y \\ i\sigma_y & \end{pmatrix} \right]^T = \begin{pmatrix} \psi_k \\ \psi_{-k}^\dagger(-i\sigma_y) \end{pmatrix} = \begin{pmatrix} \psi_k \\ i\sigma_y \psi_{-k}^\dagger \end{pmatrix}, \tag{5.32}$$

and under the representation we could let $C = \varsigma_y \sigma_y \mathscr{K}$ with $\mathscr{K}$ complex-conjugate operation so that

$$C h(k) C^{-1} = -h(-k), \tag{5.33}$$

and it can be verified that out H_{BdG} satisfies this by

$$\begin{aligned} U^\dagger H_{\mathrm{BdG}}^T(-k) U &= \begin{pmatrix} & -i\sigma_y \\ i\sigma_y & \end{pmatrix} \begin{pmatrix} H_{\mathrm{Film}}^T(-k) & \Delta_{-k}^* \\ \Delta_{-k} & -H_{\mathrm{Film}}^T(-k) \end{pmatrix} \begin{pmatrix} & -i\sigma_y \\ i\sigma_y & \end{pmatrix} \\ &= \begin{pmatrix} -\sigma_y H_{\mathrm{Film}}^T(-k)\sigma_y & -\sigma_y \Delta_{-k} \sigma_y \\ -\sigma_y \Delta_{-k}^* \sigma_y & \sigma_y H_{\mathrm{Film}}^T(-k) \sigma_y \end{pmatrix} = -H_{\mathrm{BdG}}(k). \end{aligned}$$

Still, our strategy is to solve the $m_0(\boldsymbol{k})$-dependent 1-D Hamiltonian inside, and now it becomes

$$H_{\mathrm{BdG}}^{1d}(\boldsymbol{k}) = \begin{pmatrix} H_{1d}(\boldsymbol{k}) & \\ & -H_{1d}(\boldsymbol{k}) \end{pmatrix}, \tag{5.34}$$

which owns both time-reversal, particle hole above and their combination as emergent chiral symmetry. However, by the highly symmetric form, the 1-D Hamiltonian owns an additional chiral symmetry τ_y which relates its basic solutions

$$\chi(s) = i\tau_y \varphi(s), \ \ s = \pm, \tag{5.35}$$

where

$$\varphi(s,l_z,\boldsymbol{k})=C\begin{pmatrix}-is\lambda_\perp f_+(l_z,\boldsymbol{k})\\ t_\perp\eta(\boldsymbol{k})f_-(l_z,\boldsymbol{k})\end{pmatrix}, \tag{5.36}$$

with parity even/odd functions $f_\pm$ as

$$\begin{cases} f_+(l_z)=\dfrac{\cos(\xi_1 l_z)}{\cos(\xi_1 l/2)}-\dfrac{\cos(\xi_2 l_z)}{\cos(\xi_2 l/2)} \\ f_-(l_z)=\dfrac{\sin(\xi_1 l_z)}{\sin(\xi_1 l/2)}-\dfrac{\sin(\xi_2 l_z)}{\sin(\xi_2 l/2)} \end{cases}, \tag{5.37}$$

where $l=L_z+1$ with L_z the layer number, and the dependence on wavevector is introduced by $\xi_\alpha(\boldsymbol{k})$, $\alpha=1,2$, which again, can be self-consistently obtained by solving simultaneous equations

$$m=M_0(\boldsymbol{k})+2t_\perp\frac{\cos\xi_1 g(\xi_1)-\cos\xi_2 g(\xi_2)}{g(\xi_1)-g(\xi_2)}, \tag{5.38a}$$

$$\cos\xi_\alpha=\frac{-M_0(\boldsymbol{k})t_\perp+(-1)^{\alpha-1}\sqrt{M_0^2(\boldsymbol{k})t_\perp^2-(t_\perp^2-\lambda_\perp^2/4)(M_0^2(\boldsymbol{k})+\lambda_\perp^2-m^2)}}{2(t_\perp^2-\lambda_\perp^2/4)}, \tag{5.38b}$$

with m eigenvalue of H_{1d} with defined function

$$g(\xi)=\frac{\tan(\xi l)/2}{\sin\xi}. \tag{5.39}$$

The factor η reads

$$\eta=\frac{-2(\cos\xi_1-\cos\xi_2)}{\sin\xi_1\cot(\xi_1 l/2)-\sin\xi_2\cot(\xi_2 l/2)}. \tag{5.40}$$

The solutions of $H^{1d}_{\mathrm{BdG}}(\boldsymbol{k})$ can be constructed easily. By denoting solutions of $H_{1d}(\boldsymbol{k})$ as

$$\phi_1=\begin{pmatrix}\varphi(+)\\0\end{pmatrix},\ \phi_2=\begin{pmatrix}0\\\chi(-)\end{pmatrix},\ \phi_3=\begin{pmatrix}\chi(+)\\0\end{pmatrix},\ \phi_4=\begin{pmatrix}0\\\varphi(-)\end{pmatrix}, \tag{5.41}$$

whose eigenvalues are $(m,-m,-m,m)$, the solution of $H^{1d}_{\mathrm{BdG}}(\boldsymbol{k})$ can be assigned as

$$\Phi_{1\sim4}=\begin{pmatrix}\phi_{1\sim4}\\0\end{pmatrix},\ \Phi_{5\sim8}=\begin{pmatrix}0\\\phi_{1\sim4}\end{pmatrix}, \tag{5.42}$$

with eigenvalues $(m,-m,-m,m,-m,m,m,-m)$, respectively. In principle $m=m_n(\boldsymbol{k})$ with $n=1,2,\cdots,L_z$, denoting L_z solutions, and for our purpose to focus on the lowest band, we let

$$m=m_1(\boldsymbol{k})=\Theta(-m_0(\boldsymbol{k}))m_0(\boldsymbol{k}), \tag{5.43}$$

which reveals the surface state mass of a strong topological insulator.

The projection procedure then follows that $(H_{\mathrm{Pro}})_{ij}=\langle\Phi_i|H_{\mathrm{BdG}}|\Phi_j\rangle$, and the film and Zeeman parts are projected to be

$$\begin{aligned}&\langle\Phi|\begin{pmatrix}H_{\mathrm{Film}}(\boldsymbol{k})+V_Z & \\ & -H_{\mathrm{Film}}(\boldsymbol{k})+V_Z\end{pmatrix}|\Phi\rangle\\ &=\lambda_\parallel\varsigma_z\tau_0(\sin(k_x a)\sigma_x+\sin(k_y b)\sigma_y)-\mu\varsigma_z\tau_0\sigma_0+m(\boldsymbol{k})\varsigma_z\tau_z\sigma_z+I_S(\boldsymbol{k})\varsigma_0\tau_0\sigma_z-I_A(\boldsymbol{k})\varsigma_0\tau_y\sigma_z.\end{aligned} \tag{5.44}$$

To project the superconducting part out, we note that in ς space, the superconducting term is purely off-diagonal, then the only conserving projection happens between $1 \sim 4$ to $5 \sim 8$, where two quantities appear as

$$\Delta_S = |C|^2 \sum_{l_z} \Delta_k(l_z) \left[\lambda_\perp^2 |f_+|^2 + t_\perp \eta^2 |f_-|^2\right], \tag{5.45a}$$

$$\Delta_A = |C|^2 \sum_{l_z} \Delta_k(l_z) 2\lambda_\perp t_\perp \eta \,\mathrm{Re}(f_+^* f_-), \tag{5.45b}$$

assigned as

$$\langle \Phi_n|\Delta|\Phi_{n+4}\rangle = \langle \phi_n|\Delta_k|\phi_n\rangle = \Delta_S,\ n = 1,2,3,4, \tag{5.46a}$$

$$\langle \Phi_n|\Delta|\Phi_{n+6}\rangle = i\Delta_A,\ n = 1,2. \tag{5.46b}$$

The projected superconducting Hamiltonian will be

$$\begin{aligned}\langle \Phi|\Delta|\Phi\rangle &= \begin{pmatrix} & \Delta_S\tau_0\sigma_0 - \Delta_A\tau_y\sigma_0 \\ \text{h.c.} & \end{pmatrix} \\ &= \mathrm{Re}\,\Delta_S\varsigma_x\tau_0\sigma_0 - \mathrm{Re}\,\Delta_A\varsigma_x\tau_y\sigma_0 - \mathrm{Im}\,\Delta_S\varsigma_y\tau_0\sigma_0 + \mathrm{Im}\,\Delta_A\varsigma_y\tau_y\sigma_0.\end{aligned} \tag{5.47}$$

The projected total Hamiltonian will become

$$\begin{aligned}H_{\mathrm{Pro}} &= \langle \Phi|H_{\mathrm{BdG}}|\Phi\rangle \\ &= \lambda_\parallel\varsigma_z\tau_0(\sin(k_xa)\sigma_x + \sin(k_yb)\sigma_y) - \mu\varsigma_z\tau_0\sigma_0 + m(\boldsymbol{k})\varsigma_z\tau_z\sigma_z + I_S(\boldsymbol{k})\varsigma_0\tau_0\sigma_z - I_A(\boldsymbol{k})\varsigma_0\tau_y\sigma_z \\ &\quad + \mathrm{Re}\,\Delta_S\varsigma_x\tau_0\sigma_0 - \mathrm{Re}\,\Delta_A\varsigma_x\tau_y\sigma_0 - \mathrm{Im}\,\Delta_S\varsigma_y\tau_0\sigma_0 + \mathrm{Im}\,\Delta_A\varsigma_y\tau_y\sigma_0.\end{aligned} \tag{5.48}$$

Written more compactly,

$$\begin{aligned}H_{\mathrm{Pro}} &= \lambda_\parallel\varsigma_z(\sin(k_xa)\sigma_x + \sin(k_yb)\sigma_y) - \mu\varsigma_z + m(\boldsymbol{k})\varsigma_z\tau_z\sigma_z + [I_S(\boldsymbol{k})\tau_0 - I_A(\boldsymbol{k})\tau_y]\sigma_z \\ &\quad + \varsigma_x(\mathrm{Re}\,\Delta_S\tau_0 - \mathrm{Re}\,\Delta_A\tau_y) - \varsigma_y(\mathrm{Im}\,\Delta_S\tau_0 - \mathrm{Im}\,\Delta_A\tau_y).\end{aligned} \tag{5.49}$$

5.3 Basis Consideration

The above utilizes mirror-symmetric basis, whose physics is deeper but may be lack of intuitive understanding. Nevertheless, we can introduce a unitary transformation $U = U_1U_2$, where

$$U_1 = e^{-i\tau_y\sigma_z\pi/4}, \tag{5.50a}$$

$$U_2 = e^{-i\tau_x\sigma_0\pi/4}, \tag{5.50b}$$

so that we can transform projected Hamiltonian $H_4 = \langle\phi|H_{\mathrm{Film}} + V_Z|\phi\rangle$ without superconducting part into

$$\begin{aligned}H_{SS} &= U^\dagger H_4 U = \lambda_\parallel\tau_z(\sin(k_xa)\sigma_y - \sin(k_yb)\sigma_x) - m(\boldsymbol{k})\tau_x\sigma_0 + I_S(\boldsymbol{k})\tau_0\sigma_z + I_A(\boldsymbol{k})\tau_z\sigma_z \\ &= \begin{pmatrix} \lambda_\parallel(\sin(k_xa)\sigma_y - \sin(k_yb)\sigma_x) + I_+(\boldsymbol{k})\sigma_z & -m(\boldsymbol{k})\sigma_0 \\ -m(\boldsymbol{k})\sigma_0 & -\lambda_\parallel(\sin(k_xa)\sigma_y - \sin(k_yb)\sigma_x) + I_-(\boldsymbol{k})\sigma_z \end{pmatrix},\end{aligned} \tag{5.51}$$

with chemical potential ignored and $I_\pm = I_S \pm I_A$. This transformation brings the Hamiltonian into the surface-state representation labelled by τ, and clearly the low-energy Zeeman gap is directly described by $I_\pm$. Act the same transformation $U \to \varsigma_0 U$ on the projected BdG Hamiltonian by noticing that

$$U^\dagger \varsigma_{x/y}\tau_0\sigma_0 U = \varsigma_{x/y}, \tag{5.52a}$$
$$U^\dagger \varsigma_{x/y}\tau_y\sigma_0 U = -\varsigma_{x/y}\tau_z, \tag{5.52b}$$

we obtain the transformed Hamiltonian as

$$\begin{aligned}
H^{SS}_{\rm Pro} &= U^\dagger H_{\rm Pro} U \\
&= \lambda_\parallel \varsigma_z\tau_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) - m(\boldsymbol{k})\varsigma_z\tau_x\sigma_0 - \mu\varsigma_z + \varsigma_0[I_S(\boldsymbol{k})\tau_0 + I_A(\boldsymbol{k})\tau_z]\sigma_z \\
&\quad + (\varsigma_x \,\mathrm{Re} - \varsigma_y \,\mathrm{Im})(\Delta_S(\boldsymbol{k})\tau_0 + \Delta_A(\boldsymbol{k})\tau_y)\sigma_0 \\
&= \begin{pmatrix}
\lambda_\parallel\tau_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) & \\
\quad -m(\boldsymbol{k})\tau_x\sigma_0 - \mu & (\Delta_S(\boldsymbol{k})\tau_0 + = \Delta_A(\boldsymbol{k})\tau_y)\sigma_0 \\
\quad +[I_S(\boldsymbol{k})\tau_0 + I_A(\boldsymbol{k})\tau_z]\sigma_z & \\
& -\lambda_\parallel\tau_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) \\
(\Delta^*_S(\boldsymbol{k})\tau_0 + \Delta^*_A(\boldsymbol{k})\tau_y)\sigma_0 & \quad +m(\boldsymbol{k})\tau_x\sigma_0 + \mu \\
& \quad +[I_S(\boldsymbol{k})\tau_0 + I_A(\boldsymbol{k})\tau_z]\sigma_z
\end{pmatrix} \\
&= \begin{pmatrix}
\lambda_\parallel(k\times\sigma)_z + I_+\sigma_z - \mu & -m & \Delta_+ & \\
-m & -\lambda_\parallel(k\times\sigma)_z + I_-\sigma_z - \mu & & \Delta_- \\
\Delta^*_+ & & -\lambda_\parallel(k\times\sigma)_z + I_+\sigma_z + \mu & m \\
& \Delta^*_- & m & \lambda_\parallel(k\times\sigma)_z + I_-\sigma_z + \mu
\end{pmatrix}
\end{aligned} \tag{5.53}$$

where in the last line, we use the continuum marks to represent the Hamiltonian for simplicity. On the other hand, we may write by putting surface index at the outside as

$$\begin{aligned}
H^{SS}_{\rm Pro} &= \lambda_\parallel\tau_z\varsigma_z(\sin(k_x a)\sigma_y - \sin(k_y b)\sigma_x) - m(\boldsymbol{k})\tau_x\varsigma_z - \mu\varsigma_z + [I_S(\boldsymbol{k})\tau_0 + I_A(\boldsymbol{k})\tau_z]\sigma_z \\
&\quad + (\Delta_S(\boldsymbol{k})\tau_0 + \Delta_A(\boldsymbol{k})\tau_y)(\overleftarrow{\mathrm{Re}}\,\varsigma_x - \overleftarrow{\mathrm{Im}}\,\varsigma_y) \\
&= \begin{pmatrix}
\lambda_\parallel\varsigma_z(\sin(k_x a)\sigma_y - \sin(\overleftarrow{k_y} b)\sigma_x) - \mu\varsigma_z & \\
\quad +I_+(\boldsymbol{k})\varsigma_0\sigma_z + \Delta_+(\boldsymbol{k})(\overleftarrow{\mathrm{Re}}\,\varsigma_x - \overleftarrow{\mathrm{Im}}\,\varsigma_y)\sigma_0 & -m(\boldsymbol{k})\varsigma_z\sigma_0 \\
& -\lambda_\parallel\varsigma_z(\sin(k_x a)\sigma_y - \sin(\overleftarrow{k_y} b)\sigma_x) - \mu\varsigma_z \\
-m(\boldsymbol{k})\varsigma_z\sigma_0 & \quad +I_-(\boldsymbol{k})\varsigma_0\sigma_z + \Delta_-(\boldsymbol{k})(\overleftarrow{\mathrm{Re}}\,\varsigma_x - \overleftarrow{\mathrm{Im}}\,\varsigma_y)\sigma_0
\end{pmatrix} \\
&= \begin{pmatrix}
\lambda_\parallel(k\times\sigma)_z + I_+\sigma_z - \mu & \Delta_+ & -m & \\
\Delta^*_+ & -\lambda_\parallel(k\times\sigma)_z + I_+\sigma_z + \mu & & m \\
-m & & -\lambda_\parallel(k\times\sigma)_z + I_-\sigma_z - \mu & \Delta_- \\
& m & \Delta^*_- & \lambda_\parallel(k\times\sigma)_z + I_-\sigma_z + \mu
\end{pmatrix}
\end{aligned} \tag{5.54}$$

with physical interpretation as: left-top blocks represent Hamiltonian for the **top** surface with surface Zeeman and superconducting terms, and right-bottom for **bottom** surface; terms off-diagonal represent top-bottom coupling, which vanishes at low-energy while mixing their high-energy bulk states as a requirement of the parity symmetry.

5.4 Case I: Uniform Zeeman and SC-Pairing

This case equals considering a magnetic topological insulator (MTI) thin film with long coherent length superconducting pairing immersed in.

5.5 Case II: Top Zeeman and Bottom SC-Pairing

For the FM-TI-SC heterostructure case, considering magnetism and superconducting pairing exist only at top and bottom surfaces separately leads to a good approximation that

$$\Delta_+ = 0,\ \Delta_- = \Delta, \tag{5.55a}$$
$$I_+ = V,\ I_- = 0, \tag{5.55b}$$

so that

$$H_{\text{Pro}}^{SS} = \begin{pmatrix} \lambda_\parallel(k\times\sigma)_z + V\sigma_z - \mu & & -m & \\ & -\lambda_\parallel(k\times\sigma)_z + V\sigma_z + \mu & & m \\ -m & & -\lambda_\parallel(k\times\sigma)_z - \mu & \Delta \\ & m & \Delta^* & \lambda_\parallel(k\times\sigma)_z + \mu \end{pmatrix}. \tag{5.56}$$

It should be stressed that both magnetism and superconducting pairing only modifies behavior near Fermi surface, which serves as low-energy mass term at most, and the ultimate topology will be determined by the high-energy coupling induced mass m.

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Appendix A

Discrete Symmetry

Parity

A parity transformation (also called parity inversion) is the flip in the sign of one spatial coordinate, which we could assign with a unitary operator $\hat{P}$. Sometimes we would like to assign parity operation as total action on whole space like inversion (point reflection) if space dimension is odd, and mirror reflection if it is even, and we just assume usual 3-D Euclidean space thus no need to try to distinguish in detail. Intuitively,

$$\hat{P}\hat{\boldsymbol{x}}\hat{P}^\dagger = -\hat{\boldsymbol{x}} \Longrightarrow \{\hat{P}, \hat{\boldsymbol{x}}\} = 0, \tag{A.1}$$

where super-operating form is bounded by

$$|\psi\rangle \to \hat{P}\,|\psi\rangle \tag{A.2}$$

and probability conservation.

From

$$\hat{\boldsymbol{x}}\hat{P}\,|\boldsymbol{x}\rangle = -\hat{P}\hat{\boldsymbol{x}}\,|\boldsymbol{x}\rangle = -\boldsymbol{x}\hat{P}\,|\boldsymbol{x}\rangle \Longrightarrow \hat{P}\,|\boldsymbol{x}\rangle = |-\boldsymbol{x}\rangle \tag{A.3}$$

where possible phase is chosen to be one. Then

$$\hat{P}^2\,|\boldsymbol{x}\rangle = |\boldsymbol{x}\rangle \Longrightarrow \hat{P}^2 = \mathbb{I}, \tag{A.4}$$

with eigenvalues + and − as even or odd parity. Using unitary condition,

$$\hat{P}^\dagger = \hat{P}^{-1} = \hat{P}. \tag{A.5}$$

Furthermore, it is also Intuitive

$$\hat{P}\hat{\boldsymbol{p}}\hat{P}^\dagger = -\hat{\boldsymbol{p}},\ \hat{P}\hat{\boldsymbol{J}}\hat{P}^\dagger = \hat{\boldsymbol{J}}, \tag{A.6}$$

where the latter is positive since $L = r \times p$. More strictly,

$$\hat{P}\hat{T}(\delta x)\,|x\rangle = \hat{P}\,|x + \delta x\rangle = |-x - \delta x\rangle = \hat{T}(-\delta x)\hat{P}\,|x\rangle \Longrightarrow \hat{P}\hat{T}(\delta x)\hat{P}^\dagger = \hat{T}(-\delta x),$$

which means that

$$\hat{P}\hat{\boldsymbol{p}}\hat{P}^\dagger = -\hat{\boldsymbol{p}},$$

as $\delta\boldsymbol{x}$ is just number. Meanwhile, for rotations generated by angular momentum, in 3-D Euclid representation

$$[R, P] = 0 \Longrightarrow P = \mathrm{diag}(-1, -1, -1), \tag{A.7}$$

where the minus sign reflects action on position.

For wavefunction in position space, assume that ket

$$\hat{P}\left|\psi\right\rangle = \pm\left|\psi\right\rangle, \tag{A.8}$$

then

$$\begin{aligned}\langle x|\hat{P}|\psi\rangle &= \langle x|\pm|\psi\rangle = \pm\psi(x)\\ &= \langle -x|\psi\rangle = \psi(-x)\\ \implies \psi(x) &= \pm\psi(-x).\end{aligned} \tag{A.9}$$

For Hamiltonian satisfying $[\hat{H},\hat{P}]=0$, i.e.,

$$H(x) = \langle x|\hat{H}|x\rangle = \langle x|\hat{P}^\dagger\hat{P}\hat{H}\hat{P}^\dagger\hat{P}|x\rangle = H(-x), \tag{A.10}$$

then if $H(x)\psi(x) = E\psi(x)$, one must have $H(x)\psi(-x) = H(-x)\psi(-x) = E\psi(-x)$, i.e., $\psi(-x)$ is also a solution with the same energy, and one may construct manually

$$\psi_E(x)\sim\psi(x)+\psi(-x),\ \psi_O(x)\sim\psi(x)-\psi(-x). \tag{A.11}$$

However, if $|\psi\rangle$, i.e. $\psi(x)$ itself is even, then $\psi_O = 0$ and loses meaning, vice versa.

By even and odd parity we may let *polar vector* (operator) to be those odd under parity like position and momentum; *axial/pseudo vector* to be even like angular momentum and magnetic field; and *pseudo scalar* to be scalar (operator) odd under parity like $\hat{\boldsymbol{S}}\cdot\hat{\boldsymbol{x}}$.

Parity Selection

For two states being eigenkets of parity operator

$$\hat{P}\left|\alpha\right\rangle = s_\alpha\left|\alpha\right\rangle,\ \hat{P}\left|\beta\right\rangle = s_\beta\left|\beta\right\rangle, \tag{A.12}$$

the scattering process $\langle\beta|\hat{O}|\alpha\rangle$ with intermediate operator

$$\hat{P}\hat{O}\hat{P}^\dagger = s_o\hat{O} \tag{A.13}$$

is non-zero unless $s_\alpha s_\beta s_o = +$, since

$$\langle\beta|\hat{O}|\alpha\rangle = \langle\beta|\hat{P}^\dagger\hat{P}\hat{O}\hat{P}^\dagger\hat{P}|\alpha\rangle = s_\alpha s_\beta s_o\left\langle\beta|\hat{O}|\alpha\right\rangle. \tag{A.14}$$

Time Reversal

Lorentz force is written as

$$m\ddot{\boldsymbol{r}} = q(\boldsymbol{E}+\dot{\boldsymbol{r}}\times\boldsymbol{B}).$$

Intuitively, under time reversal $t\to -t$ so that only $\dot{\boldsymbol{r}}$ changes sign, then we see that *external* magnetic field would break time reversal invariance.

Now given time reversal operator $\hat{T}$, intuitively,

$$\hat{T}\hat{x}\hat{T}^{-1} = \hat{x},\ \hat{T}\hat{p}\hat{T}^{-1} = -\hat{p},\ \hat{T}\hat{J}\hat{T}^{-1} = -\hat{J}, \tag{A.15}$$

which serves as a definition for time reversal operation and corresponding operator. And one may also admit if we have 'time operator'

$$\hat{T}\hat{t}\hat{T}^{-1} = -\hat{t}.$$

Then, similar to the parity case one has

$$\begin{aligned} & \hat{x}\hat{T}\left|x\right\rangle = \hat{T}\hat{x}\left|x\right\rangle = x\hat{T}\left|x\right\rangle, \ \hat{p}\hat{T}\left|p\right\rangle = -p\hat{T}\left|p\right\rangle, \ \hat{J}\hat{T}\left|J\right\rangle = -J\hat{T}\left|J\right\rangle, \\ \Longrightarrow \quad & \hat{T}\left|x\right\rangle = \left|x\right\rangle, \qquad \hat{T}\left|p\right\rangle = \left|-p\right\rangle, \qquad \hat{T}\left|J\right\rangle = \left|-J\right\rangle, \end{aligned}$$

where a possible phase factor is chosen to be identity. And when it acts on some general ket, also intuitively we have

$$\hat{T}\left|\psi(\delta t)\right\rangle = \hat{T}\left(\mathbb{I} - \frac{i}{\hbar}\hat{H}\delta t\right)\left|\psi(0)\right\rangle \sim \left|\psi(-\delta t)\right\rangle = \left(\mathbb{I} + \frac{i}{\hbar}\hat{H}\delta t\right)\left|\psi(0)\right\rangle,$$

then confusingly we have $\{\hat{T}, \hat{H}\} = 0$? This is not the symmetry we want. Then we have to assume that $\hat{T} = \hat{U}\mathscr{K}$ is anti-unitary with

$$\mathscr{K}i = -i\mathscr{K}, \tag{A.16}$$

so that

$$[\hat{T}, \hat{H}] = 0. \tag{A.17}$$

And our mission reduces to find $\hat{U}$.

Before that, some general properties need discussion. First of all, from the definition one could see it that this symmetry operation is related to certain representation, and the anti-unitarity further leads to more sensitive choice for a representation. Say, given state $\left|\psi\right\rangle$ and representation $\hat{A}$, one would have

$$\begin{aligned} \left|T\psi\right\rangle = \hat{T}\left|\psi\right\rangle &= \hat{U}\mathscr{K}\sum_n \left|a_n\right\rangle\left\langle a_n|\psi\right\rangle = \sum_n \hat{U}\mathscr{K}\left\langle a_n|\psi\right\rangle\left|a_n\right\rangle \\ &= \sum_n \left\langle a_n|\psi\right\rangle^* \hat{U}\mathscr{K}\left|a_n\right\rangle = \sum_n \left\langle\psi|a_n\right\rangle\hat{U}\left|a_n\right\rangle, \end{aligned} \tag{A.18}$$

where in the last step we have used the fact that we are under representation of $\hat{A}$ so that action of $\mathscr{K}$ on its eigenket $\left|a_n\right\rangle$ just changes nothing. One needs to notice it that $\mathscr{K}$ is an operator so it could not just pass a ket and says nothing, however, on the other hand, it only acts on complex field, i.e., the expansion coefficient of a ket. One suitable example here is spin-half case. Say now we are under representation of $\hat{S}_z$, then

$$\mathscr{K}\left|z\pm\right\rangle = \left|z\pm\right\rangle,$$

as the expansion coefficient

$$\left\langle z\pm|z\pm\right\rangle = 1, \ \left\langle z\pm|z\mp\right\rangle = 0,$$

and there is nothing to change. However, consider that

$$\left|y\pm\right\rangle = \frac{1}{\sqrt{2}}(\left|z+\right\rangle \pm i\left|z-\right\rangle),$$

and the action of complex conjugate would lead to

$$\mathscr{K}\left|y\pm\right\rangle=\frac{1}{\sqrt{2}}(\mathscr{K}\left|z+\right\rangle\pm\mathscr{K}i\left|z-\right\rangle)=\frac{1}{\sqrt{2}}(\mathscr{K}\left|z+\right\rangle\mp i\mathscr{K}\left|z-\right\rangle)=\left|y\mp\right\rangle.$$

For the action of time reversal on a bra, we need to utilize the dual correspondence as Hermitian conjugate

$$\langle T\psi|=(|T\psi\rangle)^{\dagger}=\sum_{n}\langle a_{n}|\,\hat{U}^{\dagger}\,\langle a_{n}|\psi\rangle\,. \tag{A.19}$$

One may formally introduce the complex conjugate of a ket as

$$|\psi\rangle^{*}=\mathscr{K}\,|\psi\rangle=\sum_{n}\langle a_{n}|\psi\rangle^{*}\,|a_{n}\rangle\,, \tag{A.20}$$

and one may formally write

$$\begin{cases}|T\psi\rangle=\hat{U}\,|\psi\rangle^{*}\\ \langle T\psi|=\langle\psi|^{*}\,\hat{U}^{\dagger}\end{cases}. \tag{A.21}$$

However, we would better avoid writing in this way to get away from the right action of anti-unitary operator.

Using expressions above, one finds

$$\begin{aligned}\langle T\phi|T\psi\rangle&=\sum_{mn}\langle a_{m}|\,\hat{U}^{\dagger}\,\langle a_{m}|\phi\rangle\,\langle\psi|a_{n}\rangle\,\hat{U}\,|a_{n}\rangle\\&=\sum_{mn}\langle\psi|a_{n}\rangle\,\delta_{mn}\,\langle a_{m}|\phi\rangle=\langle\psi|\phi\rangle\\&=\langle\phi|\psi\rangle^{*}\,,\end{aligned}$$

which just conforms the definition of anti-unitary operator. What is more,

$$\begin{aligned}\langle\phi|\hat{A}|\psi\rangle&=\langle\phi A^{\dagger}|\psi\rangle=\langle T\psi|TA^{\dagger}\phi\rangle=\langle T\psi|\,\hat{T}\hat{A}^{\dagger}\,|\phi\rangle\\&=\langle T\psi|\,\hat{T}\hat{A}^{\dagger}\hat{T}^{-1}\,|T\phi\rangle\,.\end{aligned} \tag{A.22}$$

A Glance of Spin-half particle

By representation under $\hat{\sigma}_z$, we may let $T=U\mathscr{K}$, and from $T\boldsymbol{\sigma}T^{-1}=-\boldsymbol{\sigma}$ as time reversal operation on spin-angular momentum, one find

$$U\sigma_{x/z}U^{-1}=-\sigma_{x/z},\ U\sigma_{y}U^{-1}=\sigma_{y}, \tag{A.23}$$

then by construction one could let

$$U=i\sigma_{y},\ T=i\sigma_{y}\mathscr{K}=\exp\Big(i\sigma_{y}\frac{\pi}{2}\Big)\mathscr{K},\ \hat{T}=\exp\left(\frac{i}{\hbar}\hat{S}_{y}\pi\right)\mathscr{K}, \tag{A.24}$$

which could be generalized to common angular momentum

$$\hat{T}=\exp\left(\frac{i}{\hbar}\hat{J}_{y}\pi\right)\mathscr{K}. \tag{A.25}$$

Now since

$$\begin{aligned}\hat{T}^2 |\psi\rangle &= \exp\left(\frac{i}{\hbar}\hat{J}_y 2\pi\right)\mathscr{K}^2 \sum_m |j,m\rangle \langle j,m|\psi\rangle \\ &= (-1)^{2j}\sum_m |j,m\rangle \langle j,m|\psi\rangle = (-1)^{2j}|\psi\rangle ,\end{aligned} \tag{A.26}$$

we have

$$\hat{T}^2 = \begin{cases} +\mathbb{I}, \text{ integer spin} \\ -\mathbb{I}, \text{ half-integer spin} \end{cases} . \tag{A.27}$$

Kramers Degeneracy

For system with time reversal invariance,

$$\hat{H}\hat{T}|n\rangle = \hat{T}\hat{H}|n\rangle = E_n\hat{T}|n\rangle . \tag{A.28}$$

If assuming $\hat{T}|n\rangle = e^{i\theta}|n\rangle$,

$$\hat{T}^2|n\rangle = \hat{T}(e^{i\theta}|n\rangle) = e^{-i\theta}\hat{T}|n\rangle = |n\rangle , \tag{A.29}$$

which holds only when $\hat{T}^2 = \mathbb{I}$. And in this way we say that states electronic system with time reversal invariance must be at least doubly degenerated as $\hat{T}^2 = -\mathbb{I}$.

Appendix B

Model of the Topological Insulator

The model of the topological insulator we utilize originates from the $\boldsymbol{k}\cdot\boldsymbol{p}$ model of the lowest four p orbitals in Bi_2Se_3, which is deduced solely from symmetry considerations. In principle, we need to consider actual symmetries, as the time-reversal symmetry $\mathscr{T}$, the inversion symmetry $\mathscr{P}$, the 3-fold rotational symmetry C_{3z} and mirror symmetry $\mathscr{M}_x$. Since the latter can be decoupled into inversion symmetry $\mathscr{P}$ and 2-fold rotational symmetries C_{2x}, we split it, which are more convenient to work with. The Hamiltonian is the obtained by restricting possible terms using symmetries, which can be solved most generally using the invariant theory.

By letting σ, τ to represent the spin and orbital degrees of freedom, the most general Hamiltonian up to the second order in $\boldsymbol{k}$ reads

$$H(\boldsymbol{k}) = \sum_{\mu\nu=0,1,2,3} \sum_{ij=1,2,3} \left(A^i_{\mu\nu} k_i + B^{ij}_{\mu\nu} k_i k_j + C_{\mu\nu}\right) \sigma_\mu \tau_\nu, \tag{B.1}$$

with A, B, Cs real coefficients, and the symmetries are represented as $\mathscr{T} = i\sigma_y \mathscr{K}$, $\mathscr{P} = \tau_z$, $C_{3z} = e^{-i\pi\sigma_z/3}$, $C_{2x/y} = e^{-i\pi\sigma_{x/y}/2}$, respectively. These symmetry requirements lead to the following constraints

$$\mathscr{T} H(-\boldsymbol{k}) \mathscr{T}^{-1} = H(\boldsymbol{k}), \tag{B.2a}$$
$$\mathscr{P} H(-\boldsymbol{k}) \mathscr{P}^{-1} = H(\boldsymbol{k}), \tag{B.2b}$$
$$C_{3z} H(R_{3z}\boldsymbol{k}) C_{3z}^{-1} = H(\boldsymbol{k}), \tag{B.2c}$$
$$C_{2x/y} H(R_{2x/y}\boldsymbol{k}) C_{2x/y}^{-1} = H(\boldsymbol{k}), \tag{B.2d}$$

and we shall figure them out one by one.

Time-reversal symmetry

This leads to the equation for each $\sigma_\mu\tau_\nu$ (as they compose a complete basis for 4×4 Hermitian matrices) (sum over i, j ignored)

$$(-A^i_{\mu\nu} k_i + B^{ij}_{\mu\nu} k_i k_j + C_{\mu\nu})(\sigma_y \sigma^*_\mu \sigma_y)\tau^*_\nu = (A^i_{\mu\nu} k_i + B^{ij}_{\mu\nu} k_i k_j + C_{\mu\nu})\sigma_\mu\tau_\nu,$$

and since $\sigma_y \sigma_i^* \sigma_y = -\sigma_i$, $\tau_y^* = -\tau_y$, we have the following constraints that

$$\begin{cases} A: & -(-1)^{(1-\delta_{\mu 0})+\delta_{j2}} = 1 \\ B/C: & (-1)^{(1-\delta_{\mu 0})+\delta_{j2}} = 1 \end{cases} \Longrightarrow \begin{cases} \delta_{i0} = \delta_{j2} \\ \delta_{i0} = \delta_{j2} \pm 1 \end{cases},$$

and leads to the possible terms as

$$A: \ \sigma_0\tau_y \text{ or } \sigma_{x/y/z}\tau_{0/x/z}, \tag{B.3a}$$
$$B/C: \ \sigma_0\tau_{0/x/z} \text{ or } \sigma_{x/y/z}\tau_y. \tag{B.3b}$$

Inversion symmetry

This leads to the equation for each $\sigma_\mu\tau_\nu$

$$(-A^i_{\mu\nu}k_i + B^{ij}_{\mu\nu}k_ik_j + C_{\mu\nu})\sigma_\mu(\tau_z\tau_\nu\tau_z) = (A^i_{\mu\nu}k_i + B^{ij}_{\mu\nu}k_ik_j + C_{\mu\nu})\sigma_\mu\tau_\nu,$$

which leads to the constraints that

$$\begin{cases} A: & -\tau_z\tau_\nu\tau_z = \tau_\nu \\ B/C: & \tau_z\tau_\nu\tau_z = \tau_\nu \end{cases} \Longrightarrow \begin{cases} \tau_{x/y} \\ \tau_{0/z} \end{cases}.$$

The above two constraints lead to the possible terms as

$$\begin{cases} A: & \sigma_0\tau_y \text{ or } \sigma_{x/y/z}\tau_x \\ B/C: & \sigma_0\tau_{0/z} \end{cases},$$

by taking their intersection meaning that two symmetries are simultaneously satisfied.

Rotational symmetry

3-fold rotational symmetry

We have $R_{3z}\boldsymbol{k} = \left(-\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y, -\frac{\sqrt{3}}{2}k_x - \frac{1}{2}k_y, k_z\right)$ by counter-clockwise $2\pi/3$ rotation, and together with $C_{3z} = (1 - i\sqrt{3}\sigma_z)/2$, the equation for each $\sigma_\mu\tau_\nu$ reads

$$\left(A^i_{\mu\nu}(R_{3z}\boldsymbol{k})_i + B^{ij}_{\mu\nu}(R_{3z}\boldsymbol{k})_i(R_{3z}\boldsymbol{k})_j + C_{\mu\nu}\right) C_{3z}\sigma_\mu\tau_\nu C^{-1}_{3z} = (A^i_{\mu\nu}k_i + B^{ij}_{\mu\nu}k_ik_j + C_{\mu\nu})\sigma_\mu\tau_\nu,$$

Notice that since $C_{3z}\sigma_xC^{-1}_{3z} = (-\sigma_x + \sqrt{3}\sigma_y)/2$, $C_{3z}\sigma_yC^{-1}_{3z} = (-\sqrt{3}\sigma_x - \sigma_y)/2$ and $C_{3z}\sigma_{0/z}C^{-1}_{3z} = \sigma_{0/z}$, the rotation mixes x and y, and we can not separate the two indices.

We firstly consider $\mu = z$, which naturally excludes B/C terms by the above classification, and for A, we have the equation

$$A^i_{\mu\nu}(R_{3z}\boldsymbol{k})_i = A^i_{\mu\nu}k_i,$$

where k_z term directly cancels on the two sides thus imposing no constraints, while for x, y, requiring equal coefficients before $k_{x/y}$ separately leads to the equation that

$$\begin{pmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{pmatrix} \begin{pmatrix} A^x_{z\nu} \\ A^y_{z\nu} \end{pmatrix} = 0,$$

which has no non-trivial solutions. Thus, we conclude that there are no $A^{x/y}_{z\nu}$ terms.

We then consider $\mu = 0$, and based on the same reason, there are no $A_{0\nu}^{x/y}$ terms, while for $A_{0\nu}^{z}$ term, there is no constraint. Turning to B/C, we see also that $B_{0\nu}^{zz}$ and $C_{0\nu}$ are not constrained, and for $B_{0\nu}^{ij}$ with $ij \neq zz$, we firstly consider one of i, j equals to z (say $j = z$), which then leads to the equation that

$$B_{0\nu}^{xz}\frac{-k_x+\sqrt{3}k_y}{2}+B_{0\nu}^{yz}\frac{-\sqrt{3}k_x-k_y}{2}=B_{0\nu}^{xz}k_x+B_{0\nu}^{yz}k_y,$$

which is impossible to satisfy, and we conclude that there are no $B_{0\nu}^{x/y,z}$ or $B_{0\nu}^{z,x/y}$ terms. For the rest with $i, j \neq z$, we have the equation that

$$\begin{aligned}&B_{0\nu}^{xx}\left(\frac{-k_x+\sqrt{3}k_y}{2}\right)^2+B_{0\nu}^{yy}\left(\frac{-\sqrt{3}k_x-k_y}{2}\right)^2+B_{0\nu}^{xy}\left(\frac{-k_x+\sqrt{3}k_y}{2}\right)\left(\frac{-\sqrt{3}k_x-k_y}{2}\right)\\&+B_{0\nu}^{yx}\left(\frac{-k_x+\sqrt{3}k_y}{2}\right)\left(\frac{-\sqrt{3}k_x-k_y}{2}\right)=B_{0\nu}^{xx}k_x^2+B_{0\nu}^{yy}k_y^2+B_{0\nu}^{xy}k_xk_y+B_{0\nu}^{yx}k_xk_y,\end{aligned}$$

and by comparing coefficients before k_x^2, k_y^2, k_xk_y separately, we have the equations that

$$\begin{cases}(B_{0\nu}^{xx}+3B_{0\nu}^{yy})/4+\sqrt{3}(B_{0\nu}^{xy}+B_{0\nu}^{yx})/4=B_{0\nu}^{xx}\\(3B_{0\nu}^{xx}+B_{0\nu}^{yy})/4-\sqrt{3}(B_{0\nu}^{xy}+B_{0\nu}^{yx})/4=B_{0\nu}^{yy}\\-\sqrt{3}(B_{0\nu}^{xx}-B_{0\nu}^{yy})/2-(B_{0\nu}^{xy}+B_{0\nu}^{yx})/2=B_{0\nu}^{xy}+B_{0\nu}^{yx}\end{cases},$$

which leads to the constrains that

$$\begin{cases}B_{0\nu}^{xx}=B_{0\nu}^{yy}\\B_{0\nu}^{xy}=-B_{0\nu}^{yx}\end{cases},$$

where the second constraint leads to vanishing k_xk_y cross terms, and thus can be thrown.

Finally, we turn to $\mu = x, y$, where there are also no B/C terms. We then write down equations for A terms as

$$\begin{aligned}&\left(A_{x\nu}^{x}\frac{-k_x+\sqrt{3}k_y}{2}+A_{x\nu}^{y}\frac{-\sqrt{3}k_x-k_y}{2}+A_{x\nu}^{z}k_z\right)\frac{-\sigma_x+\sqrt{3}\sigma_y}{2}\\&+\left(A_{y\nu}^{x}\frac{-k_x+\sqrt{3}k_y}{2}+A_{y\nu}^{y}\frac{-\sqrt{3}k_x-k_y}{2}+A_{y\nu}^{z}k_z\right)\frac{-\sqrt{3}\sigma_x-\sigma_y}{2}\\&=\left(A_{x\nu}^{x}k_x+A_{x\nu}^{y}k_y+A_{x\nu}^{z}k_z\right)\sigma_x+\left(A_{y\nu}^{x}k_x+A_{y\nu}^{y}k_y+A_{y\nu}^{z}k_z\right)\sigma_y,\end{aligned}$$

and by comparing coefficients before $k_i\sigma_{x/y}$, we have

$$\begin{cases}k_x\sigma_x: & \left(A_{x\nu}^{x}+\sqrt{3}A_{x\nu}^{y}+\sqrt{3}A_{y\nu}^{x}+3A_{y\nu}^{y}\right)/4=A_{x\nu}^{x}\\k_y\sigma_x: & \left(-\sqrt{3}A_{x\nu}^{x}+A_{x\nu}^{y}-3A_{y\nu}^{x}+\sqrt{3}A_{y\nu}^{y}\right)/4=A_{x\nu}^{y}\\k_z\sigma_x: & \left(-A_{x\nu}^{z}-\sqrt{3}A_{y\nu}^{z}\right)/2=A_{x\nu}^{z}\\k_x\sigma_y: & \left(-\sqrt{3}A_{x\nu}^{x}-3A_{x\nu}^{y}+A_{y\nu}^{x}+\sqrt{3}A_{y\nu}^{y}\right)/4=A_{y\nu}^{x}\\k_y\sigma_y: & \left(3A_{x\nu}^{x}-\sqrt{3}A_{x\nu}^{y}-\sqrt{3}A_{y\nu}^{x}+A_{y\nu}^{y}\right)/4=A_{y\nu}^{y}\\k_z\sigma_y: & \left(\sqrt{3}A_{x\nu}^{z}-A_{y\nu}^{z}\right)/2=A_{y\nu}^{z}\end{cases},$$

which leads to

$$\begin{cases} A^x_{x\nu} = A^y_{y\nu} \\ A^y_{x\nu} = -A^x_{y\nu} \\ A^z_{x\nu} = A^z_{y\nu} = 0 \end{cases}.$$

We see that by the 3-fold z-rotation symmetry, the possible terms are

$$\begin{cases} A: & A^x_{xx} = A^y_{yx},\ A^y_{xx} = -A^x_{yx},\ A^z_{zx},\ A^z_{0y}, \\ B: & B^{xx}_{0\nu} = B^{yy}_{0\nu},\ B^{zz}_{0\nu}, \\ C: & C_{0\nu}, \end{cases}$$

which leads to a stage-I $\boldsymbol{k}\cdot\boldsymbol{p}$ model as

$$\begin{aligned} H_I(\boldsymbol{k}) = &(D_0 + D_1\boldsymbol{k}_\parallel^2 + D_2 k_\perp^2)\sigma_0\tau_0 + (C_0 + C_1\boldsymbol{k}_\parallel^2 + C_2 k_\perp^2)\sigma_0\tau_z \\ &+ \left[A\boldsymbol{k}_\parallel\cdot\boldsymbol{\sigma} + \tilde{A}(\boldsymbol{k}_\parallel\times\boldsymbol{\sigma})_z\right]\tau_x + A_z k_z\sigma_z\tau_x + \tilde{A}_z k_z\sigma_0\tau_y, \end{aligned}$$

where $\boldsymbol{k}_\parallel = (k_x, k_y)$. This Hamiltonian, when compared with the usual TI model, contains additional Rashba spin-orbital coupling terms and a chiral-symmetry breaking term linear in k_z.

2-fold rotational symmetry

We consider $C_{2x} = e^{-i\sigma_x\pi/2} = -i\sigma_x$, with $R_{2x}\boldsymbol{k} = (k_x, -k_y, -k_z)$, and by imposing this symmetry over $H_I(\boldsymbol{k})$, it is required that $\tilde{A} = 0 = \tilde{A}_z$, and the same for C_{2y}.

Finally, we obtain the stage-II $\boldsymbol{k}\cdot\boldsymbol{p}$ model as our final effective Hamiltonian as

$$H(\boldsymbol{k}) = \varepsilon(\boldsymbol{k}) + \left(A\boldsymbol{k}_\parallel\cdot\boldsymbol{\sigma} + A_z k_z\sigma_z\right)\tau_x + \mathscr{M}(\boldsymbol{k})\tau_z, \tag{B.4}$$

where $\varepsilon(\boldsymbol{k}) = D_0 + D_1\boldsymbol{k}_\parallel^2 + D_2 k_\perp^2$, $\mathscr{M}(\boldsymbol{k}) = C_0 + C_1\boldsymbol{k}_\parallel^2 + C_2 k_\perp^2$. This Hamiltonian owns time-reversal $\mathscr{T}$, inversion $\mathscr{P}$, three-fold rotational C_{3z} and two-fold rotational $C_{2x/y}$ symmetries, together with their combinational symmetries as $\mathscr{P}\mathscr{T}$, mirror symmetries $M_{x/y} = \mathscr{P}C_{2x/y}$, and $M_{x/y}\mathscr{T}$. One can further deduce that the Hamiltonian is planar-rotational symmetric, i.e., homogeneous in $x-y$ plane.

B.1 Replace $\mathscr{P}$ with $\mathscr{M}_z$

In a twin boundary structure, the inversion symmetry $\mathscr{P}$ is broken, while the mirror symmetry $\mathscr{M}_z$ is preserved. After constraining with the time-reversal symmetry $\mathscr{T}$, we similarly get

$$\begin{cases} A: & \sigma_0\tau_y \text{ or } \sigma_{x/y/z}\tau_{0/x/z} \\ B/C: & \sigma_0\tau_{0/x/z} \text{ or } \sigma_{x/y/z}\tau_y \end{cases}.$$

Then by imposing the mirror symmetry $\mathscr{M}_z = i\sigma_z\tau_z$, which turns k_z into $-k_z$, we have the constraints that

$$\begin{cases} A^i/B^{ij}, B^{zz}/C: & (\sigma_z\sigma_\mu\sigma_z)(\tau_z\tau_\nu\tau_z) = \sigma_\mu\tau_\nu,\ i,j\in\{x,y\} \\ A^z/B^{zi}: & (\sigma_z\sigma_\mu\sigma_z)(\tau_z\tau_\nu\tau_z) = -\sigma_\mu\tau_\nu,\ i\in\{x,y\} \end{cases},$$

which gives that

$$\begin{cases} A^i/B^{ij}, B^{zz}/C: & \sigma_{0/z}\tau_{0/z},\ \sigma_{x/y}\tau_{x/y} \\ A^z/B^{zi}: & \sigma_{0/z}\tau_{x/y},\ \sigma_{x/y}\tau_{0/z} \end{cases},$$

which, when combined with the time-reversal symmetry, leads to the possible terms as

$$\begin{cases} A^i: & \sigma_{x/y}\tau_x,\ \sigma_z\tau_{0/z} \\ A^z: & \sigma_0\tau_y,\ \sigma_{x/y}\tau_{0/z},\ \sigma_z\tau_x \\ B^{ij}, B^{zz}/C: & \sigma_0\tau_{0/z},\ \sigma_{x/y}\tau_y \\ B^{zi}: & \sigma_0\tau_x,\ \sigma_z\tau_y \end{cases},\ i,j \in \{x,y\}.$$

Similar consideration with $M_x = i\sigma_x\tau_z$ symmetry with $k_x \to -k_x$ leads to the constraints that

$$\begin{cases} A^i/B^{ij}, B^{xx}/C: & \sigma_{0/x}\tau_{0/z},\ \sigma_{y/z}\tau_{x/y},\ i,j \in \{y,z\} \\ A^x/B^{xi}: & \sigma_{0/x}\tau_{x/y},\ \sigma_{y/z}\tau_{0/z},\ i \in \{y,z\} \end{cases},$$

and by taking the intersection, we finally obtain the possible terms as

$$\begin{cases} A^x: & \sigma_x\tau_x,\ \sigma_z\tau_{0/z} \\ A^y: & \sigma_y\tau_x \\ A^z: & \sigma_x\tau_{0/z},\ \sigma_z\tau_x \\ B^{xx/yy/zz}/C: & \sigma_0\tau_{0/z},\ \sigma_y\tau_y \\ B^{xy}, B^{yx}: & \sigma_x\tau_y \\ B^{xz}, B^{zx}: & \sigma_0\tau_x \\ B^{yz}, B^{zy}: & \sigma_z\tau_y \end{cases}.$$

The three-fold rotational symmetry similarly restricts, according to the above discussion, by eliminating $A^{x/y}_{z\nu}$ $A^z_{x/y,\nu}$, $B^{xy}_{0\nu}$ and constraining

$$\begin{cases} B^{xx}_{0\nu} = B^{yy}_{0\nu} \\ A^x_{x\nu} = A^y_{y\nu} \end{cases},$$

the terms. It turns out that there are still

$$\begin{cases} B^{xx/yy/zz}/C: & \sigma_y\tau_y \\ B^{xy}, B^{yx}: & \sigma_x\tau_y \\ B^{xz}, B^{zx}: & \sigma_0\tau_x \\ B^{yz}, B^{zy}: & \sigma_z\tau_y \end{cases}$$

terms that are not discussed above. Since for the first line with B^{zz}/C, single σ_y will be rotated to include σ_x and can not be compensated, while in the third and fourth line, $k_x(k_y)$ will be rotated to $k_y(k_x)$ and can not be compensated. If we ignore $B^{xx/yy}: \sigma_y\tau_y$ and $B^{xy}, B^{yx}: \sigma_x\tau_y$ terms first, we get

$$\begin{cases} A^x: \sigma_x\tau_x \\ A^y: \sigma_y\tau_x \\ A^z: \sigma_z\tau_x \\ B^{ii}/C: \sigma_0\tau_{0/z} \\ B^{xx}_{0,0/z} = B^{yy}_{0,0/z} \\ A^x_{xx} = A^y_{yx} \end{cases}, \tag{B.5}$$

which is the same as the above result. However, as derived below, one more term $\tilde{B}[(k_x^2 - k_y^2)\sigma_y + 2k_xk_y\sigma_x]\tau_y$ is also allowed, and we write

$$H(\boldsymbol{k}) = \varepsilon(\boldsymbol{k}) + \left(A\boldsymbol{k}_\parallel \cdot \boldsymbol{\sigma} + A_z k_z \sigma_z\right)\tau_x + \mathscr{M}(\boldsymbol{k})\tau_z + \tilde{B}[(k_x^2 - k_y^2)\sigma_y + 2k_xk_y\sigma_x]\tau_y, \quad \text{(B.6)}$$

where $\varepsilon(\boldsymbol{k}) = D_0 + D_1\boldsymbol{k}_\parallel^2 + D_2k_\perp^2$, $\mathscr{M}(\boldsymbol{k}) = C_0 + C_1\boldsymbol{k}_\parallel^2 + C_2k_\perp^2$.

B.2 Model of Surface States

Following the projection procedure in the main text, we can obtain the effective Hamiltonian for the surface states. Nevertheless, it can also be argued directly from the symmetry constraints. Now we take τ and σ to be the top-bottom and spin degrees of freedom, respectively, which does not change the time-reversal operator $\mathscr{T} = i\sigma_y\mathscr{K}$, but alters the inversion operator to $\mathscr{P} = \tau_x$, reading as the exchange of top and bottom layers. We let the surface states to inherit the three-fold rotational symmetry $C_{3z} = \exp(-i\sigma_z\pi/3)$ and mirror symmetry $\mathscr{M}_x = \mathscr{P}C_{2x} = i\sigma_x\tau_0$. (This comes from that, in TI bulk model, $\mathscr{M}_x = i\sigma_x\tau_z$, which will be projected into $i\sigma_y\tau_y$ under our basis, and by the rotations into surface state representation it becomes $i\sigma_x$.) Again, there are two chances as whether we adopt the inversion or $\mathscr{M}_z$ symmetries, but from the above discussion, it does not matter for the bulk, which should also be true for the surface states. And we adopt $\mathscr{M}_z = i\sigma_z\tau_x$ here.

Again, the most general Hamiltonian up to the second order in $\boldsymbol{k}$ reads

$$H(\boldsymbol{k}) = \sum_{\mu\nu=0,1,2,3}\sum_{ij=1,2}\left(A^i_{\mu\nu}k_i + B^{ij}_{\mu\nu}k_ik_j + C_{\mu\nu}\right)\sigma_\mu\tau_\nu, \quad \text{(B.7)}$$

which is greatly simplified by throwing k_z. The time-reversal symmetry, as usual, leads to that

$$\begin{cases} A: & \sigma_0\tau_y \text{ or } \sigma_{x/y/z}\tau_{0/x/z} \\ B/C: & \sigma_0\tau_{0/x/z} \text{ or } \sigma_{x/y/z}\tau_y \end{cases}.$$

And by considering $\mathscr{M}_z$ which does not change the wavevector, we have the constraints that

$$A/B/C : (\sigma_z\sigma_\mu\sigma_z)(\tau_x\tau_\nu\tau_x) = \sigma_\mu\tau_\nu \Rightarrow \sigma_{0/z}\tau_{0/x},\ \sigma_{x/y}\tau_{y/z},$$

combined with the time-reversal symmetry, we have the possible terms as

$$\begin{cases} A: & \sigma_z\tau_{0/x},\ \sigma_{x/y}\tau_z \\ B/C: & \sigma_0\tau_{0/x},\ \sigma_{x/y}\tau_y \end{cases}.$$

The $\mathscr{M}_x$ symmetry makes $k_x \to -k_x$ and satisfies that

$$\begin{cases} A^x/B^{xy}: & \sigma_{y/z}\tau_\nu \\ A^y/B^{ii}/C: & \sigma_{0/x}\tau_\nu \end{cases},$$

which further leads to

$$\begin{cases} A^x: & \sigma_y\tau_z,\ \sigma_z\tau_{0/x} \\ A^y: & \sigma_x\tau_z \\ B^{ii}/C: & \sigma_0\tau_{0/x},\ \sigma_x\tau_y \\ B^{xy}: & \sigma_y\tau_y \end{cases}.$$

The action of C_{3z} symmetry constrains that

$$\begin{cases} A^x_{x\nu} = A^y_{y\nu} \\ A^y_{x\nu} = -A^x_{y\nu} \\ B^{xx}_{0\nu} = B^{yy}_{0\nu} \\ B^{xy}_{0\nu} = -B^{yx}_{0\nu} = 0 \end{cases}.$$

To find the constraint imposed over $B : \sigma_{x/y}\tau_y$ by C_{3z} symmetry, we have to exploit smarter idea from the invariant theory. In the subspace, the constraint equation can be written as

$$B^{ij}_\alpha (R_{3z}\boldsymbol{k})_i (R_{3z}\boldsymbol{k})_j (C_{3z}\sigma^\alpha C^{-1}_{3z}) = B^{ij}_\alpha k_i k_j \sigma^\alpha,$$

with $\alpha, i, j \in \{x, y\}$ identified. Utilizing the self-adjoint representation of $SU(2)$, we have $C_{3z}\sigma^\alpha C^{-1}_{3z} = (R_{3z}\boldsymbol{\sigma})^\alpha$, and the equation reads

$$\boldsymbol{k}^T R^T_{3z} B_\alpha R_{3z} \boldsymbol{k} R^{\alpha\beta}_{3z} \sigma^\beta = \boldsymbol{k}^T B_\alpha \boldsymbol{k} \sigma^\alpha,$$

requiring its correctness over basis σ, k leads to the constraint that

$$R^T_{3z}(R^T_{3z}\boldsymbol{B})R_{3z} = \boldsymbol{B}, \tag{B.8}$$

or in components

$$(R^T_{3z})_{pi} B^{ij}_\beta (R_{3z})_{\beta\alpha} (R_{3z})_{jq} = B^{ij}_\alpha.$$

The $B_{x/y}$ matrix should be symmetric since the antisymmetric part is cancelled with the symmetric index $k_x k_y$. Then by the real-symmetric nature of $B_{x/y}$, we consider three the trace conditions over $\sigma_{0,x,z}$, firstly the contraction with σ_0

$$\mathrm{Tr}\big(R^T_{3z}(R^T_{3z}\boldsymbol{B})_\alpha R_{3z}\big) = \mathrm{Tr}\big((R^T_{3z}\boldsymbol{B})_\alpha\big) = \mathrm{Tr}(B_x(R_{3z})_{x\alpha} + B_y(R_{3z})_{y\alpha}),$$

which, if we equal it to $\mathrm{Tr}(B_\alpha)$, becomes by $R_{3z} = \frac{1}{2}\begin{pmatrix} -1 & \sqrt{3} \\ -\sqrt{3} & -1 \end{pmatrix}$

$$\begin{cases} \mathrm{Tr}\Big(-\frac{1}{2}B_x - \frac{\sqrt{3}}{2}B_y\Big) = \mathrm{Tr}(B_x) \\ \mathrm{Tr}\Big(\frac{\sqrt{3}}{2}B_x - \frac{1}{2}B_y\Big) = \mathrm{Tr}(B_y) \end{cases} \Rightarrow \begin{cases} \mathrm{Tr}(B_y) = -\sqrt{3}\,\mathrm{Tr}(B_x) \\ \mathrm{Tr}(B_x) = \sqrt{3}\,\mathrm{Tr}(B_y) \end{cases},$$

and only gives zero trace results. Then we consider the contraction with σ_z, which leads to the equation that

$$\mathrm{Tr}\big[(R_{3z}\sigma_z R^T_{3z})(B_x(R_{3z})_{x\alpha} + B_y(R_{3z})_{y\alpha})\big] = \mathrm{Tr}(B_\alpha\sigma_z),$$

and gives the equation that

$$\begin{cases} -\frac{1}{2}(\sqrt{3}B^{12}_x - B^{11}_x) - \frac{\sqrt{3}}{2}(-B^{11}_y + \sqrt{3}B^{12}_y) = 2B^{11}_x \\ \frac{\sqrt{3}}{2}(\sqrt{3}B^{12}_x - B^{11}_x) - \frac{1}{2}(-B^{11}_y + \sqrt{3}B^{12}_y) = 2B^{11}_y \end{cases},$$

where we have used $R_{3z}\sigma_z R^T_{3z} = \frac{1}{2}\begin{pmatrix} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$, $B^{11} = -B^{22}, B^{12} = B^{21}$. And similarly for σ_x, we have the equation that

$$\mathrm{Tr}\big[(R_{3z}\sigma_x R^T_{3z})(B_x(R_{3z})_{x\alpha} + B_y(R_{3z})_{y\alpha})\big] = \mathrm{Tr}(B_\alpha\sigma_x),$$

and gives the equation that

$$\begin{cases} -\frac{1}{2}(-\sqrt{3}B_x^{11} - B_x^{12}) - \frac{\sqrt{3}}{2}(-\sqrt{3}B_y^{11} - B_y^{12}) = 2B_x^{12} \\ \frac{\sqrt{3}}{2}(-\sqrt{3}B_x^{11} - B_x^{12}) - \frac{1}{2}(-\sqrt{3}B_y^{11} - B_y^{12}) = 2B_y^{12} \end{cases}.$$

The combination of the above two equations leads to the constraints that $\mathbf{P} \cdot \tilde{\boldsymbol{B}} = 0$, $\tilde{\boldsymbol{B}} = (B_x^{11}, B_y^{11}, B_x^{12}, B_y^{12})^T$, and

$$\mathbf{P} = \begin{pmatrix} -\sqrt{3} & 1 & -1 & -\sqrt{3} \\ -1 & -\sqrt{3} & \sqrt{3} & -1 \\ 1 & \sqrt{3} & -\sqrt{3} & 1 \\ -\sqrt{3} & 1 & -1 & -\sqrt{3} \end{pmatrix},$$

which is a rank-2 matrix, thus allowing non-trivial solutions and leads to

$$\begin{cases} B_x^{11} = -B_y^{12} \\ B_y^{11} = B_x^{12} \end{cases},$$

which leads to two terms $B_x[(k_x^2 - k_y^2)\sigma_x - 2k_xk_y\sigma_y]\tau_y$ and $B_y[(k_x^2 - k_y^2)\sigma_y + 2k_xk_y\sigma_x]\tau_y$, with the latter excluded by $\mathscr{M}_x$ symmetry. The former term will also be forbidden if we require inversion and C_{2x} symmetries individually.

The surface state Hamiltonian is

$$H_S = C_0 + C_1k^2 + A\tau_z(\boldsymbol{k} \times \boldsymbol{\sigma})_z + (B_0 + B_1k^2)\tau_x + \tilde{B}[(k_x^2 - k_y^2)\sigma_x - 2k_xk_y\sigma_y]\tau_y. \quad \text{(B.9)}$$

Appendix C

Hall Conductance

Hall effect is a classical electro-magnetic effect as follows: say apply I_x along longitudinal and B_z along perpendicular direction, then one is able to measure the transverse voltage V_y to obtain the Hall effect (and longitudinal voltage V_x for magnetoresistance). This result is just some linear response, and its physical core is balance of Lorentz force,

$$0 = \boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) \Longrightarrow E_y = v_x B_z,$$

then for transverse conductance

$$\sigma_{xy} = \frac{j_x}{E_y} = \frac{nqv_x}{E_y} = \frac{nq}{B_z},$$

which varies linear to $1/B_z$ and changes sign for electrons and holes. However, from 1980s, people find this quantity quantized over a universal unit (for electrons)

$$\sigma_{xy} = \nu \frac{e^2}{h},\ \nu = 1, 2, \cdots, \tag{C.1}$$

where $h = 2\pi\hbar$. And this effect is known as Integer Quantum Hall Effect (IQHE), and the corresponding (insulating) system holding the phenomenon is called Chern insulator. By insulating we in fact mean that longitudinal conductance vanishes as $\rho_{xx} = 0$, and using the conductance matrix

$$\begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix}^{-1},$$

one immediately finds that

$$\rho_{xx} = \frac{\sigma_{xx}}{\sigma_{xx}^2 + \sigma_{xy}^2} = 0 = \rho_{yy},\ \rho_{xy} = -\rho_{yx} = -\frac{\sigma_{xy}}{\sigma_{xx}^2 + \sigma_{xy}^2} = -\frac{e^2}{\nu h}.$$

(*Streda Formula*) With differential form, we conclude easily that

$$\sigma_{xy} = q\frac{\delta n}{\delta B}, \tag{C.2}$$

which, however, needs further examination. Say consider definition $j_x = \sigma_{xy} E_y$, where we have set zone to insulating phase so that longitudinal conductance equals zero, then to

relate this formula with charge density and magnetic field, one notices that by continuity equation and gauge potential form of electro-magnetic field,

$$\begin{cases} \partial_t \rho + \partial_x j_x = 0 \\ E_y = -\partial_t A_y \\ B_z = (\nabla \times \boldsymbol{A})_z = \partial_x A_y \end{cases},$$

one obtains

$$\partial_t \rho = -\partial_x j_x = \sigma_{xy} \partial_x \partial_t A_y = \sigma_{xy} \partial_t B_z,$$

i.e., Streda formula we write above.

C.1 Quantization from Landau Levels

This phenomenon could firstly be explained by Landau levels of the electron, where one finds the key is to figure out the electron density n for each Landau level (LL) occupied, if one believes that each fully filled LL could contribute to the conductance in some way — extended states and edge states are all ok. One could try to view from symmetric gauge where one finds the wavefunction

$$\psi_{m,0}(x,y) \propto \exp\left(-\frac{z\bar{z}}{4l_B^2}\right) = \exp\left(-\frac{x^2+y^2}{4l_B^2}\right),\ z = x + iy,$$

while the second quantum number is lifted by $b^\dagger = -i\sqrt{2}(l_B \partial_z - \bar{z}/4l_B)$ as

$$\psi_{m,n}(x,y) = (b^\dagger)^n \psi_{m,0}(x,y) \propto \left(\frac{z}{l_B}\right)^n \exp\left(-\frac{z\bar{z}}{4l_B^2}\right);$$

then by angular momentum operator

$$J = J_z = (r \times p)_z = -i\hbar \left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) = \hbar(z\partial_z - \bar{z}\partial_{\bar{z}}),$$

one finds that the wavefunction under symmetric gauge is eigenstate of angular momentum

$$J\psi_{m,n}(x,y) = m\hbar\psi_{m,n}(x,y).$$

Then one may say that each state may occupy an area with radius $r_n = \sqrt{2m} l_B$, where the seemingly extra 2 factor comes from choice of physical momentum, say now roughly speaking,

$$rp = r(mv - qA) = m\hbar,$$

and using $qvB = mv^2/r$ to have $mv = qrB$, substituting which and integrating over the circle (unit one) one has

$$2q\pi r^2 B - q(2\pi r A) = 2\pi m\hbar \Longrightarrow 2q\Phi - q\Phi = q\Phi = mh,$$

which is just semi-classical angular momentum quantization. Now,

$$r_m = \sqrt{\frac{\Phi}{\pi B}} = \sqrt{\frac{mh}{q\pi B}} = \sqrt{2m} l_B.$$

On the contrary, if one just use mv as momentum for quantization, one would find the factor 2 lost. Now given an area A with radius R, density of state, i.e., number of state it could hold per unit area (when $r_m = R$) averagely is

$$\Omega = \frac{m}{A}\bigg|_{r_n=R} = \frac{1}{A}\frac{R^2}{2l_B^2} = \frac{1}{2\pi l_B^2} = \frac{q}{Bh}. \tag{C.3}$$

And one yields

$$\sigma_{xy} = \frac{\Omega q}{B} = \frac{q^2}{h},$$

as one quantum unit conductance. Notice that we did not consider inner DoFs for more degeneracy and this is only for one Landau level, considering all would give

$$\sigma_{xy} = gN\frac{q^2}{h},\ N = 1, 2, \cdots, \tag{C.4}$$

where g describes inner DoFs like spin and N counts for occupied LLs.

On the other hand, one could obtain exactly the same degeneracy for each LL using Landau gauge. Clearly, the degeneracy now comes from varying of center of one state, then

$$\Omega = \frac{1}{S}\frac{L_y}{\Delta y} = \frac{1}{L_xL_y}\frac{L_y}{l_B^2\Delta k_x} = \frac{1}{L_xL_y}\frac{L_yqB}{\hbar 2\pi/L_x} = \frac{qB}{h},$$

where a rectangular area is exploited and periodic boundary condition (PBC) is used which forces that

$$k_x \cdot L_x = 2\pi \Longrightarrow k_x = n\frac{2\pi}{L_x},$$

respecting the periodicity of plane wave.

C.2 Quantization from Berry Terminology

A quick conclusion is, using wave-packet dynamics,

$$\dot{\boldsymbol{r}}_c = \boldsymbol{v}_n(\boldsymbol{k}_c) - \dot{\boldsymbol{k}}_c \times \boldsymbol{\Omega}_n(\boldsymbol{k}_c), \tag{C.5}$$

where c for centre and n for band index, and Berry curvature, mathematically equivalent to Gaussian curvature, is generally defined as a 2nd-order tensor

$$\Omega_n^{\mu\nu}(k) \equiv \partial^\mu A_n^\nu(k) - \partial^\nu A_n^\mu(k), \tag{C.6}$$

from which one immediately yields A^μ and $F^{\mu\nu}$ are just $U(1)$ vector field and electromagnetic tensor, but in k-space. Then, by considering pure-magnetic part, we could transform 3×3 anti-symmetric tensor into vector magnetic field, and transform

$$\boldsymbol{k} \cdot \Omega \to \boldsymbol{k} \times \boldsymbol{\Omega}$$

correspondingly.

Then, current is defined through

$$\boldsymbol{J} = -e\sum \dot{\boldsymbol{r}}_c = -e\sum_{n\boldsymbol{k}} \left(\boldsymbol{v}_n(\boldsymbol{k}_c) - \dot{\boldsymbol{k}}_c \times \boldsymbol{\Omega}_n(\boldsymbol{k}_c)\right), \tag{C.7}$$

and for Hall conductance given to an 2-D insulator, we have

$$\begin{aligned} J_y &= -e\sum_{n\boldsymbol{k}}(\dot{\boldsymbol{k}}_c)_x\boldsymbol{\Omega}_n(\boldsymbol{k}_c)_z = -e\sum_{n\boldsymbol{k}}\dot{k}_x\Omega_n^{xy}(\boldsymbol{k}),\ \hbar\dot{k}_x = -eE_x \\ &= \frac{e^2}{\hbar}\sum_n \frac{S}{(2\pi)^2}\int_{\mathrm{BZ}}\mathrm{d}^2k\Omega_n^{xy}(k)E_x, \end{aligned}$$

where one needs to notice that summation over k is in Brillouin zone and S is area of 2-D system. Then one sees

$$\sigma_{yx} = \frac{j_y}{E_x} = \frac{J_y}{SE_x} = \frac{e^2}{h}c_1 = -\sigma_{xy}, \tag{C.8}$$

where

$$c_1 = \frac{1}{2\pi}\sum_n\int_{\mathrm{BZ}}\mathrm{d}^2k\Omega_n^{xy}(k) = \frac{1}{2\pi}\sum_n\int_{\mathrm{BZ}}\mathrm{d}^2k\Omega_n^{z}(k) \tag{C.9}$$

is known as first Chern number. Be aware that there seems to exist a minus sign difference with Chern number mathematically.

C.3 Kubo Formula

Given a perturbative Hamiltonian

$$H = H_0 + V,$$

we could express quantum dynamics in interaction picture as

$$V_I(t) = U_0^\dagger(t)V_SU_0(t),\ \ |\psi_I(t)\rangle = U_0^\dagger(t)\,|\psi_S(t)\rangle\,,\ U_0(t) = e^{-iH_0t/\hbar},$$

where S means Schrödinger picture. Essentially the LHS of second equation is equivalent to

$$|\psi_I(t)\rangle = U_I(t)\,|\psi_I(0)\rangle\,,$$

where

$$U_I(t) = \mathscr{T}\exp\left(-\frac{i}{\hbar}\int_0^t\mathrm{d}t'V_I(t')\right) \approx \mathrm{I} - \frac{i}{\hbar}\int_0^t\mathrm{d}t_1V_I(t_1)$$

up to first linear order. Now given arbitrary observable, its expectation value could be approximated up to linear order by

$$\begin{aligned} \langle O(t)\rangle &= \langle\psi_I(t)|O_I(t)|\psi_I(t)\rangle = \langle\psi_I(0)|\,U_I^{-1}(t)O_I(t)U_I(t)\,|\psi_I(0)\rangle \\ &\approx \langle\psi_I(0)|\left(\mathrm{I}+\frac{i}{\hbar}\int_0^t\mathrm{d}t_1V_I(t_1)\right)O_I(t)\left(\mathrm{I}-\frac{i}{\hbar}\int_0^t\mathrm{d}t_1V_I(t_1)\right)|\psi_I(0)\rangle \\ &\approx \langle\psi_I(0)|\,O_I(t) + \frac{i}{\hbar}\int_0^t\mathrm{d}t'[V_I(t'),O_I(t)]\,|\psi_I(0)\rangle\,. \end{aligned}$$

Then, if we push initial time 0 to t_0 where we recognize $|\psi_I(0)\rangle$ as eigenstate of H_0 in equilibrium, and replace single-state average with statistical equilibrium average, we get Kubo formula as linear response

$$\langle O(t)\rangle = \langle O_I(t)\rangle_0 + \frac{i}{\hbar}\int_{t_0}^t\mathrm{d}t'\,\langle[V_I(t'),O_I(t)]\rangle_0\,. \tag{C.10}$$

Now we come back to our conductance. From single-particle Lagrangian term $q\dot{x}\cdot A$ which generates basically Lorentz force with vector potential, we see that when written in Hamiltonian, its generalized form as current-potential interaction would become

$$H = H_0 - J\cdot A,\ H_0\left|n\right\rangle = E_n\left|n\right\rangle. \tag{C.11}$$

And we want its linear response from Kubo formula. Consider an electrical field added, by letting $A\propto e^{-i\omega t}$ as an alternating current (AC) potential, we get a AC field by

$$E = -\partial_t A = i\omega A \Rightarrow A = \frac{E}{i\omega}e^{-i\omega t}.$$

Later on, we will let ω go to zero so the normal direct current (DC) in experiment is restored. This may sound tedious but could save us from infrared divergence problem. Now by pushing $t_0\to-\infty$ so that A is viewed as added adiabatically, and also use single state average as an example which we would restore afterwards, we have, for the response part (i.e. V included part),

$$\begin{aligned}\left\langle J_\alpha(t)\right\rangle &= \frac{i}{\hbar}\int_{-\infty}^{t}\mathrm{d}t'\left\langle[-J_I(t')\cdot A(t'), J_{I,\alpha}(t)]\right\rangle_0\\ &= -\frac{i}{\hbar}\int_{-\infty}^{t}\mathrm{d}t'\left\langle 0\right|\left[J_{I,\beta}(t'), J_{I,\alpha}(t)\right]\left|0\right\rangle\frac{E_\beta}{i\omega}e^{-i\omega t'},\ t''=t-t'\\ &= -\frac{1}{\hbar\omega}\left(\int_0^{+\infty}\mathrm{d}t'' e^{i\omega t''}\left\langle 0\right|\left[J_{I,\beta}(t-t''), J_{I,\alpha}(t)\right]\left|0\right\rangle\right)E_\beta e^{-i\omega t}.\end{aligned}$$

Now, the averaged part is basically some correlation which attach time translational invariance to so we could minus $t-t''$ for each part; then, t-dependent part is just embedded in the last AC oscillating part sharing the same frequency with external A, which reflects linear response. Since J is current, we have current density defined as it divided by general volume, then we have frequency-dependent conductance in 2-D case:

$$\begin{aligned}\sigma_{\alpha\beta}(\omega) \equiv \frac{\left\langle j_\alpha(t)\right\rangle}{E_\beta e^{-i\omega t}} &= \frac{1}{S}\frac{\left\langle J_\alpha(t)\right\rangle}{E_\beta e^{-i\omega t}}\\ &= -\frac{1}{S\hbar\omega}\int_0^{+\infty}\mathrm{d}t e^{i\omega t}\left\langle 0\right|\left[J_{I,\beta}(0), J_{I,\alpha}(t)\right]\left|0\right\rangle\\ &= -\frac{S}{\hbar\omega}\int_0^{+\infty}\mathrm{d}t e^{i\omega t}\left\langle 0\right|j_\beta e^{iH_0t}j_\alpha e^{-iH_0t} - e^{iH_0t}j_\alpha e^{-iH_0t}j_\beta\left|0\right\rangle\\ &= -\frac{S}{\hbar\omega}\sum_{n\neq 0}\int_0^{+\infty}\mathrm{d}t e^{i\omega t}\left[\left\langle 0|j_\beta|n\right\rangle\left\langle n|j_\alpha|0\right\rangle e^{i(E_n-E_0)t/\hbar} - \left\langle 0|j_\alpha|n\right\rangle\left\langle n|j_\beta|0\right\rangle e^{i(E_0-E_n)t/\hbar}\right],\end{aligned}$$

where we have again replaced current with density and inserted identity, $n\neq 0$ is attributed to general ground state property as with no net current. Then by doing integral upon t where retarded nature should be introduced as $\omega\to\omega+i\epsilon$ for convergence, as

$$\begin{aligned}\int_0^{+\infty}\mathrm{d}t e^{i(\omega+i\epsilon)t}e^{\pm i(E_n-E_0)t/\hbar} &= \frac{1}{i(\omega+i\epsilon)\pm i(E_n-E_0)/\hbar}e^{-\epsilon t+\cdots}\Big|_0^{+\infty}\\ &= \frac{i\hbar}{\hbar\omega\pm(E_n-E_0)},\end{aligned}$$

we have

$$\sigma_{\alpha\beta}(\omega) = -\frac{iS}{\omega}\sum_{n\neq 0}\left[\frac{\langle 0|j_\beta|n\rangle\langle n|j_\alpha|0\rangle}{\hbar\omega + (E_n - E_0)} - \frac{\langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle}{\hbar\omega - (E_n - E_0)}\right],$$

then by letting ω goes to zero, we see

$$\begin{aligned}\lim_{\omega\to 0}\frac{1}{\omega}\frac{1}{\hbar\omega \pm (E_n - E_0)} &= \pm\frac{1}{E_n - E_0}\lim_{\omega\to 0}\frac{1}{\omega}\frac{1}{1 \pm \hbar\omega/(E_n - E_0)} \\ &= \pm\frac{1}{E_n - E_0}\frac{1}{\omega}\left(1 \mp \frac{\hbar\omega}{E_n - E_0} + O(\omega^2)\right) \\ &\overset{\text{P.V.}}{=} -\frac{\hbar}{(E_n - E_0)^2},\end{aligned}$$

where higher order together with infrared divergent terms are abandoned. Indeed, by noticing that $\operatorname{Im}\langle 0|j_\beta|n\rangle\langle n|j_\alpha|0\rangle = -\operatorname{Im}\langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle$ and requiring $\sigma_{\alpha\beta}$ to be real, we see that the infrared divergence is exactly canceled out. Now we get

$$\begin{aligned}\sigma_{\alpha\beta} &= iS\hbar\sum_{n\neq 0}\left[\frac{\langle 0|j_\beta|n\rangle\langle n|j_\alpha|0\rangle - \langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle}{(E_n - E_0)^2}\right] \\ &= 2S\hbar\sum_{n\neq 0}\frac{\operatorname{Im}(\langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle)}{(E_n - E_0)^2}\end{aligned}$$

Generically, we have multi-states occupied as total ground state to replace single $|0\rangle$, then it is easy to write

$$\sigma_{\alpha\beta} = -iS\hbar\sum_{E_m<\mu<E_n}\frac{\langle m|j_\alpha|n\rangle\langle n|j_\beta|m\rangle - \langle m|j_\beta|n\rangle\langle n|j_\alpha|m\rangle}{(E_n - E_m)^2},$$

where we have introduced chemical potential μ. And when temperature enters, the relation $E_m < \mu < E_n$ should be replaced by general Fermi-Dirac distribution counting as

$$\sigma_{\alpha\beta} = -iS\hbar\sum_{m,n}[f_z(E_m) - f_z(E_n)]\frac{\langle m|j_\alpha|n\rangle\langle n|j_\beta|m\rangle}{(E_n - E_m)^2},$$

where

$$f_z(\varepsilon) = \frac{1}{z^{-1}e^{\beta\varepsilon} + 1},\ z = e^{\beta\mu}.$$

Notice it that this formula is still upon single-particle response, as

$$J = -ev,\ e > 0,$$

with canonical velocity

$$v = \dot{x} = \frac{1}{i\hbar}[x, H] = \frac{1}{\hbar}\nabla_k H,$$

which leads further to single particle response

$$\sigma_{\alpha\beta} = -i\frac{e^2}{S\hbar}\sum_{E_m<\mu<E_n}\frac{\langle m|\partial_{k_\alpha}H|n\rangle\langle n|\partial_{k_\beta}H|m\rangle - \langle m|\partial_{k_\beta}H|n\rangle\langle n|\partial_{k_\alpha}H|m\rangle}{(E_n - E_m)^2},$$

where we have gone back to zero-temperature to see topological feature for ground state.

To account for many-body state, one needs to sum over contribution from all particles, which is equivalent to summing over k thanks to its good-quantum-number property as a (quasi-)particle label. This leads to

$$\begin{aligned}\sigma_{\alpha\beta} &= -i\frac{e^2}{S\hbar}\sum_{E_m<\mu<E_n}\sum_{k}\frac{\langle m|\partial_{k_\alpha}H|n\rangle\langle n|\partial_{k_\beta}H|m\rangle - \langle m|\partial_{k_\beta}H|n\rangle\langle n|\partial_{k_\alpha}H|m\rangle}{(E_n-E_m)^2}\\ &= -i\frac{e^2}{S\hbar}\sum_{E_m<\mu<E_n}\frac{S}{(2\pi)^2}\int \mathrm{d}^2k\frac{\langle m|\partial_{k_\alpha}H|n\rangle\langle n|\partial_{k_\beta}H|m\rangle - \langle m|\partial_{k_\beta}H|n\rangle\langle n|\partial_{k_\alpha}H|m\rangle}{(E_n-E_m)^2}\\ &= \frac{e^2}{h}\left[-\frac{i}{2\pi}\sum_{E_m<\mu<E_n}\int \mathrm{d}^2k\frac{\langle m|\partial_{k_\alpha}H|n\rangle\langle n|\partial_{k_\beta}H|m\rangle - \langle m|\partial_{k_\beta}H|n\rangle\langle n|\partial_{k_\alpha}H|m\rangle}{(E_n-E_m)^2}\right].\end{aligned}$$

Now we want to show that this formula is equivalent to that we give above using Berry terminology, and to achieve which we need to utilize an equation parallel to Feynman-Hellman theorem, which we have also used when deriving linear response before. By taking derivative upon eigen-equation,

$$(\partial_k H)\,|n\rangle + H\,|\partial_k n\rangle = (\partial_k E_n)\,|n\rangle + E_n\,|\partial_k n\rangle\,,$$

then by left-multiplying $\langle n|$, we get F-H theorem, but if we multiply $\langle m\neq n|$ similar to the case when we do series expansion perturbation, we have

$$\langle m|\,\partial_k H\,|n\rangle + E_m\langle m|\partial_k n\rangle = 0 + E_n\langle m|\partial_k n\rangle\,,$$

which leads to

$$\langle m|\,\partial_k H\,|n\rangle = (E_n-E_m)\langle m|\partial_k n\rangle\,,$$

bring which back to conductance would leads to

$$\sigma_{\alpha\beta} = \frac{e^2}{h}\left[\frac{i}{2\pi}\int \mathrm{d}^2k\sum_{E_m<\mu<E_n}\big(\langle m|\partial_{k_\alpha}n\rangle\langle n|\partial_{k_\beta}m\rangle - \langle m|\partial_{k_\beta}n\rangle\langle n|\partial_{k_\alpha}m\rangle\big)\right].$$

In fact, RHS could already be viewed as some special summation form with non-Abelian Berry curvature, which, however, is much tedious for this simple case. By noticing that $\langle n|m\rangle = 0$ and re-arranging partial-derivative, we obtain

$$\sigma_{\alpha\beta} = \frac{e^2}{h}\sum_{m}^{\text{occ}}\left[-\frac{i}{2\pi}\int \mathrm{d}^2k\sum_{n}^{\text{un-o}}\big(\langle\partial_{k_\alpha}m|n\rangle\langle n|\partial_{k_\beta}m\rangle - \langle\partial_{k_\beta}m|n\rangle\langle n|\partial_{k_\alpha}m\rangle\big)\right],$$

then by introducing projection operator on occupied states as

$$P_{\text{occ}} = \sum_{m}^{\text{occ}}|m\rangle\langle m|\,,$$

we have

$$\begin{aligned}&\sum_{n}^{\text{un-o}}\big(\langle\partial_{k_\alpha}m|n\rangle\langle n|\partial_{k_\beta}m\rangle - \langle\partial_{k_\beta}m|n\rangle\langle n|\partial_{k_\alpha}m\rangle\big)\\ &\quad= \langle\partial_{k_\alpha}m|\,(\mathbb{I}-P_{\text{occ}})\,|\partial_{k_\beta}m\rangle - \langle\partial_{k_\beta}m|\,(\mathbb{I}-P_{\text{occ}})\,|\partial_{k_\alpha}m\rangle\end{aligned}$$

$$= \langle \partial_{k_\alpha} m | \partial_{k_\beta} m \rangle - \langle \partial_{k_\beta} m | \partial_{k_\alpha} m \rangle - \sum_{m'}^{\text{occ}} \left(\langle \partial_{k_\alpha} m | m' \rangle \langle m' | \partial_{k_\beta} m \rangle - \langle \partial_{k_\beta} m | m' \rangle \langle m' | \partial_{k_\alpha} m \rangle \right)$$
$$= \langle \partial_{k_\alpha} m | \partial_{k_\beta} m \rangle - \langle \partial_{k_\beta} m | \partial_{k_\alpha} m \rangle = \Omega_m^{\alpha\beta}/i,$$

where the last summation disappears because $\langle \partial_{k_\alpha} m | m' \rangle$ is purely imaginary like shown for Berry phase. And finally, we come back to the so-called Thouless-Kohmoto-Nightingale-den Nijs (TKNN) formula:

$$\sigma_{\alpha\beta} = -\frac{e^2}{h} \sum_n^{\text{occ}} \frac{1}{2\pi} \int \mathrm{d}^2 k \Omega_n^{\alpha\beta}. \tag{C.12}$$

And especially in 3-D (or lower dimension), we could map the 2-tensor $\Omega^{\alpha\beta}$ to a vector $\boldsymbol{\Omega}$, viewed as k-space magnetic field by

$$\boldsymbol{\Omega} = \nabla_k \times \boldsymbol{A}, \ A_n^\alpha = i \langle n | \partial_{k_\alpha} n \rangle . \tag{C.13}$$

C.3.1 Two-Band System

General two-band system as discussed above could be written as

$$H = \boldsymbol{d} \cdot \boldsymbol{\sigma}, \ \hat{H} = \hat{\boldsymbol{d}} \cdot \boldsymbol{\sigma},$$

where we have also introduced reduced Hamiltonian with $\hat{\boldsymbol{d}} \equiv \boldsymbol{d}/d$, where $d = |\boldsymbol{d}|$ as absolute eigenvalue of original Hamiltonian, whose eigenvalue now is ± 1 (we tend to not include the tricky 0 energy as gapless case, which might be considered using limit). Anyway, in the continuum model case when

$$\boldsymbol{d} = (k_x, k_y, m(\boldsymbol{k})),$$

it could be shown after somehow tedious calculation that

$$\Omega_z = i(\langle \partial_{k_x} u_- | \partial_{k_y} u_- \rangle - \text{c.c}) = -\frac{m(k)}{(k^2 + m^2(k))^{3/2}},$$

then admittedly that $m(0) \neq 0$, and it is at least 2nd order of k so that dominates when $k \to \infty$, one is advised to be convinced that

$$C = \frac{1}{2\pi} \int \mathrm{d}^2 k \Omega_z(k) = \frac{1}{2}[\text{sgn}(m(k \to \infty)) - \text{sgn}(m(k = 0))],$$

or when written in some standard regulated form $m(k) = m_0 - Bk^2$, we have

$$C = -\frac{\text{sgn}(B) + \text{sgn}(m_0)}{2} = -\text{sgn}(B)\Theta(m_0 B),$$

which is requiring a kind soliton solution in 1-D to achieve non-trivial property.

On the other hand, we may offer a formula with more geometric meaning. Notice that for our two band system,

$$\sigma_{xy} = -\frac{e^2}{h} \frac{1}{\pi} \int_{\text{BZ}} \mathrm{d}^2 k \frac{\text{Im}(\langle + | \partial_{k_x} H | - \rangle \langle - | \partial_{k_y} H | + \rangle)}{(E_+ - E_-)^2},$$

with $H|\pm\rangle = E_\pm|\pm\rangle$. Then if we continuously deform $H = d\cdot\sigma$ to a unit form $\hat{H} = \hat{d}\cdot\sigma$ as shown above, topological property should be conserved, while giving convenience that

$$\hat{H}|\pm\rangle = \pm|\pm\rangle,$$

where $|\pm\rangle$ would also be modified accordingly (but due to normalization there should actually no difference). This procedure provides that

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{4\pi}\int_{\mathrm{BZ}} \mathrm{d}^2k\,\mathrm{Im}(\langle+|\partial_{k_x}\hat{H}|-\rangle\langle-|\partial_{k_y}\hat{H}|+\rangle), \tag{C.14}$$

which could be further modified since

$$\langle+|\partial_{k_x}\hat{H}|-\rangle = \partial_{k_x}\hat{d}_i\langle+|\sigma_i|-\rangle,$$

and

$$\langle+|\partial_{k_x}\hat{H}|-\rangle\langle-|\partial_{k_y}\hat{H}|+\rangle = \partial_{k_x}\hat{d}_i\partial_{k_y}\hat{d}_j\langle+|\sigma_i|-\rangle\langle-|\sigma_j|+\rangle.$$

Now, using complete relation (or from point view of projection operator) by $P_- = \mathbb{I} - P_+$, we have

$$\begin{aligned}\langle+|\sigma_i|-\rangle\langle-|\sigma_j|+\rangle &= \langle+|\sigma_i\sigma_j|+\rangle - \langle+|\sigma_i|+\rangle\langle+|\sigma_j|+\rangle \\ &= i\epsilon_{ijk}\langle+|\sigma_k|+\rangle + (\langle+|\delta_{ij}|+\rangle - \langle+|\sigma_i|+\rangle\langle+|\sigma_j|+\rangle),\end{aligned}$$

where the second term does not contribute by operation of Im( ) since σ_i as Pauli matrix is Hermitian thus owns real expectation value; and the first term could be further modified using eigenfunction with $\hat{E}_+ = +1$ as

$$\begin{aligned}\langle+|\sigma_k|+\rangle &= \langle+|\sigma_k(\hat{d}\cdot\sigma)|+\rangle = \hat{d}_m\langle+|(\delta_{km} + i\epsilon_{kmn}\sigma_n)|+\rangle \\ &= \hat{d}_k + i\epsilon_{kmn}\hat{d}_m\langle+|\sigma_n|+\rangle,\end{aligned}$$

and the second term again does not contribute since we have extra factor $i\epsilon_{ijk}$ ahead and Im( ) operation. Then totally we find

$$\mathrm{Im}(\langle+|\partial_{k_x}\hat{H}|-\rangle\langle-|\partial_{k_y}\hat{H}|+\rangle) = \epsilon_{ijk}\partial_{k_x}\hat{d}_i\partial_{k_y}\hat{d}_j\hat{d}_k,$$

which gives us final result

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{4\pi}\int_{\mathrm{BZ}}\mathrm{d}^2k(\partial_{k_x}\hat{\boldsymbol{d}}\times\partial_{k_y}\hat{\boldsymbol{d}})\cdot\hat{\boldsymbol{d}} \tag{C.15}$$

written in a compact form. Equivalently, we can write

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{4\pi}\int_{\mathrm{BZ}}\mathrm{d}^2k\frac{(\partial_{k_x}\boldsymbol{d}\times\partial_{k_y}\boldsymbol{d})\cdot\boldsymbol{d}}{d^3}, \tag{C.16}$$

by direct expansion of $\hat{\boldsymbol{d}}$, since $(\cdots\times\boldsymbol{d})\cdot\boldsymbol{d}$ terms vanish. Notice that the integral form just serves as solid angle, whose geometric meaning is most straightforward when considering integral on sphere. This offers with some 'winding' like formula, whose 1-D partner is written as

$$w_\alpha = \frac{1}{2\pi}\int_0^{2\pi}\mathrm{d}k[(\partial_k\hat{\boldsymbol{d}})\times\hat{\boldsymbol{d}}]_\alpha, \tag{C.17}$$

as usual winding number for 1-D system (there seems to be several winding numbers but only when calculated with chiral symmetry makes it invariant and robust), and may be generalized into arbitrary dimension

$$I = \frac{1}{S_n}\int \mathrm{d}^n k \epsilon_{\alpha_1\alpha_2\cdots\alpha_{n+1}} (\partial_{k_1}\hat{d}_{\alpha_1})(\partial_{k_2}\hat{d}_{\alpha_2})\cdots(\partial_{k_n}\hat{d}_{\alpha_n})\hat{d}_{\alpha_{n+1}},$$

with S_n area of S^n, super-sphere surface of dim-n (n-sphere). In fact, the n-th winding number is equivalent to n-th Chern number defined in $(2n+1)$D space-time, i.e., even spatial dimension. And what makes the definition of winding number possible instead of a more general Chern number lies in the special structure of Hamiltonian, constructed upon Clifford algebra, which effectively maps curvature integral on T^n into winding angle on S^n.

C.3.2 The Sum Rule

There is a famous sum rule in the context of Hall conductance, which we start from the general zero-temperature frequency-dependent conductance formula

$$\begin{aligned}\sigma_{\alpha\beta}(\omega) &= \frac{e^2}{h^2} i \sum_{E_m<\mu<E_n} \int \mathrm{d}^2 k \frac{1}{\omega}\left[\frac{\langle m|\partial_\alpha H|n\rangle\langle n|\partial_\beta H|m\rangle}{\hbar\omega + E_n - E_m} - \frac{\langle m|\partial_\beta H|n\rangle\langle n|\partial_\alpha H|m\rangle}{\hbar\omega - (E_n - E_m)}\right] \\ &= \frac{e^2}{h^2}(-i)\sum_{E_m<\mu<E_n}\int \mathrm{d}^2 k \frac{\omega_{nm}^2}{\omega}\left[\frac{\langle m|\partial_\alpha n\rangle\langle n|\partial_\beta m\rangle}{\hbar\omega + \omega_{nm}} - \frac{\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle}{\hbar\omega - \omega_{nm}}\right],\end{aligned}$$

where we have introduced $\omega_{nm} = E_n - E_m$, and utilize a derived formula $\langle m|\,\partial_k H\,|n\rangle = (E_n - E_m)\langle m|\partial_k n\rangle$. Now by considering the retarded nature of the virtual process, we let $\omega \to \omega + i\epsilon$ with $\epsilon > 0$ a small positive quantity counting for the finite life-time of the quasi-excitation, and then by Dirac identity $1/(x+i\epsilon) = \mathrm{P.V.}1/x - i\pi\delta(x)$, we take the dissipation part represented by the delta-function as virtual excitation as

$$\begin{aligned}\sigma_{\mathrm{dis},\alpha\beta}(\omega) &= \frac{e^2}{h^2}\pi \sum_{E_m<\mu<E_n}\int \mathrm{d}^2 k \frac{\omega_{nm}^2}{\omega}\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle\,\delta(\hbar\omega - \omega_{nm}) \\ &= \frac{e^2}{2h}\sum_{E_m<\mu<E_n}\int \mathrm{d}^2 k \omega_{nm}\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle\,\delta(\hbar\omega - \omega_{nm}),\end{aligned}$$

where we have abandoned a $\delta(\hbar\omega + \omega_{nm})$ term since $\omega > 0$ is required for the frequency. In the second line we have brought into a $\omega_{nm} = \hbar\omega$. It is also easy to check that $\sigma_{\mathrm{dis},\alpha\beta}(\omega) = \sigma^*_{\mathrm{dis},\beta\alpha}(\omega)$, which indicates that the considered part of conductance matrix is Hermitian thus dissipative.

Now we introduce the sum rule integral as

$$\begin{aligned}\int_0^{+\infty}\mathrm{d}\omega\frac{\sigma_{\mathrm{dis},\alpha\beta}(\omega)}{\omega} &= \frac{e^2}{2h}\sum_{E_m<\mu<E_n}\int \mathrm{d}^2 k\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle\int_0^{+\infty}\mathrm{d}\omega\frac{\omega_{nm}}{\omega}\delta(\hbar\omega-\omega_{nm}) \\ &= \frac{e^2}{2h}\sum_{E_m<\mu<E_n}\int \mathrm{d}^2 k\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle,\end{aligned}$$

and immediately, one finds that a sum rule is satisfied as (refer to Eq. (**??**))

$$\mathrm{Im}\int_0^{+\infty}\mathrm{d}\omega\frac{\sigma_{\mathrm{dis},\alpha\beta}(\omega)}{\omega} = -\frac{\pi}{2}\sigma_{\alpha\beta}(\omega=0). \tag{C.18}$$

The rule reveals that weighted dissipative transverse conductance is topologically determined by the static Hall conductance.

C.3.3 From k-space to Real Space

A rewriting to the Chern number in the general non-Abelian case reads

$$C = \frac{1}{2\pi}\int \mathrm{d}^2k\,\mathrm{Tr}\,\Omega_{xy} = \frac{i}{2\pi}\int \mathrm{d}^2k\,\mathrm{Tr}\left\{P_k[\partial_{k_x}P_k, \partial_{k_y}P_k]\right\}, \tag{C.19}$$

where P_k is the projection operator onto occupied states. An additional P_k is added in the trace to insure that only occupied states contribute to the Chern number. To transform the formula into real space, we introduce a lattice $\mathrm{d}k \to \Delta = 2\pi/N$ for a square lattice. On the mesh, we expect that

$$\partial_{k_i}P_k \to \delta_{k_i}P_{k_n} = \sum_{m=1}^{Q} c_m[P_{k_n+m\Delta_i} - P_{k_n-m\Delta_i}],$$

where we give a lower index to Δ for its direction. Importantly, we have expected an expansion for the finite difference in k-space for better approximation. The form is indeed Taylor series like, but we can choose what function series we would like to utilize as the basis as long as it ensures the convergence. Meanwhile, by also expecting that P_k to be a smooth function, we can write $P_k = \sum_n b_n e^{ikn}$. Then by noticing that

$$\begin{aligned}(\partial_{k_i} - \delta_{k_i})e^{ikx} &= (ix_i e^{ikx} - \sum_{m=1}^{Q} c_m[e^{i(kx+m\Delta_i x_i)} - e^{i(kx-m\Delta_i x_i)}]) \\ &= i(x_i - 2\sum_{m=1}^{Q} c_m \sin m\Delta x_i)e^{ikx},\end{aligned}$$

we can choose c_m to let Qth order difference between ∂_i and δ_i to vanish, when acting on P_k. This is done by notice that for n, we need to require $(n - 2\sum_{m=1}^{Q} c_m \sin m\Delta n) = 0$, and by taking the Taylor expansion of $\sin x = \sum_{n=1}^{+\infty} \frac{(-1)^{n-1}}{(2n-1)!}x^{2n-1}$, we write

$$\begin{aligned}0 = n - 2\sum_{m=1}^{Q} c_m \sin m\Delta n &= n - 2\sum_{m=1}^{Q} c_m \sum_{l=1}^{+\infty} \frac{(-1)^{l-1}}{(2l-1)!}(m\Delta n)^{2l-1} \\ = n\left(1 - 2\sum_{m=1}^{Q} c_m m\Delta\right) &- 2\sum_{l=2}^{+\infty} \frac{(-1)^{l-1}}{(2l-1)!}(\Delta n)^{2l-1}\sum_{m=1}^{Q} c_m m^{2l-1},\end{aligned}$$

and one can require order-by-order vanishment up to Qth in the Taylor expansion by

$$\begin{cases} 2\sum_{m=1}^{Q} c_m m\Delta = 1 \\ \sum_{m=1}^{Q} c_m m^{2l-1} = 0,\ l = 2, 3, \cdots, Q \end{cases}, \tag{C.20}$$

which is equivalent to solving a linear equation system $Ac = (\frac{1}{2\Delta}, \cdots, 0)^T$, $A_{ij} = j^{2i-1}$. Since we expand the difference and sin function all up to Qth order, we expect the error

to be $O(\Delta^{2Q})$, which is vanishingly small by choosing $Q \lesssim N/2$. Notice that this matrix-equation indeed has an analytical solution that

$$c_m = (-1)^{m+1}\frac{1}{\Delta j}\prod_{n=1}^{m}\frac{Q-m+n}{Q+n}, \tag{C.21}$$

which can avoid memory collapse when Q is large.

By this setup, we can write

$$C = \frac{i}{2\pi}\Delta^2\sum_n \mathrm{Tr}\left\{P_{k_n}[\delta_{k_x}P_{k_n}, \delta_{k_y}P_{k_n}]\right\},$$

with error $O(N^2\Delta^{2Q}) \sim O(N^{2-N})$. Notice that the coarse-graining process tells that we still accept loosely a k-space, which in turn requires a periodic boundary condition for the lattice system. Now consider the unitary discrete anti-Fourier transformation back to real space, we have

$$\begin{aligned}
&\sum_n \delta_{k_x}P_{k_n} = \sum_n\sum_m c_m[P_{k_n+m\Delta_x} - P_{k_n-m\Delta_x}]\\
\to&\sum_m c_m\left[e^{-im\Delta_x x_i}P_{ij}e^{im\Delta_x x_j} - e^{im\Delta_x x_i}P_{ij}e^{-im\Delta_x x_j}\right]\\
&\equiv -i[x,P]_{ij},
\end{aligned}$$

where

$$P_{ij} = \frac{1}{N^2}\sum_n e^{ik_n x_i}P_{k_n}e^{-ik_n x_j} \tag{C.22}$$

is read from the transformation. In detail, we consider one of the terms in the trace as

$$\begin{aligned}
&\sum_n \mathrm{Tr}\, P_{k_n}P_{k_n+p\Delta}P_{k_n+q\Delta}\\
=&\frac{1}{N^3}\mathrm{Tr}\sum_n\sum_{jkl} e^{-ik_n(x_a-x_j)}P_{aj}e^{-i(k_n+p\Delta)(x_b-x_k)}P_{bk}e^{-i(k_n+q\Delta)(x_c-x_l)}P_{cl}\\
=&\frac{1}{N^2}\mathrm{Tr}\sum_{jkl} P_{aj}e^{-ip\Delta(x_b-x_k)}P_{bk}e^{-iq\Delta(x_c-x_l)}P_{cl}\delta(x_a+x_b+x_c-x_j-x_k-x_l)\\
=&\frac{1}{N^2}\mathrm{Tr}\sum_{jkl} P_{aj}e^{-ip\Delta(x_b-x_k)}P_{j,k-b+j}e^{-iq\Delta(x_c-x_l)}P_{k-b+j,l-c+k-b+j}\delta(x_a+x_b+x_c-x_j-x_k-x_l)\\
=&\frac{1}{N^2}\mathrm{Tr}\sum_{jk} P_{aj}e^{-ip\Delta x_j}P_{j,k-b+j}e^{ip\Delta(x_k-x_b+x_j)}e^{-iq\Delta(x_k-x_b+x_j)}P_{k-b+j,a}e^{iq\Delta x_a}\\
=&\frac{1}{N^3}\mathrm{Tr}\sum_{aj\tilde{k}} P_{aj}e^{-ip\Delta x_j}P_{j,\tilde{k}}e^{ip\Delta\tilde{k}}e^{-iq\Delta\tilde{k}}P_{\tilde{k},a}e^{iq\Delta x_a}\\
=&\frac{1}{N^3}\mathrm{Tr}(PP_pP_q),
\end{aligned}$$

where in the last line, the trace turns from summing over inner DoF to inner + real-space DoFs. This gives the discrete Fourier transformation and turns the Prodan formula into real space. Notice that P is written under a basis $|k\rangle$, and when performing the

transformation we have to include it. Also notice that the commutator is defined in an adjoint representation way. Then we have

$$C = \frac{2\pi i}{N^2} \operatorname{Tr}\{P[-i[x,P],[y,P]]\}, \tag{C.23}$$

where

$$[x,P] = i\sum_m c_m \left[e^{-im\Delta_x x} P e^{im\Delta_x x} - e^{im\Delta_x x} P e^{-im\Delta_x x}\right]. \tag{C.24}$$

For numerical calculation, we may use a simplified form by explicitly carrying out this commutator as

$$\begin{aligned}[x,P]_{ij} &= i\sum_m c_m \left[e^{-im\Delta_x x_i} P_{ij} e^{im\Delta_x x_j} - e^{im\Delta_x x_i} P_{ij} e^{-im\Delta_x x_j}\right] \\ &= 2\sum_m c_m P_{ij} \sin(m\Delta_x(x_i - x_j)).\end{aligned}$$

Solution to c_m: Let $2\Delta = 1$ for simplicity (one only needs rescale all c_m). Notice that by Cramer's rule, we have:

Theorem C.3.1. *(Cramer's rule) For equation $Ac = b$, $c_m = \det(A_m)/\det(A)$, where A_m is the matrix with its mth column replaced by $b = (1, \cdots, 0)^T$.*

Proof. Consider that $\det(A) = \sum_i C_i^m A_{im}$, where A_{im} is elements in mth column, while C_i^m is the some cofactor of A_{im} (fully determined by the rest of A). Then by multiplying C_i^m to each row of equation $Ac = b$ and add-up all rows, we get $\sum_j (\sum_i C_i^m A_{ij}) c_j = \sum_i C_i^m b_i$. However, due to that two same columns leads to vanishing determinant, we have $\sum_i C_i^m A_{ij} = 0$ for all $j \neq m$, thus $\det(A) c_m = \sum_i C_i^m b_i = \det(A_m)$. □

Thus our aim becomes getting the two determinants. Since $A_{ij} = j^{2i-1}$ is a Vandermonde-like matrix, we have

$$\det(A) = Q! \prod_{1\le i<j\le Q} (j^2 - i^2) = Q! \prod_{j=1}^{Q-1} \frac{(Q+j)!(Q-j)!}{(2j)!},$$

where $Q!$ comes from extracting a j from each column. To derive the latter equality, we firstly notice that for a Vandermonde matrix $B_{ij} = j^{i-1}$,

$$\det(B) = \prod_{1\le i<j\le Q} (j-i) = \prod_{k\equiv j-i=1}^{Q-1} k^{Q-k} = 1^{Q-1} 2^{Q-2} \cdots (Q-1)^1 = \prod_{j=1}^{Q-1} j!,$$

and that gives the $(Q-j)!$ part in the equality with inverse order. For the $(j+i)$ part from $j^2 - i^2$, we have

$$\begin{aligned}\prod_{1\le i<j\le Q} (j+i) &= \prod_{k=1}^{Q-1} (k+2)(k+4)\cdots(2Q-k) \\ &= 3\cdot 5\cdots(2Q-3)\cdot(2Q-1) \\ &\quad 4\cdot 6\cdots(2Q-2) \\ &\quad 5\cdot 7\cdots \\ &\quad \cdots \\ &= \prod_{k=1}^{Q-1} \frac{(Q+k)!}{(2k)!}.\end{aligned}$$

Thus totally we have $\det(A) = Q! \prod_{j=1}^{Q-1} \frac{(Q+j)!(Q-j)!}{(2j)!}$.

To solve $\det(A_m)$, we notice that A_m is a Vandermonde matrix with mth column replaced by $b = (1, \cdots, 0)^T$, thus by doing the determinant from the mth column, we meet cubic power (since the first row does not work due to zeros of b) at the top, and the rest is a Vandermonde matrix with $Q-1$ columns, thus

$$\det(A_m) = (-1)^{m+1} \frac{(Q!)^3}{j^3} \prod_{1 \le i < j \le Q} (b_j - b_i), \quad b_i = \begin{cases} i^2, & i < m \\ (i+1)^2, & i \ge m \end{cases}.$$

To calculate this formula, we separate the product into the following cases:

$$b_j - b_i = \begin{cases} j^2 - i^2, & 1 \le i < j \le m-1 \\ (j+1)^2 - (i+1)^2, & m \le i < j \le Q-1 \\ (j+1)^2 - i^2, & 1 \le i \le m-1, m \le j \le Q-1 \end{cases}.$$

For the first case, which is the same as the determinant of A but with a smaller cut-off, we have

$$\prod_{1 \le i < j \le m-1} (j^2 - i^2) = \prod_{k=1}^{m-2} \frac{(m-1+k)!(m-1-k)!}{(2k)!}. \tag{C.25}$$

For the second case, it also follows similarly with $\det(A)$. Since $(j+1)^2 - (i+1)^2 = (j-i)(j+i+2)$, we have that the product of $(j-i)$ part is

$$\prod_{m \le i < j \le Q-1} (j-i) = \prod_{k=1}^{Q-m-1} k! = \prod_{k=m+1}^{Q-1} (k-m)! = \prod_{k=m+1}^{Q-1} (Q-k)!,$$

while for the $(j+i+2)$ part, we also list some terms and observe the re-organization as

$$\begin{aligned} \prod_{m \le i < j \le Q-1} (j+i+2) &= \prod_{d \equiv l-k=1}^{Q-m-1} (2m+2+d)(2m+2+d+2) \cdots (2Q-d) \\ &= (2m+3) \cdot (2m+5) \cdots (2Q-3) \cdot (2Q-1) \\ &\quad (2m+4) \cdot (2m+6) \cdots (2Q-2) \\ &\quad \cdots \\ &\quad (Q+m+1) \\ &= \frac{(Q+m+1)!}{(2m+2)!} \cdot \frac{(Q+m+2)!}{(2m+4)!} \cdots \frac{(2Q-1)!}{(2Q-2)!} \\ &= \prod_{d=1}^{Q-m-1} \frac{(Q+m+d)!}{(2(m+d))!} \\ &= \prod_{k=m+1}^{Q-1} \frac{(Q+k)!}{(2k)!}, \end{aligned}$$

then totally, the second case also similarly leads to

$$\prod_{m \le i < j \le Q-1} ((j+1)^2 - (i+1)^2) = \prod_{k=m+1}^{Q-1} \frac{(Q+k)!(Q-k)!}{(2k)!}. \tag{C.26}$$

Finally, for the third case, since there is no order restriction between i and j, we can directly calculate the product as

$$\begin{aligned}\prod_{1\le i\le m-1, m\le j\le Q-1}((j+1)^2-i^2) &= \prod_{i=1}^{m-1}\left[\prod_{j=m}^{Q-1}(j+1-i)\cdot\prod_{j=m}^{Q-1}(j+1+i)\right] \\ &= \prod_{k=1}^{m-1}\frac{(Q-k)!}{(m-k)!}\cdot\frac{(Q+k)!}{(m+k)!}.\end{aligned} \tag{C.27}$$

Combine Eq. (C.25), Eq. (C.26) and Eq. (C.27), we have

$$\begin{aligned}\prod_{1\le i<j\le Q}(b_j-b_i) &= \prod_{k=1}^{m-2}\frac{(m-1+k)!(m-1-k)!}{(2k)!}\prod_{k=m+1}^{Q-1}\frac{(Q+k)!(Q-k)!}{(2k)!}\prod_{k=1}^{m-1}\frac{(Q-k)!}{(m-k)!}\cdot\frac{(Q+k)!}{(m+k)!} \\ &= \frac{(Q-m+1)!(Q+m-1)!}{(2m-1)!}\prod_{k=1}^{m-2}\frac{(Q+k)!(Q-k)!}{(m^2-k^2)(2k)!}\prod_{k=m+1}^{Q-1}\frac{(Q+k)!(Q-k)!}{(2k)!} \\ &= \prod_{k=1}^{m-1}\frac{(Q+k)!(Q-k)!}{(m^2-k^2)(2k)!}\prod_{k=m+1}^{Q-1}\frac{(Q+k)!(Q-k)!}{(2k)!} \\ &= \prod_{k=1}^{m-1}\frac{1}{m^2-k^2}\cdot\frac{(2m)!}{(Q+m)!(Q-m)!}\cdot\prod_{k=1}^{Q-1}\frac{(Q+k)!(Q-k)!}{(2k)!},\end{aligned} \tag{C.28}$$

where we use color-labeling to show the cancellation of the same terms in the first line, notice $(2m-1)! = [m^2-(m-1)^2](2m-2)!$ in the second line, and make-up the $k=m$ term in the third line. Notice that the last product term is just the determinant of A aside a $Q!$.

Finally, we have the coefficient

$$\begin{aligned}c_m &= \frac{\det(A_m)}{\det(A)} \\ &= (-1)^{m+1}\frac{(Q!)^2}{m^3}\frac{(2m)!}{(Q+m)!(Q-m)!}\prod_{k=1}^{m-1}\frac{1}{m^2-k^2} \\ &= (-1)^{m+1}\frac{(Q!)^2}{m^3}\frac{(2m)!}{(Q+m)!(Q-m)!}\frac{m}{(2m-1)!} \\ &= (-1)^{m+1}\frac{2}{m}\frac{(Q!)^2}{(Q+m)!(Q-m)!} \\ &= (-1)^{m+1}\frac{2}{m}\prod_{k=1}^{m}\frac{Q-m+k}{Q+k}.\end{aligned} \tag{C.29}$$

C.3.4 Hall Conductance within Parity Invariant Regime

As discussed in the half quantized mirror Hall effect, here parity invariant regime refers to an area in the 2-D BZ with parity symmetry, which is indeed mirror symmetry. By, say, x-mirror operation we have

$$\mathscr{M}_x\hat{r}\mathscr{M}_x^{-1} = (\hat{x},-\hat{y}),\ \mathscr{M}_x\hat{p}\mathscr{M}_x^{-1} = (\hat{p}_x,-\hat{p}_y),\ \mathscr{M}_x\hat{L}_z\mathscr{M}_x^{-1} = \hat{L}_z,$$

which indeed can be viewed as the definition. Correspondingly, when we deal with multi-component Fermionic operators, we have

$$\mathscr{M}\hat{\psi}_k\mathscr{M}^{-1} = M\hat{\psi}_{D_M k},$$

where we do not specify which direction the mirror acts on. Notice that there are two changes of particle annihilation operator under mirror transformation, i.e., wavevector change and inner degrees of freedom change, reflected by $k \to D_M k$ and M, respectively.

Now, for a non-interacting quadratic Fermionic Hamiltonian written using second-quantized language as $\mathscr{H} = \sum_k \mathscr{H}_k = \sum_k \hat{\psi}_k^\dagger h(k)\hat{\psi}_k$, the existence of mirror symmetry requires that $[\mathscr{M}, \mathscr{H}] = 0$, which offers us with

$$\mathscr{M}\mathscr{H}_k\mathscr{M}^{-1} = \mathscr{H}_{D_M k} \Leftrightarrow M^\dagger h(k) M = h(D_M k), \tag{C.30}$$

and notice that the equivalent form reads $M^\dagger h(D_M^{-1}k)M = h(k)$ with an inverse mirror action on wavevector (distinguish M acting on inner degrees of freedom of matrix form of $h(k)$ and general transformation D_M on wavevector k, although they are both generated by $\mathscr{M}$).

By mirror symmetry, if we have $\mathscr{H}_k \left|u_k\right\rangle = E_k \left|u_k\right\rangle$,

$$\mathscr{M} E_k \left|u_k\right\rangle = \mathscr{M}\mathscr{H}_k\mathscr{M}^{-1}\mathscr{M}\left|u_k\right\rangle = \mathscr{H}_{D_M k}\mathscr{M}\left|u_k\right\rangle,$$

where the last equivalence holds only if the system is mirror-symmetric, and if this is true, we immediately see that $\mathscr{M}\left|u_k\right\rangle$ is also an eigenstate of the Hamiltonian with energy E_k. To see what $\mathscr{M}\left|u_k\right\rangle$ is, let us assume that $h(k)U_k = U_k\varepsilon_k$ as the eigen-solution of the first-quantized Hamiltonian, then

$$\mathscr{H} = \sum_k \hat{\psi}_k^\dagger h(k)\hat{\psi}_k = \sum_k \hat{c}_k^\dagger \varepsilon_k \hat{c}_k,$$

where $\hat{c}_k = U_k^\dagger \hat{\psi}_k$. And the mirror transformation on it becomes

$$\mathscr{M}\hat{c}_k\mathscr{M}^{-1} = U_k^\dagger \mathscr{M}\hat{\psi}_k\mathscr{M}^{-1} = U_k^\dagger M\hat{\psi}_{Mk} = U_k^\dagger M U_{Mk}\hat{c}_{Mk},$$

and we will redefine $U_k^\dagger M U_{Mk} \to U_k$. Now the observation between two eigen-equations leads to conclusion that $c_k\left|u_k\right\rangle = \left|0\right\rangle$. However, there should be a more rigorous correspondence since we may write $\varepsilon_k = \mathrm{diag}(E_k^1, \cdots, E_k^n)$ with n counting inner degrees of freedom, and in that case, $\hat{c}_k^n\left|u_k^n\right\rangle = \left|0\right\rangle$ is written without fuzziness. Then

$$\mathscr{M}\left|u_k^n\right\rangle = \mathscr{M}(\hat{c}_k^n)^\dagger\mathscr{M}^{-1}\mathscr{M}\left|0\right\rangle = (\hat{c}_{Mk}^\dagger U_k^\dagger)^n\left|0\right\rangle = \sum_m (U_k^*)^{nm}\left|u_{Mk}^m\right\rangle,$$

as a gauge transformation where the gauge uncertainty in hidden in the diagonalization process. Rename the gauge transformation matrix

$$\mathscr{M}\left|u_k^n\right\rangle = \sum_m U_k^{nm}\left|u_{Mk}^m\right\rangle. \tag{C.31}$$

Now, to calculate the Hall conductance, we need integral of Berry curvature over occupied bands as $\sigma_H/(e^2/h) = \frac{1}{2\pi}\int_{\mathrm{occ}} \mathrm{d}^2k\,\mathrm{Tr}(F_{xy})$, where the general non-Abelian Berry quantities read

$$F_{xy}^{ab} = \partial_{k_x}A_y^{ab} - \partial_{k_y}A_x^{ab} + [A_x, A_y]^{ab},\ A_\mu^{ab}(k) = i\left\langle u_k^a\right|\partial_{k_\mu}\left|u_k^b\right\rangle.$$

To see how they are transformed under mirror transformation, we may firstly consider the Abelian case, where F_{xy} is gauge-invariant thus independent of U_k. Since mirror transformed quantity is defined as the quantities calculated by mirror transformed states, in the Abelian case, we find the transformed Berry connection as

$$\begin{aligned} A^n_\mu(k) &\xrightarrow{\mathscr{M}} i\left\langle \mathscr{M}u^n_k\right| \partial_{k_\mu} \left|\mathscr{M}u^n_k\right\rangle = i\sum_{l,m}\left\langle u^l_{D_Mk}\right| (U^\dagger_k)^{ln}\partial_{k_\mu}U^{nm}_k \left|u^m_{D_Mk}\right\rangle \\ &= i\sum_{l,m}(U^\dagger_k)^{ln}\partial_{k_\mu}U^{nm}_k \left\langle u^l_{D_Mk}|u^m_{D_Mk}\right\rangle + i\sum_{l,m}(U^\dagger_k)^{ln}U^{nm}_k \left\langle u^l_{D_Mk}\right| \partial_{k_\mu} \left|u^m_{D_Mk}\right\rangle \\ &= i[(U^*_k)\partial_{k_\mu}(U^T_k)]^{nn} + \sum_\nu\sum_l U^{nl}_k(U^\dagger_k)^{ln}A^n_\nu(D_Mk)J_{\nu\mu}, \end{aligned}$$

where $J_{\nu\mu} = \partial(D_Mk)_\nu/\partial_{k_\mu}$ is the Jacobian matrix with $\det(J) = -1$. In the above formula, definition $A(D_Mk) = i\left\langle u_{D_Mk}|\partial_{D_Mk}|u_{D_Mk}\right\rangle$ should be recognized. Notice that the first long arrow is in fact an equal, since it does not dependent on k. It will be much clearer if one directly uses the wavefunction u_k to write down the connection, and the k-invariant matrix M defined can be made out explicitly. With the belief that F is gauge-invariant in Abelian case, we then abandon U_k-related term and conclude that

$$F^n_{xy}(k) \xrightarrow{\mathscr{M}} F^n_{xy}(Mk).$$

Then for system with mirror symmetry, say with both $\mathscr{M}_x$ and $\mathscr{M}_y$, we see that

$$F(k_x,k_y) \overset{\mathscr{M}_x}{=} F(k_x,-k_y),\ F(k_x,k_y) \overset{\mathscr{M}_y}{=} F(-k_x,k_y),$$

where the k change is due to mirror operation while equal due to possession of symmetry. Notice that the simultaneous possession of x and y mirror symmetry also leads to the conclusion that

$$F(k_x,k_y) \overset{\mathscr{M}_x\mathscr{M}_y}{=} F(-k_x,-k_y),$$

as the same to have inversion symmetry, a C_2 rotation in 2-D case. Usually, $\mathscr{P}\mathscr{T}$ symmetry as the combination of inversion and time-reversal is usually required to be unbroken for isolated systems, and in this case since inversion in unbroken, so does the time-reversal, which leads to

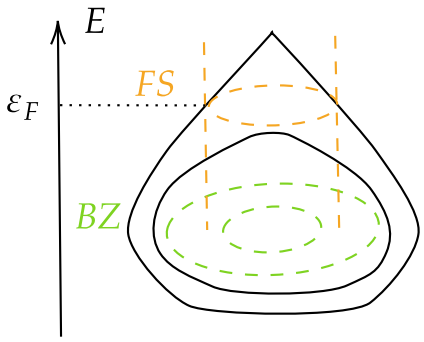
$$F(k) \overset{\mathscr{T}}{=} -F(-k),$$

and in total we find $F(k) = 0$ in the parity-invariant regime.

Now let us turn to the more general non-Abelian case, and for the Hall conductance we have

$$\sigma_H/(e^2/h) = \frac{1}{2\pi}\int_{\text{occ}} \mathrm{d}^2k\,\mathrm{Tr}(F_{xy}) = \frac{1}{2\pi}\int_{\text{occ}} \mathrm{d}^2k[\nabla\times\mathrm{Tr}(A)]_{xy},$$

where the last equality comes from the fact that trace over a commutator is zero. At this step, one has to figure out some additional geometric structures upon the bands. As shown in the schematic diagram, by assuming the existence parity symmetry regime, it is natural to assume that at the Fermi surface, the total N bands are separable into two sets: one sets with $\varepsilon_i(k_F) = \varepsilon_F,\ i = 1,2,\cdots,n$ respecting mirror symmetry; while the other sets $\varepsilon_j(k)\big|_{k\in\text{BZ}} < \varepsilon_F,\ j = n+1,\cdots,N$ with energy lower than given chemical potential in the whole Brillouin zone. Let us further label the first set with M by mirror and the second set with L as lower energy. Notice that the separable nature of bands near Fermi surface tells that there exists a gap locally in whole Brillouin zone between M and L sets



while globally on Fermi surface, which allows us to perform local gauge transformation upon two sets near Fermi surface without changing physical states, respectively. The allowed local gauge transformation then enables us to treat the contribution of two sets separately, thanks to the trace form appeared in the integral. At the time, we separate the whole Brillouin zone into two areas bounded by Fermi wavevector, and we simply denote the normally orientated boundary as FS. And the Hall conductance upon integration over two sets reads

$$\sigma_H/(e^2/h) = \frac{1}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big(A^M\big) + \frac{1}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big(A^L\big) + \frac{1}{2\pi}\oint_{\overline{\mathrm{FS}}} \mathrm{d}k\cdot\mathrm{Tr}\Big(\tilde{A}^L\Big),$$

utilizing the Stokes theorem, where $\overline{\mathrm{FS}}$ means boundary given with opposite orientation. Notice that we have used $\tilde{A}^L$ to indicate possible non-trivial condition where A^L and $\tilde{A}^L$ extended to two charts covering the Brillouin zone are related by a gauge transformation near FS as

$$\tilde{A}^L = (U^L)^\dagger A^L U^L - i(U^L)^\dagger \nabla_k U^L,$$

where U_k^L is a unitary matrix performing local gauge transformation. Then the last two terms could be organized into

$$\begin{aligned}
&\frac{1}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big(A^L\big) + \frac{1}{2\pi}\oint_{\overline{\mathrm{FS}}} \mathrm{d}k\cdot\mathrm{Tr}\Big(\tilde{A}^L\Big)\\
=&\frac{1}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\Big[\mathrm{Tr}\big(A^L - (U^L)^\dagger A^L U^L\big) + \mathrm{Tr}\big(i(U^L)^\dagger\nabla_k U^L\big)\Big]\\
=&\frac{i}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\nabla_k \,\mathrm{Tr}\ln\big(U^L\big) = N_1.
\end{aligned}$$

For the first term in evaluating Hall conductance, one could firstly find similar mirror transformation formula of Berry connection as (M ignored here)

$$A_\mu^{ab}(k) \overset{\mathscr{M}}{=\!=} i[(U_k^*)\partial_{k_\mu}(U_k^T)]^{ab} + \sum_\nu \sum_{l,m} (U_k^*)^{al} A_\mu^{lm}(D_M k)(U_k^T)^{mb} J_{\nu\mu}.$$

Now bring mirror transformed Berry connection in and require the symmetry, and we see

$$\begin{aligned}
\frac{1}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big(A^M\big) &= \frac{i}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big((U_k^M)^\dagger\nabla_k U_k^M\big) + \frac{1}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big[(U^M)^* A^M(Mk)(U^M)^T\big]\\
&= \frac{i}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big((U_k^M)^\dagger\nabla_k U_k^M\big) + \det(J)\frac{1}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big(A^M\big)\\
&= \frac{i}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big((U_k^M)^\dagger\nabla_k U_k^M\big) - \frac{1}{2\pi}\oint_{\mathrm{FS}} \mathrm{d}k\cdot\mathrm{Tr}\big(A^M\big),
\end{aligned}$$

i.e., we find that

$$\frac{1}{2\pi}\oint_{\mathrm{FS}}\mathrm{d}k\cdot\mathrm{Tr}\big(A^M\big)=\frac{1}{2}\frac{i}{2\pi}\oint_{\mathrm{FS}}\mathrm{d}k\cdot\mathrm{Tr}\big((U_k^M)^\dagger\nabla_k U_k^M\big)=\frac{1}{2}\frac{i}{2\pi}\,\mathrm{Tr}\ln\big(U^M\big)\Big|_{k_0-0^+}^{k_0+0^+}=\frac{N_2}{2}.$$

For the last equality (and for N_1), we may consider U as diagonal near k_0 so that

$$\mathrm{Tr}\ln U_{k_0}=i\sum_i\theta_i(k_0),$$

with $\mathrm{e}^{i\theta_i}$ eigenvalues of U, then the single-valueness of U requires that the loop change of θ has to be $2n\pi$ with n some integer.

In sum, we prove that

$$\sigma_H/(e^2/h)=N_1+\frac{N_2}{2},\tag{C.32}$$

composed of an integer and a half-integer.

C.3.5 Quantization from Topological Field Theory

From minimal coupling

$$H(k)\longrightarrow H\left(k-\frac{q}{\hbar}A\right)\Leftrightarrow H(k)-j\cdot A,\tag{C.33}$$

with $q=-e$ the charge of electron, we identify

$$j=-\partial_A H=\frac{q}{\hbar}\partial_k H=-\frac{q}{\hbar}\partial_k G^{-1}(\omega,k),\tag{C.34}$$

where Green's function reads

$$G(\omega,k)=\frac{1}{\omega-H(k)}.\tag{C.35}$$

In $(2+1)$-D spacetime, for a Dirac fermion we write $H(k)=\lambda k\cdot\sigma+m(k)\sigma_z$, which we assume to have corresponding lattice version without need of regularization. We choose retarded Green's function so that

$$G(\omega,k)=\frac{1}{\omega-d\cdot\sigma(1-i\eta)}.\tag{C.36}$$

Notice that $d=(\lambda k_x,\lambda k_y,m(k))$, and one should distinguish σ appeared.

Put the theory on lattice in $(2+1)$-D spacetime, from the linear expansion of Peierls substitution or the analogy from QED, where (classic field assumed)

$$S_c^{\mathrm{QED}}=i\int\mathrm{d}x\bar{\psi}\gamma^\mu A_\mu\psi(x)=i\int\mathrm{d}q\mathrm{d}kA_\mu(q)\bar{\psi}(k+\frac{q}{2})\gamma^\mu\psi(k-\frac{q}{2}),\tag{C.37}$$

and substitute $i\gamma^\mu\to\frac{q}{\hbar}\partial_{k_\mu}G^{-1}(k)$ (Euclidean), one has the action

$$S=\int_k\psi_k^\dagger G^{-1}(k)\psi_k+\frac{q}{\hbar}\int_k\int_q A^\mu\psi^\dagger_{k+\frac{q}{2}}\partial_{k^\mu}G^{-1}(k)\psi_{k-\frac{q}{2}}\equiv\psi^\dagger\mathscr{L}\psi,\tag{C.38}$$

where

$$\int_k=\int\frac{\mathrm{d}\omega}{2\pi}\int\frac{\mathrm{d}^2k}{(2\pi)^2},\tag{C.39}$$

and $d = (\lambda \sin k_x, \lambda \sin k_y, m(k))$. Integrate out fermions in partition function ($q = \hbar = 1$),

$$\begin{aligned}\mathscr{Z} &= \int D\psi^\dagger D\psi DA e^{-S} = \int DA e^{\ln\det \mathscr{L}(A)} \\ &= \int DA e^{\mathrm{Tr}\ln \mathscr{L}(A)} = \int DA e^{\mathrm{Tr}\ln[G^{-1}+A\partial_k G^{-1}]} \\ &= \int DA e^{\mathrm{Tr}\left[\ln G^{-1}+\ln\left(1+GA\partial_k G^{-1}\right)\right]} = N\int DA e^{\mathrm{Tr}\ln\left(1+GA\partial_k G^{-1}\right)} \\ &= N\int DA e^{-\mathrm{Tr}\left(GA\partial_k G^{-1}\right)^2/2+O(A^3)},\end{aligned} \tag{C.40}$$

where one notices that this is a pure Gaussian integral and order expansion counting on linear response has been used. Now

$$S_{eff} = \frac{e^2}{2\hbar}\mathrm{Tr}\left(GA\partial_k G^{-1}\right)^2 = \frac{e^2}{2\hbar}\int \frac{\mathrm{d}^3 q}{(2\pi)^3} A^\mu(-q)\Pi_{\mu\nu}(q)A^\nu(q), \tag{C.41}$$

where the lattice vacuum polarization operator reads

$$\Pi_{\mu\nu}(q) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3}\,\mathrm{tr}\,\partial_{k^\mu} G^{-1}(k)G(k+\frac{q}{2})\partial_{k^\nu}G^{-1}(k)G(k-\frac{q}{2}), \tag{C.42}$$

where the trace only acts on σ DoF. The DC linear response current will be proportional to the antisymmetric part of $\Pi_{\mu\nu}$.

Now to evaluate the polarization operator, notice that

$$\partial_\mu G^{-1} = \delta_{\mu 0} - \partial_\mu d\cdot\sigma, \tag{C.43}$$

while

$$G = \frac{1}{\omega - d\cdot\sigma(1-i\eta)} = \frac{\omega + d\cdot\sigma}{\omega^2 - d^2(1-i\eta)^2}, \tag{C.44}$$

which leads to

$$\Pi_{\mu\nu}(q) = \int_k \frac{\mathrm{tr}[(\delta_{\mu 0}-\partial_\mu d(k)\cdot\sigma)(\omega+q_0/2+d(k+q/2)\cdot\sigma)(\delta_{\nu 0}-\partial_\nu d(k)\cdot\sigma)(\omega-q_0/2+d(k-q/2)\cdot\sigma)}{[(\omega+q_0/2)^2-d(k+q/2)^2(1-i\eta)^2][(\omega-q_0/2)^2-d(k-q/2)^2(1-i\eta)^2]} \tag{C.45}$$

Now,

$$\begin{aligned}\Pi_{0\nu}(q) &= \int_k \frac{-\mathrm{tr}[(\omega+q_0/2+d(k+q/2)\cdot\sigma)\partial_\nu d(k)\cdot\sigma(\omega-q_0/2+d(k-q/2)\cdot\sigma)]}{[(\omega+q_0/2)^2-d(k+q/2)^2(1-i\eta)^2][(\omega-q_0/2)^2-d(k-q/2)^2(1-i\eta)^2]} \\ &= \int_k \frac{-\mathrm{tr}[d(k+q/2)\cdot\sigma\partial_\nu d(k)\cdot\sigma d(k-q/2)\cdot\sigma]}{[(\omega+q_0/2)^2-d(k+q/2)^2(1-i\eta)^2][(\omega-q_0/2)^2-d(k-q/2)^2(1-i\eta)^2]} \\ &\approx \int_k \frac{-2i\epsilon_{ijk}d_i(k+q/2)\partial_\nu d_j(k)d_k(k-q/2)}{[\omega-d(1-i\eta)]^2[\omega+d(1-i\eta)]^2} \\ &\approx -\int \frac{\mathrm{d}^2 k}{(2\pi)^2}\frac{\epsilon_{ijk}d_i(k+q/2)\partial_\nu d_j(k)d_k(k-q/2)}{2|d(k)|^3},\end{aligned} \tag{C.46}$$

where by

$$\begin{aligned}&\epsilon_{ijk}d_i(k+q/2)\partial_\nu d_j(k)d_k(k-q/2) = \epsilon_{ijk}[d_i+\frac{q^\alpha}{2}\partial_\alpha d_i + O(q^2)]\partial_\nu d_j[d_k - \frac{q^\beta}{2}\partial_\beta d_k + O(q^2)] \\ &= (d\times d)\cdot\partial_\nu d + (q\cdot\partial)d\cdot(\partial_\nu d\times d) + [(q\cdot\partial)d\times(q\cdot\partial)d]\cdot\partial_{nu}d + O(q^2) \\ &= (q\cdot\partial)d\times\partial_\nu d\cdot d + O(q^2) = \epsilon_{0\varsigma\nu}q^\varsigma\partial_{k_x}d\times\partial_{k_y}d\times d + O(q^2),\end{aligned} \tag{C.47}$$

one gets

$$\begin{aligned}\Pi_{0\nu}(q) &\approx -\int \frac{\mathrm{d}^2k}{(2\pi)^2}\frac{\epsilon_{0\varsigma\nu}q^\varsigma \partial_{k_x}d\times\partial_{k_y}d\times d}{2|d(k)|^3}\\ &= -\epsilon_{0\varsigma\nu}q^\varsigma\frac{1}{\pi}\int\frac{\mathrm{d}^2k}{4\pi}\frac{\partial_{k_x}d\times\partial_{k_y}d\times d}{2|d(k)|^3} = -\epsilon_{0\varsigma\nu}q^\varsigma\frac{C}{2\pi},\end{aligned} \tag{C.48}$$

with C identified as Chern number. One also recognizes that when Fermi surface is finite, i.e., the Fermi level lies inside some band, the integral should be modified by a Fermi-Dirac distribution. What is more, the approximate sign becomes strict for DC response, i.e., when $q\to 0$. In the deduction, the residue theorem has been used for ω-integral. On the other hand, counting on non-vanishing antisymmetric part,

$$\begin{aligned}\Pi_{ij}(q) &= \int_k \frac{-\operatorname{tr}[\partial_i d(k)\cdot\sigma(\omega+q_0/2+d(k+q/2)\cdot\sigma)\partial_j d(k)\cdot\sigma(\omega-q_0/2+d(k-q/2)\cdot\sigma)]}{[(\omega+q_0/2)^2-d(k+q/2)^2(1-i\eta)^2][(\omega-q_0/2)^2-d(k-q/2)^2(1-i\eta)^2]}\\ &= \int_k \frac{-q_0\operatorname{tr}[\partial_i d(k)\cdot\sigma\partial_j d(k)\cdot\sigma d(k-q/2)\cdot\sigma-\partial_j d(k)\cdot\sigma\partial_i d(k)\cdot\sigma d(k+q/2)\cdot\sigma]}{2[(\omega+q_0/2)^2-d(k+q/2)^2(1-i\eta)^2][(\omega-q_0/2)^2-d(k-q/2)^2(1-i\eta)^2]},\\ &= \int_k \frac{-iq_0\epsilon_{mnk}[\partial_i d_m(k)\partial_j d_n(k)d_k(k-q/2)-\partial_j d_m(k)\partial_i d_n(k)d_k(k+q/2)]}{[(\omega+q_0/2)^2-d(k+q/2)^2(1-i\eta)^2][(\omega-q_0/2)^2-d(k-q/2)^2(1-i\eta)^2]},\end{aligned} \tag{C.49}$$

where by

$$\begin{aligned}&\epsilon_{mnk}[\partial_i d_m(k)\partial_j d_n(k)d_k(k-q/2)-\partial_j d_m(k)\partial_i d_n(k)d_k(k+q/2)]\\ =&\epsilon_{mnk}[\partial_i d_m\partial_j d_n(1-\frac{q}{2}^\alpha\partial_\alpha)d_k-\partial_j d_m\partial_i d_n(1+\frac{q}{2}^\beta\partial_\beta)d_k]+O(q^2)\\ =&2(\partial_i d\times\partial_j d)\cdot d+O(q^2),\end{aligned} \tag{C.50}$$

one gets

$$\begin{aligned}\Pi_{ij}(q) &\approx \int_k \frac{-iq_0 2(\partial_i d\times\partial_j d)\cdot d}{[\omega-d(1-i\eta)]^2[\omega+d(1-i\eta)]^2}\\ &\approx -q_0\int\frac{\mathrm{d}^2k}{(2\pi)^2}\frac{(\partial_i d\times\partial_j d)\cdot d}{2|d(k)|^3} = -\epsilon_{0ij}q_0\frac{C}{2\pi}.\end{aligned} \tag{C.51}$$

In short,

$$\Pi_{\mu\nu}(q) = -\epsilon_{\mu\nu\varsigma}q^\varsigma\frac{C}{2\pi}+O(q^2). \tag{C.52}$$

Now by ignoring $O(q^2)$ terms,

$$\begin{aligned}S_{eff} &= -\frac{e^2}{2\hbar}\int\frac{\mathrm{d}^3q}{(2\pi)^3}A^\mu(-q)\epsilon_{\mu\nu\varsigma}q^\varsigma\frac{C}{2\pi}A^\nu(q)\\ &= -C\frac{e^2}{2h}\frac{\mathrm{d}^3q}{(2\pi)^3}\epsilon_{\mu\nu\varsigma}A^\mu(-q)q^\varsigma A^\nu(q)\\ &= iC\frac{e^2}{2h}\int\mathrm{d}^3x\epsilon^{\mu\nu\varsigma}A_\mu\partial_\varsigma A_\nu,\end{aligned} \tag{C.53}$$

and the homogeneous DC linear response current reads ($\frac{\delta}{\delta A_\mu}=\frac{\partial}{\partial A_\mu}-\partial_\alpha\frac{\partial}{\partial(\partial_\alpha A_\mu)}$)

$$j^\mu = \lim_{q\to0}\frac{i}{q}\frac{\delta S_{eff}}{\delta A_\mu(q)} = iC\frac{e^2}{h}\epsilon^{\mu\nu\varsigma}\partial_\varsigma A_\nu, \tag{C.54}$$

e.g.,

$$j_x = iC\frac{e^2}{h}[\epsilon_{120}\partial_0 A_2+\epsilon_{102}\partial_2 A_0] = -C\frac{e^2}{h}E_y, \tag{C.55}$$

where $x^0=\tau=it$ is identified with corresponding vector potential.

C.3.6 Kubo-Streda-Bastin Approach

Restoring static ($\omega = 0$) conductance matrix component with arbitrary temperature with counting band and k-index leads to

$$\sigma_{\alpha\beta} = -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k},m,n}[f_z(E_m) - f_z(E_n)]\frac{\langle m|v_\alpha|n\rangle\langle n|v_\beta|m\rangle}{(E_m - E_n)^2}, \tag{C.56}$$

with $\boldsymbol{v} = \frac{1}{i\hbar}[\boldsymbol{x}, H] = \frac{1}{\hbar}\nabla_k H$ denoting the single-particle velocity operator in momentum representation, and notice that generally we do not have correspondence $x \leftrightarrow i\partial_k$, and the sum over $\boldsymbol{k}, m, n$ should be replaced by general quantum index. Also notice that we have indeed abandoned a small converging factor, which could be restored as

$$\sigma_{\alpha\beta} = -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k},m,n}[f_z(E_m) - f_z(E_n)]\frac{\langle m|v_\alpha|n\rangle\langle n|v_\beta|m\rangle}{(E_m - E_n + i\delta)^2},$$

with $\delta > 0$ with a physical meaning usually referred to as band broadening by disorder. Now observation leads to the conclusion that we could manually insert Dirac-δ function integral over certain energy variable as

$$\begin{aligned}
\sigma_{\alpha\beta} &= -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k},m,n}\int \mathrm{d}\varepsilon\left[f_z(\varepsilon)\delta(\varepsilon - E_m)\frac{\langle m|v_\alpha|n\rangle\langle n|v_\beta|m\rangle}{(\varepsilon - E_n + i\delta)^2} - f_z(\varepsilon)\delta(\varepsilon - E_n)\frac{\langle m|v_\alpha|n\rangle\langle n|v_\beta|m\rangle}{(E_m - \varepsilon + i\delta)^2}\right] \\
&= -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon f_z(\varepsilon)\left[\sum_n \mathrm{Tr}\left(\frac{\delta(\varepsilon - H)v_\alpha\,|n\rangle\langle n|\,v_\beta}{(\varepsilon - E_n + i\delta)^2}\right) - \sum_m \mathrm{Tr}\left(\frac{\delta(\varepsilon - H)v_\beta\,|m\rangle\langle m|\,v_\alpha}{(E_m - \varepsilon + i\delta)^2}\right)\right] \\
&= -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon f_z(\varepsilon)\left[\mathrm{Tr}\left(\delta(\varepsilon - H)v_\alpha\frac{1}{(\varepsilon - H + i\delta)^2}v_\beta\right) - \mathrm{Tr}\left(\delta(\varepsilon - H)v_\beta\frac{1}{(H - \varepsilon + i\delta)^2}v_\alpha\right)\right] \\
&= -i\frac{e^2\hbar}{\pi S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon f_z(\varepsilon)\left[\mathrm{Tr}\big(\mathrm{Im}\,G^R v_\alpha \partial_\varepsilon G^R v_\beta\big) - \mathrm{Tr}\big(\mathrm{Im}\,G^R v_\beta \partial_\varepsilon G^A v_\alpha\big)\right] \\
&= -i\frac{e^2\hbar}{\pi S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon f_z(\varepsilon)\left[\mathrm{Tr}\big(v_\alpha \partial_\varepsilon G^R v_\beta\,\mathrm{Im}\,G^R\big) - Tr(v_\alpha\,\mathrm{Im}\,G^R v_\beta \partial_\varepsilon G^A)\right],
\end{aligned}$$

which is the plausible Kubo-Streda-Bastin formula. Notice that sequence inside trace is important, since all quantities inside are operators, and in order to formally simplify the expression, we have introduced the retarded/advanced Green function as

$$G^{R/A} = \frac{1}{\varepsilon - H \pm i\delta},\ \delta \to 0^+,$$

and

$$\mathrm{Im}\,G^R = -\frac{\delta}{(\varepsilon - H)^2 + \delta^2} = -\pi\delta(\varepsilon - H)$$

is identified with Dirac identity

$$\frac{1}{x + i0^+} = \mathrm{P.V}\frac{1}{x} - i\pi\delta(x).$$

To show how this formula works, we may try to attempt at one famous phenomenon as minimal conductivity of one single Dirac cone, whose Hamiltonian and eigen-quantities are

$$H = \lambda(k_x\sigma_x + k_y\sigma_y),\ E_\pm = \pm\lambda k,\ |\pm\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ \pm e^{i\phi}\end{pmatrix},\ \tan\phi = \frac{k_y}{k_x},$$

and our purpose is to obtain the DC longitudinal conductivity of this band at $\mu = 0$ ($z = 1$). With formula above, the calculation of this problem is quite direct as

$$\begin{aligned}
\sigma_{xx} &= \mathrm{Im}\,\frac{e^2\hbar}{\pi S}\sum_k \int n_F(\epsilon-\mu)\mathrm{d}\epsilon \mathrm{Tr}\left[v_x\partial_\epsilon G^R v_x\,\mathrm{Im}\,G^R - v_x\,\mathrm{Im}\,G^R v_x\partial_\epsilon G^A\right] \\
&= \mathrm{Im}\,\frac{e^2\hbar}{\pi S}\sum_{k,mn} \int n_F(\epsilon-\mu)\mathrm{d}\epsilon\left[\langle v_x\rangle_{mn}\partial_\epsilon G^R_n\langle v_x\rangle_{nm}\,\mathrm{Im}\,G^R_m - \langle v_x\rangle_{mn}\,\mathrm{Im}\,G^R_n\langle v_x\rangle_{nm}\partial_\epsilon G^A_m\right] \\
&\overset{\text{I.B.P}}{=} \frac{1}{2}\frac{e^2\hbar}{\pi S}\sum_{k,mn} \int -\partial_\epsilon n_F(\epsilon-\mu)\mathrm{d}\epsilon\left[\langle v_x\rangle_{mn}\,\mathrm{Im}\,G^R_n\langle v_x\rangle_{nm}\,\mathrm{Im}\,G^R_m - \langle v_x\rangle_{mn}\,\mathrm{Im}\,G^R_n\langle v_x\rangle_{nm}\,\mathrm{Im}\,G^A_m\right] \\
&\overset{T=0}{=} \frac{e^2\hbar}{\pi S}\sum_{k,mn} \int \delta(\epsilon-\mu)d\epsilon\langle v_x\rangle_{mn}\,\mathrm{Im}\,G^R_n\langle v_x\rangle_{nm}\,\mathrm{Im}\,G^R_m \\
&\overset{\mu=0}{=} \frac{e^2\hbar}{\pi S}\sum_{k,mn}\frac{\delta^2|\langle v_x\rangle_{mn}|^2}{(\epsilon_n^2+\delta^2)(\epsilon_m^2+\delta^2)} \\
&= 2\frac{e^2\hbar}{\pi S}\frac{S}{(2\pi)^2}\int d^2k\left[\frac{\delta^2|\langle v_x\rangle_{++}|^2}{(\lambda^2k^2+\delta^2)^2} + \frac{\delta^2|\langle v_x\rangle_{+-}|^2}{(\lambda^2k^2+\delta^2)^2}\right] \\
&= \frac{e^2}{2\pi^3\hbar}\int d^2k\frac{\lambda^2\delta^2}{(\lambda^2k^2+\delta^2)^2} \\
&\overset{t=(\lambda k/\delta)^2}{=} \frac{e^2}{h}\frac{1}{\pi}\int_0^{+\infty} dt\frac{1}{(t+1)^2} = \frac{e^2}{h}\frac{1}{\pi}.
\end{aligned}$$

Note that we have used $n_F(\varepsilon-\mu)$ to re-write Fermi-Dirac distribution with explicit chemical potential dependence. In the first step, Im in the formula is equivalent to multiplying $-i$ in the front. In the third line for integration by part, the factor $1/2$ comes from the fact that we have two Green functions in each part, as

$$n_F(\partial G)G = \partial(n_F GG) - (\partial n_F)GG - n_F G(\partial G),$$

where on RHS, the first term is integrated out to be zero, and the minus third term is exactly the same as the original, with only indices symmetrically permuted, and that accounts for a one-half factor. After taking zero temperature and neutral limit, the sum m, n runs over all two states and gives double counting in the sixth line. Direct calculation leads to $|\langle v_x\rangle_{++}| = (\lambda\cos\phi/\hbar)^2$, $|\langle v_x\rangle_{+-}| = (\lambda\sin\phi/\hbar)^2$, bringing which inside integral formula gives the final result presented, as $1/\pi$ fraction of quantum unit conductance. Take caution that δ as disorder broadening is not thrown in the whole calculation while the final answer is independent of it, which gives a hint on the existence of 'intrinsic disorder'; also be careful on that our calculation is based on limit sequence $\lim_{\delta\to 0}\lim_{\omega\to 0}$, i.e., firstly a DC limit with clean limit afterwards, while inverse limit sequence may cause a different result, and experiment compare may have to depend on whether system is clean enough.

Appendix D

Other Bound States

Here, we mainly focus the the 1D Schrödinger equation with Hamiltonian as

$$h(x) = v(-i\partial_x)\sigma_x + m(x)\sigma_z,\ v > 0, \tag{D.1}$$

and search for its zero-energy bound states near $x = 0$.

We shall have basically four types of solutions, and in any case, we can always assume $\varphi(x)$ as its solution, and then we multiply the equation by σ_x to find that

$$v\partial_x\varphi(x) + m(x)\sigma_y\varphi(x) = 0, \tag{D.2}$$

which hints us that we can take $\varphi_s(x) = \phi(x)\eta_s$ with $\sigma_y\eta_s = s\eta_s$ ($s = \pm$), such that

$$v\partial_x\phi(x) + sm(x)\phi(x) = 0. \tag{D.3}$$

D.1 Jackiw-Rebbi Bound State

In this picture, we have $m(x) = m\mathrm{sgn}(x)$ with m a constant, and by assuming $\phi(x) \propto e^{-\lambda|x|}$ with $\mathrm{Re}\,\lambda > 0$ as bound state condition, we have

$$\mathrm{sgn}(x)\,(v\lambda + sm) = 0, \tag{D.4}$$

by

$$\partial_x e^{-\lambda|x|} = -\lambda\mathrm{sgn}(x)e^{-\lambda|x|}.$$

We thus obtain $\lambda^s = -sm/v$, while the $\mathrm{Re}\,\lambda > 0$ condition constrains to only one solution with $s = -\mathrm{sgn}(m)$, which gives us the bound state solution as

$$\varphi_{-\mathrm{sgn}(m)}(x) = C\eta_{-\mathrm{sgn}(m)}e^{-\frac{|mx|}{v}}. \tag{D.5}$$

The above solution enlightens us to search for the bound state with a more general form of $m(x)$, with condition $\mathrm{sgn}(m(\infty)m(-\infty)) < 0$ satisfied. We write the solution as

$$\phi(x) \propto \exp\left(-\left|\int^x \mathrm{d}x' m(x')\right|/v\right), \tag{D.6}$$

and finds that $\partial_x\phi(x) = -|m(x)|/v$ for $x \to +\infty$, which leads to

$$\lim_{x\to+\infty} |m(x)| + sm(x) = 0, \tag{D.7}$$

and the equation stands when $s = -\mathrm{sgn}(m(+\infty))$, thus the bound state solution is given by

$$\varphi_{-\mathrm{sgn}(m(+\infty))}(x) = C\eta_{-\mathrm{sgn}(m(+\infty))} \exp\left(-\left|\int^x \mathrm{d}x' m(x')\right|/v\right). \tag{D.8}$$

This is a single solution.

D.2 Modified Dirac Equation Edge State

Here we consider $m(x) = m - Bk_x^2 = m + \partial_x^2$ with $mB > 0$, and the Hamiltonian is only for $x > 0$. By assuming $\phi(x) \propto e^{-\lambda x}$ with $\mathrm{Re}\,\lambda > 0$, we have

$$\partial_x^2 e^{-\lambda x} = \lambda^2 e^{-\lambda x},\ \partial_x e^{-\lambda x} = -\lambda e^{-\lambda x},$$

and thus

$$-v\lambda + s(m + B\lambda^2) = 0 \Rightarrow B\lambda^2 - sv\lambda + m = 0, \tag{D.9}$$

which gives us the solution as

$$\lambda_\pm^s = \frac{sv \pm \sqrt{v^2 - 4mB}}{2B}. \tag{D.10}$$

Thus, by $\mathrm{Re}\,\lambda > 0$, we require $s = \mathrm{sgn}(B)$, and the bound state reads

$$\varphi_{\mathrm{sgn}(B)}(x) = C\eta_{\mathrm{sgn}(B)} \left(e^{-\lambda_+^{\mathrm{sgn}(B)}x} - e^{-\lambda_-^{\mathrm{sgn}(B)}x}\right), \tag{D.11}$$

with respect to the boundary condition at $x = 0$. This is also a single solution.

D.3 Modified Dirac Equation Domain Wall State

Here we consider $m(x) = m - \mathrm{sgn}(x)Bk_x^2 = m + \mathrm{sgn}(x)B\partial_x^2$ with $mB > 0$, which give us a non-trivial phase for $x > 0$ and a trivial phase for $x < 0$. In such belief, there should be a domain wall state at $x = 0$, and we claim that it is

$$\varphi_{\mathrm{sgn}(B)}(x) = \Theta(x)C\eta_{\mathrm{sgn}(B)} \left(e^{-\lambda_+^{\mathrm{sgn}(B)}x} - e^{-\lambda_-^{\mathrm{sgn}(B)}x}\right), \tag{D.12}$$

i.e., it is just the modified Dirac equation edge state. It is easy to see that the solution satisfies the equation at $x \neq 0$, while for $x = 0$, let us define $\mathrm{sgn}(x = 0) = 0$, $\Theta(x = 0) = 0$ and things go smoothly.

The solution is also a single solution.

D.4 Trivial Domain Wall State

Here we consider $m(x) = \Theta(x)(m + Bk_x^2) = \Theta(x)(m - B\partial_x^2)$ with $mB > 0$, which gives two trivial phases for $x > 0$ and $x < 0$, but the masses near $x = 0$ change sign.

Here, we consider the trivial state $\phi(x) \propto e^{-\lambda|x|}$ with $\mathrm{Re}\,\lambda > 0$, and as

$$\begin{aligned} \partial_x e^{-\lambda|x|} &= -\lambda \mathrm{sgn}(x) e^{-\lambda|x|}, \\ \partial_x^2 e^{-\lambda|x|} &= \lambda^2 e^{-\lambda|x|} - 2\lambda\delta(x) e^{-\lambda|x|}, \end{aligned} \tag{D.13}$$

we obtain the equation as

$$\mathrm{sgn}(x)\left[-v\lambda + s(m - B(\lambda^2 - 2\lambda\delta(x)))\right] = 0. \tag{D.14}$$

Notice that here, rather than a form factor, the $\mathrm{sgn}(x)$ function helps us to avoid the annoying delta function. Note that even in the integral form, this equation has no singularity since

$$\int_{0^-}^{0^+} \mathrm{d}x \mathrm{sgn}(x) = |x|\Big|_{0^-}^{0^+} = 0,$$
$$\int_{0^-}^{0^+} \mathrm{d}x \mathrm{sgn}(x)\delta(x) = \frac{1}{2}\int_{0^-}^{0^+} \mathrm{sgn}(x)\mathrm{dsgn}(x) = \frac{1}{4}\mathrm{sgn}^2(x)\Big|_{0^-}^{0^+} = 0.$$

Now we can consider $x \neq 0$ and remove the sign and delta functions:

$$B\lambda^2 + vs\lambda - m = 0, \tag{D.15}$$

which leads to

$$\lambda_\pm^s = \frac{-vs \pm \sqrt{v^2 + 4mB}}{2B}. \tag{D.16}$$

For our $mB > 0$ case, we have two solutions with $\lambda^s_{\mathrm{sgn}(B)}$ with $\mathrm{Re}\,\lambda > 0$, thus

$$\varphi_s(x) = C\eta_s e^{-\lambda^s_{\mathrm{sgn}(B)}|x|}. \tag{D.17}$$

It is interesting that there are two zero-energy bound state solutions between the two trivial phases, and they are something similar to Jackiw-Rebbi bound states as solitons in the trivial phase.

We may further consider the $mB < 0$ case, where now both sides are non-trivial but with inverse winding numbers. In this case, since we always have

$$|v| > |\,\mathrm{Re}\sqrt{v^2 + 4mB}|, \tag{D.18}$$

the choice depends on s rather than $\pm$, and we need to choose $s = -\mathrm{sgn}(B)$. In this case, we still obtain two zero-energy bound states

$$\varphi^\pm_{-\mathrm{sgn}(B)} = C\eta_{-\mathrm{sgn}(B)} e^{-\lambda_\pm^{-\mathrm{sgn}(B)}|x|}. \tag{D.19}$$

What is the difference between the two cases? To answer this, we have to introduce an extra dimension, say along y, and we add simply $\delta H = vk_y\sigma_y + \mathrm{sgn}(x)Bk_y^2$. We focus only near $k_y \approx 0$ and consider the first term, then the trivial phase domain wall with $mB > 0$ will project out two chiral boundary modes in a helical manner $vk_y\sigma_z$, while the non-trivial phase domain wall with $mB < 0$ will project out two chiral boundary modes with the same chirality $vk_y\sigma_0$.

Stranger things happen when we include the k_y^2 term in solving the 1D Dirac equation, i.e., we have

$$\begin{cases} H_{1D} = v(-i\partial_x)\sigma_x + \mathrm{sgn}(x)(m + Bk_y^2 - B\partial_x^2)\sigma_z \\ \delta H = vk_y\sigma_y \end{cases}. \tag{D.20}$$

In this case, for trivial domain wall with $mB > 0$, the 1D zero energy boundary mode near $x = 0$ always exist, and we obtain projected helical modes extending at all k_y, i.e.,

$$H_{\mathrm{Pro}}(k_y) = vk_y\sigma_z,\ \forall k_y. \tag{D.21}$$

Or on a lattice, this leads to pure Dirac fermion with doubled Dirac cones:

$$H^{\text{lat}}_{\text{Pro}}(k_y) = v \sin(k_y)\sigma_z. \tag{D.22}$$

On the other hand, for non-trivial domain wall with $mB < 0$, the 1D zero energy boundary mode evolves from non-trivial phase ($k_y^2 < -m/B$) to trivial phase ($k_y^2 > -m/B$), and we obtain the boundary modes with discontinuity at $k_y^2 = -m/B$

$$H_{\text{Pro}}(k_y) = \begin{cases} vk_y\sigma_z, & k_y^2 < -m/B \\ vk_y\sigma_0, & k_y^2 > -m/B \end{cases}. \tag{D.23}$$

However, this should be false appearance, since we know that the chiral edge modes will evolve into bulk states with non-zero energy when $k_y^2 > -m/B$, and we expect similar things happen for the helical edge modes.